

केन्द्रीय भूमि जल बोर्ड

जल संसाधन, नदी विकास और गंगा संरक्षण

विभाग, जल शक्ति मंत्रालय

भारत सरकार

Central Ground Water Board

Department of Water Resources, River Development and Ganga Rejuvenation,

Ministry of Jal Shakti

Government of India

AQUIFER MAPPING AND MANAGEMENT OF GROUND WATER RESOURCES

Prayagraj District
Uttar Pradesh

उत्तरी क्षेत्र, लखनऊ

Northern Region, Lucknow

NATIONAL AQUFER MAPPING AND MANAGEMENT PLAN

(NAQUIM)

PRAYAGRAJ DISTRICT, UTTAR PRADESH

(AAP: 2018-19)

Government of India
Ministry of Jal Shakti
Central Ground Water Board
Northern Region
Lucknow, Uttar Pradesh

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Groundwater has emerged as the preferred water source as a poverty reduction tool in India's rural areas due to its near-universal availability, dependability, and low capital cost. It has made significant contributions to the growth of Indian economy and acted as an important catalyst for its socio-economic development. Its importance as a precious natural resource in the Indian context can be gauged from the fact that more than 85 % of India's rural domestic water requirements, more than 50 % of its urban water requirements; and more than 50 % of its irrigation requirements are being met from groundwater resources.

Increasing dependence on groundwater as a reliable source of water has resulted in its large-scale and indiscriminate exploitation in various parts of the country, without due regard to the recharging capacities of aquifers and other environmental factors. The unplanned and non-scientific exploitation of groundwater resources, mostly driven by individual initiatives has led to increased stress on the available resources which adverse impacts can be observed in the form of the long-term decline of groundwater levels, de-saturation of aquifer zones. It will also increase the energy consumption for the lifting of water from progressively deeper depth.

Central Ground Water Board has taken up the task of Aquifer Mapping and Ground Water Management Plan under NAQUIM in the state of Uttar Pradesh with an objective of (i) Delineation and characterization of aquifers in three dimensions (ii) Identification and quantification of issues and (iii) Development of management plans to ensure the sustainability of groundwater resources. Under this initiative, different management plans/interventions have been suggested for each aquifer system and also for the quality affected areas. The management options also include supply side management (which includes identification of the feasible areas for the artificial recharge and water conservation structures, which are helpful to arrest the declining water levels), and demand-side management options (which include crop diversification, water use efficiency practices in irrigation etc.).

The sincere efforts of Sri. Sant Lal (Retd. Sc-D), Smt. Richa Gautam (Sc-B) and Dr. Arjun Ram (Young Professional) Central Ground Water Board, Northern Region, Lucknow are highly appreciated in preparing the NAQUIM report of Prayagraj District. I am sure that the Ground Water Management Plan of Prayagraj district and its aquifer maps would be an immense help to formulate the scientifically viable implementable strategies for efficient management of groundwater resources to ensure groundwater sustainability.

P.K.Tripathi
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National Aquifer Mapping and Management Plan
Prayagraj District
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AQUIFER MAPPING AND GROUND WATER MANAGEMENT PLAN

(NAQUIM)

PRAYAGRAJ DISTRICT, UTTAR PRADESH

(AAP 2018-19)

DISTRICT AT A GLANCE

1.	GENERAL INFORMATION			
	i.	Geographical Area (Sq. Km.)		5482
	ii.	Administrative Divisions Number of Block Number of Panchayat/Villages		 20 3065
	iii.	Population (2011 census)		5959798
	iv.	Average Annual Rainfall (mm)		959.1
2.		GEOMORPHOLOGY		Middle Ganga Plain
		Major Physiographic Units		a. Flood Plain b. Younger alluvial plain c. Older alluvial plain d. Kaimur Sandstone of Vindhyan Supergroup
		Major Drainages		Ganga, Yamuna, Tons
3.		LAND USE (Ha.), 2017-18		
		Net Irrigated Area (Thousand Ha.)	:	310
		Gross Irrigated Area (Thousand Ha.)		511
		Net Area Sown (Thousand Ha.)	:	353
		Area Sown More Than Once (Thousand Hectares)		221
		Forest (Thousand Ha.)		21.454
4.		MAJOR SOIL TYPES	:	Clay loam to sandy loam, Loam & Sandy Loam , Sandy loam to sodic, Sandy loam & clay
5.		AREA UNDER PRINCIPAL CROPS (Thousand Ha.) Jayad Kharif Rabi	 : : :	 10.283 229.439 259.296
6.		IRRIGATION BY DIFFERENT SOURCES (2018 – 19)		
		Tube wells / Bore wells	:	48628(in numbers)
		Canals	:	2145 km
		Other Sources	:	3887(in numbers)
7.		NUMBER OF GROUND WATER MONITORING WELLS OF CGWB (As on 31 3-2020)		
		No. of Dug wells	:	37
		No. of Piezometers	:	Nil
8.		PREDOMINANT GEOLOGICAL FORMATIONS	:	Quaternary Alluvium

9.		HYDROGEOLOGY AND AQUIFER GROUP		Quaternary alluvium deposited by Ganga and Yamuna river systems.
		Major water bearing formation	:	Sand, silt and gravel
		Pre-monsoon Depth to water level during May 2019	:	1.32 – 20.36 mbgl
		Post-monsoon Depth to water level during Nov. 2019	:	0.59- 15.69 mbgl
		Long term water level trend in 10 years (2010-2019) in m/yr	: :	Pre-monsoon Rise: 0.492 cm/yr. Fall: 0.3658 cm/yr. Post-monsoon Rise: 0.5156 cm/yr. Fall: 0.91 cm/yr.
10.		GROUND WATER EXPLORATION BY CGWB (As on 31-3-2019)		
		No of wells drilled (EW, OW, PZ, SH, Total)	:	EW-21, SH-3
		Depth range (m)	:	146.00-218.00
		Discharge	:	1000-3050lpm
		Storativity (S)	:	$1.79\times10^{-4} – 5.10\times10^{-4}$
		Transmissivity (m^2/day)	:	857 – 3330 m^2/day
11.		GROUND WATER QUALITY		
		Presence of chemical constituents more than permissible limit (e.g. EC, F, As, Fe)	:	No constituents above permissible limit
		Type of water Quality	:	Good and potable
12.		DYNAMIC GROUND WATER RESOURCES (Ham) as on 31 March, 2020		
		Annual Replenishable Ground Water Resources	:	135052.1
		Gross Annual Ground Water Draft	:	100192.68
		Stage of Ground Water Development	:	74.2 %
13.		GROUND WATER CONTROL AND REGULATION	:	
		Number of Over Exploited Blocks	:	1
		Number of Critical Blocks		2
		Number of Semi Critical Blocks	:	6
		Number of Safe Blocks		12
14.		MAJOR GROUND WATER PROBLEMS AND ISSUES	:	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose

1.0 INTRODUCTION

Each and every human being in this world should get clean and safe drinking water. It is one of the most important components that enormously influence the life of human as well as other beings that are animals and plants. According to information 97% of earth's water supply is in the sea, which is not fit for human uptake. 2% of water is captured in the polar ice-caps and only 1% is available as fresh water in river, lakes, streams, reservoir & ground water which is available for irrigation, industries & household purposes.

Ground water is playing a vital role in the fulfilment of drinking, irrigational and industrial needs of the district. Although the ground water was in use since the ancient times, but with the population explosion, multi cropping practices in agriculture sector and rapid industrial development enhanced the need of water in the district. In recent time the dependency on ground water has increased manifold for socio-economic and agricultural development of the region to a great extent on easily availability with low capital cost. The scarcity of this vital resource can hamper the overall development of the area. The unplanned and unscientific development in various part of the state has led to stress on the availability of ground water resources with proper management. Due to paradigm shift from "ground water development" to "groundwater management" in the past two decades in the country, it is imperative to generate an accurate and comprehensive micro-level picture of groundwater through aquifer mapping in different hydrogeological settings. It would enable robust groundwater management plans in an appropriate scale. Aquifer mapping is a process wherein a combination of geologic, geophysical, hydrologic and chemical field and laboratory analyses are applied to characterize the quantity, quality and sustainability of ground water in aquifers. This would help achieving drinking water security, improved irrigation facility and sustainability in ground water resources development in Prayagraj District Central Ground Water Board (CGWB) implemented the aquifer Mapping Programme in Prayagraj District, Uttar Pradesh with broad objective of preparing an aquifer wise management plan for the district. The present report is based upon the integration of existing data of CGWB & Department of State government during A.A.P. 2018-19. It will be very useful for the planners and various executive agencies engaged in the development and management of ground water for agricultural, industrial and drinking purposes.

1.1 OBJECTIVE

The broad objective of the study is to establish the geometry of the underlying aquifer systems in horizontal and vertical domain and characterize them, so as to work out the development potential and prepare block wise management plan. It is envisaged to assess the aquifer wise availability, utilization and water quality, especially in problematic/vulnerable area. Finally, the aim of this study is to prepare block wise management plan to facilitate the suitable development and management of ground water resources.

1.2 SCOPE OF STUDY

The scope of the present study is broadly outlined within the framework of National Aquifer Management programme (NAQUIM) being implemented by CGWB. There are four major activities viz.: (i) data collection/compilation (ii) Data gap analysis (iii) Data generation and (VI) Preparation of aquifer maps and management plan to achieve the primary objective. Data compilation includes collection of data and all maps from concerned Agencies, such as the Survey of India, Geological Survey of India, State Ground Water Department, U.P. Jal Nigam, Revenue Department., computerization and analyses of all acquired data, and preparation of a data base. Identification of Data Gap included ascertaining requirement for further data generation in respect of hydrometeorological, hydrogeological, geophysical, chemical, studies. In continuity of data gap analysis, Data generation includes those of hydrometeorology, soil infiltration, and subsurface geophysics, chemical quality of ground water, litho logs and aquifer parameters. Generation of ground water chemical quality data was accomplished by collection of water samples and their laboratory analyses for all major parameters, heavy metals, pesticides and bacteria. Sub-surface geophysical studies are incorporated vertical electrical sounding, two-dimensional image profiling, and borehole logging. Additional data pertaining to sub-surface lithology and aquifer parameters are obtained through drilling of additional exploratory wells and slim holes, pumping tests at the drilling sites and slug tests in a number as required and their analyses.

1.3 APPROACH AND METHODOLOGY

An approach and methodology adopted to achieve the major objective are Compilation of existing data collected from different sources and agencies and identification of data gaps. Based on existing data various thematic layers and maps have been prepared in GIS environment and aquifer maps incorporating the data and management plans are prepared.

1.4 STUDY AREA (ADMINISTRATIVE DETAILS)

Prayagraj district is located in the southern part of Uttar Pradesh. It forms a part of central Ganga plain and lies between latitudes 24º47' and 25º47' North and longitudes 81º21' and 82º21' falling in survey of India toposheet nos. 63 G, H, K and L. Total geographical area of the district is 5482 sq.km.

The district in general has excellent communication facilities. Prayagraj, the district headquarter is connected to other places and the surrounding districts through a network of railway lives and all-weather metalled roads.

The district is divided into 8 Tehsils and 20 blocks. The rivers of the district belong to the main drainage system of the Ganga. The most important tributaries are Yamuna and Tons.

Systematic hydrogeological surveys, reappraisal hydrogeological surveys, many short term investigation and scientific exploration for ascertaining the ground water potential of various formations were carried out from time to time.

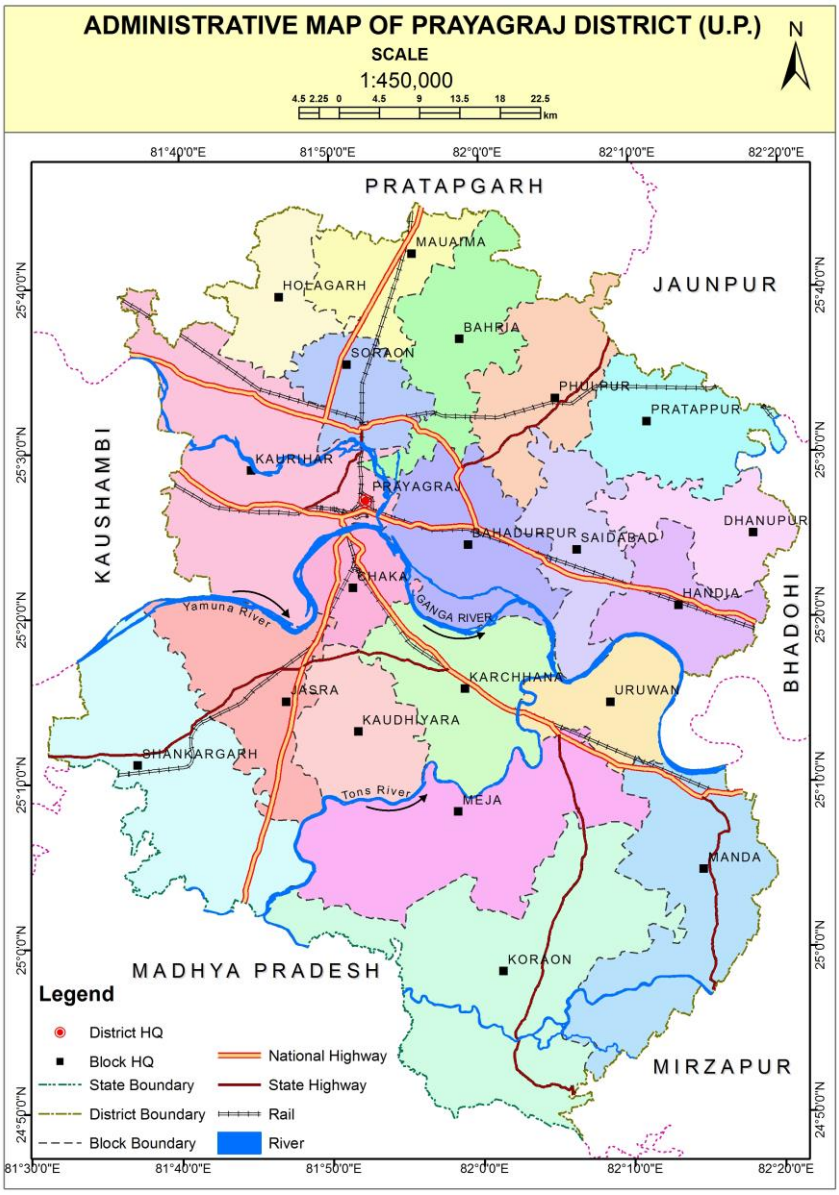


Figure 1.1: Administrative Map of Prayagraj District, U.P.

1.5 DEMOGRAPHY

Total population of Prayagraj district is 4936110 as per 2001 census. The grow rate is 20.74% in the last 10 years. Population of Prayagraj district as per 2011 census is 5959798.

Table 1.1: Demography of Prayagraj District as per census 2011

S.No.	Block Name	Population			
		Male	Female	Total	Growth Rate
1	Kaudrihar	209426	174866	384292	16.82
2	Holagarh	93802	88901	182703	18.39
3	Mauaima	95784	90067	185851	24.35
4	Soraon	101105	92078	193183	14.52
5	Bahria	140887	131892	272779	19.92
6	Phulpur	115683	109338	225021	17.65
7	Bahadurpur	155410	140091	295501	19.49
8	Pratappur	114639	110090	224729	20.18
9	Saidabad	132839	120327	253166	20.76
10	Dhanupur	119438	114405	233843	21.8
11	Handia	109197	101984	211181	23.35
12	Jasra	90543	81159	171702	19.88
13	Shankargarh	86100	77410	163510	20.68
14	Chaka	90687	83179	173866	2.96
15	Ladle	123630	111995	235625	19.47
16	Kaudhiara	83808	75892	159700	27.63
17	Uruva	105405	98222	203627	22.8
18	Majora	102828	94258	197086	23.98
19	Koraon	166226	150508	316734	27.3
20	Manda	103522	93897	197419	24.56

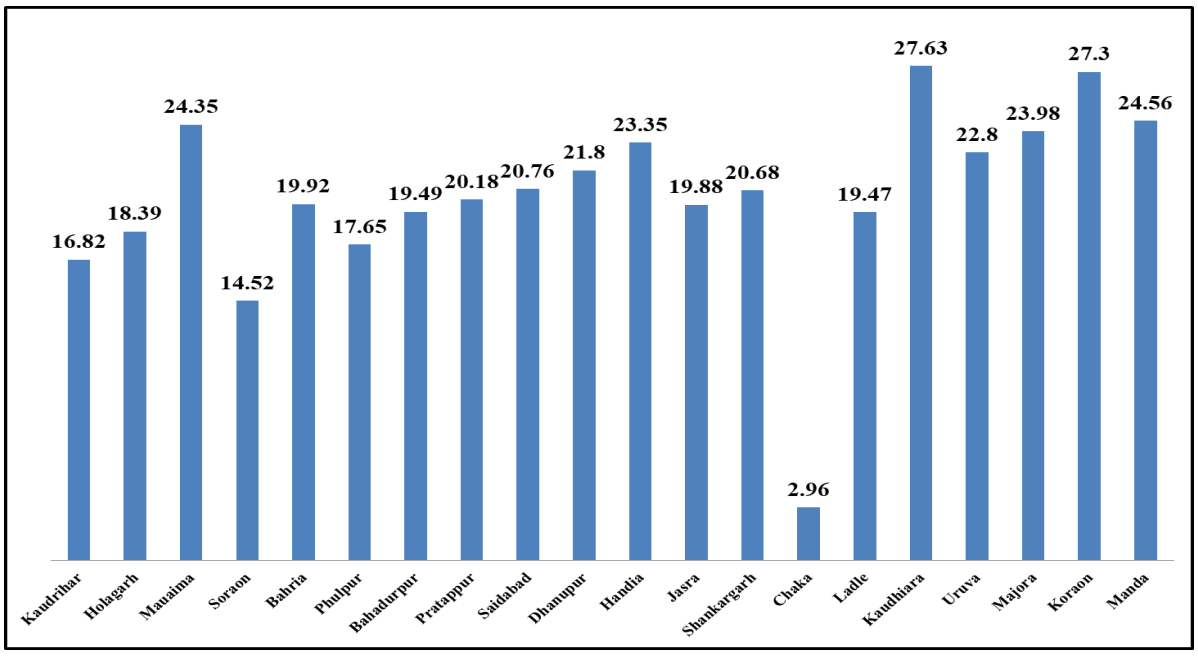


Figure 1.2: Block-Wise Population Growth Rate (as on census 2011)

1.6 Drainage / Basin:

The Northern half of the district occupies part of Ganga basin and Sothern part of the district occupies the part of Yamuna and Tons Sub- basin. The rivers of the district belong to the main drainage system of the Ganga. The most important tributaries are Yamuna and Tons while those of minor systems include Sai and Yamuna.

The Ganga enters the district in the Kaurihar Block. It maintains a meandering course. Yamuna, the second major river, enters the district in the extreme west in the Sankargarh block and flows narrowly before joining to the Ganga at Sangam in the proximity of Prayagraj city. Yamuna has more constant channel and steeper banks.

Yamuna has a more rapid stream and flowing fast down a greater bed slope than the Ganga. Its water is comparatively cleaner than Ganga water most of the streams joining the Yamuna are mere drainage channels or ravines.

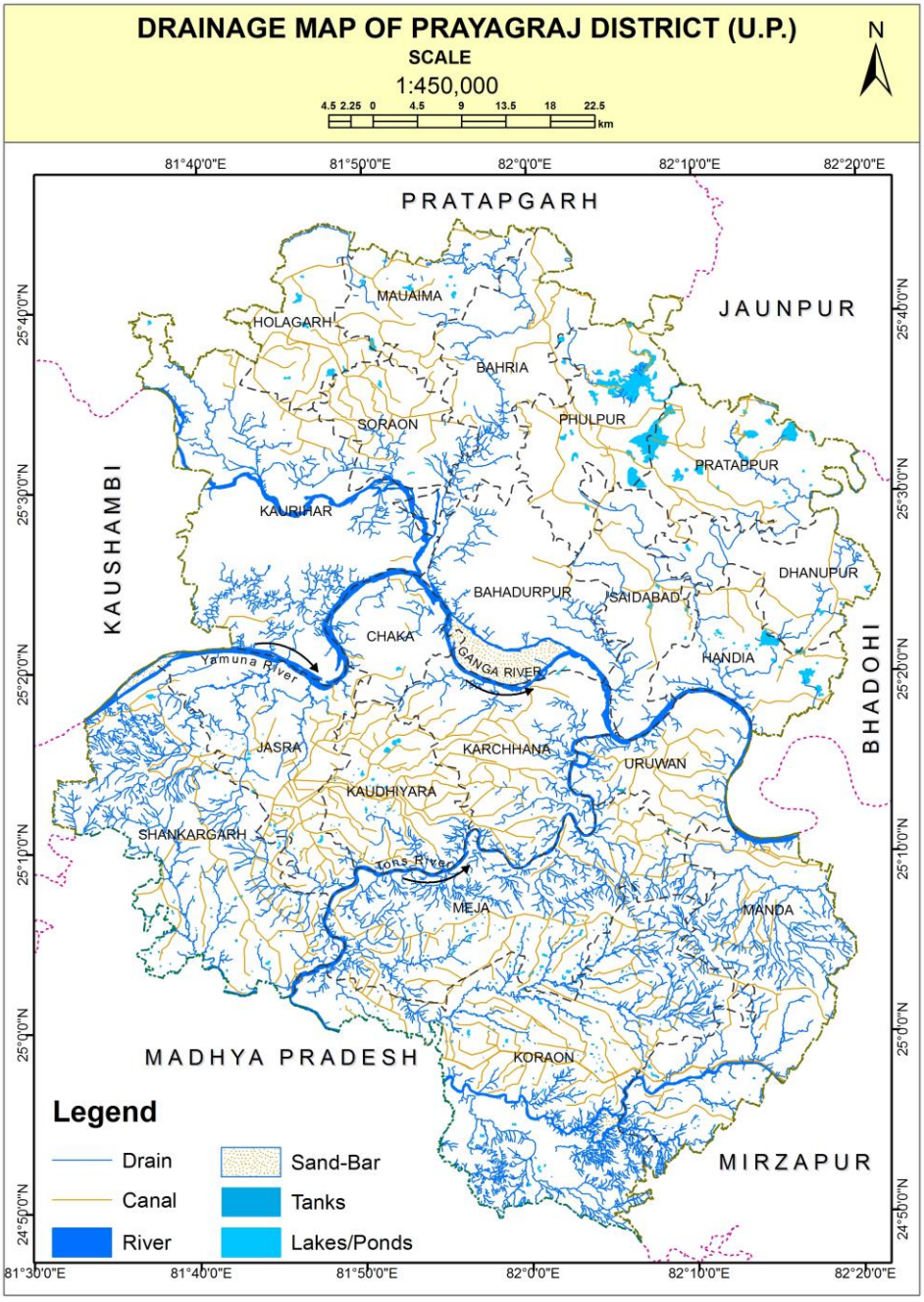


Figure 1.3: Drainage Map of Prayagraj District, U.P

1.7 DATA AVAILABILITY & DATA GAP ANALYSIS

The data pertaining to various attributes of ground water were collected from available literatures of Central Ground Water Board, State Departments and other agencies. The compiled data were plotted on 1:50,000 scale map and analysis of Data Gap was carried out for ascertaining additional requirement of Hydrogeological, Hydrological, Hydrochemical, and Geophysical Studies. Data Requirement, Data Availability and Data Gap Analysis are summarized in Table-2.

Table 1.2: Data Requirement, Data Availability and Data Gap Analysis for Aquifer Mapping

S. No.	Study Aspect	Data Requirement	Data Availability	Data Gap
1	Rainfall and Other Climatic data	IMD Meteorological Station	Rainfall data of Study area Available	No
2	Soil	Soil Map and Soil infiltration rate	Soil Map	Soil infiltration test requires for infiltration rate.
3	Land Use	Latest Land use Pattern in GIS Environment	Land Use available in Satellite data and UP Statistics Department- 2018- 19	No
4	Geomorphology	Digitized Geomorphological Map	Digitized Geomorphological Map available	No
5	Exploration	Data of Exploratory well along with aquifer parameters	21 Exploratory wells and 3 Slim Hole exist Aquifer parameters not available	No
6	Recharge Parameters	Recharge parameters of different soil and aquifer types based on field studies	Recharge parameters are available in Ground Water Resource Estimation	No
7	Discharge Parameters	Discharge parameters for different Ground Water abstraction structure	Discharge parameters are available in Ground Water Resource Estimation	No

1.8 CROPPING PATTERN AND IRRIGATION PRACTICES:

Entire district of Prayagraj falls between Ganga, Yamuna and Tos rivers. The loamy soils of the area are very fertile. About 64.36 % of the total geographical area of the district is cultivated area. The main *rabi* crops are wheat and oil seeds while paddy and pulses are the main *kharif* crops. The sparsely produced sugarcane is a non-perennial crop.

Table 1.3: Details of the Cropping Pattern (area in Ha) Prayagraj District 2017-18

S. No.	Block	Net area sown	Area sown more than once	Gross area sown				Sugarcane land	Net irrigated area	Gross irrigated area
				The total	Rabi	Kharif	Zayed			
1	Kaurihar	19374	13274	32648	15296	16110	1242	0	17565	27798
2	Holagarh	12477	7949	20426	9124	9901	1401	0	12082	20844
3	Mau-Aima	12777	8647	21424	9578	10456	1390	0	12712	21721
4	Soraon	10769	7908	18677	8540	8537	1600	0	9576	16437
5	Baheria	18173	16328	34501	16118	16942	1437	4	18126	33732
6	Phoolpur	17707	6866	24573	10786	12464	1316	7	11018	21451
7	Bahadurpur	21075	6149	27224	13058	13398	722	46	11229	20012
8	Pratappur	14463	14514	28977	14448	14124	405	0	13262	24635
9	Saidabad	14016	8338	22354	11294	10901	159	0	11724	20601
10	Dhanupur	12645	9618	22263	11066	10744	427	26	12479	23433
11	Handia	13123	6345	19468	9619	9846	3	0	9764	17665
12	Jasra	18310	13573	31883	16764	15102	17	0	18201	30286
13	Shankergarh	22452	21839	44291	23207	20920	164	0	22279	37956
14	Chaka	9399	1513	10912	4413	6484	13	2	5242	7455
15	Karchhana	16082	11503	27585	13761	13789	20	15	15825	23502
16	Kaundhiyara	16083	5842	21925	9199	12658	10	58	15637	23699
17	Urwa	11900	7990	19890	10018	9842	30	0	9646	15855
18	Meja	24165	12434	36599	17731	18856	11	1	24075	35873
19	Koraon	45315	29215	74530	36629	37841	42	18	45082	66063
20	Manda	15923	10909	26832	15224	11592	16	0	13774	21323
Village Total		**346228**	**220754**	**566982**	**275873**	**280507**	**10425**	**177**	**309298**	**510341**
Urban Total		**6462**	**55**	**6517**	**3650**	**2785**	**45**	**37**	**263**	**327**
District Total		**352690**	**220809**	**573499**	**279523**	**283292**	**10470**	**214**	**309561**	**510668**

Table 1.4: Details of Area under Different Crops (area in ha)

S.No.	Block Name	Rice Kharif		Rice Jayad		Rice		Wheat		Sugarcane	
		Total	Irrigated	Total	Irrigated	Total	Irrigated	Total	Irrigated	Total	Irrigated
1	Kaurihar	6377	6372	0	0	6377	6372	12494	12360	0	0
2	Holagarh	8119	8114	0	0	8119	8114	9460	9420	2	2
3	Mau-Aima	8407	8402	0	0	8407	8402	8165	8120	5	5
4	Soraon	5401	5396	0	0	5401	5396	6219	6184	22	22
5	Baheria	9350	8947	0	0	9350	8947	12941	12902	24	24
6	Phoolpur	9787	9750	0	0	9787	9750	16896	16470	37	37
7	Bahadurpur	4277	3966	0	0	4277	3966	8456	8415	0	0
8	Pratappur	8157	7475	0	0	8157	7475	11695	11655	74	74
9	Saidabad	6426	6421	0	0	6426	6421	9395	9375	30	30
10	Dhanupur	10507	8772	0	0	10507	8772	8480	8430	145	145
11	Handia	5560	5205	0	0	5560	5205	8105	8070	52	52
12	Jasra	7655	7300	0	0	7655	7300	12195	12145	0	0
13	Shankergarh	5815	5460	0	0	5815	5460	10965	10900	0	0
14	Chaka	2478	1377	0	0	2478	1377	4065	3990	0	0
15	Karchhana	4885	4880	0	0	4885	4880	10765	10720	171	171
16	Kaundhiyara	7811	7365	0	0	7811	7365	10325	10262	26	26
17	Urwa	3863	2863	0	0	3863	2863	6855	6820	10	10
18	Meja	11547	10035	0	0	11547	10035	29390	29255	10	10
19	Koraon	19926	19021	0	0	19926	19021	41650	40626	38	38
20	Manda	8441	7245	0	0	8441	7245	9723	9669	18	18
Village Total		**154789**	**144366**	**0**	**0**	**154789**	**144366**	**248239**	**245788**	**664**	**664**
Urban Total		**866**	**849**	**0**	**0**	**866**	**849**	**1663**	**1663**	**2**	**2**
District Total		**155655**	**145215**	**0**	**0**	**155655**	**145215**	**249902**	**247451**	**666**	**666**

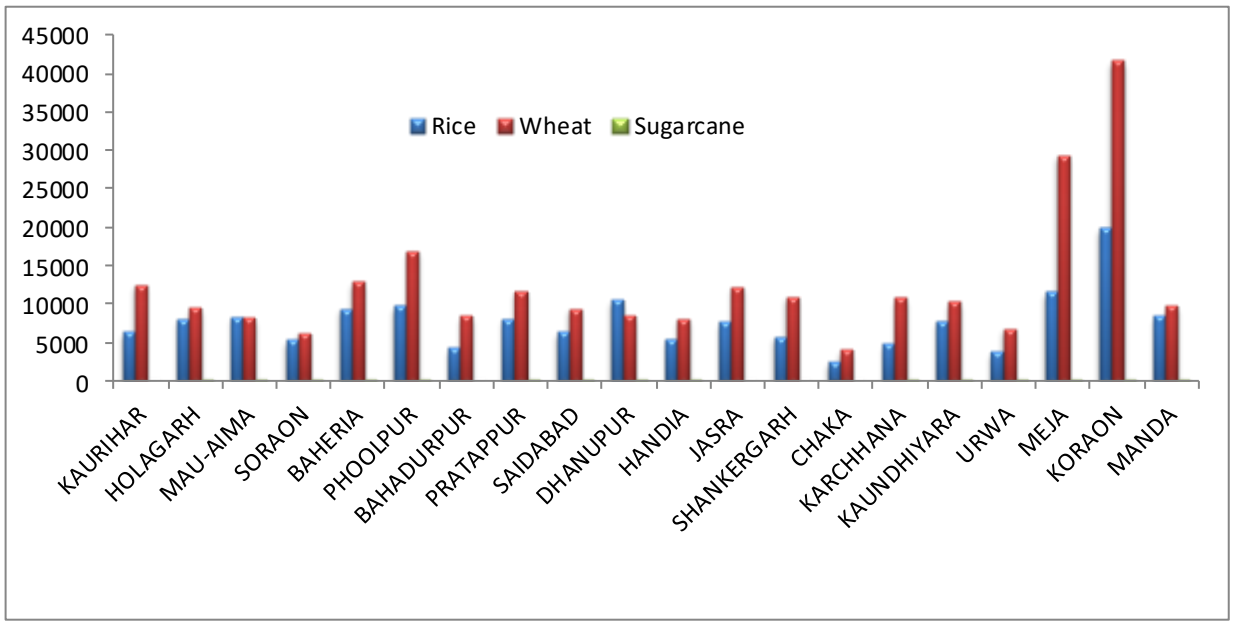


Figure 1.4: Block wise areal distribution of Crops (Area in ha)

Table 1.5: Details of the distribution of Surface and Ground Water for Irrigation (Area in ha)

S. No.	Block	Canal	Tube Well		Dugwell	Pond	Other	Total	SW Source	GW Source
			State	Private						
1	Kaurihar	3021	101	14443	0	0	0	17565	3021	14544
2	Holagarh	5954	348	5780	0	0	0	12082	5954	6128
3	Mau-Aima	6382	256	6074	0	0	0	12712	6382	6330
4	Soraon	2799	537	6240	0	0	0	9576	2799	6777
5	Baheria	5079	77	12462	508	0	0	18126	5079	13047
6	Phoolpur	3451	329	7139	78	19	2	11018	3472	7546
7	Bahadurpur	474	371	10016	361	3	4	11229	481	10748
8	Pratappur	939	1201	11122	0	0	0	13262	939	12323
9	Saidabad	165	312	11227	20	0	0	11724	165	11559
10	Dhanupur	1591	875	10013	0	0	0	12479	1591	10888
11	Handia	1437	643	7676	8	0	0	9764	1437	8327
12	Jasra	11122	728	6141	8	139	63	18201	11324	6877
13	Shankergarh	17909	223	2778	219	1097	53	22279	19059	3220
14	Chaka	2705	1135	1402	0	0	0	5242	2705	2537
15	Karchhana	9821	1887	3958	139	0	20	15825	9841	5984
16	Kaundhiyara	13339	552	1726	20	0	0	15637	13339	2298
17	Urwa	2628	1225	5793	0	0	0	9646	2628	7018
18	Meja	18171	87	5293	377	115	32	24075	18318	5757
19	Koraon	37130	0	2633	4860	452	7	45082	37589	7493
20	Manda	4589	501	6852	1269	505	58	13774	5152	8622
Village Total		148706	11388	138768	7867	2330	239	309298	151275	158023
Urban Total		14	0	138	111	0	0	263	14	249
District Total		148720	11388	138906	7978	2330	239	309561	151289	158272

Table 1.6: Irrigation by Surface and Ground Water in 2017-18, Prayagraj District

Block	Area Irrigated by Surface water/Canal (ha)	Area Irrigated by Ground Water (ha)	Contribution of GW (%)
Kaudrihar	3021	14544	82.8
Holagarh	5954	6128	50.72
Mouaima	6382	6330	49.79
Soraon	2799	6777	70.77
Bahria	5079	13047	71.97
Phulpur	3451	7487	68.44
Bahadurpur	474	10748	95.77
Pratappur	939	12323	92.91
Saidabad	165	11559	98.59
Dhanupur	1591	10888	87.25
Handia	1437	8327	85.28
Jasra	11122	6877	38.2
Shankargarh	17909	3220	15.23
Chaka	2705	2543	48.45
Karchhana	9821	5984	37.86
Kaudhiyara	13339	2298	14.69
Uruva	2628	7018	72.75
Meja	18171	5757	24.05
Koraon	37130	7493	16.79
Manda	4589	8622	65.26
Total Rural	**148706**	**158023**	**51.51**
Total Urban	**14**	**249**	**94.67**
Total Districts	**148720**	**316242**	**68.01**

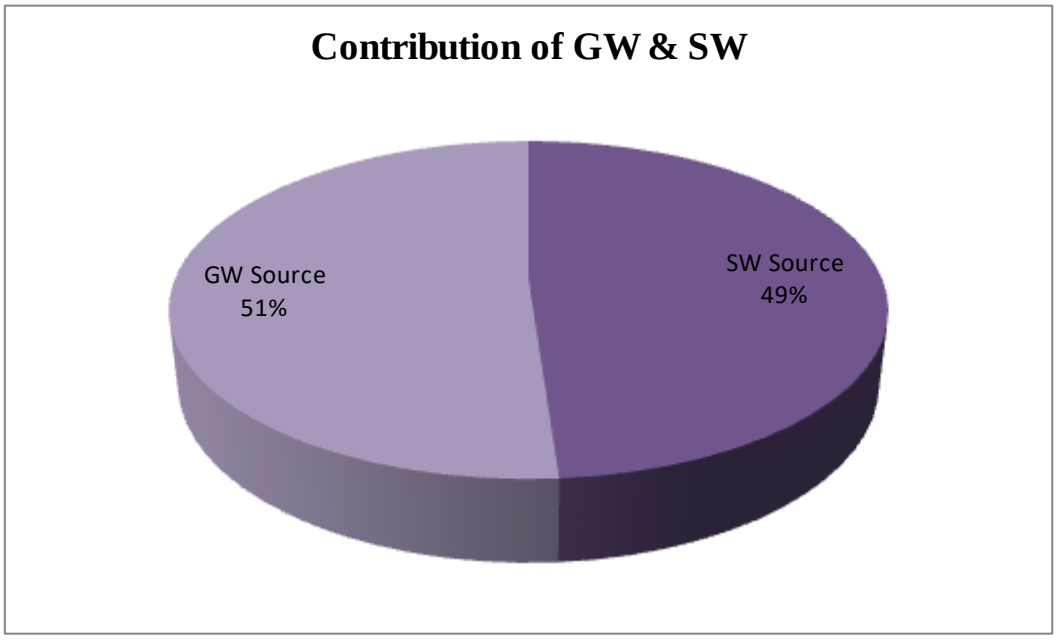


Figure 1.5: Distribution of Surface and Ground Water use

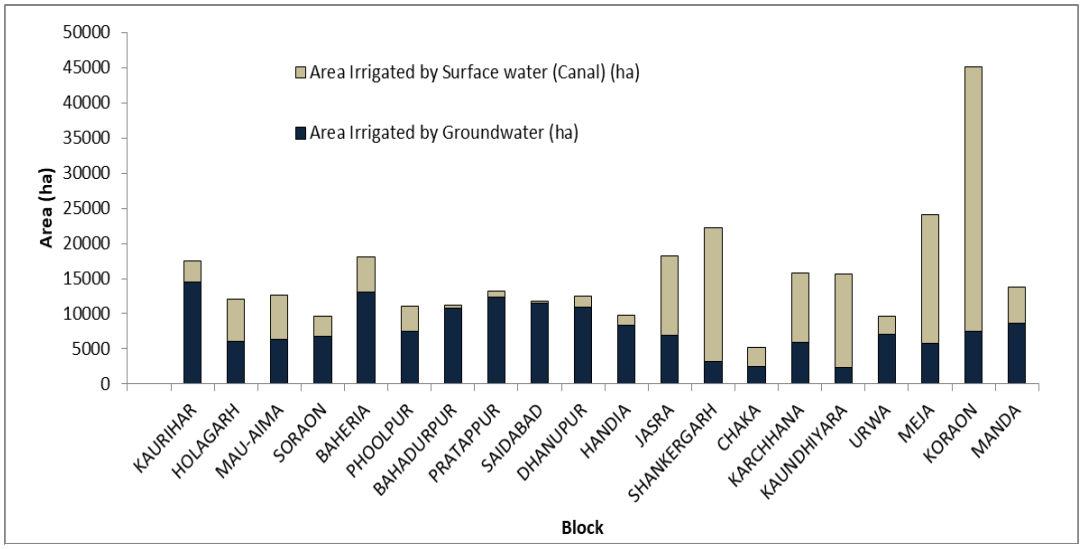


Figure 1.6: Block wise irrigated area (ha) by Surface and Ground Water

Table 1.7: Details of the Irrigation Sources in Prayagraj District 2018-19

Block	Length of canals (km)	State tube well (number)	Pucca well (number)	Geostationary Pumpset (Number)	Shallow tube wells				Medium Tube (Number)	Deep Tube (Number)
					Electrically operated	Diesel powered	other	Yoga		
Kaudrihar	99	84	0	0	682	2644	7	3333	342	236
Holagarh	84	1	0	0	226	3328	45	3599	50	30
Mouaima	82	6	128	0	476	3364	7	3847	48	54
Soraon	56	18	436	0	390	2104	6	2500	126	69
Bahria	119	33	4	0	872	3018	18	3908	275	117
Phulpur	112	51	0	4	562	2488	52	3102	256	145
Bahadurpur	10	77	0	0	784	595	4	1383	35	343
Pratappur	109	66	0	0	543	1388	21	1952	310	71
Saidabad	55	94	0	0	766	1098	3	1867	273	258
Dhanupur	19	69	3	0	1570	884	6	2460	186	22
Handia	28	76	0	23	543	634	2	1179	50	232
Jasra	139	60	62	13	175	1800	34	2009	118	103
Shankargarh	182	2	275	254	10	69	0	79	13	22
Chaka	15	84	0	0	231	255	1	487	185	139
Karchhana	102	172	0	121	658	1095	3	1756	276	280
Kaudhiyara	132	30	0	0	133	2398	4	2535	49	29
Uruva	71	161	44	0	368	727	0	1095	396	188
Meja	240	4	350	150	14	652	3	669	28	98
Koraon	308	0	1469	240	25	2181	16	2222	9	9
Manda	183	19	1116	0	62	1189	2	1253	10	28
Total Rural	**2145**	**1107**	**3887**	**805**	**9090**	**31911**	**234**	**41235**	**3035**	**2473**
Total urban	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total forest area	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total District	**2145**	**1107**	**3887**	**805**	**9090**	**31911**	**234**	**41235**	**3035**	**2473**

1.9 RAINFALL AND CLIMATE

Prayagraj features the typical version of a humid subtropical climate that is common to cities in north-central India. The average annual rainfall is 1207.00 mm. November to February is followed by summer season from March to June. About 90% of rainfall takes place after south-west monsoon season commences from middle of June to September. During monsoon surplus water is available for deep percolation to ground water.

January is the coldest month with mean daily minimum temperature at 9.1°C and mean daily maximum temperature at 23.7°C. May is the hottest month with mean daily maximum temperature at 42.1°C and mean daily minimum temperature at 27.4°C with the advance of the south-west monsoon about middle of June the the Day temperature drops appreciably, but night temperature is a little higher than May. The mean monthly maximum temperature is 32.8°C and means monthly minimum temperature is 19.5°C.

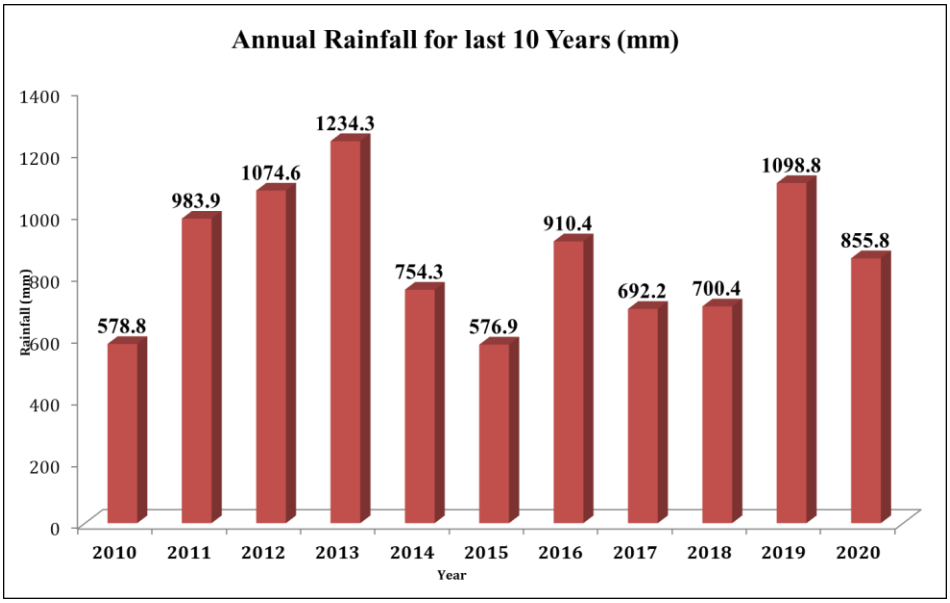


Figure 1.7: Average Annual Rainfall (mm) in Prayagraj District

During the monsoon season the air is very humid and after onset of the monsoon season humidity decreases progressively. The mean monthly morning relative humidity is 64% and means monthly evening relative humidity is 48%.

Winds are generally high throughout the year with some increase in force in summer and monsoon season. The mean wind velocity is 5.1 kmph. The potential evapotranspiration is 1537.5 mm.

Table 1.8: Rainfall Data (mm) and Departure from Normal Rainfall (mm), Prayagraj District, U.P. (2010-2020)

YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	Total RF	Monsoon RF	Non-Monsoon	Departure from Normal Rainfall (mm)	% Departure from Normal Rainfall (mm)	Status
2010	3.6	19.2	0.0	0.5	7.5	8.5	201.7	160.6	126.4	45.5	5.3	0.0	578.8	497.2	81.6	-323.8	-35.87	Deficient
2011	0.9	4.5	0.0	9.3	12.2	237.3	171.0	282.3	266.4	0.0	0.0	0.0	983.9	957	26.9	81.3	9.01	Normal
2012	43.7	0.0	0.8	9.2	0.1	23.1	385.4	246.1	363.5	2.7	0.0	0.0	1074.6	1018.1	56.5	172	19.06	Normal
2013	0.0	68.1	3.2	0.3	0.0	245.0	294.8	337.9	64.1	220.3	0.0	0.6	1234.3	941.8	292.5	331.7	36.75	Excess
2014	81.8	47.6	28.7	0.0	2.9	38.2	208.1	149.2	73.6	114.5	0.0	9.7	754.3	469.1	285.2	-148.3	-16.43	Normal
2015	59.9	1.0	48.3	33.8	0.0	81.9	160.1	156.6	14.2	19.9	0.0	1.2	576.9	412.8	164.1	-325.7	-36.08	Deficient
2016	9.5	0.8	23.8	0.0	4.6	51.7	229.7	398.6	162.3	29.4	0.0	0.0	910.4	842.3	68.1	7.8	0.86	Normal
2017	0.2	0	0.1	1.2	4.7	40.2	466.7	117.8	60.9	0.4	0	0	692.2	685.6	6.6	-210.4	-23.31	Deficient
2018	0	6.6	0	15.4	27.4	24.3	244.1	292.9	89.5	0	0.2	0	700.4	650.8	49.6	-202.2	-22.40	Deficient
2019	10.5	7	2.2	1.7	0.9	17.2	306.8	279.5	385.8	68	0	19.2	1098.8	989.3	109.5	196.2	21.74	Excess
2020	16.5	5.4	34.8	21.9	35.6	147	228.5	223	135.2	0	2.5	5.4	855.8	733.7	122.1	-46.8	-5.19	Normal

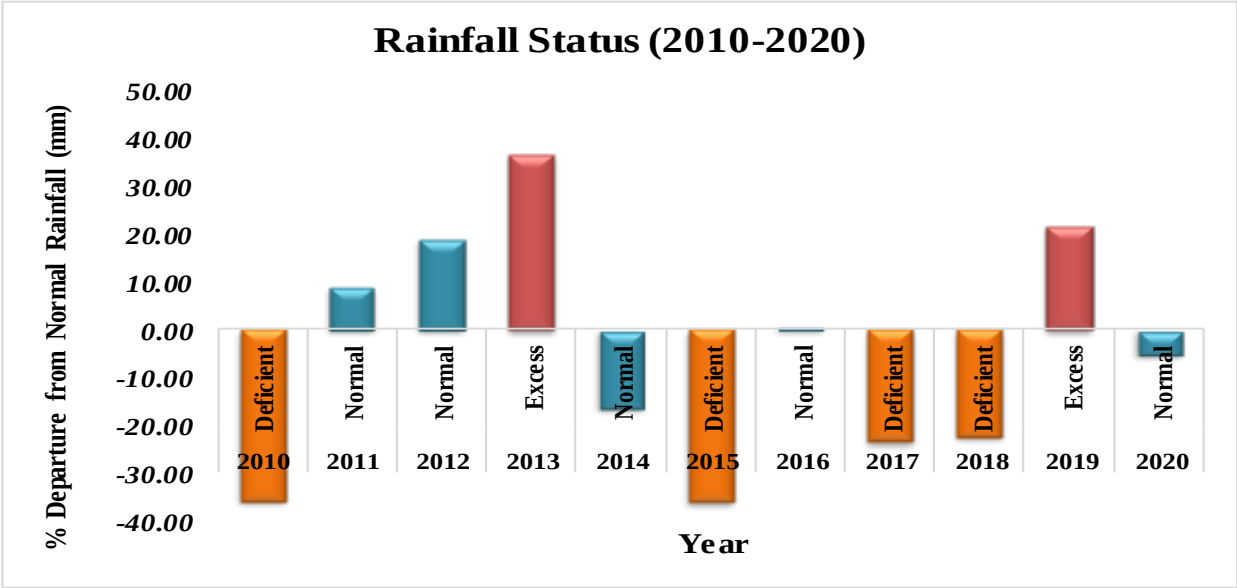


Figure 1.8: Departure of rainfall from normal rainfall, Prayagraj District

1.10 GEOMORPHOLOGY AND SOIL TYPES

The district is drained by river Ganga and its right bank tributary Yamuna and Tons, broadly represents following geomorphic units-

a) Ganga alluvial plains.
b) Yamuna alluvial plains and
c) Vindhyan plateau

The alluvial plains may be further sub-divided into newer alluvial plain and older alluvial plain. Newer alluvial plains are confined to present day flood plain region all along the rivers while the older alluvial occupy the higher parts within the northern portions of the district. No significant alluvial deposits are present over the Yamuna region. The important land forms observed in the alluvial plain are the meanders, scrolls, point bars, back swamps etc.

In the Vindhyan plateau region the important forms are mesa, butte and glacies. From ground Water occurrence point of view and its yield prospects, the alluvial plains sediments are the most potential ground water repositories.



Figure1. 9: Geomorphological map of Prayagraj District

In Vindhyan plateau region the formation though hard sediments are very well jointed giving rise to promising level of secondary porosity to hold and yield water. The northern part of Prayagraj district popularly known as Gangapar provides rich loam soil for cultivation of food grains, pulses, oilseeds and vegetables. The Southern part of Prayagraj district also known as Yamunapar is partly rocky and somewhat agriculturally backward.

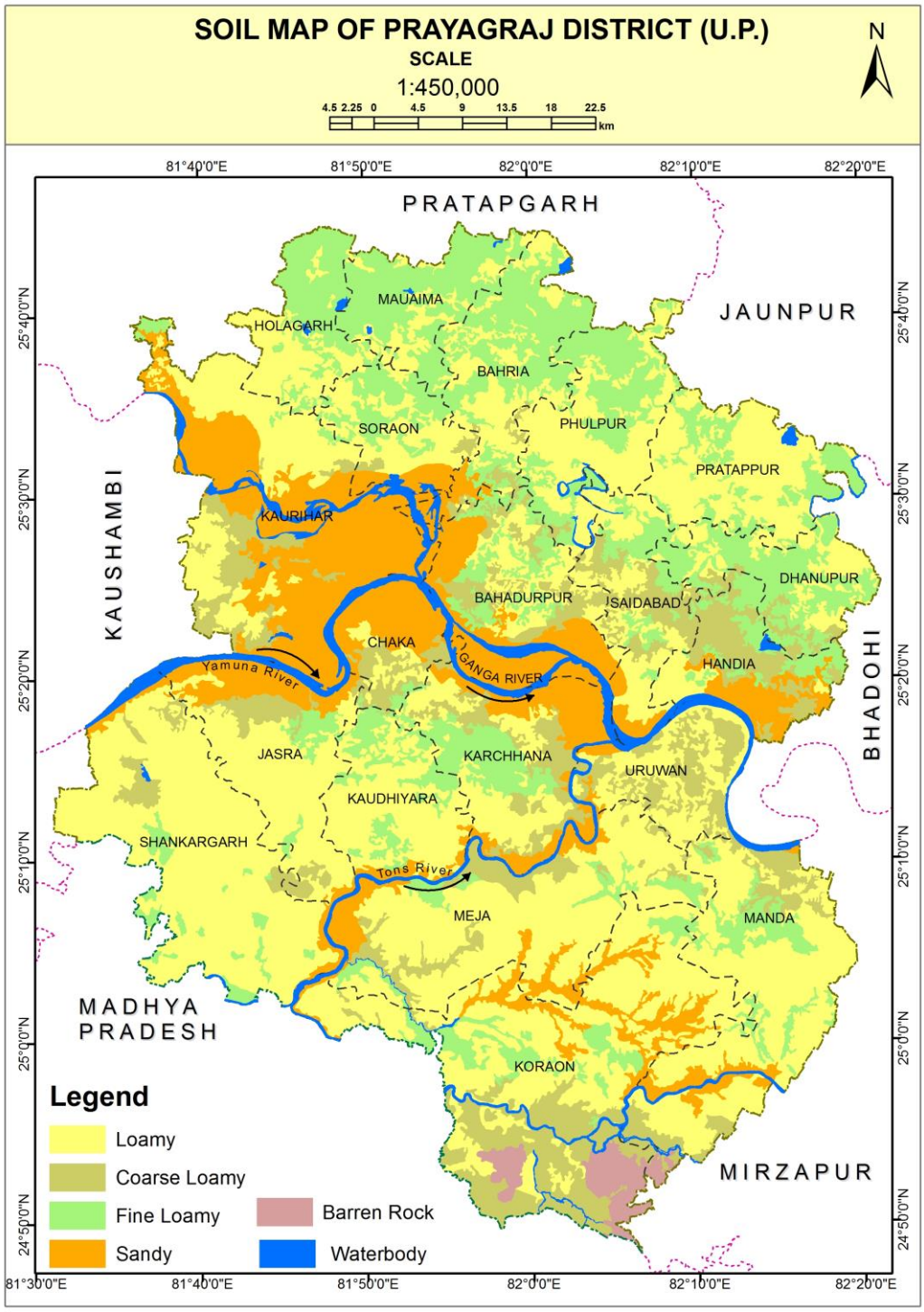


Figure 1.10: Soil Map of Prayagraj District

1.11 LAND USE/LAND COVER

A major part of the land in the district is utilized for agriculture purpose. As per the latest statistical data available for the year 2017-18, following land utilization pattern has been observed in the district (Table 9). Land use distribution and map of Prayagraj district is shown in Figure-11 and Figure-12. A perusal of the Table-9 and figure-11 and figure-12 shows that 63% of land in the district is used for agricultural activities, 17 % land other than agriculture, 6% current fallow and 4% other fallow, 4% is forest cover, 3% barren land, 2% cultivable waste land and rest are trees and grasses.

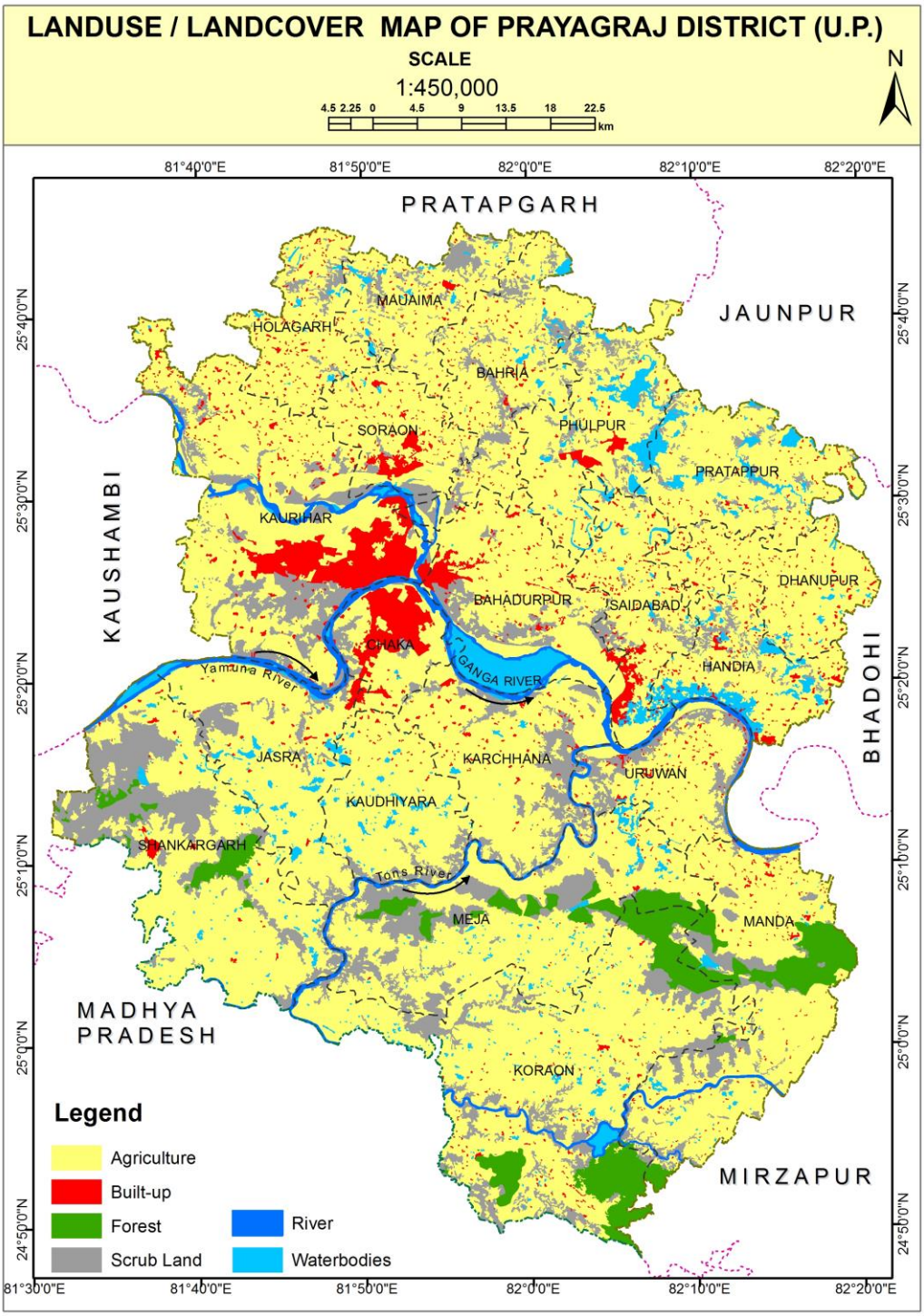


Figure 1.11: Landuse and Landcover Map, Prayagraj

Table 1.9: Block wise Land use pattern of Prayagraj District 2017-18 (Source: Statistical Diary, Directorate of Economics and Statistics, Govt. of U.P.

S.No.	Block Name		Block - wise Land Use in the district in Hectare							
		Agricultural land	Forest	Cultivable Waste Land	Current fallow	Another Fallow	Barren Land	Land other than agriculture	Grasslands	Area of garden, trees & shrubs
1	Kaudrihar	19374	132	392	9303	3290	1225	9755	24	685
2	Holagarh	12477	0	71	765	532	299	2462	110	475
3	Mouaima	12777	12	152	477	404	511	1249	120	49
4	Soraon	10769	10	126	437	1009	114	2535	24	242
5	Bahria	18173	0	88	937	225	1079	6024	114	174
6	Phulpur	17707	0	275	1707	707	1220	6135	107	407
7	Bahadurpur	21075	0	239	2165	353	1577	5899	81	61
8	Pratappur	14463	1	48	1851	534	320	5215	16	264
9	Saidabad	14016	2	34	2119	335	359	3120	1	744
10	Dhanupur	12645	18	70	986	403	155	4912	15	344
11	Handia	13123	0	151	1063	335	239	2818	22	246
12	Jasra	18310	0	321	1384	562	1210	3735	10	73
13	Shankargarh	22452	5218	3708	2969	2184	1376	1871	12	394
14	Chaka	9399	0	293	387	360	1603	3677	58	231
15	Karchhana	16082	0	162	856	1029	113	7028	0	570
16	Kaudhiyara	16083	0	212	414	416	170	2889	5	237
17	Uruva	11900	17	44	648	338	252	3739	0	235
18	Meja	24165	4383	859	2408	1520	1774	1131	13	70
19	Koraon	45315	6415	762	365	480	240	3062	877	20
20	Manda	15923	5264	2173	1646	1035	378	5407	30	52
	District Total	**352690**	**21476**	**11340**	**33910**	**20728**	**14947**	**94690**	**1639**	**5654**

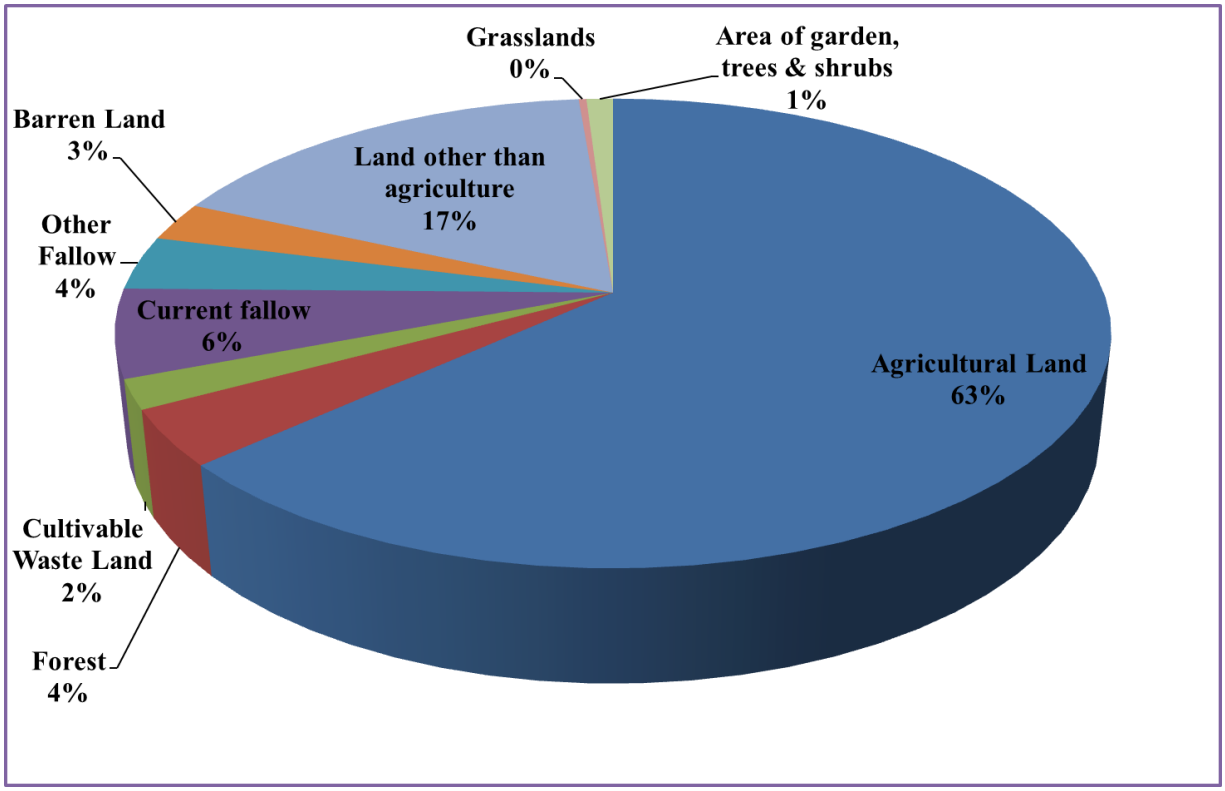


Figure 1.12: Land use and agricultural activities, Prayagraj District (2017-18)

CHAPTER-02

DATA COLLECTION, GENERATION, INTERPRETATION, INTEGRATION AND AQUIFER MAPPING

2.1 GEOLOGY

The district represents a complex geology. The formation belonging to Quaternary period covers larger part of the district which directly overlies over the Vindhyan formations which covers bulk of the district area. The southern plateau area constitutes the Vindhyan sediments whereas the Quaternary covers the northern part of the district (Figure-2.2). Following is the generalized geological sequence:

Quaternary: Alluvium – Sand, silt, clay and Kankar, Laterite.

Pre Cambrian-Vindhyans: Sand stone and shale (Kaimur sandstone and Bijaigarh Shale).

The Distribution of geological formations in Prayagraj district is described as under-

Vindhyans: Different stages of Kaimaur formations, belonging to upper Vindhyans, exposed in the extreme Sothern part of the district. The oldest formations exposed are Bijaigarh Shale in the edge of the district around Hanumangunj in Koraon block. Further to the north the sandstone and quartzites , representing the uppermost part of the Kaimur series, are seen as isolated part of the district covering the blocks of Meja,Manda, Koraon and Shankargarh, The Vindhyans forms the floor of the younger sediments in the Cis-Yamuna area.

Laterite: The Laterite normally overlies Vindhyans at certain places .it is reddish brown to chocolate in color, highly ferruginous, and perforated mass.

Alluvium: The whole of Trans-Ganga, Doab and Part of Trans – Yauna tract is covered by the Ganga alluvium of Quaternary age. The alluvium detritus of Vindhyan is found at some places in the southern part of the Doab. In Trans – Yamuna tract, The Vindhyan detritus merges with the Yamuna sand and silt.

The thickness of the alluvial sediments gradually increases in the north and maximum thickness is encountered within the flood plain area of river Ganga and Yamuna.

Greater thickness is encountered within the Trans – Ganga area where it exceeds 300meters followed by doab region where it is less than 250meters and ultimately less than 50 meters in the Trans-Yamuna region. The alluvial sediments essentially composed sands of various grades, clay and silts within the unconsolidated granular mass assemblage of nodular concretions of Calcium Carbonate have been recorded at different depths.

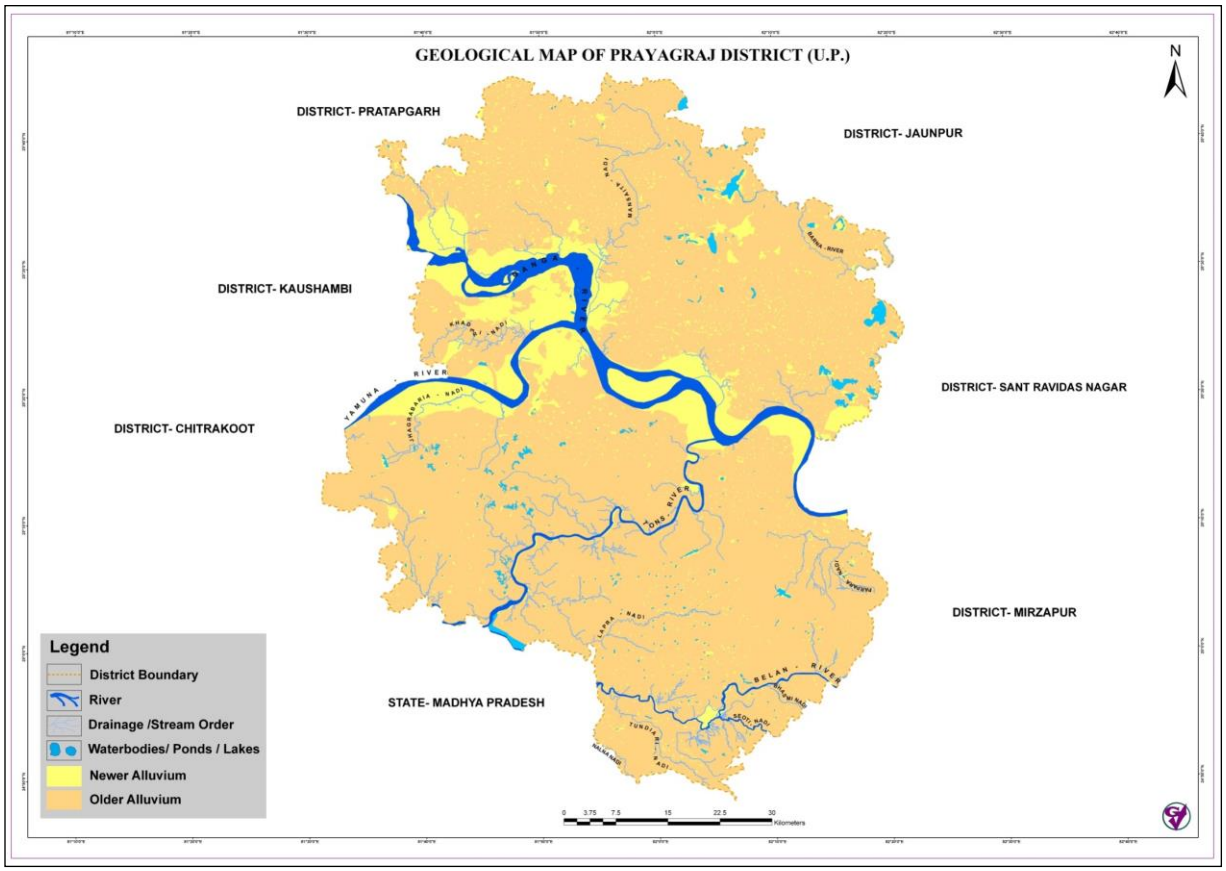


Figure 2.1: Geological Map of Prayagraj District

2.2 Hydrogeological Framework:

Ground water in the district occurs both in alluvium and the weathered and jointed sandstone in area which are underlain by hard rocks. Two-broad based hydrogeological units namely unconsolidated (Alluvium) and consolidated (Hard Rock) have been identified.

The unconsolidated or alluvial formations occur in the Trans- Ganga and Doab region. Localized patches of Trans Yamuna region are also covered by the unconsolidated formations.

Occurrence of consolidated formations is restricted primarily Trans – Yamuna tract and this sizable area flanks the southernmost segment of the district.

2.2.1 Alluvium Area: Ground water occurs within the primary porosity of alluvial sediments in the north, the aquifer materials are medium to coarse grained sands. The shallow aquifer materials are medium to coarse grained sands. The shallow aquifer occurs under unconfined condition while deeper aquifers are under semi confined to confined conditions.

2.2.2 Ground water flow: Water table elevation map prepared, available data indicates that ground water flow is towards the river Ganga in the north and river Yamuna in the south i.e. both the rivers are effluent in nature.



Figure 2.2: Hydrogeological Map of Prayagraj District

2.2.3 Hard Rock Area: The Ground water in the widely covered Vindhyan Plateau region primarily occurs under unconfined condition within the secondary porosity of the formation. However, exploration data indicates that Kaimur sandstone found at depths does have enough potentiality. These sandstones after leaching of cementing material get disintegrated and reduce to silica sand which are loose and act as promising repository of ground water. Below the sandstone particularly in the western part of the plateau region shales have been encountered which possess reasonable development of secondary porosity and projects moderate prospect of occurrence of ground water.

Earlier studies indicated presence of four sets of vertical joints of variable dimensions in the sandstone and have reasonable yield prospects.

Sand Horizons are conspicuously exposed in the extreme NW part where silica sand are being mined and used in ceramics. These sand beds may be termed as **Marker Horizon."**

Above the Marker Horizon, friable sandstone or loose sandy matrix occurs which are the weathering products of the Kaimur Sandstone. Below the marker horizon occurs compact sandstone which is normally fractured and yield fairly good quantity of fresh water.

Hydrogeological Scenarios in the hard rock region is highly variable. Geomorphology, besides geology, plays an important role in the occurrence and movement of ground water.

2.3 Sub-Surface Geology: Sub – surface geology has been deciphered from the study of lithological logs and electrical logs of the boreholes drilled by central ground water board. In order to decipher the sub – surface disposition of aquifer system, 5 numbers of linear sections and two numbers of Fence Diagrams were prepared. Location of tube wells and section lines have shown in the index map (Figure 2.3).

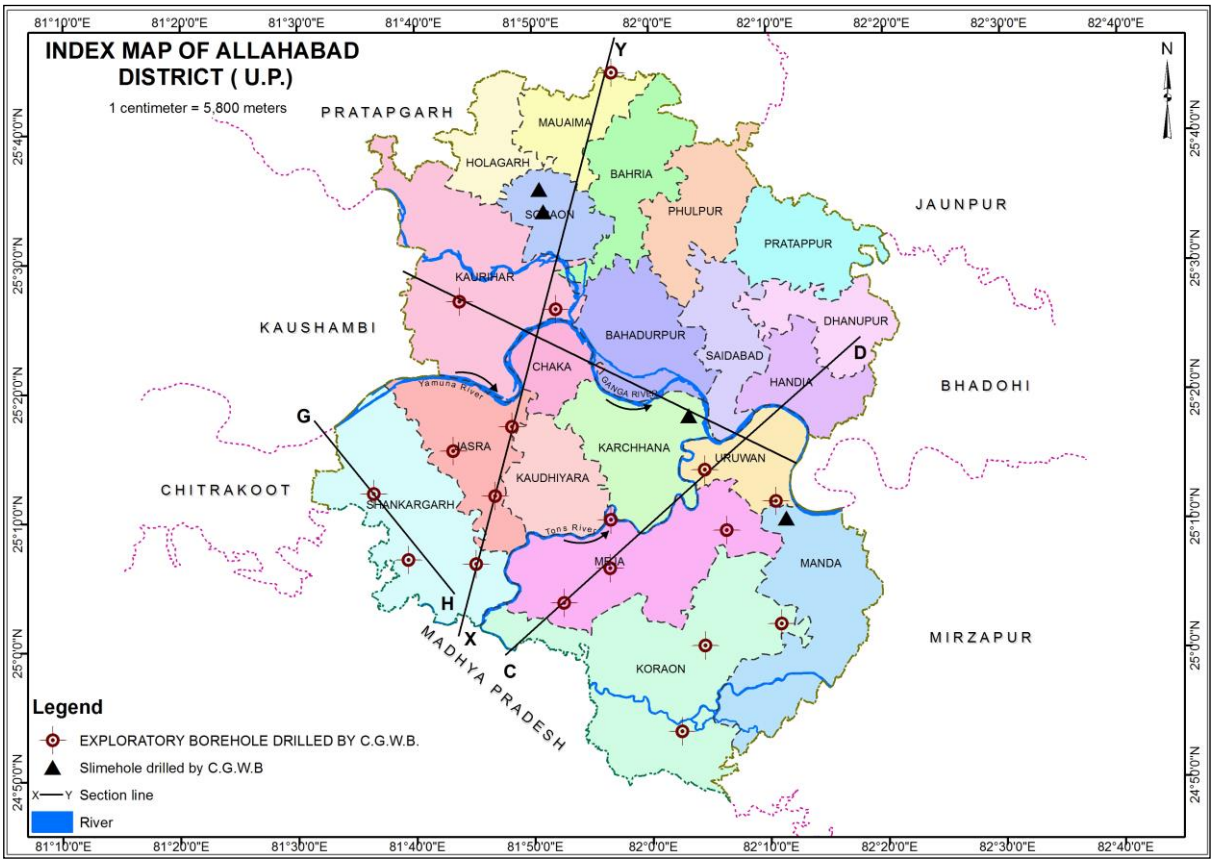


Figure 2.3: Index Map depicting Location of tube wells and section lines

2.3.1 Geological cross-section along A-B line

This section runs in the NW-SE direction of the district covering the Sirathu, Chail and Karachana blocks covering a distance of 70kms. Boreholes depth ranges from 113.68 m (Basaha-I) to 278.12 m (Sulem Sarai).The top clay, which is sandy in nature, at places ranges in thickness from 2m to 20m. Thickness of clay is maximum at Sulem Serai and minimum at Basahai. Below this clay blanket, lies the 1st aquifer which is continuous and extends upto 120 m. This aquifer is inter spaced with 4 minor clay beds which are 2-10 m in thickness and at places it remains admixed with sands , fine to medium grained .the clay and sand beds so described occurs as alternates layers. In the top or first aquifer, the sandy horizon is encountered at 20m depth below the top clay blanket and continues at 120m. Its thickness progressively increases towards the Ganga and Yamuna and attains at its maximum at Basahi. The intercalated minor clay beds having the maximum thickness at Nighawawan.

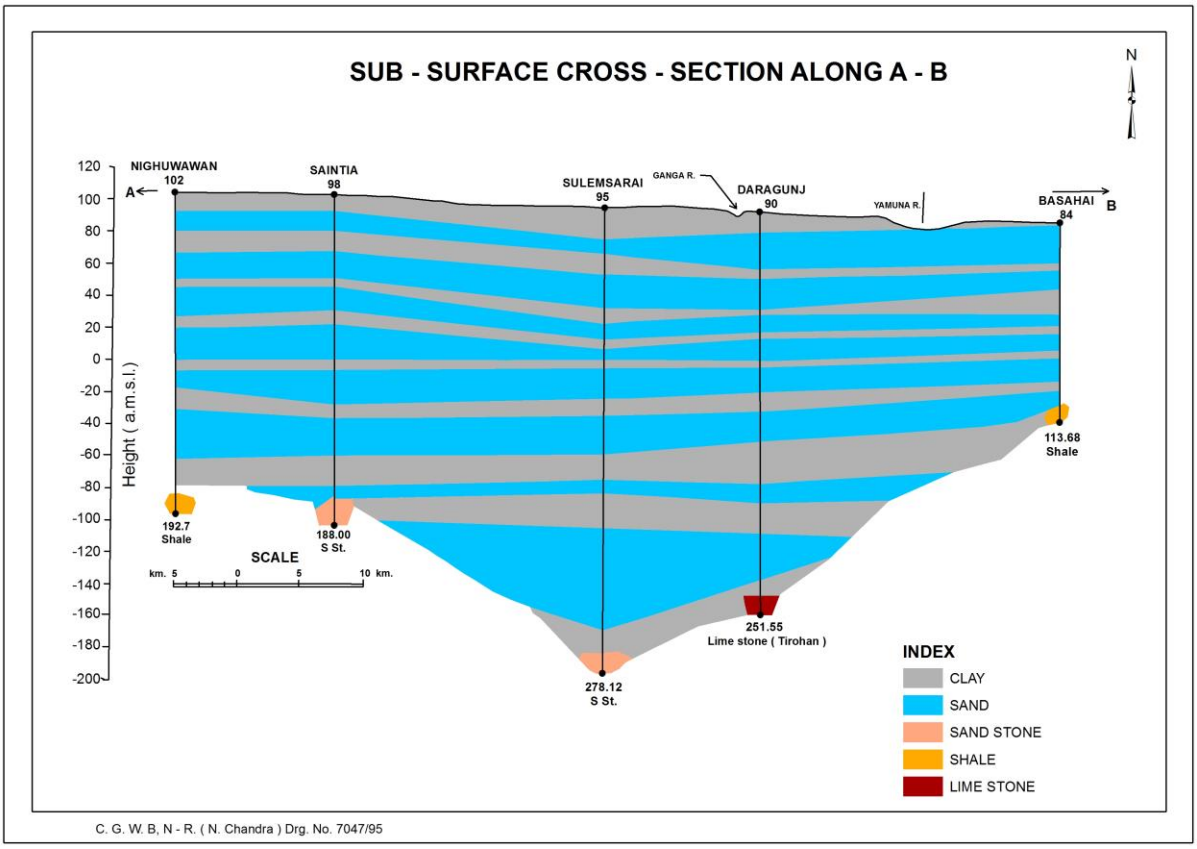


Figure-2.4: Geological cross-section along A-B line

The alternate sandy horizon has 10-25 m of granular material which is generally medium to fine grained greyish sand essentially composed of quartz, feldspars and ferromagnesian minerals. The coarse sediments are rounded to sub rounded and well sorted. The size of the sand grains increases towards Ganga and Yamuna. Mostly, state tube wells are confined to first aquifer only and yielding good discharge. The second aquifer starts below 150 m and

continues up to basement. Its thickness varies at places. The first and second aquifers are separated by a thin clay layer 10-15 m in thickness. This clay is sticky and hard yellowish. Below this impermeable zone the sandy horizon has 30-40 meters of granular material which is generally medium to coarse grained and mixed with gravel sorted to well sort comprised of smoky quartz, feldspars and ferromagnesian minerals. This aquifer has very high potentiality. Below this sandy horizon (second aquifer) a 20-30m thick clay bed exists, whose thickness is maximum at Basahai, overlying the bed rock (shale) which has been encountered at Nighuwan at a depth of 192.7m. Depth of basement sediments is maximum at salem sarai 279.12m (sandstone) and shallowest at Basahai 113.68m (shale) phenomenal presence of Vindhyan formation exist in the entire section drawn.

2.3.2 Geological cross – section along C-D line

This section runs along due SW-NE in the southern part of the district. This section depicts the subsurface disposition from Itwa kalan to Bandhwaka-lan, and is approximately 41.6kms.long depth of boreholes along this section ranges from 105.25 to 153.95m.Thickness of surface clay blankets is 16m at Itawa kalan which progressively decreases

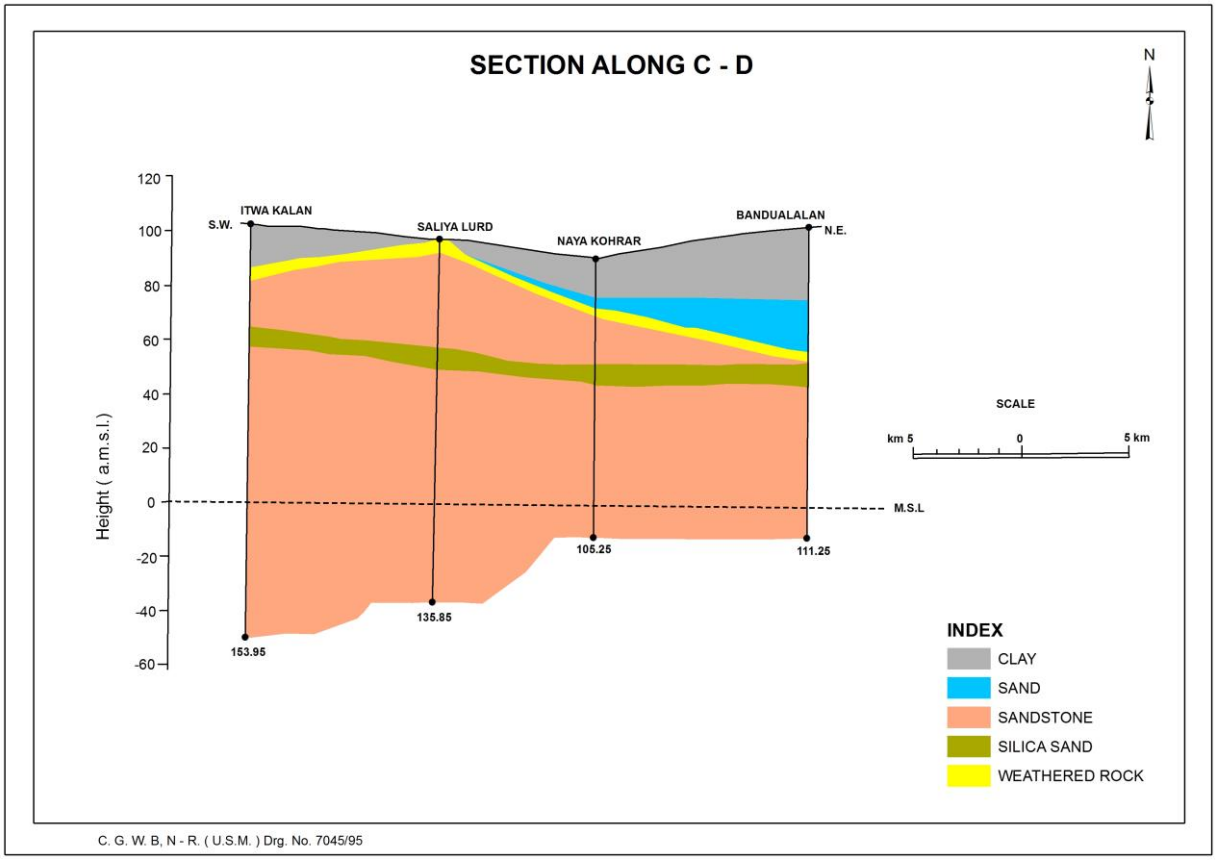


Figure 2.5: Geological cross – section along C-D line

towards Saliya Khurd and at Saliya Khurd it becomes negligible where the weathered rock is exposed . From Saliya Khurd to Bandhwakalan where it is found to occurs and mixed with

Kankars and fine grained sand at places. The top sand horizon occurs below this clay layer at Naya Kohrar and the thickness of the granular horizon increases towards Bandhwakalan where it attains up to 20m, and forms the aquifer in the alluvial portion.

Alluvial mantle is maximum at Bandhwakalan (50 m) and minimum at Itwakalan and zero at Saliya Khurd underlying the alluvium weathered sandstone occurs which is rather lateritic . Below the weathered zone sandstone occurs. This typical friable sandstone thickness is maximum at Saliya Khurd whereas minimum at Bandhwakalan. Below this sandstone a highly friable silica sand bed of 5 to 8 m thickness occurs at a depth of 36m at Itwa Kalan. The sand is medium grained, well sorted and has pronouncedly developed due to leaching of the cementing material. This sand is locally knows as 'Shankargarh Sand". Below the shankargarh sand the friable and less compact sandstone occur whose compactness increases with depth. Generally the secondary porosity gives rise to cracks, fissures and joints up to 100m. At Itwa-kalan secondary openings are found upto 73m and this characteristics is pronounced up to 100 m at Nayakohrar and Bandhwa Kalan

2.3.3 Geological section along line E-F

This section runs in W-E direction and depicts the sub surface disposition of geological formation in the southern part of Allahabad district extending from Benipur (Shankargarh) to Newaria. This section is approximately 57kms long and depth of borehole ranges from 111.7

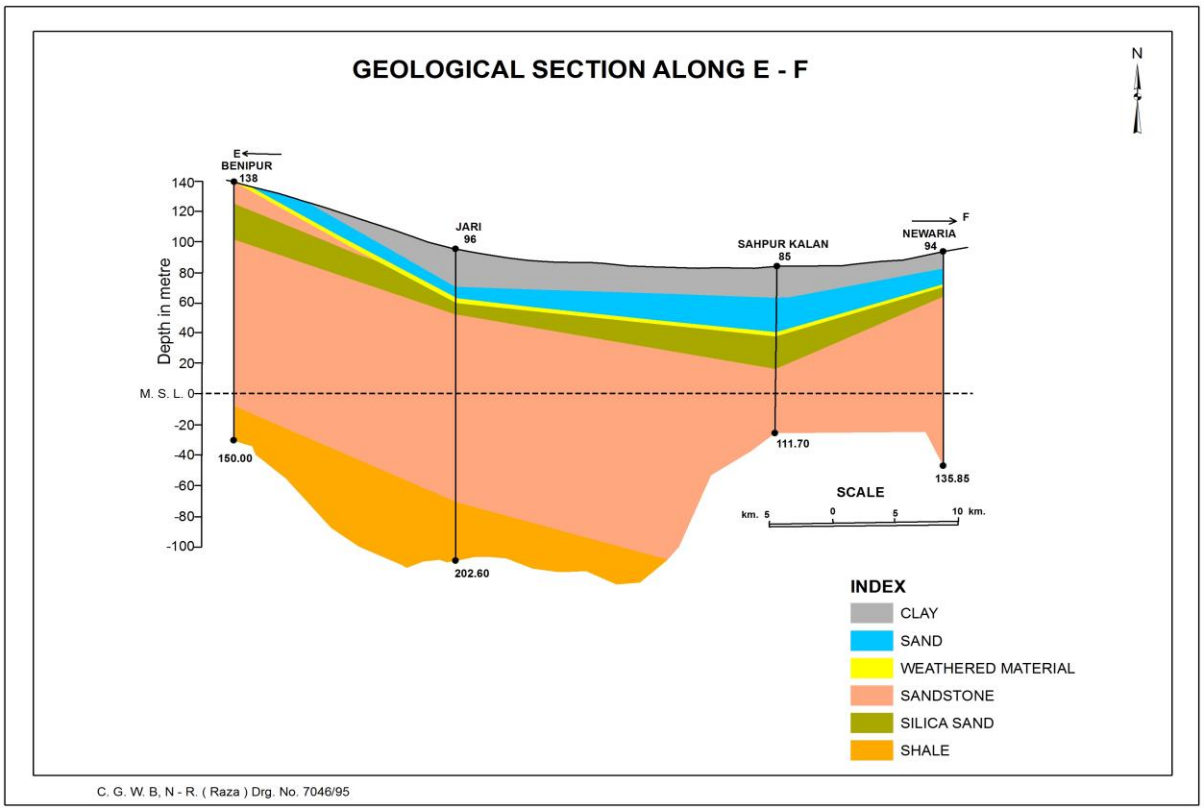


Figure 2.6: Geological cross section along E-F line

to 202.30m. The drawn section depicts that thickness of alluvium is almost negligible at Benipur and maximum at Sahpur Kalan (45.75). Thickness of clay blanket is 20-22m at Jari and Sahpur Kalan and 10m at Newaria. Below this clay capping the alluvial aquifer exists, having thickness ranging from 9-14m at Jari and Sahpur Kalan. This alluvial aquifer essentially composed of medium to coarse grained sand, well sorted, rounded to sub-rounded and brownish in colour. Shallow tubewells tapping the granular zones in the alluvium can be constructed.

Below the alluvium a thin 1-2m thick weathered mantle occurs. Friable and whitish sandstone occurs below weathered mantle. This friable sandstone has become highly friable at Benipur giving rise to loose whitish, fine and medium grained sand locally known as "Shankargarh Sand". Thickness of this Shankargarh sand is 25m at Benipur and is being exposed due west of Benipur towards Yamuna River. Its thickness about 20m at Sahpur Kalan near Tons river and in rest of the places namely, at Jari and Newaia it is about 5m. Below this Shankargarh sand white and purple colour, fine to medium grained sandstone exists. Ground water occurs in the weathered and jointed sandstone. Cracks, joints and fissures have developed upto a maximum depth of 110m. These sandstones overlie Bijaigarh shale which is greyish in colour and impermeable. These shales were encountered at benipur and Jari exploratory boreholes and their local extension and thickness could not be making out.

2.3.4 Geological section along G-H line

This section runs N to S direction in south- western part of Prayagraj district. It

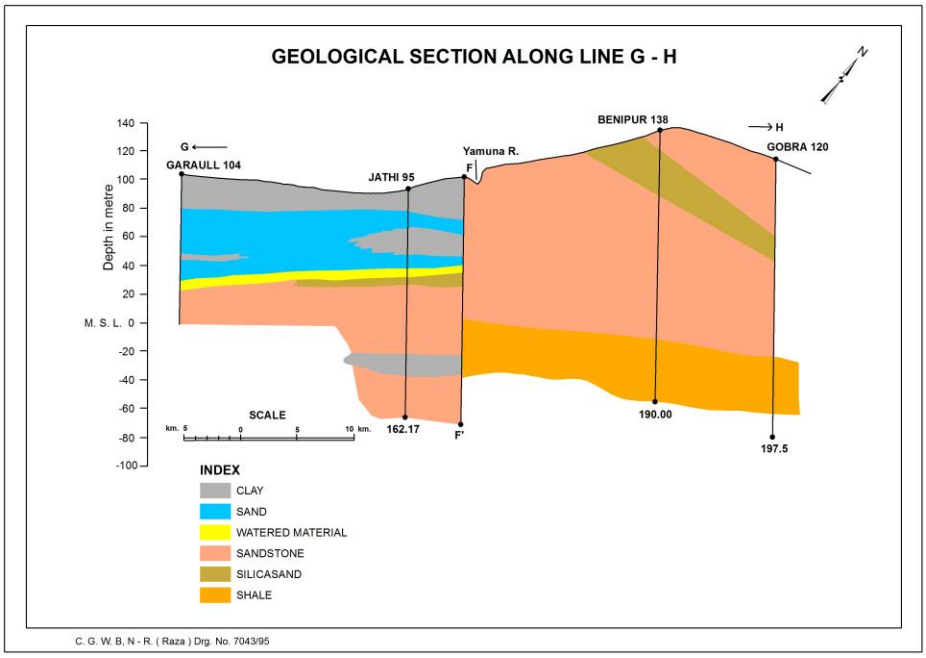


Figure 2.7: Geological cross-section along G-H line

Depicts subsurface disposition of formations from Garauli to Gobra and is approx. 43kms long. Depth of boreholes considered for depiction varies from 101.66 to 190.00 m.

It could be seen thickness of top clay is maximum at Carauli and gradually increases towards the river Yamuna. Below this clay blanket the major aquifer exists which is 40-45 m thick intercalated with minor clay lenses. The aquifer material comprises of fine to medium grained sand inter mixed with gravels at places. Below this aquifer horizon weathered material do exist followed by friable sandstone of Yamuna par (trans-Yamuna) area.

A major fault does exist along the river Yamuna which separates the sub-surface features in Trans-Yamuna area.

2.3.5 Geological cross-section along X-Y line

This section, which is of 73km long runs due SW to NE and depicts the sub-surface disposition of aquifer system in the southern part of the district extending from Nari Bari to Delhupur. The depth of boreholes considered ranges from 78.6m to 447.96m. It is evident from the section configuration that there is a marked litho-characteristic difference in the sub-surface formations encountered north of the river Yamuna. A major fault lies along the river Yamuna which separates the sub-surface features. It is conjectured west from Nari-Bari up to Yamuna River the alluvial thickness is maximum near Yamuna and minimum at Nari Bari (3m). In the alluvial (40m) mass generally a thick clay bed occurs at places intermixed with sand, medium to fine grained. Weathered rock exists below this clay layer which acts as a

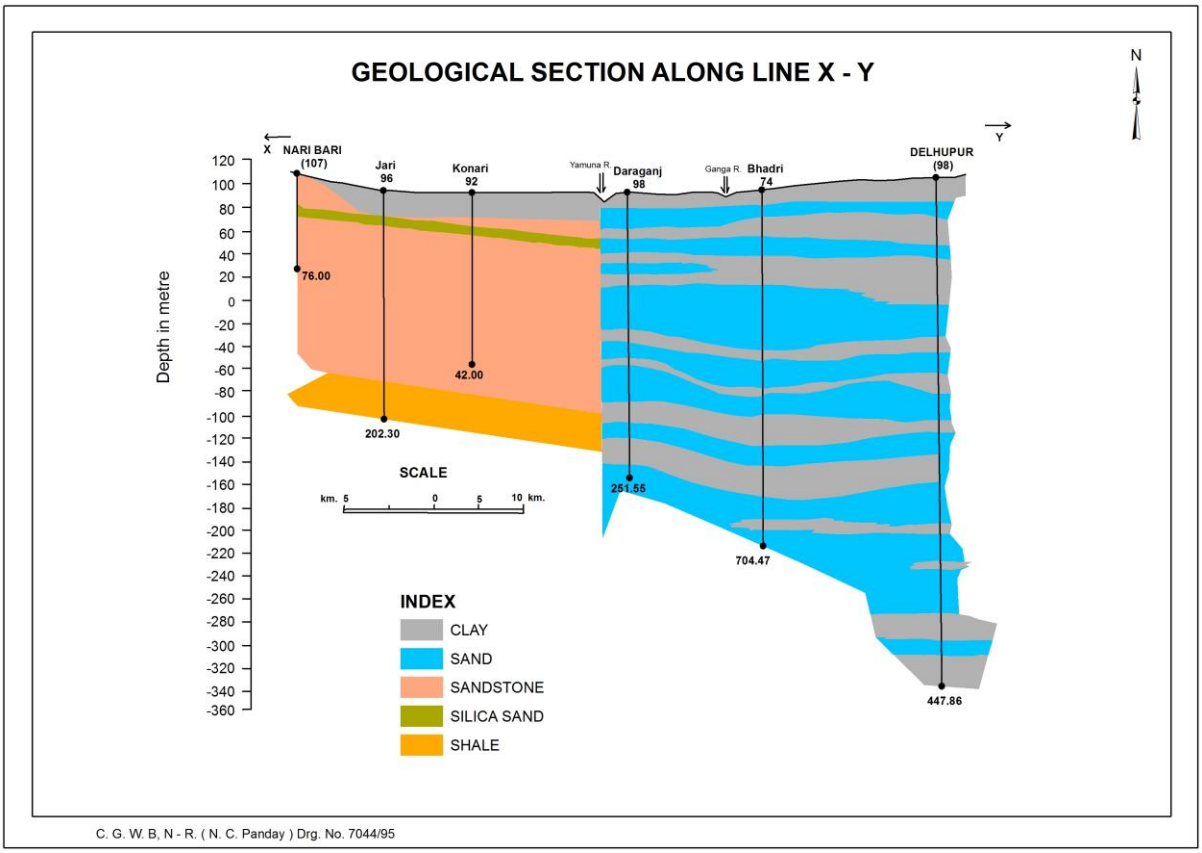


Figure 2.8: Geological cross-section along X-Y line

Capping over the Vindhyan sandstone. It varies from 0.3m to 0.5m. Below this weathered rock purple to whitish colour medium to fine grained sandstone exists which is friable in nature. Below this silica sand occurs which is good ground water repository. Thickness of this highly friable fine to medium grained white silica sand is 5-10m. Rock masses become more compact and impermeable at depth. Bijaigarh shale exists at depth of 168.75m at Jari. Ground water in this hard rock occurs in the weathered and jointed sandstones. Secondary porosity has developed cracks, joints and fissures bedding planes along Nari Bari to Konri serve as a potential ground water repository. In this alluvial area from Daraganj to Delhupur section a clay layer 10-20m thick exists at the top whose thickness is maximum at Delhupur. Below this clay blanket the first aquifer upto a depth of 120m exists which is separated by minor clay beds of 5-10m thickness. Sand and clay beds occur alternatively. Thicknesses of sand horizons are maximum at Bhadri. The sands are medium to coarse grained and found occasionally associated with gravels which is composed of quartz and feldspar and little amount of ferromagnesian minerals. Below 120m prominent sticky and hard yellow clay exists which is 10-12m thick and its thickness increases towards Delhupur. Below this clay layers of sand bed occurs which is 10-30m thick. Sand layers comprised of medium to coarse sand and mixed with gravel. These sands are well sorted with angular to sub – angular sediments. Vindhyan basement (Lime stone) was encountered at a depth of 250.55m at Daragunj. Thickness of bed rock increases towards Delhupur and the same has been encountered in Bhadri and Delhupur exploratory boreholes.

2.3.6 Fence Diagram of Alluvial and Hard Rock Area

To decipher the sub-surface configuration of the aquifer system over the district fence diagram has been drawn utilizing lithological logs of exploratory boreholes drilled by central ground water board. Review of the fence diagram reveals that two tier aquifer systems, beside the phreatic zone, exist in the Cis-Ganga-Yamuna area. The top aquifer disposition ranges from 25-120 m below ground level in general. In the eastern part of the district the thickness of this aquifer increases gradually in the central part of district, however, covering Prayagraj city area its extension traceable down to the depth of 200m below ground level. At Balkaranpur this granular horizon attains as much as 180 m in thickness while at Delhupur (in Pratapgarh) its thickness is approximately 230m. At Nibhapur (Jaunpur) the thickness of this granular zone is 200m. Overlying this extensive aquifer horizon a persistent clay bed occurs whose thickness is variable ranges between 25 and 125m. Thickness of clay horizon

increases in the north – eastern part of the district. This top aquifer horizon is extensively tapped in shallow state and private tube wells.

The underlying second aquifer lies in the depth ranges 150-200 m below ground level. The granular materials are medium to coarse grained sand and mixed with gravels. The sand is well sorted and has assemblage of heavy minerals. Because of coarser nature of aquifer materials yield prospect from this repository is better in comparison to top system.

Studies have revealed that both the above described aquifer groups are under confined conditions. The piezometric heads in these aquifers varies between 7-20m bgl. In the western part of the district however the potentiometric head of the auto flow zone is 1.5 to 3.00 m above ground level.

The Fence Diagram was prepared on the basis of lithological logs of exploratory boreholes drilled by Central Ground Water Board in the hard rock area of the district. The review of the Fence Diagram (Figure D) in the trans-Yamuna region that thickness of the superficial alluvial sediments is limited over the major part except along the river Ganga-Yamuna. These Quaternary sediments, deposited by the rivers Ganga & Yamuna, constitute of sands of different grades intermixed with clays & Kankar. These alluvial formations support multitudes of open wells & shallow tube wells which are capable of yielding 1000-1500lpm of water at economic drawdown.

Fractured Vindhyan sandstone which are considered to be relatively promising aquifer over the entire Ganga-Yamuna region. These sedimentary rocks at the top (capping) is generally laterites which serve as recharge horizon.

The productive horizons within vindhyan sandstones are encountered between 30 and 100m bgl. These sandstone are highly weathered at the top thickness of weathered zone is rather variable. At Bara and Konri the thickness of weathered material is rather less and maximum at Bhawanipatti. Just below the weathered residuum there occurs a marker horizon “Silica Sand” which is characterized by its poor cementation and highly friable nature. This silica sand bed is regionally extensive, locally known as “Shankargarh Sandstone.”

Fence diagram demonstrated regional extensive deposition of silica sand bed in the depth range of 13.63-90 m bgl.

Thickness of this sand bed is variable ranges from 3m to as much as 36m (Benipur). Just below this silica sand horizon the productive horizons within Vindhyan sandstones have been encountered invariably over the entire region. The hard rock aquifers are recorded to yield 500-100lpm of ground water at reasonably high drawdown. These sandstones are underlain by Bijaigarh shale which has been encountered in the depth ranges of range of 145-

170m below ground level in the western part while in the western part while in the eastern part the shale has not been struck even up to the depth of 190meter below ground level.

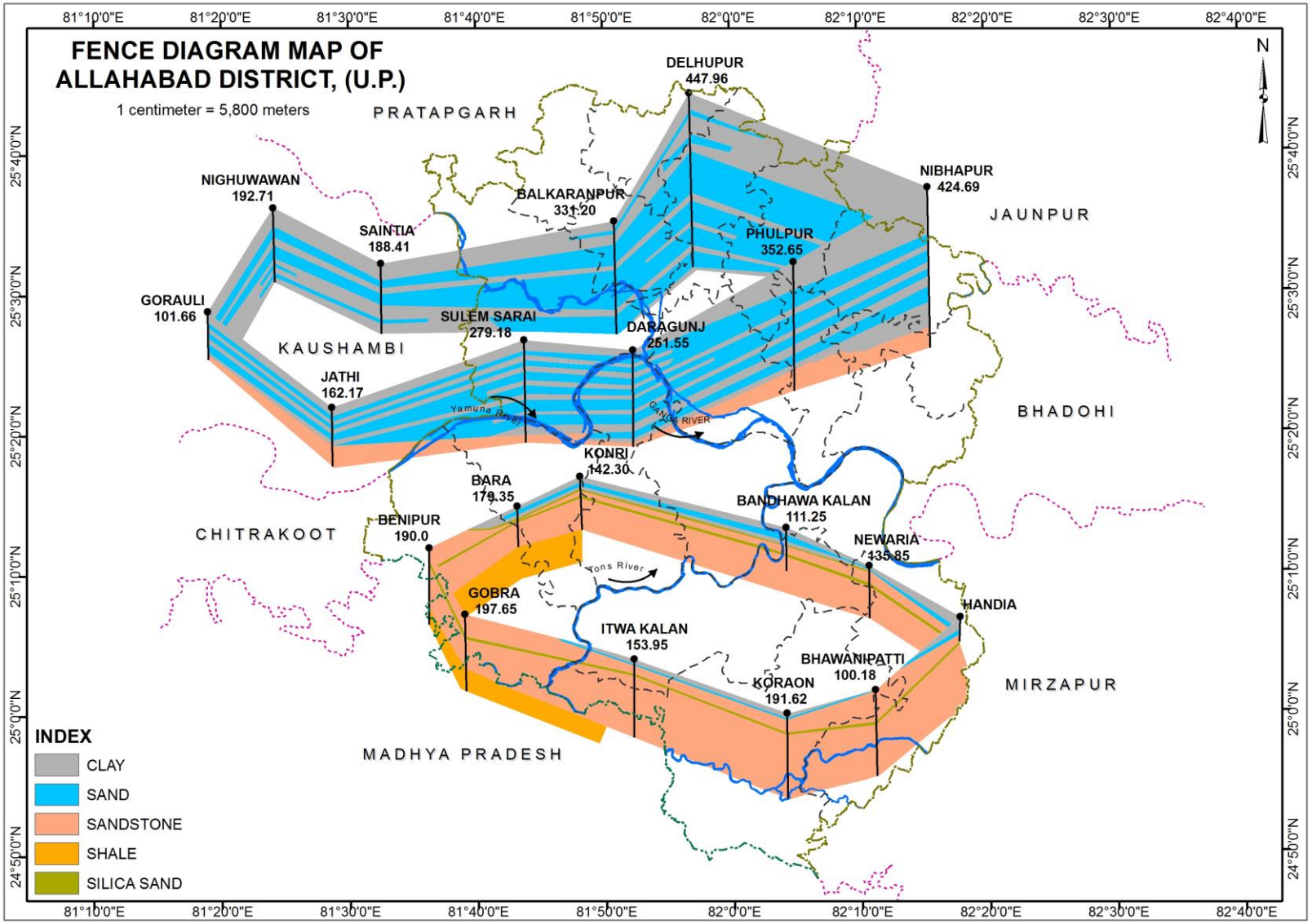


Figure 2.9: Fence Diagram of Alluvial area and Hard Rock

Thickness of sandstones is minimum in the western part and maximum in the south-eastern segments of Prayagraj where it exceeds 200m. The shally horizon below this sandstone is impervious and do not hold any prospect for ground water development. Thickness of this shale is not known as it has not been completely pierced in any of the boreholes so far drilled.

2.3.7 Depth to bed rock

The geological setting of the district represents primarily two hydrogeological units, namely the alluvial sediments in the north of river Ganga and Yamuna. Vindhyans are encountered 100 meters to as much as 400m. It depicts that the vindhyan topography gradually deepens northerly and progressively goes down to depth of more than 400 meters along the northern part of Prayagraj District. Vindhyan basement is shallower in the south- eastern part of the district in comparison to northern part, shallowest Vindhyan basement is encountered in the western part of Prayagraj district. Vindhyans are found exposed in isolated patches in the south of Ganga and Yamuna; and depth of basement in this region varies between 0-50 meters below ground level. The secondary porosity within the Vindhyan formations, which are basically sandstone, form the main ground water depository in the trans-Ganga belt.

Table 2.1: Lithological Data of the Prayagraj District

Location	Lat	Long	Rock Type	Location	Lat	Long	Rock Type
Bandhwa Kalan	25.20417	82.075	Hardrock	Nighwawan	25.6	81.42889	Alluvium
Newaria	25.15833	82.16667	Hardrock	Saintia	25.55333	81.53333	Alluvium
Itwa Kalan	25.04333	81.88167	Hardrock	Daragunj	25.44167	81.89167	Alluvium
Konri	25.16667	82.765	Hardrock	Sulem Sarai	25.45	81.76667	Alluvium
Bhawani Patti	25.05	82.2	Hardrock	Lehderi	25.76667	81.38333	Alluvium
Paithia	24.87194	82.06667	Hardrock	Jathi	25.36333	81.47	Alluvium
Koraon	24.99167	82.07667	Hardrock	Bandhwa Kalan	25.20417	82.075	Alluvium
Gobra	25.16667	81.58333	Hardrock	Chorbana	25.14722	82.20139	Alluvium
Nari Bari	25.08333	81.73333	Hardrock	Bhadri	25.5625	81.83333	Alluvium
Benipur	25.20556	81.60972	Hardrock	Balkaranpur	25.60778	81.83333	Alluvium
Bara	25.24167	81.725	Hardrock	Bashai	25.29639	82.0375	Alluvium
Jari	25.20278	81.78056	Hardrock	Garauli	25.46667	81.31167	Alluvium
Meja	25.14167	82.10833	Hardrock				
Naya Kohrar	25.15	81.99167	Hardrock				
Saliya Khurd	25.09722	81.93056	Hardrock				
Sahpur Kalan	25.175	81.94167	Hardrock				

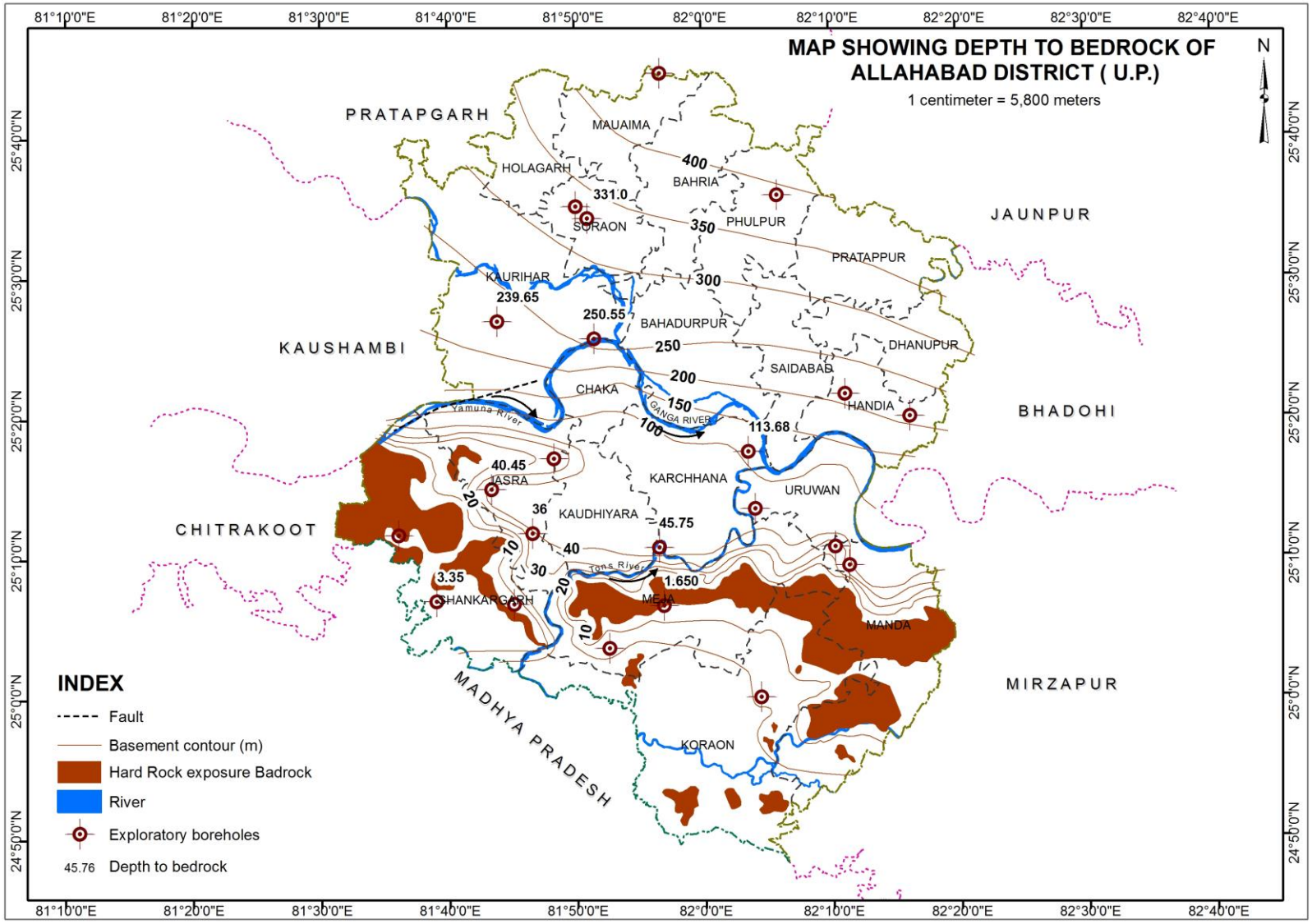


Figure 2.10: Depth to bed rock, Prayagraj District

2.4 GROUND WATER SCENARIO

2.4.1 Mode of occurrence of Ground water

Ground water occurs in the pore spaces of the unconsolidated alluvial sediments in the zone of saturation. The sediments consist of fine to medium sand with intercalated pockets of clayey material within, and a thin layer of clayey sediments at the top. Alluvial formations has also been encountered in Trans Yamuna area but has limited extension adjacent to river Yamuna.

Vindhyan plateau is characterised by jointed and faulted hard rock formation which have limited ground water potential. Fissured formations are only found in Trans Yamuna area and particularly dominant in southern part of Trans Yamuna area. It has developed secondary porosity and has limited yield prospects. Sandstone and Quartzites are important formations with layer of silica sand which forms the fissured formations for ground water withdrawal and act as potential horizons.

Infiltration from rainfall is the main source of ground water recharge in the area. Influent seepage from the surface water bodies is also responsible for recharge to the ground water reservoir.

2.4.2 Ground Water Condition

Ground water occurs under phreatic to semi confined and confined conditions. The near surface aquifer is under unconfined / water table condition. The shallow phreatic aquifer is tapped by dugwells. The depth to water level of the CGWB monitoring stations ranges from 1.32 to 20.36 mbgl during pre-monsoon whereas it ranges from 0.59 to 15.69 mbgl in post monsoon. The pre & post monsoon water level fluctuation varies from 0.46 to 7.54 m. Depth to water level trend for long term for 10 years average water level rise 0.492 cm/year and fall 0.3658 cm/year during pre-monsoon and rise 0.5156 cm/year and fall 0.91cm/year during post-monsoon. Annual rise rate 0.3587cm/year and fall 0.7232 cm/yr.

2.4.3 Pre-monsoon Depth to water level

Perusal of map showing deeper water level observed in parts of Bahadurpur, Baharia and Chaka blocks (Figure-2.11). Almost fifty percent northern part of the district having the water level ranges from 5 to 10 mbgl and southern part of the district having water level ranges from 5 to 10 mbgl except a few pocket ranges from 2 to 5 mbgl during the pre-monsoon season.

2.4.4 Post-monsoon Depth to water level

During Post monsoon period depth to water level in central part to northern part of the district ranges from 2-5m bgl and 5 to 10 mbgl respectively (Figure-2.11). In Southern part of the

District depth to water level ranges from 2 to 5mbgl except a few pockets of less than 2 mbgl.

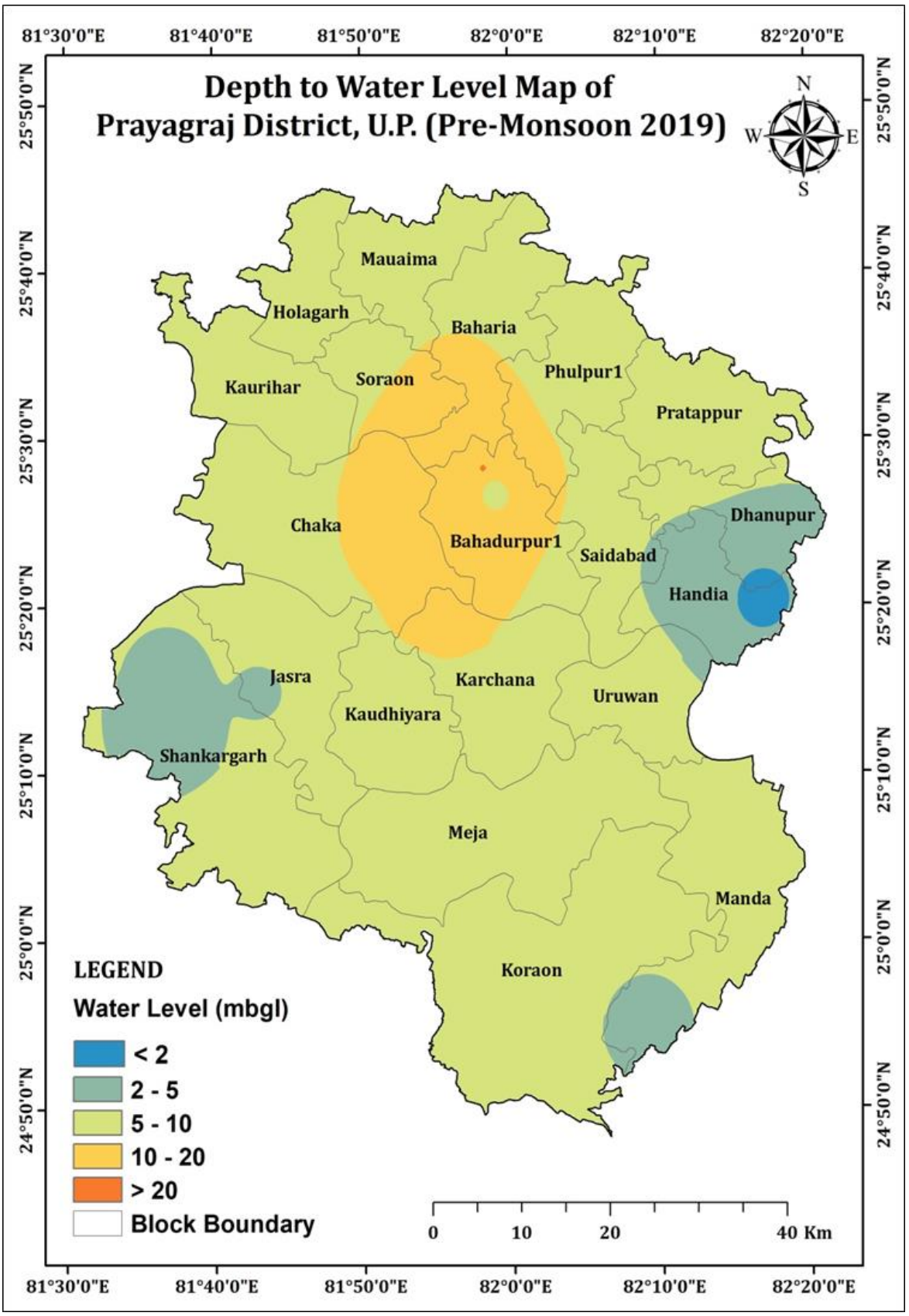


Figure 2.11: Depth to water level map (Pre-Monsoon 2019)

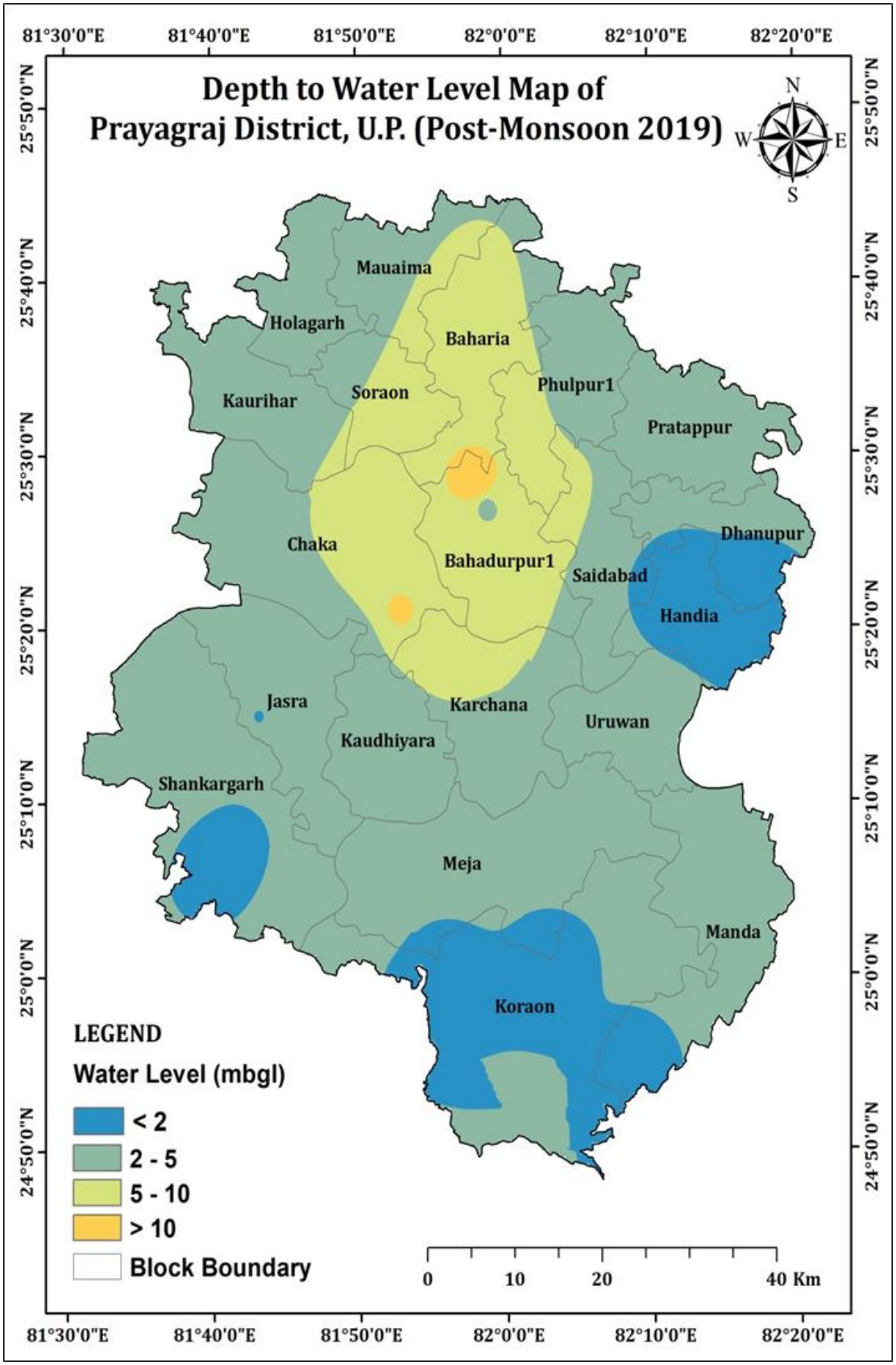


Figure 2.12: Depth to water level map (Post-Monsoon 2019)

2.4.5 Water Level Fluctuation

Maximum water level fluctuation is observed in the north-western part whereas minimum fluctuation observed in patches of Shankargarh, Meja, Handia,and Dhanupur blocks of the district. Rising water level trend has been observed in the study area ranges between 0.46 m to 6.42 m.

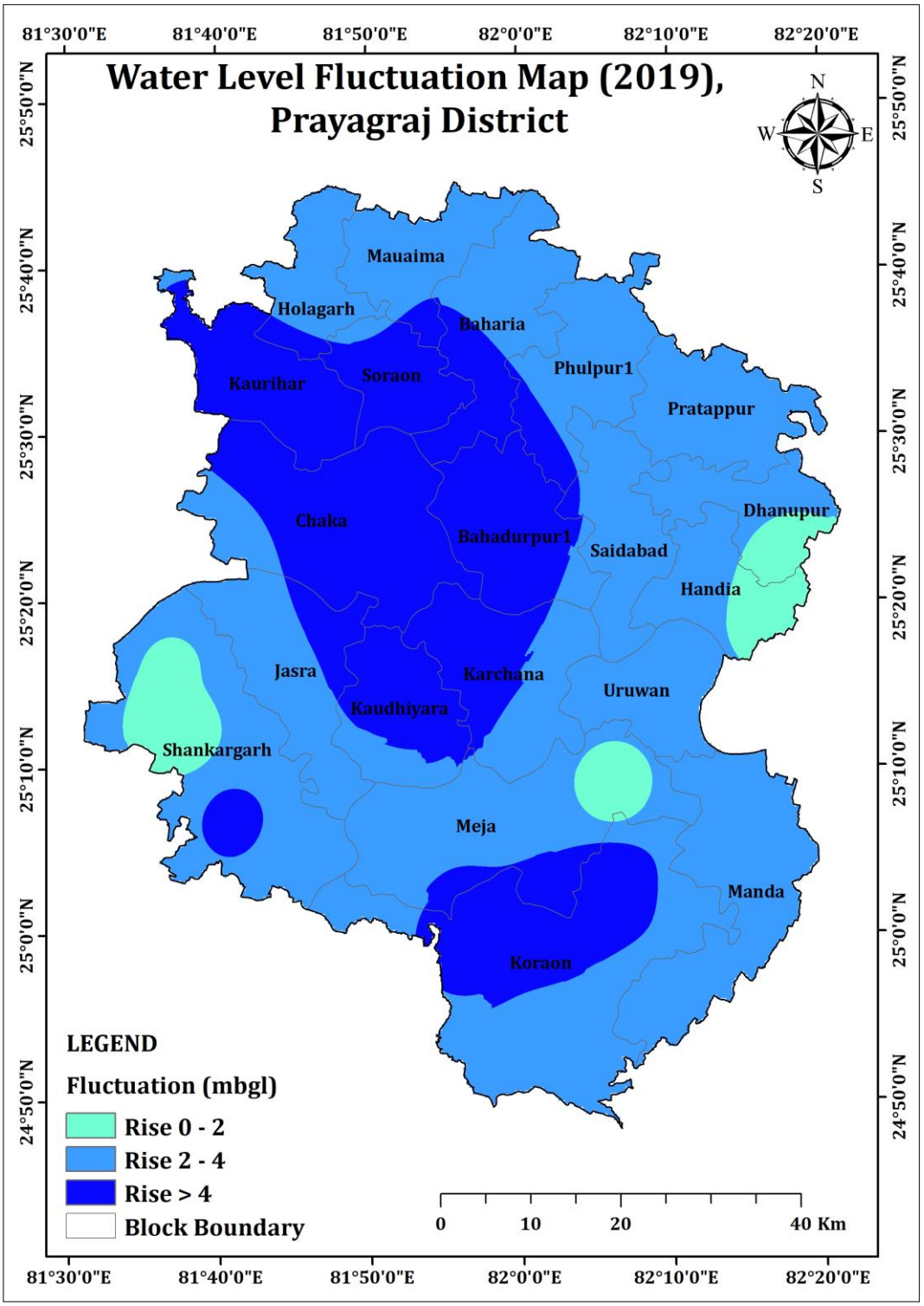


Figure 2.13: Water Level Fluctuation Map (2019), Prayagraj District

Table 2.2: Water level of Pre & Post Monsoon-2019, Prayagraj District

Block	Location	Type of well	Lat.	Long.	Pre-19	Post-19	Fluctuation
Bahadurpur	Allahabad	Dugwell	25.45	81.98	7.59	3.47	4.12
Kaurihar	Atrampur	Dugwell	25.57	81.74	9.01	3.64	5.37
Shankargarh	Bairi	Dugwell	25.12	81.68	5.45	0.85	4.60
Jasra	Bara	Dugwell	25.25	81.72	4.51	1.98	2.53
Handia	Baraut1	Dugwell	25.34	82.28	1.32	0.86	0.46
Manda	Bawarikalaan	Dugwell	24.93	82.15	4.11	1.49	2.62
Shankargarh	Gadwa Fort	Dugwell	25.20	81.63	3.99	3.28	0.71
Jasra	Gauhania	Dugwell	25.32	81.83	7.84	2	5.84
Handia	Handia	Dugwell	25.37	82.19	3.82	0.55	3.27
Holagarh	Holagarh	Dugwell	25.64	81.77	6.23	3.31	2.92
Bahadurpur	Kasarua	Dugwell	25.47	81.97	20.36	14.52	5.84
Shankargarh	Khaptia Kheri	Dugwell	25.07	81.78	6.11	3.03	3.08
Karaon	Koraon	Dugwell	25.02	82.06	5.34	1.13	4.21
Karaon	Ladiri Bazar	Dugwell	25.00	81.93	5.66	1.44	4.22
Jasra	Lalpur	Dugwell	25.27	81.62	4.31	2.52	1.79
Phulpur	Mallhan	Dugwell	25.61	82.10	7.17	4.39	2.78
Meja	Meja	Dugwell	25.14	82.10	5.04	4.37	0.67
Chaka	Naini	Dugwell	25.35	81.88	15.56	10.81	4.75
Karaon	Pasna	Dugwell	25.05	82.10	8.43	2	6.43
Phulpur	Phulpur2	Dugwell	25.55	82.09	7.52	4.36	3.16
Manda	Rampur tulapur	Dugwell	25.04	82.20	6.98	4.44	2.54
Kaudhiyara	Saraon	Dugwell	25.27	81.84	7.63	3.09	4.54

2.4.6 Decadal Fluctuation of Groundwater (2010-2019), Prayagraj District

The range of fluctuation shows minimum rise is 0.012 m in Bara block and maximum rise is 0.3587 m in Mooratganj block whereas minimum fall is 0.022m in Mauaima block and maximum fall if 0.7232m in Koraon block.

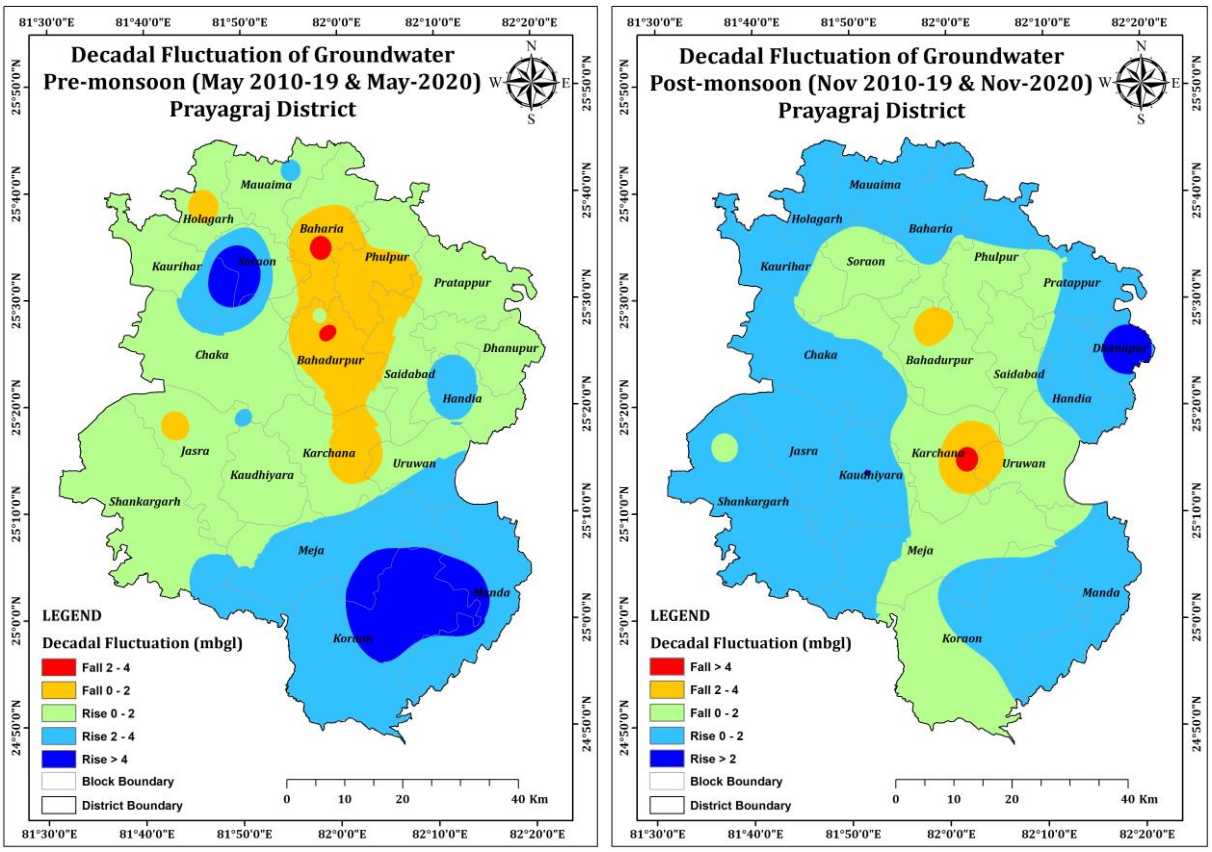


Figure 2.14: Decadal Ground water Level Fluctuation Map, Prayagraj District

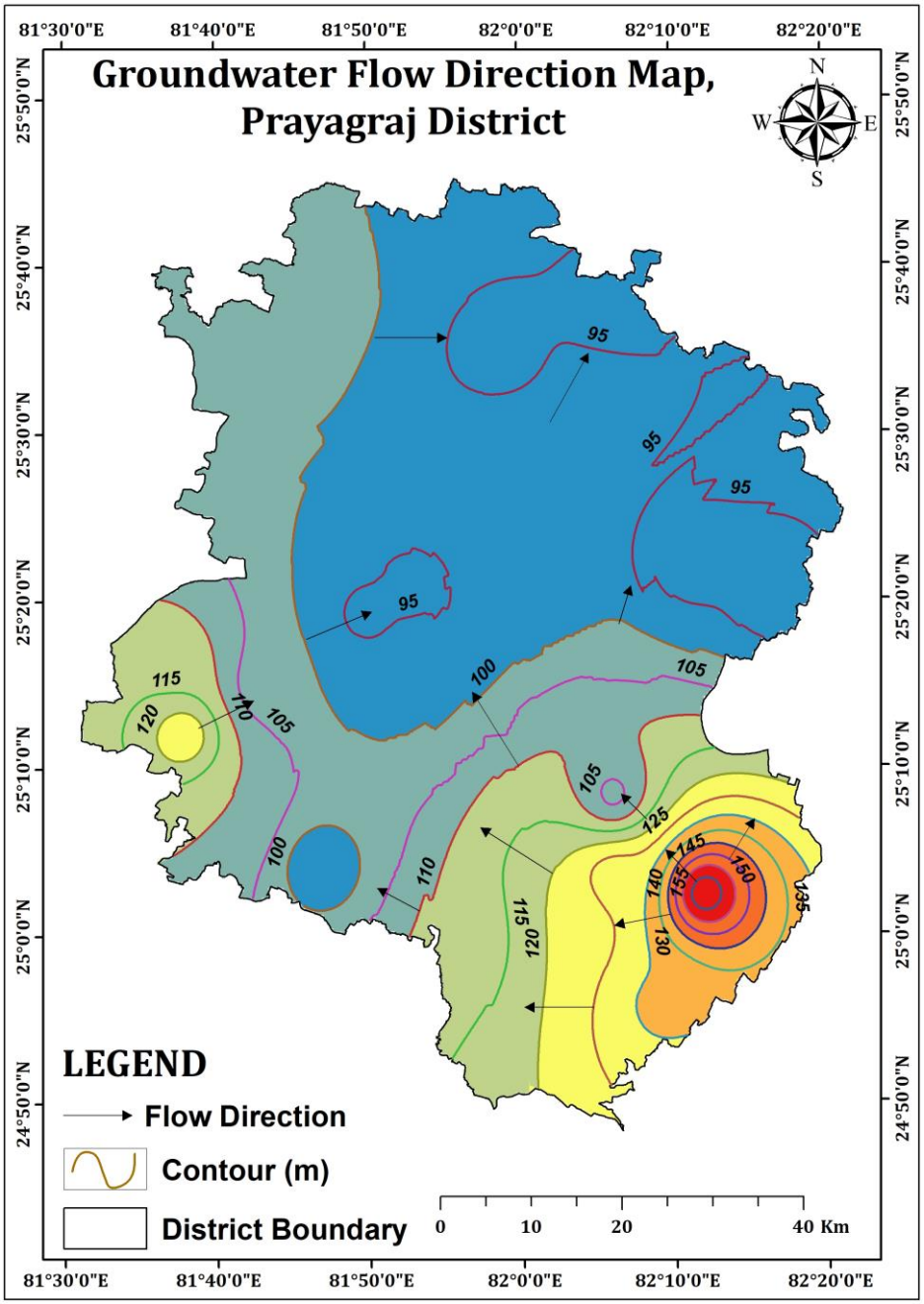


Fig.2.15.Ground water flow direction Map, Prayagraj District

2.4.7 Long-term Water Level Trend

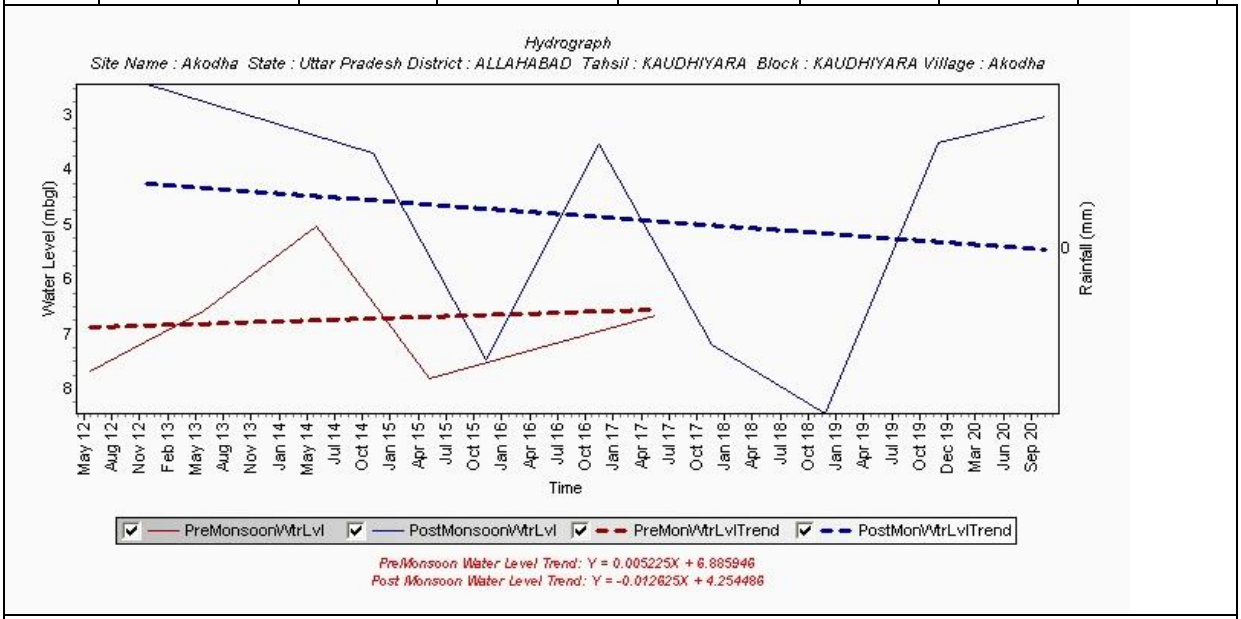
A perusal of long term water level trend of 44 observation wells for ten years (2010-2019) show that there is a rising trend of 0.012 to 0.243 m/year and the falling trend is 0.0206 to 0.3658 m/year during the pre-monsoon period while there is a rising trend of 0.0064 to

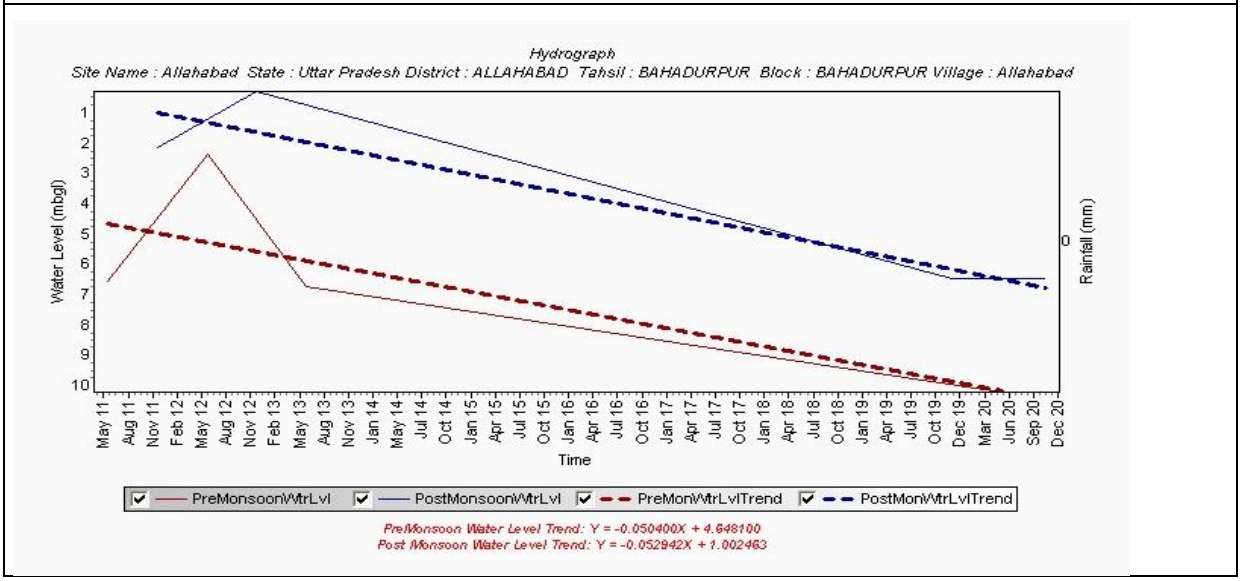
0.5156 m/year and the falling trend is 0.05 to 0.553 m/year in the post-monsoon period (Table 2.3)

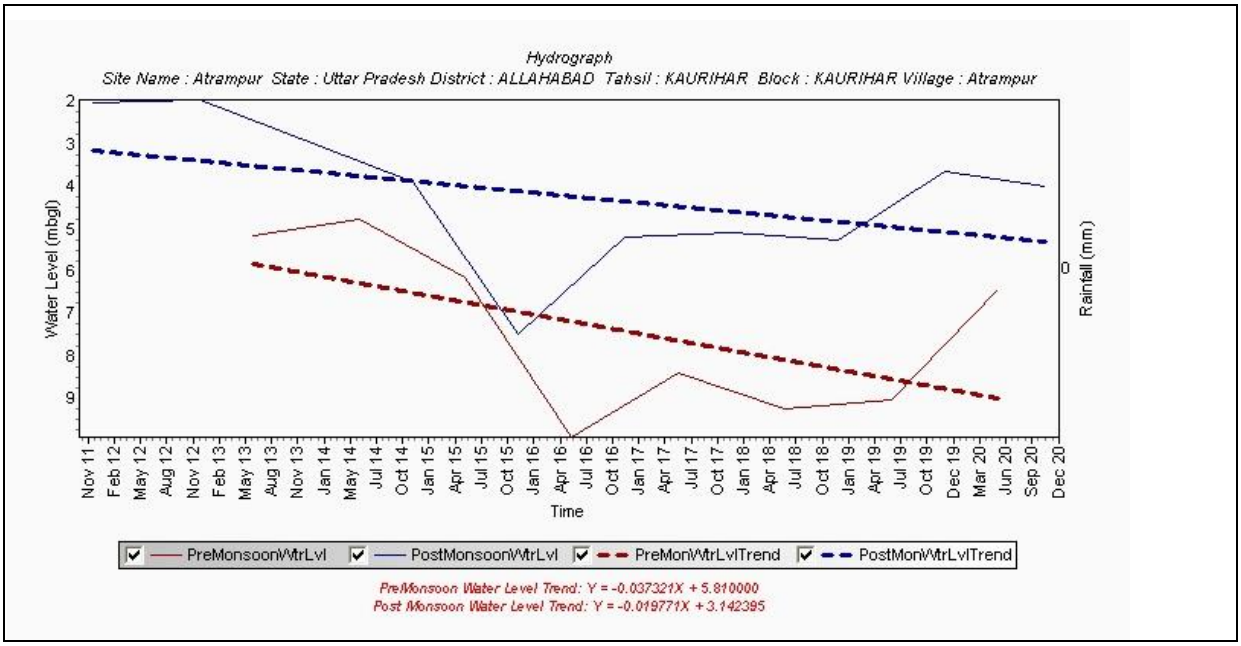
Table 2.3: Water Level Trend (2010-2019), Prayagraj District,

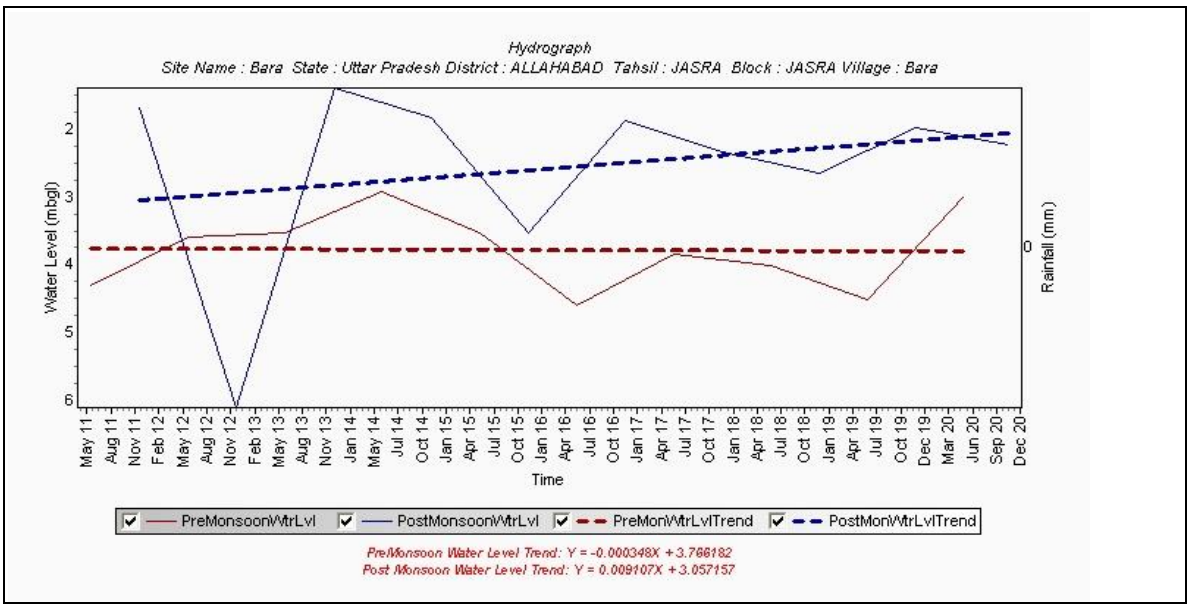
Sl No.	Location	Pre-Monsoon		Post-Monsoon		Annual	
		Rise (m/year)	Fall (m/year)	Rise (m/year)	Fall (m/year)	Rise (m/year)	Fall (m/year)
1	Seorajpur						
2	Meja	0.104		0.1686		0.1666	
3	Narayan crossin						
4	Pasna		0.0441	0.052		0.1084	
5	Barwarikalaan		0.0776	0.0614		0.0291	
6	Rampur tulapur			0.150		0.2046	
7	Naribari			0.1672		0.1922	
8	Khaptia kheri	0.103		0.0628		0.0673	
9	Ladiri bazar		0.0578		0.05	0.0165	
10	Akodha	0.186			0.292		0.2018
11	Koraon		0.0522	0.1857		0.1045	
12	Barokhar				0.91		0.7232
13	Ghoose						
14	Meja road			0.2006		0.0356	
15	Bairi		0.0475	0.0064			0.0244
16	Mau-Aima		0.0206	0.0281			0.022
17	Holagarh		0.2674		0.195		0.2674
18	Aandwa1						
19	Lalpur		0.0214				
20	Gauhania						
21	Madraha						
22	Gadwa Fort		0.0718				
23	Bida						
24	Prayagraj						
25	Hanumanganj						
26	Bikapur						
27	Baraut1	0.061		0.0287		0.0845	
28	Malka Harhar				0.553		0.4518
29	Aandwa				0.144		0.0171
30	Mugersan						
31	Atrampur		0.3658		0.146		0.2733
32	Sikandra1	0.243		0.024		0.0653	
33	Mallhan	0.012			0.131		0.0728
34	Imamganj			0.5156		0.3587	
35	Phulpur2		0.2101		0.113		0.1698
36	Handia		0.1615		0.055		0.0634
37	Kasarua		0.0052		0.335		0.0408
38	Chilla						
39	Saraon		0.1531		0.191		0.186
40	Karchhana				0.501		
41	Iradat nagar	0.492			0.255		0.1371

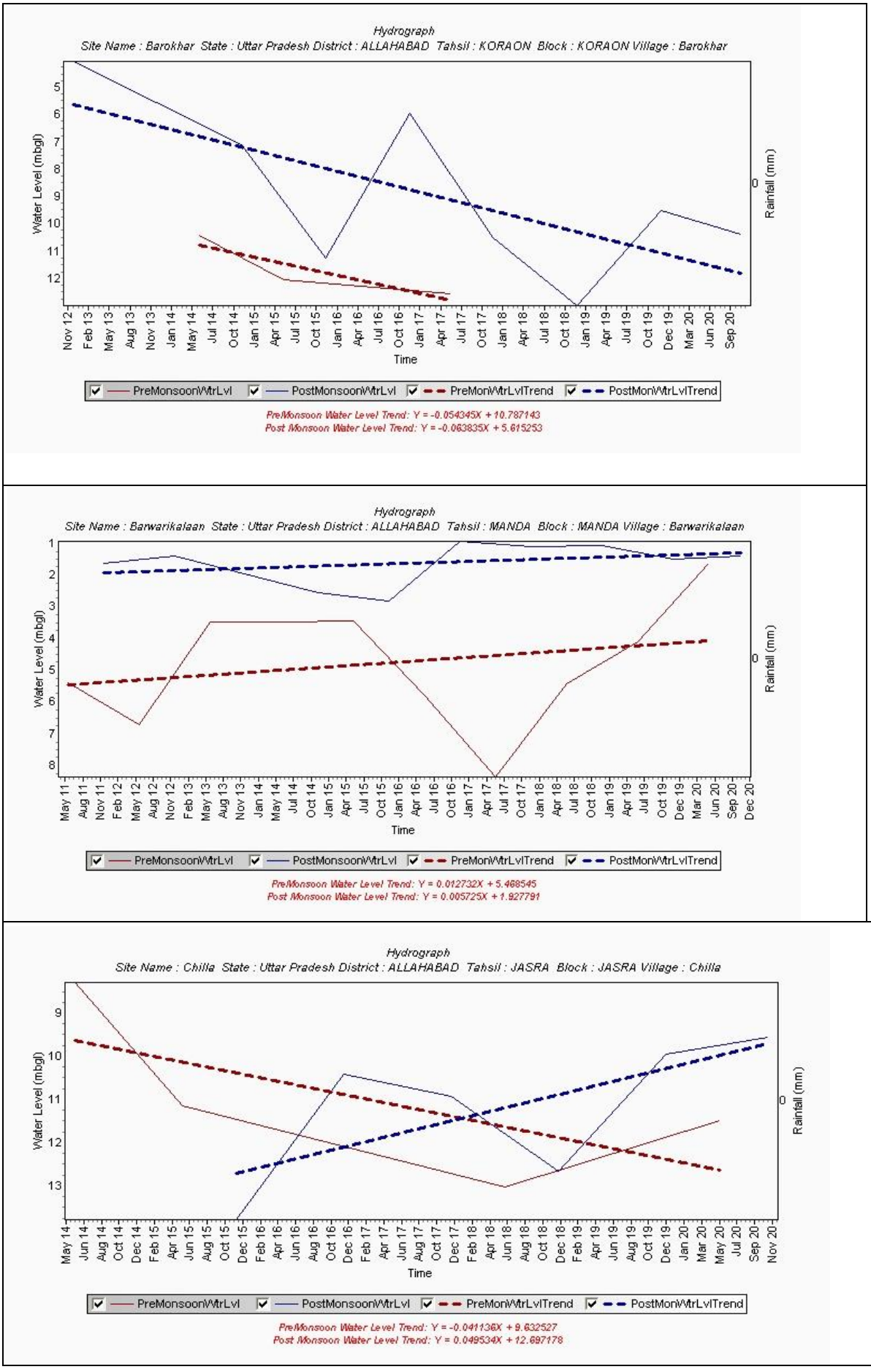
42	Jasra						
43	Bara		0.0024	0.1021		0.012	
44	Naini	0.014		0.1674			0.0406

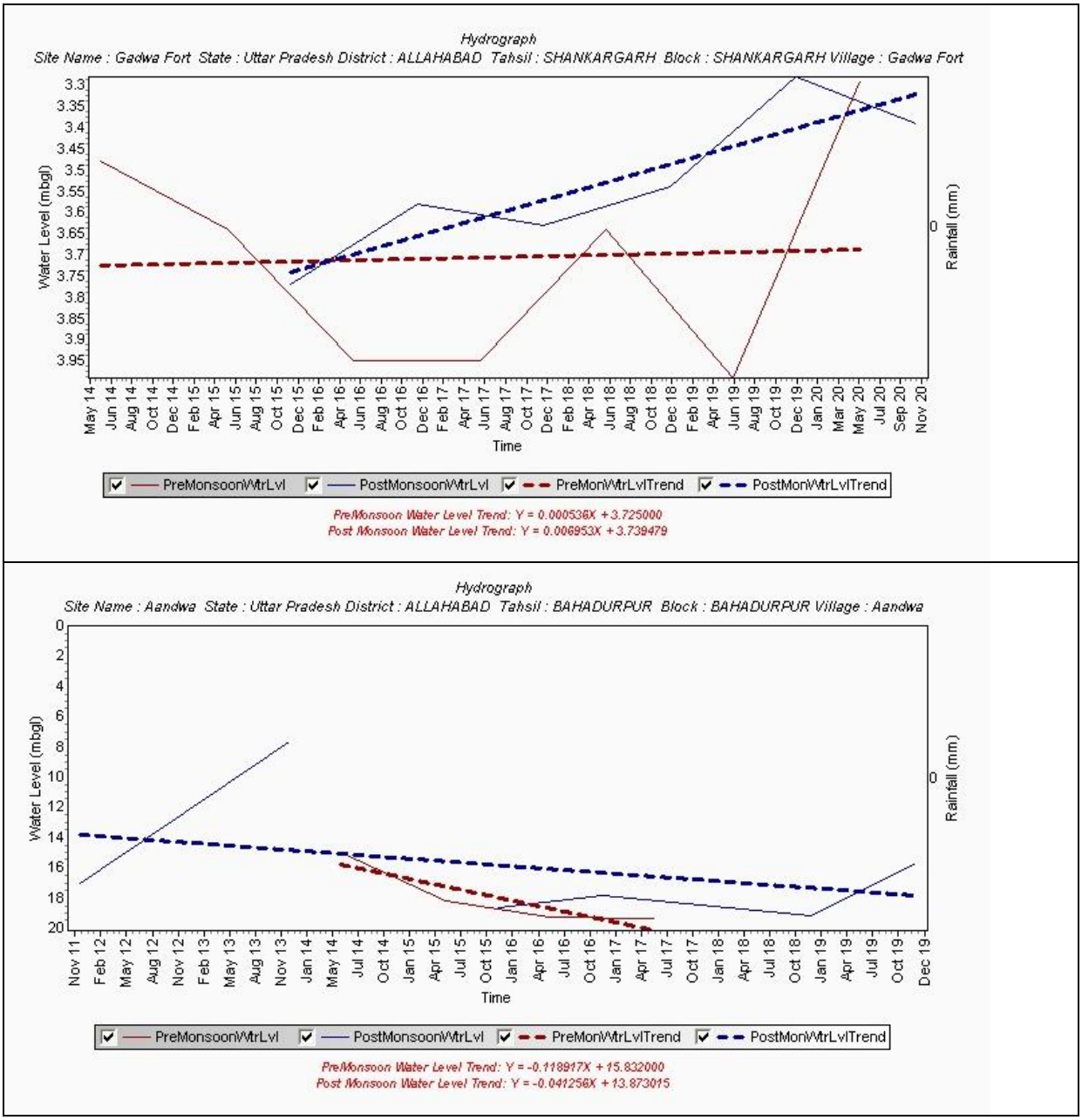


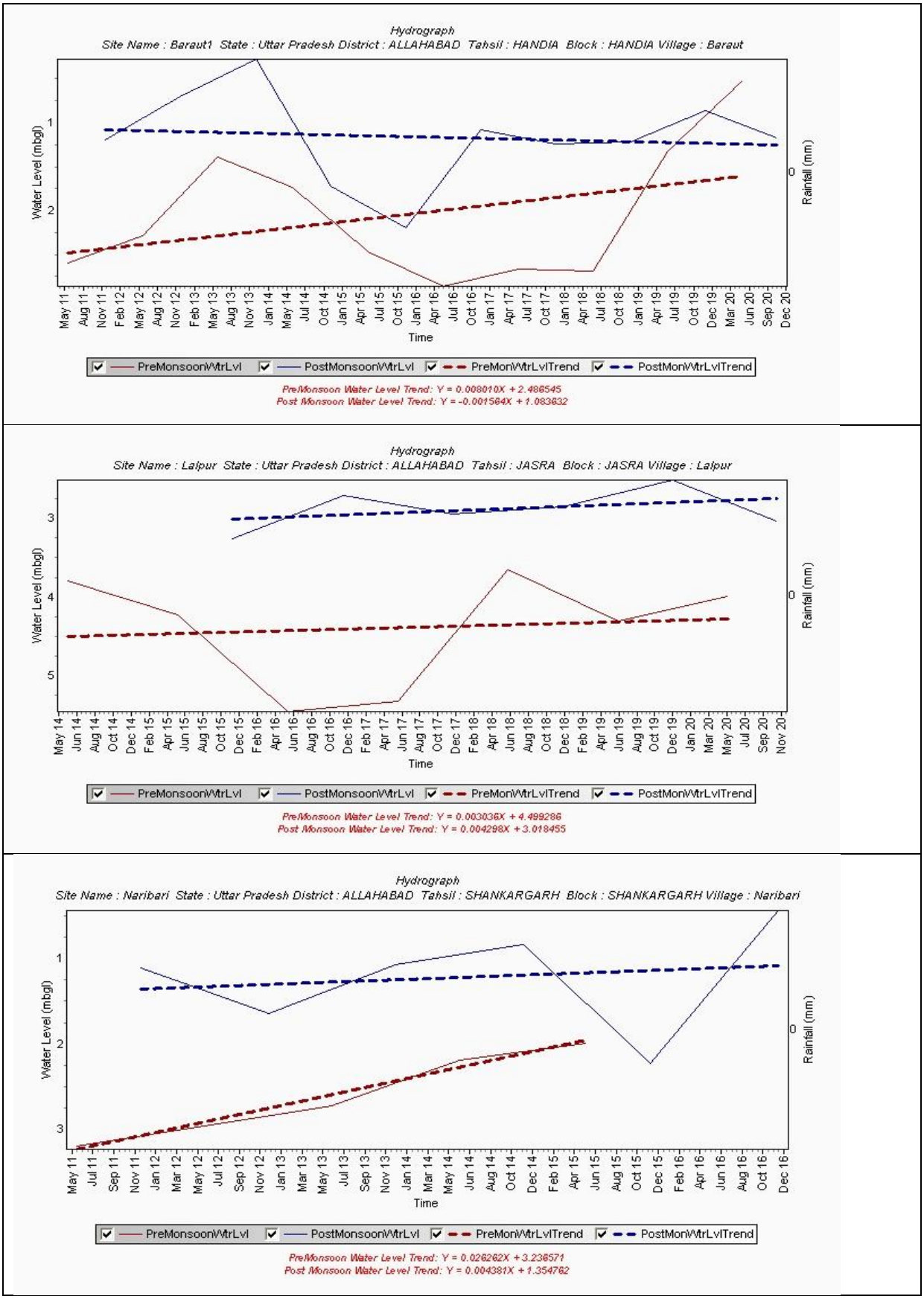


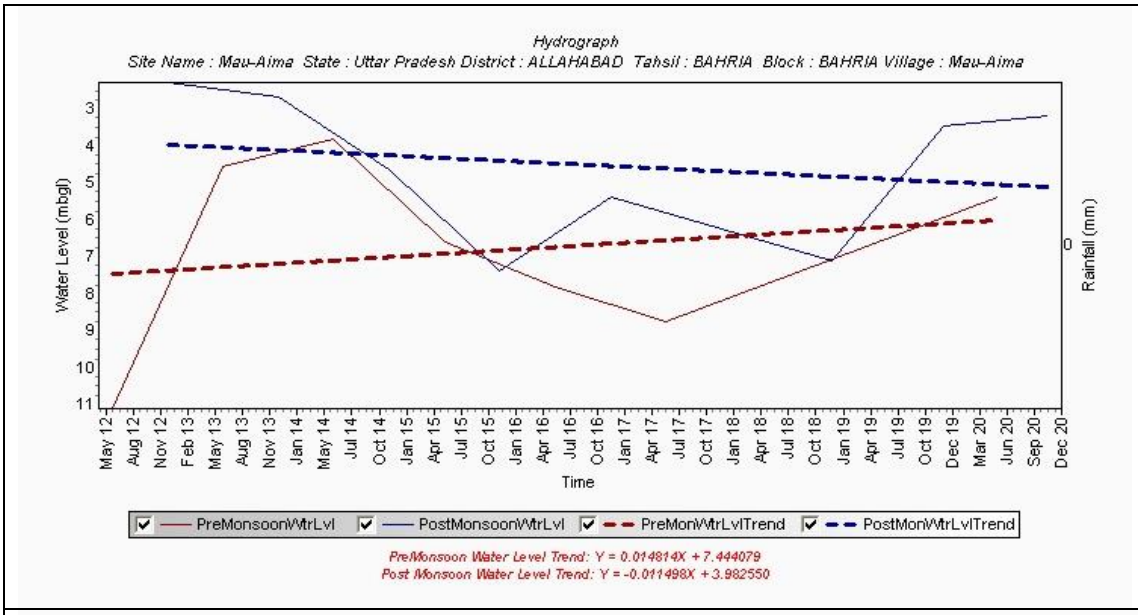

Hydrograph
Site Name : Atrampur State : Uttar Pradesh District : ALLAHABAD Tahsil : KAURIHAR Block : KAURIHAR Village : Atrampur
Water Level (mbgl)
Raintall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.037321X + 5.810000
Post Monsoon Water Level Trend: Y = -0.019771X + 3.142395

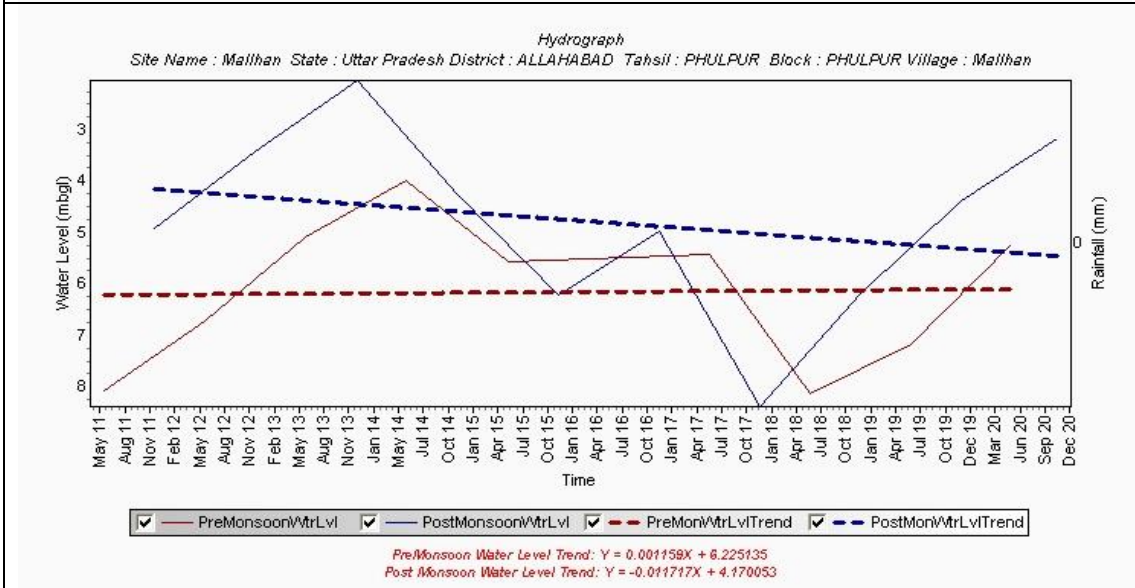

Hydrograph
Site Name : Bara State : Uttar Pradesh District : ALLAHABAD Tahsil : JASRA Block : JASRA Village : Bara
Water Level (mbgl)
Raintall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.000348X + 3.766182
Post Monsoon Water Level Trend: Y = 0.009107X + 3.057157

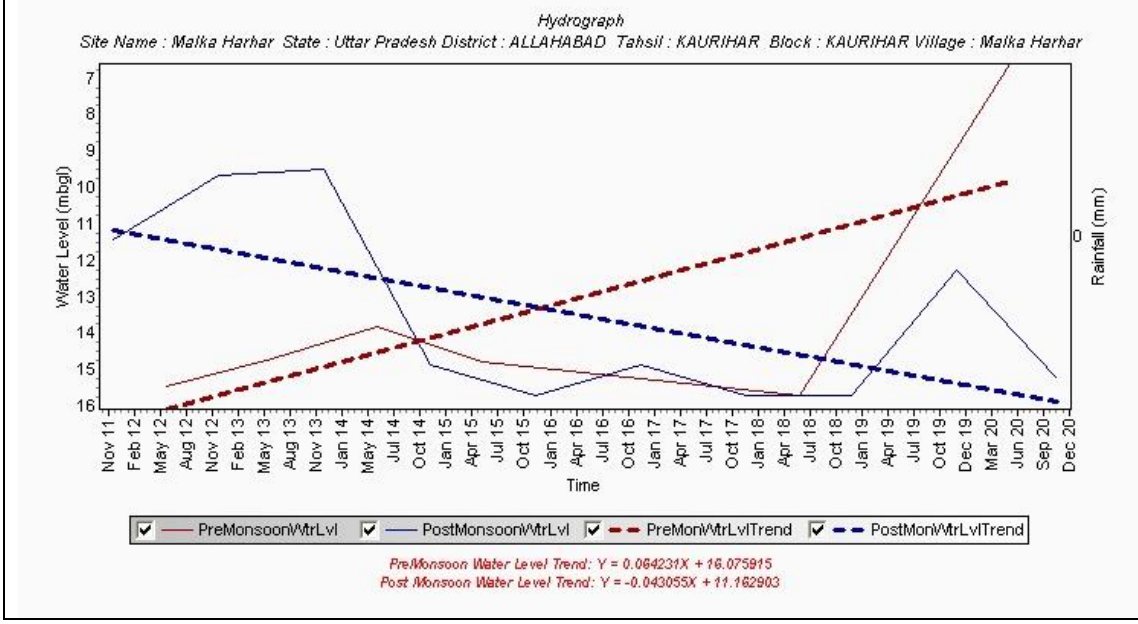

Hydrograph
Site Name : Barokhar State : Uttar Pradesh District : ALLAHABAD Tahsil : KORAON Block : KORAON Village : Barokhar
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.054345X + 10.787143
Post Monsoon Water Level Trend: Y = -0.063835X + 5.615253
Hydrograph
Site Name : Barwarikalaan State : Uttar Pradesh District : ALLAHABAD Tahsil : MANDA Block : MANDA Village : Barwarikalaan
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.012732X + 5.468545
Post Monsoon Water Level Trend: Y = 0.005725X + 1.927791
Hydrograph
Site Name : Chilla State : Uttar Pradesh District : ALLAHABAD Tahsil : JASRA Block : JASRA Village : Chilla
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.041136X + 9.632527
Post Monsoon Water Level Trend: Y = 0.049534X + 12.697178

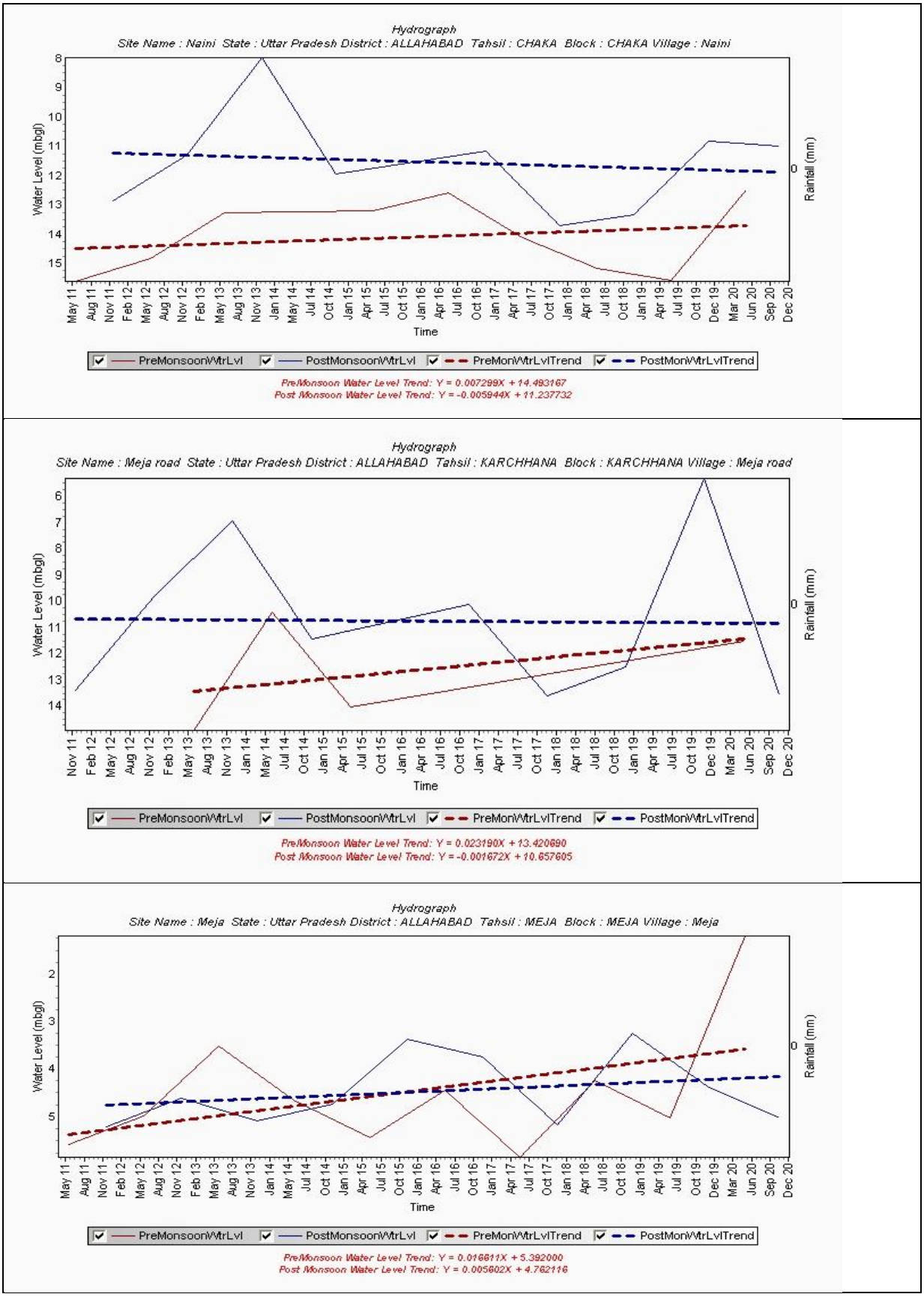

Hydrograph
Site Name : Gadwa Fort State : Uttar Pradesh District : ALLAHABAD Tahsil : SHANKARGARH Block : SHANKARGARH Village : Gadwa Fort
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.000536X + 3.725000
Post Monsoon Water Level Trend: Y = 0.006953X + 3.739479
Hydrograph
Site Name : Aandwa State : Uttar Pradesh District : ALLAHABAD Tahsil : BAHADURPUR Block : BAHADURPUR Village : Aandwa
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.118917X + 15.832000
Post Monsoon Water Level Trend: Y = -0.041256X + 13.873015

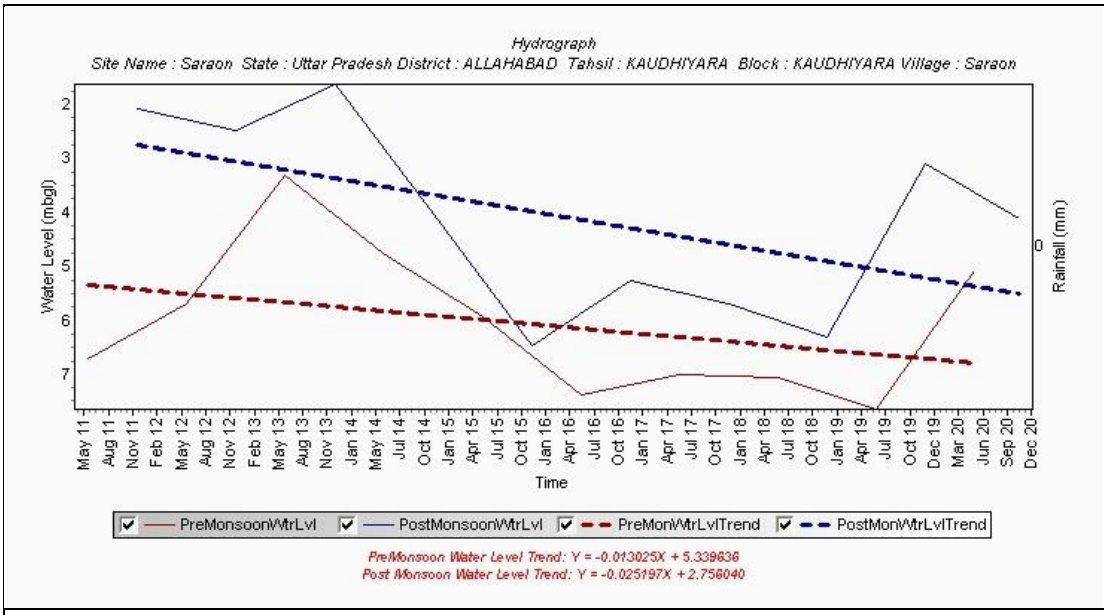

Hydrograph
Site Name : Baraut1 State : Uttar Pradesh District : ALLAHABAD Tahsil : HANDIA Block : HANDIA Village : Baraut
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.008010X + 2.486545
Post Monsoon Water Level Trend: Y = -0.001564X + 1.083632
Hydrograph
Site Name : Lalpur State : Uttar Pradesh District : ALLAHABAD Tahsil : JASRA Block : JASRA Village : Lalpur
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.003036X + 4.499286
Post Monsoon Water Level Trend: Y = 0.004298X + 3.018455
Hydrograph
Site Name : Naribari State : Uttar Pradesh District : ALLAHABAD Tahsil : SHANKARGARH Block : SHANKARGARH Village : Naribari
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.026262X + 3.236571
Post Monsoon Water Level Trend: Y = 0.004381X + 1.354762

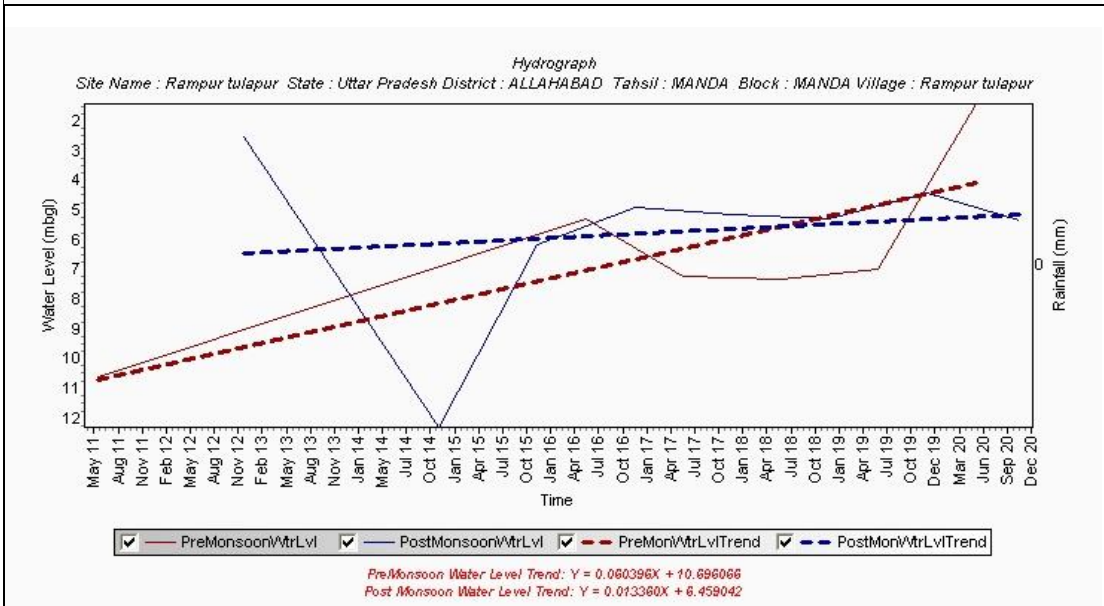

Hydrograph
Site Name : Mau-Aima State : Uttar Pradesh District : ALLAHABAD Tahsil : BAHRIA Block : BAHRIA Village : Mau-Aima
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.014814X + 7.444079
Post Monsoon Water Level Trend: Y = -0.011498X + 3.982550

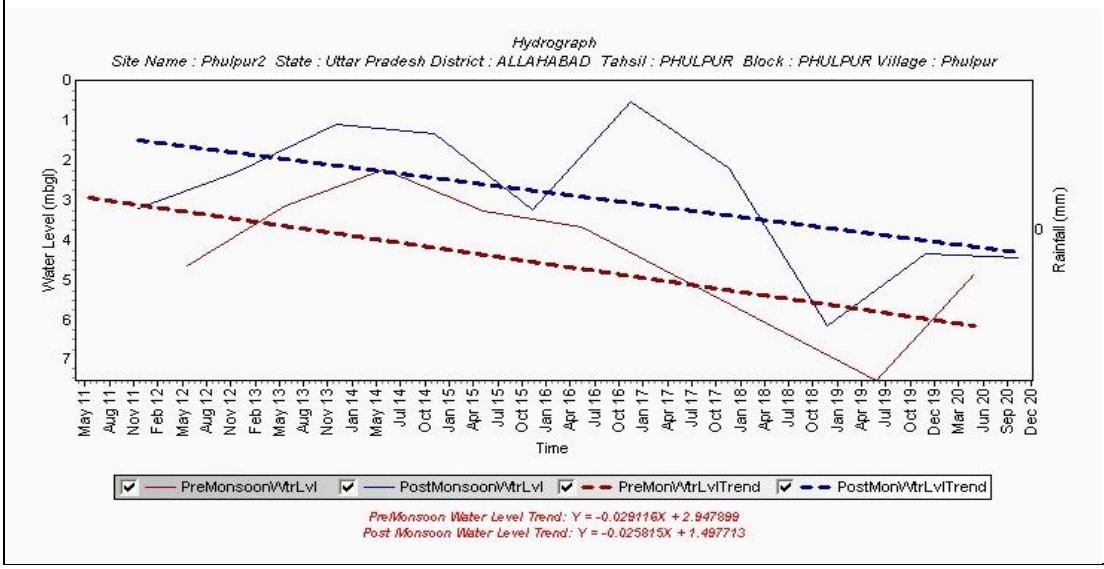

Hydrograph
Site Name : Malihan State : Uttar Pradesh District : ALLAHABAD Tahsil : PHULPUR Block : PHULPUR Village : Malihan
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.001159X + 6.225135
Post Monsoon Water Level Trend: Y = -0.011717X + 4.170053

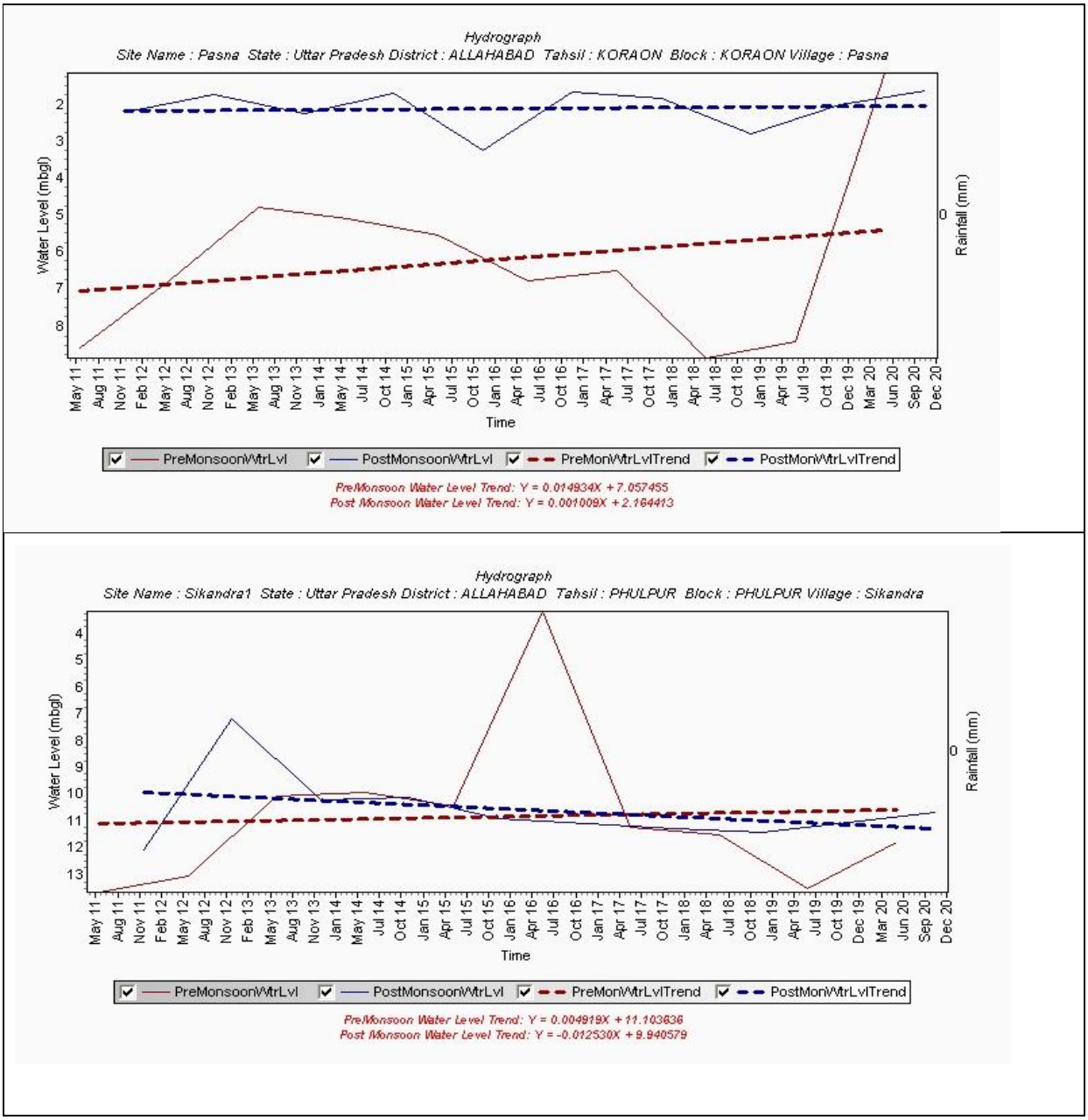

Hydrograph
Site Name : Malka Harhar State : Uttar Pradesh District : ALLAHABAD Tahsil : KAURIHAR Block : KAURIHAR Village : Malka Harhar
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.064231X + 16.075915
Post Monsoon Water Level Trend: Y = -0.043055X + 11.162903


Hydrograph
Site Name : Naini State : Uttar Pradesh District : ALLAHABAD Tahsil : CHAKA Block : CHAKA Village : Naini
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.007299X + 14.493167
Post Monsoon Water Level Trend: Y = -0.005944X + 11.237732
Hydrograph
Site Name : Meja road State : Uttar Pradesh District : ALLAHABAD Tahsil : KARCHHANA Block : KARCHHANA Village : Meja road
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.023190X + 13.420690
Post Monsoon Water Level Trend: Y = -0.001672X + 10.657605
Hydrograph
Site Name : Meja State : Uttar Pradesh District : ALLAHABAD Tahsil : MEJA Block : MEJA Village : Meja
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.016611X + 5.392000
Post Monsoon Water Level Trend: Y = 0.005602X + 4.762116


Hydrograph
Site Name : Saraon State : Uttar Pradesh District : ALLAHABAD Tahsil : KAUDHIYARA Block : KAUDHIYARA Village : Saraon
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.013025X + 5.339636
Post Monsoon Water Level Trend: Y = -0.025197X + 2.756040


Hydrograph
Site Name : Rampur tulapur State : Uttar Pradesh District : ALLAHABAD Tahsil : MANDA Block : MANDA Village : Rampur tulapur
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = 0.060396X + 10.696066
Post Monsoon Water Level Trend: Y = 0.013360X + 6.459042


Hydrograph
Site Name : Phulpur2 State : Uttar Pradesh District : ALLAHABAD Tahsil : PHULPUR Block : PHULPUR Village : Phulpur
Water Level (mbgl)
Rainfall (mm)
Time
PreMonsoonWtrLvl
PostMonsoonWtrLvl
PreMonWtrLvlTrend
PostMonWtrLvlTrend
PreMonsoon Water Level Trend: Y = -0.029116X + 2.947899
Post Monsoon Water Level Trend: Y = -0.025815X + 1.497713



2.5 GROUND WATER QUALITY

Ground water in shallow aquifers, in general, is colourless, odourless and slightly alkaline in nature. The specific electrical conductance of ground water in phreatic zone ranges from 400 to 918 µs/cm at 25ºC. Fluoride content ranges from 0.36 to 1.37 mg/l in analysed water samples. The Arsenic content ranges from 0 to 0.002 mg/l in the ground water of Prayagraj district (Table 2.4).

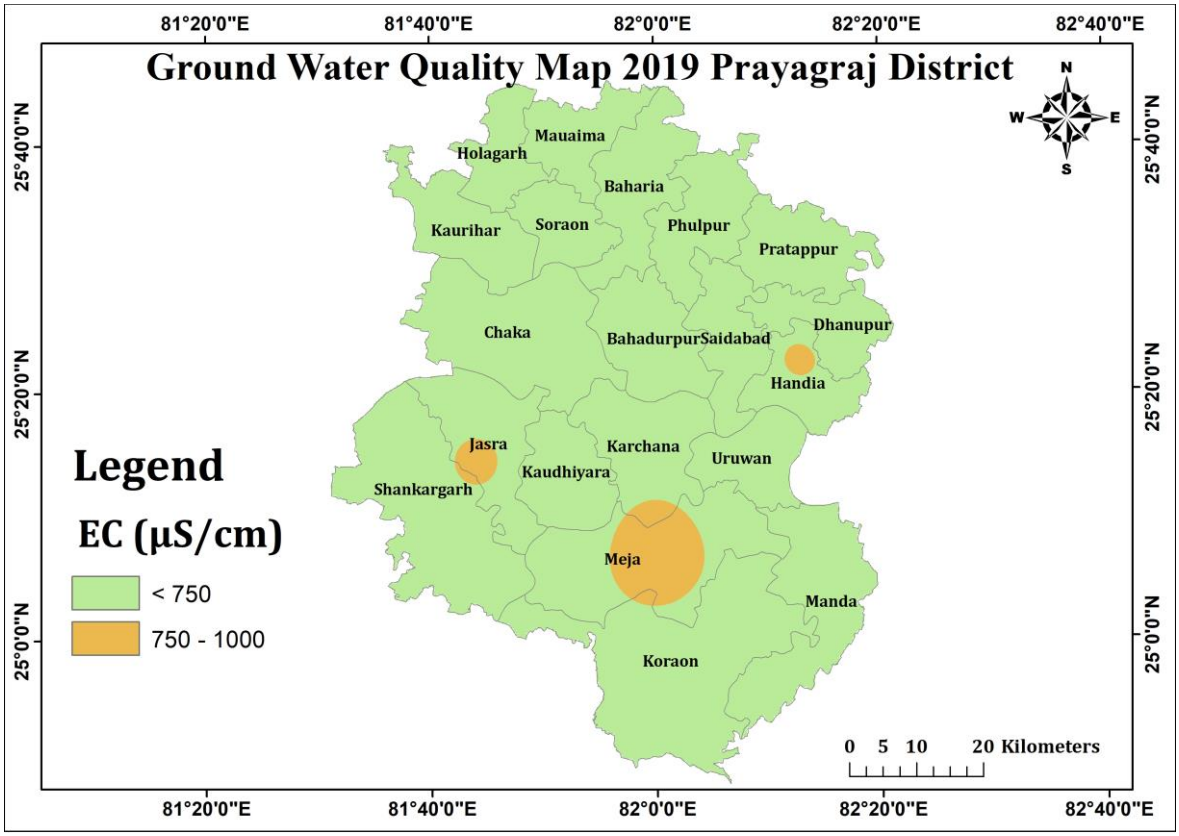


Figure 2.16: Ground Water Quality Map (2019), Prayagraj District

Table 2.4: Ground water quality parameters (2019), Prayagraj District

Sl.No.	Block	pH	EC (µs/cm)	CO3	HCO3	Cl	F	SO4	TH	Ca	Mg	Na	K	SiO2	RSC	SAR	Fe ppm	Mn ppm	Cu ppm	Zn ppm	As ppm	Pb ppm	U ppm	Cr ppm
1	Bahadurapur	7.85	450	Nil	231.8	28.4	0.239	8.731	225	40	30	9.01	1.75	35.121	-0.667105	0.262119	1.05	0.02	0.03	0.62	0	0.017	0.002	0.004
2	Baheria	7.97	664	Nil	341.6	28.4	0.77	17.44	205	20	37	68.3	4.52	31.061	1.540789	2.084429	3.84	0.05	0.01	1.32	0	0.012	0.009	0.003
3	Chaka	7.89	671	Nil	268.4	49.7	0.518	12.779	215	28	35	60.73	3.13	34.762	0.138158	1.808805	0.24	0.01	0.01	0.83	0	0.007	0.008	0.001
4	dhanpur	7.92	516	Nil	280.6	14.2	0.171	7.545	200	24	34	36	2.83	38.393	0.636842	1.111908	0.34	0.06	0.01	0.11	0	0.008	0.005	0.007
5	Handia	8.5	795	36	317.2	35.5	1.367	38.392	180	12	36	113.85	3.35	25.79	2.839474	3.709908	0	0.02	0	0	0.001	0	0.005	0.004
6	holagarh	8.09	605	Nil	341.6	14.2	0.665	7.452	195	20	35	60.16	6.29	34.506	1.738158	1.882339	0.64	0.06	0.01	0.1	0	0.02	0.009	0.003
7	Jasra	8.07	780	Nil	280.6	99.4	0.697	24.318	285	28	52	64.19	1.85	33.409	-1.043421	1.661435	0.24	0.01	0	0.11	0	0.006	0.008	0.001
8	Karchhana	7.92	672	Nil	341.6	35.5	0.475	12.593	260	64	24	37.54	4.18	27.072	0.426316	1.014802	2.38	0.15	0.02	2.9	0	0.019	0.004	0.021
9	kaundhiana	7.97	635	Nil	329.4	31.95	0.428	11.259	245	24	44	46.85	2.29	33.214	0.548684	1.307877	0.6	0.01	0	0.17	0	0.01	0.004	0.001
10	Kaurihar	8.02	627	Nil	329.4	28.4	0.429	11.763	230	16	46	50.84	3.43	35.019	0.85	1.465504	0.42	0.02	0.01	0.13	0.001	0.009	0.013	0.007
11	Koraon	8.03	584	Nil	305	28.4	0.257	7.886	225	24	40	39.99	2.08	32.876	0.543421	1.164764	0.58	0.06	0	0.2	0.002	0.013	0.003	0.058
12	Manda	7.78	918	Nil	183	106.5	0.169	21.231	360	68	46	51.39	1.72	35.439	-4.15	1.181715	0.81	0.05	0.03	0.41	0	0.02	0.001	0.033
13	Mau- aima	8.23	693	Nil	427	14.2	0.456	5.978	130	12	24	114.15	2.03	24.252	4.426316	4.375074	0.49	0.01	0	0.11	0.001	0.011	0.011	0.007
14	Meja	7.74	908	Nil	237.9	120.7	0.213	42.525	265	64	25	89.91	0	27.964	-1.372368	2.407644	0.7	0.02	0.01	1.19	0	0.013	0.008	0.029
15	Phoolpur	8.04	742	Nil	378.2	35.5	0.732	20.743	250	20	48	72.3	3.66	29.861	1.252632	1.998657	0.71	0.03	0.01	0.48	0	0.013	0.027	0.014
16	Pratappur	8.17	736	Nil	402.6	21.3	0.434	14.694	270	16	55	58.26	8.16	34.034	1.260526	1.550274	1.51	0.02	0.01	0.62	0	0.016	0.016	0.009
17	Saidabad	8.17	695	Nil	402.6	14.2	1.344	0	140	24	19	103.04	2.79	31.102	3.821053	3.800606	4.8	0.1	0.03	0.17	0.001	0.015	0.008	1.564
18	Shankargarh	7.94	450	Nil	183	49.7	0.36	18.331	145	40	11	44.46	6.06	19.688	0.111842	1.608593	3.76	0.23	0.11	0.76	0.002	0.018	0.001	0.712
19	Soraon	8.2	652	Nil	305	42.6	0.805	16.005	210	24	36	66.84	4.34	29.461	0.839474	2.014881	1.67	0.04	0.02	0.19	0.001	0.016	0.012	0.071
20	Urwa	7.9	400	Nil	207.4	14.2	0.332	7.212	120	32	10	43.24	2.34	33.163	1.010526	1.719973	0.49	0.14	0.01	0.36	0	0.01	0.001	0.064

CHAPTER-03

GROUND WATER RESOURCES

Ground water resources have been computed jointly by Central Ground Water Board and Ground Water Department, Govt. of U.P. as on 31st March 2020 according to the methodology recommended by the Ground Water Estimation Committee constituted by Government of India (GEC 2015).

3.1 DYNAMIC GROUND WATER RESOURCES

Dynamic Ground Water Resource Estimation as on 31.3.2017 has been carried out with administrative blocks as ground water assessment units by GWD & CGWB. Due to lack of data on command, non-command and poor ground water quality areas, the resources could not be estimated separately and the administrative unit (block) as a whole without sub-dividing it into sub-units has been considered. The precise estimation of ground water reserves and irrigation potential is prerequisite for proper planning and execution for socio-economic development in the area. The ground water recharge has been estimated on the basis of water level fluctuation method which has been evaluated by significant change in water level during pre and post monsoon periods. Since, the area is comprised of both alluvium and hard rock, the specific yield has been considered 16% and rainfall infiltration is 25% for alluvium formation as well as 3% (specific yield) and 12% (rainfall infiltration) for hard rock formation (GEC-1997).

3.1.1 GROUND WATER DRAFT

The ground water draft is the quantity of water withdrawn from ground water reservoirs. The principal ground water development structures for utilization of ground water in the district are open wells, dug cum borewells, private tubewells/government constructed under minor irrigation works and by other state government departments. On the basis of statistical data available on the number of various ground water structures, the block wise annual gross draft has been computed by multiplying its average discharge and annual working hours. The total draft from the district is 100192.6797 Ham. Maximum ground water draft for all uses is 8338.69 Ham in Baharia block and minimum draft of ground water for all uses is 2945 Ham in Soraon block.

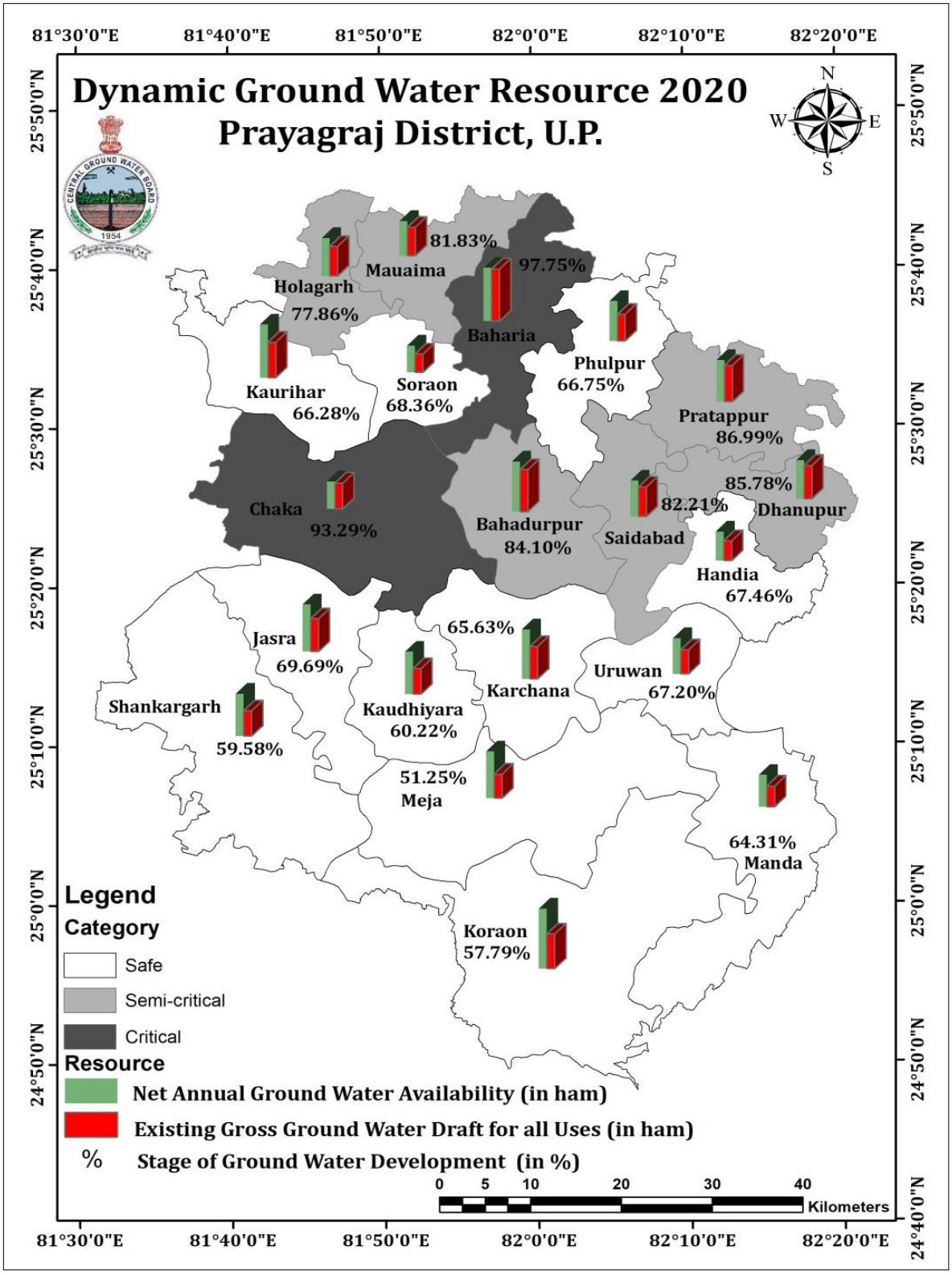


Figure 3.1: Dynamic Ground Water Resource-2020

3.1.2 STAGE OF GROUND WATER DEVELOPMENT AND CATEGORIZATION OF BLOCKS

The level of ground water development in Prayagraj district has been worked out for each block as the ratio of gross annual draft to net ground water availability. The distributions of blocks in different categories are shown in the Figure 3.1. Prayagraj City is in Over-Exploited Category. Baharia and Chaka blocks are in Critical Category. There are Six blocks Bahadurpur, Dhanupur, Holagarh, Mauaima, Pratappur, Saidabad falls under Semi-critical category whereas rest blocks fall under safe category.

Table 3.1: Dynamic Ground Water Resource-2020 of Prayagraj District

Sl. No	Assessment Unit Name	Total Area of Assessment Unit (Ha)	Recharge Worthy Area(Ha)	Recharge from Rainfall-Monsoon Season	Recharge from Other Sources-Monsoon Season	Recharge from Rainfall-Non Monsoon Season	Recharge from Other Sources-Non Monsoon Season	Total Annual Ground Water (Ham) Recharge	Total Natural Discharges (Ham)	Annual Extractable Ground Water Resource (Ham)	Ground Water Extraction for Irrigation Use (Ham)	Ground Water Extraction for Industrial Use (Ham)	Ground Water Extraction for Domestic Use (Ham)	Total Extraction (Ham)	Annual GW Allocation for for Domestic Use as on 2025 (Ham)	Net Ground Water Availability for future use (Ham)	Stage of Ground Water Extraction (%)	Categorization (Over-Exploited/ Critical/Semicritical/Safe/Saline)
1	Bahadurpur	26059	26059	4884.01	1513.2	101.88	1975.18	8474.27	423.71	8050.56	5911.2	0	859.27059	6770.47	962.23	1177.13	84.10	Semi-critical
2	Baharia	24875	24875	4707.69	1856.8	97.25	2816.64	9478.38	947.84	8530.54	7634.2	0	704.48577	8338.69	790.97	105.38	97.75	critical
3	Chaka	13575	13575	2569.12	984.67	53.07	1286.94	4893.8	489.38	4404.42	3528.55	0	580.47702	4109.03	591.99	283.88	93.29	critical
4	Dhanupur	17322	17322	2833.45	1546.44	67.72	2102.67	6550.28	327.51	6222.77	4728.81	0	608.96673	5337.78	691.66	802.3	85.78	Semi-critical
5	Handia	17052	17052	3227.15	794.24	66.66	1102.11	5190.16	519.02	4671.14	2529.91	0	621.11466	3151.02	712.28	1428.95	67.46	safe
6	Holagarh	14846	14846	2809.66	1564.32	58.04	2312.88	6744.9	674.49	6070.41	4259.88	0	466.34298	4726.22	518.76	1291.77	77.86	Semi-critical
7	Jasra	26822	26822	5076.16	1354.14	104.86	1893.76	8428.92	842.89	7586.03	4829.07	0	457.46253	5286.53	513.5	2243.46	69.69	safe
8	Karchana	23281	23281	4369.52	1655.1	91.01	2249.97	8365.6	418.28	7947.32	4609.2	0	606.44166	5215.64	679.03	2659.09	65.63	safe
9	Kaudhiyara	20046	20046	2953.7	1718.95	78.37	2445.99	7197.01	359.85	6837.16	3680.4	0	436.71447	4117.11	514.57	2642.18	60.22	safe
10	Kaurihar	41414	41414	6120.07	1027.07	161.9	1680.59	8989.63	449.48	8540.15	4691.61	0	969.00054	5660.61	1067.83	2780.7	66.28	safe
11	Koraon	72642	63842	7688.76	1200.17	158.83	1680.89	10728.65	1072.86	9655.79	4996.88	0	583.16634	5580.05	685.68	3973.23	57.79	safe
12	Manda	34943	20673	1659.82	1617.5	51.43	2365.25	5694	569.4	5124.6	2745.88	0	549.51261	3295.39	634.96	1743.76	64.31	safe
13	Mauaima	15560	15560	2944.79	1299.14	60.83	1890.11	6194.87	619.48	5575.39	4036.21	0	526.20663	4562.42	607.23	931.95	81.83	Semi-critical
14	Meja	44421	34201	3412.29	1972.58	85.09	2481	7950.96	397.54	7553.42	3198.17	0	673.20819	3871.38	775.07	3580.18	51.25	safe
15	Phulpur	23154	23154	4381.98	1033.94	90.52	1641.25	7147.69	714.77	6432.92	3723.32	0	570.33513	4293.66	631.63	2077.97	66.75	safe

16	Pratappur	21101	21101	3993.44	1354.65	82.49	1986.85	7417.43	741.74	6675.69	5152	0	655.12974	5807.13	736.72	786.97	86.99	Semi-critical
17	Prayagraj City	7005	7005	397.72	792.63	8.22	1618.28	2816.85	281.68	2535.17	0	0	4455.924439	4455.92	4831.27	0	175.76	Over-exploited
18	Saidabad	19142	19142	3465.89	1029.39	74.83	1531.52	6101.63	305.08	5796.55	4275.4	0	489.73656	4765.14	552.69	968.46	82.21	Semi-critical
19	Shankargarh	47208	33128	1519.9	2643.43	47.09	3361.49	7571.91	757.19	6814.72	3531.03	0	529.4106	4060.44	597.17	2686.52	59.58	safe
20	Soraon	14775	14775	2796.22	838.37	57.76	1094.63	4786.98	478.7	4308.28	2409.37	0	535.63239	2945.00	582.31	1316.6	68.36	safe
21	Uruwan	16890	16890	3196.49	1283.84	66.03	1808.1	6354.46	635.44	5719.02	3305.6	0	537.45009	3843.05	614.23	1799.19	67.20	safe

CHAPTER-04

GROUND WATER RELATED ISSUES

The development of ground water is 74.2% in district. The trend analysis of historical ground water level data indicates long term fall in most of the wells in the district.

4.1 GROUND WATER RESOURCES

4.1.1 AREA OF INTENSIVE GROUND WATER DEVELOPMENT

The stage of ground water development of the district is 74.2%. Prayagraj City is in Over-Exploited Category. Baharia and Chaka blocks are in Critical Category. There are Six blocks Bahadurpur, Dhanupur, Holagarh, Mauaima, Pratappur, Saidabad falls under Semi-critical category whereas rest blocks fall under safe category as shown in the Figure3.1.

The over exploited and critical blocks where ground water draft has increased many folds during past decades is a biggest challenge because these are the blocks where in storage ground water resources is depleting very fast. There is a probability for adjacent block which may also be comes in water stressed category in future if proper effective intervention is not taken by stack holder. The change in climatic pattern is also exerting withdrawal pressure on groundwater resource creating imbalance in groundwater resources.

4.2 WATER TABLE DEPLETED AREA

A perusal of the table shows that there is a rising trend of 0.012 to 0.243 m/year and the falling trend is 0.0206 to 0.3658 m/year during the pre-monsoon period while in the post-monsoon period rising trend is 0.0064 to 0.5156 m/year and falling trend is 0.05 to 0.553 m/year as shown in Table 2.3.

4.3 GROUND WATER RELATED ISSUES AND PROBLEMS

- Long term ground water level is showing declining trend at most of the places.
- Prayagraj City falls under Over-Exploited category
- Two blocks (Bahria & Chaka) under Critical category
- Six blocks under semi –critical category

4.4 PROBLEMS

- Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability
- Non-availability of Surface Water for irrigation.
- High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
- Southern part of the District overlain by hard rock area. Ground water occurs in secondary porosity (joint and fractures) .Yield of well is also low only to fulfil the drinking water need. Dug wells sustains in weathered zone as compare to northern part of the district.

CHAPTER- 05

GROUND WATER DEVELOPMENT AND MANAGEMENT STRATEGIES

5.1 GROUND WATER MANAGEMENT STRATEGY

To arrest the further decline in ground water levels and depletion of ground water resources, there is urgent need to implement both Supply side and Demand side measures which includes artificial recharge and water conservation, On-farm activities and adoption of water use efficiency measures.

5.2 GROUND WATER DEVELOPMENT

The development of groundwater resources is increasing over the years in order to meet drinking water, industrial and irrigation requirements. The stage of ground water development had been changed to 69.87 % in 2017 and 74.2 in 2020. As on 31.03.2020, it shows that block wise level of development of ground water potential in Prayagraj district varies from 51.25% in Meja to 175.76% in 'Prayagraj City'. It is observed area irrigated by Ground water was 148444 ha in 1993-94 and steadily increase to 158272ha in 2017-18. The number ground water structures are 9694 in 1994-95 and 51977 in 2017-18. With growing needs, there is an urgent need for scientific approach for proper management of the available ground water resources for sustainability of this precious natural resource without having any adverse effect on the environment. At those places where water level is gradually going down we should restrict or minimized the ground water usages for irrigation needs. Planned ground water development is possible only when the availability of the ground water potential is precisely quantified and also the demand for various uses is properly estimated for the projected development scenario of urban as well as rural area for next 25 years.

5.3 GROUND WATER MANAGEMENT OPTIONS

5.3.1 Supply side Interventions

It is proposed to adopt supply side management options only in the over-exploited and critical blocks. There is considerable scope for implementation of roof top rain water harvesting in the urban areas of the district. Check dams, cement plugs, renovation of ponds are ideal structures for rain water harvesting in rural areas. It is proposed to construct 59 Check dams, 59 nala bunds, 60 ponds and 59 stream channels.

- Construction of check dams/nala bunds

- Revival and renovation of ponds
- On farm activities like laser leveling, bench terracing, construction of farm ponds, plantation of forests etc.
- Government Irrigation and Water Supply tube-well should tap 2nd Aquifer; however, caution is to be applied to keep piezometric head maintained.
- Scope of supply side interventions is limited due to topography, land availability and also less availability of surplus water

5.3.2 Demand side Interventions

Agriculture is the major consumer of ground water. In the district, about 66% irrigation is dependent on ground water. Even in the canal command areas, enough ground water is being used to irrigate the fields. In the major parts of area, flow irrigation is being used. There is urgent need to promote piped and pressurized irrigation practices which can save 25 to 70% of water use in the agriculture. It is proposed to initiate these measures initially in 10% area of each of the over-exploited and critical blocks. It is estimated that there may be saving of 15.60 MCM of water with these measures. Such practices have the potential of saving 35-40% irrigation water thereby drastically reducing the draft for irrigation leading the change of category of block from OE to safe.

The measures adopted for supply side and demand side management in Prayagraj district will substantially bring down stage of ground water development.

- Water use efficiency through piped and pressurised irrigation (drip & Sprinkler)
- Furrow irrigation with raised bed planting in wide row crops should be practised.
- Irrigation in checks in close row crops should be practised
- Measures for reducing Evapo-transpiration losses etc.
- Diversification of cropping pattern.
- Most effective option to reduce ground water withdrawal by 50-60% specially by adopting Sprinkler and Drip Irrigation practices

5.3.3 Micro Irrigation (Sprinkler/ Drip Irrigation)

Micro irrigation is the frequent application of small quantities of water on, above or through water directly at the root zone of the plant in a uniform and effective way.

Advantages

- Water Use Efficiency can be improved from 50-60 % to 90-95%.
- The consumption of fertilizers can be reduced by 30 %.
- Weed infestation is reduced considerably.
- Can be used on undulating topography & on soils having low infiltration rates.

Table 5.1 Unit Draft, Prayagraj District

Sl No	Block	Draft For Irrigation (from Resource)	Net Irrigated Area (From Statistical Diary)	Gross Irrigated Area	Unit Draft (Total Draft/ Ne)	Ratio (Gross/ Net)
1	BAHADURPUR	5911.20	11229	20012	0.53	1.78
2	BAHARIA	7634.20	18126	33732	0.42	1.86
3	CHAKA	3528.55	5242	7455	0.67	1.42
4	DHANUPUR	4728.81	12479	23433	0.38	1.88
5	HANDIA	2529.91	9764	17665	0.26	1.81
6	HOLAGARH	4259.88	12082	20844	0.35	1.73
7	JASRA	4829.07	18201	30286	0.27	1.66
8	KARCHANA	4609.20	15825	23502	0.29	1.49
9	KAUDHIYARA	3680.40	15637	23699	0.24	1.52
10	KAURIHAR	4691.61	17565	27798	0.27	1.58
11	KORAON	4996.88	45082	66063	0.11	1.47
12	MANDA	2745.88	13774	21323	0.20	1.55
13	MAUAIMA	4036.21	12712	21721	0.32	1.71
14	MEJA	3198.17	24075	35873	0.13	1.49
15	PHULPUR	3723.32	11018	21451	0.34	1.95
16	PRATAPPUR	5152.00	13262	24635	0.39	1.86
17	SAIDABAD	4275.40	11724	20601	0.36	1.76
18	SHANKARGARH	3531.03	22279	37956	0.16	1.70
19	SORAON	2409.37	9576	16437	0.25	1.72
20	URUWAN	3305.60	9646	15855	0.34	1.64

Table 5.2: Projected Stage of Development (%) After Interventions, Prayagraj District

Sl No	Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
1	**BAHADURPUR**	3	3	3	3	1123	1123	0.21	0.21	5.27	0.21	5.48	84.10	77.10
2	**BAHARIA**	4	4	4	5	2719	2719	0.28	0.28	10.66	0.28	10.93	97.75	84.66
3	**CHAKA**	2	2	2	3	786	786	0.15	0.15	3.76	0.15	3.91	93.29	84.12
4	**DHANUPUR**	2	2	2	2	1248	1248	0.14	0.14	4.44	0.14	4.58	85.78	78.25
5	**HANDIA**	2	2	2	2	488	488	0.11	0.11	1.14	0.11	1.25	67.46	64.62
6	**HOLAGARH**	2	2	2	2	1208	1208	0.12	0.12	3.67	0.12	3.79	77.86	71.47
7	**JASRA**	3	3	3	3	910	910	0.17	0.17	2.01	0.17	2.18	69.69	66.66
8	**KARCHANA**	2	2	2	2	791	791	0.15	0.15	1.71	0.15	1.86	65.63	63.17
9	**KAUDHIYARA**	2	2	2	2	782	782	0.13	0.13	1.39	0.13	1.52	60.22	57.88
10	**KAURIHAR**	4	4	4	4	878	878	0.26	0.26	1.86	0.26	2.12	66.28	63.60
11	**KORAON**	7	7	7	7	2254	2254	0.46	0.46	1.83	0.46	2.29	57.79	55.15
12	**MANDA**	3	3	3	3	689	689	0.22	0.22	1.06	0.22	1.29	64.31	61.53
13	**MAUAIMA**	2	2	2	2	1271	1271	0.12	0.12	3.45	0.12	3.57	81.83	75.26
14	**MEJA**	4	4	4	4	1204	1204	0.28	0.28	1.19	0.28	1.47	51.25	49.12
15	**PHULPUR**	2	2	2	2	551	551	0.15	0.15	1.81	0.15	1.96	66.75	63.55
16	**PRATAPPUR**	3	3	3	3	1326	1326	0.17	0.17	4.79	0.17	4.95	86.99	79.37
17	**SAIDABAD**	2	2	2	2	1172	1172	0.15	0.15	3.76	0.15	3.91	82.21	75.26
18	**SHANKARGARH**	5	5	5	5	1114	1114	0.30	0.30	1.50	0.30	1.80	59.58	56.68
19	**SORAON**	1	1	1	1	479	479	0.09	0.09	1.03	0.09	1.13	68.36	65.59
20	**URUWAN**	2	2	2	2	482	482	0.11	0.11	1.36	0.11	1.47	67.20	64.51

Table 5.3: Projected Stage of Extraction (%) After Interventions, Prayagraj District

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
BAHADURPUR	260.59	80.51	67.70	84.10	0.21	5.48	80.71	62.23	**77.10**
BAHARIA	248.75	85.31	83.39	97.75	0.28	10.93	85.58	72.45	**84.66**
CHAKA	135.75	44.04	41.09	93.29	0.15	3.91	44.20	37.18	**84.12**
DHANUPUR	173.22	62.23	53.38	85.78	0.14	4.58	62.37	48.80	**78.25**
HANDIA	170.52	46.71	31.51	67.46	0.11	1.25	46.82	30.26	**64.62**
HOLAGARH	148.46	60.70	47.26	77.86	0.12	3.79	60.82	43.47	**71.47**
JASRA	268.22	75.86	52.87	69.69	0.17	2.18	76.03	50.69	**66.66**
KARCHANA	232.81	79.47	52.16	65.63	0.15	1.86	79.62	50.30	**63.17**
KAUDHIYARA	200.46	68.37	41.17	60.22	0.13	1.52	68.50	39.65	**57.88**
KAURIHAR	414.14	85.40	56.61	66.28	0.26	2.12	85.67	54.49	**63.60**
KORAON	726.42	96.56	55.80	57.79	0.46	2.29	97.02	53.51	**55.15**
MANDA	349.43	51.25	32.95	64.31	0.22	1.29	51.47	31.67	**61.53**
MAUAIMA	155.6	55.75	45.62	81.83	0.12	3.57	55.88	42.05	**75.26**
MEJA	444.21	75.53	38.71	51.25	0.28	1.47	75.82	37.24	**49.12**
PHULPUR	231.54	64.33	42.94	66.75	0.15	1.96	64.48	40.98	**63.55**
PRATAPPUR	211.01	66.76	58.07	86.99	0.17	4.95	66.93	53.12	**79.37**
SAIDABAD	191.42	57.97	47.65	82.21	0.15	3.91	58.12	43.74	**75.26**
SHANKARGARH	472.08	68.15	40.60	59.58	0.30	1.80	68.45	38.80	**56.68**
SORAON	147.75	43.08	29.45	68.36	0.09	1.13	43.18	28.32	**65.59**
URUWAN	168.9	57.19	38.43	67.20	0.11	1.47	57.30	36.96	**64.51**
		1325.17	**957.37**				**1328.94**	**895.89**	**67.41387458**

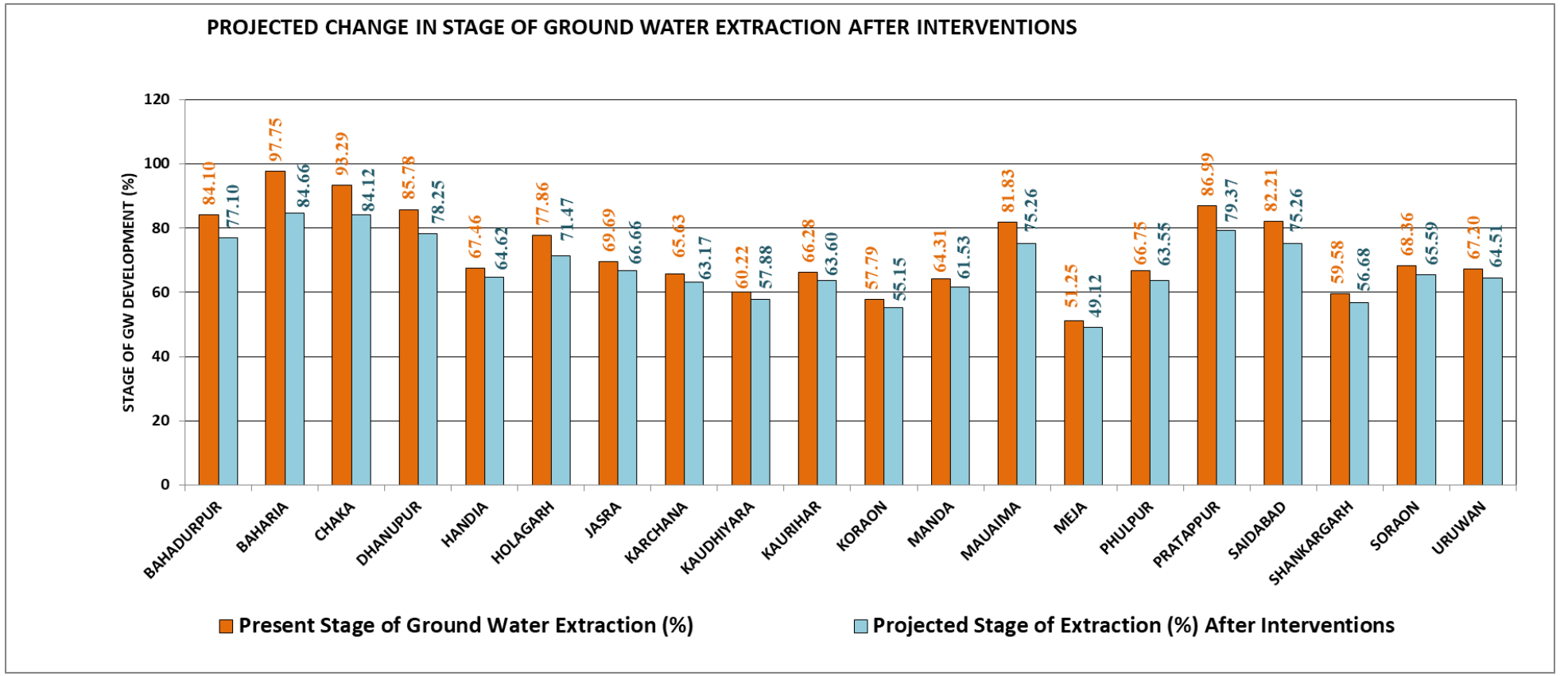


Figure 5.1: Projected change in stage of ground water extraction after interventions

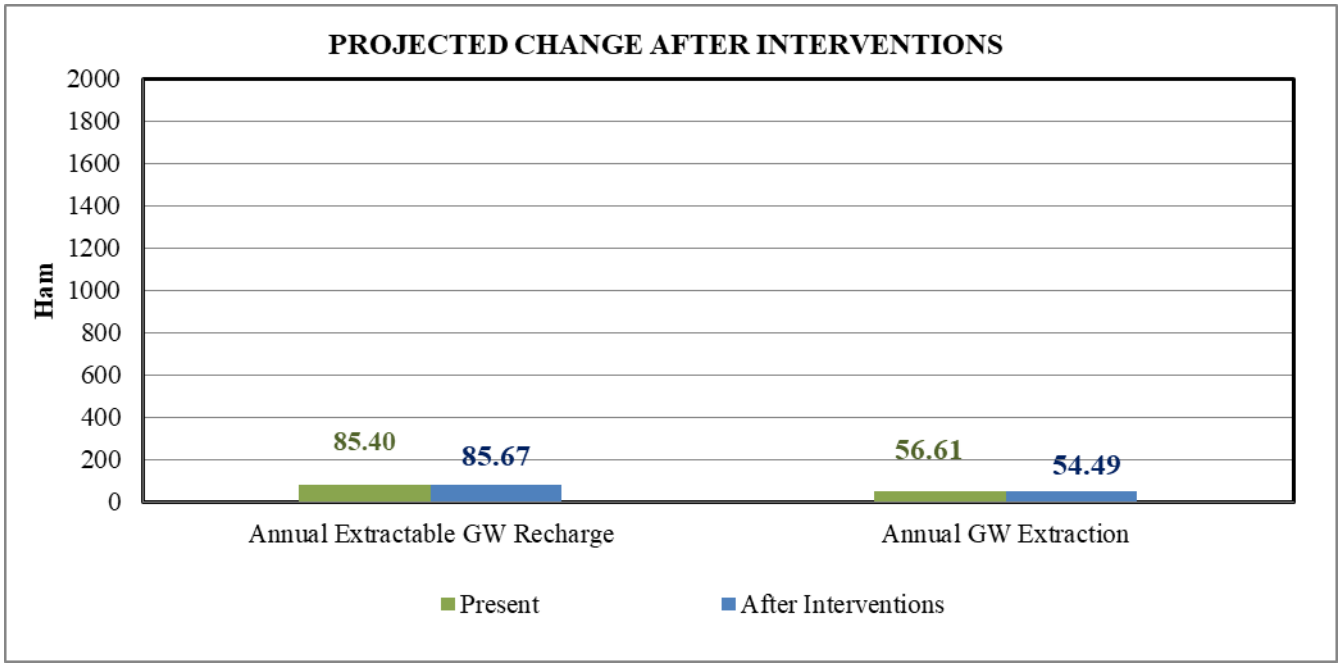


Figure 5.2: Projected change after interventions

Table 5.5: Projected change after interventions

	Annual Extractable GW Recharge	**Annual GW Extraction**
Present	85.40	56.61
After Interventions	85.67	54.49

CHAPTER-6

CONCLUSION

1. District Prayagraj is covering an area of 5482 sq. km lies in doab of River Ganga and Yamuna.

2. The area is a part of Middle Ganga alluvial plain. The gradient following the drainage lines of the principal rivers (Ganga and Yamuna).

3. The Northern half of the district occupies part of Ganga basin and Sothern part of the district occupies the part of Yamuna and Tons Sub- basin. The rivers of the district belong to the main drainage system of the Ganga. The most important tributaries are Yamuna and Tons while those of minor systems include Sai and Yamuna. The Ganga enters the district in the Kaurihar Block. It maintains a meandering course. Yamuna, the second major river, enters the district in the extreme west in the Sankargarh block and flows narrowly before joining to the Ganga at Sangam in the proximity of Prayagraj city. Yamuna has more constant channel and steeper banks.

4. The northern part of Prayagraj district popularly known as Gangapur provides rich loam soil for cultivation of food grains, pulses, oilseeds and vegetables. The Southern part of Prayagraj district also known as Yamuna par is partly rocky and somewhat agriculturally backward.

5. Net Area Sown in the district is 353 (thousand ha), Net irrigated area is 310 (thousand ha) and Gross irrigated area is 511 (thousand ha).

6. The average annual rainfall is 959.1 mm. About 85% of rainfall takes places from June to September.

7. The depth to water level ranges from 1.32 to 20.36 mbgl during pre-monsoon where as it ranges from 0.59 to 15.69 mbgl in post-monsoon. The pre-& post-monsoon water level fluctuation varies from 0.46 m to 6.42mbgl.

8. Transmissivity of aquifer varies from 857 – 3330 m^2/day, Storativity 1.79×10^{-4} – 5.10×10^{-4}. Discharge varies from1000-3050 lpm

9. The total ground water draft is 100192.68 hams, which is being used in present for domestic, irrigation & industrial purposes against the ground water availability of 135052.05 hams. Prayagraj City is in Over-Exploited Category. Baharia and Chaka blocks are in Critical Category. There are Six blocks Bahadurpur, Dhanupur, Holagarh, Mauaima, Pratappur, Saidabad falls under Semi-critical category whereas rest blocks fall under safe category.

RECOMMENDATION

1. To arrest the further decline in ground water levels and depletion of ground water resources, there is urgent need to implement both Supply side and Demand side measures which includes artificial recharge and water conservation, On-farm activities and adoption of water use efficiency measures.

2. There is considerable scope for implementation of Roof Top Rain Water Harvesting in Prayagraj City. Check dams, cement plugs, renovation of ponds are ideal structures for rain water harvesting in rural areas. Water conservation structures such as check dams, farm ponds, nala bunds etc. It is also proposed to adopt on Farm practices such as laser levelling, bench terracing, construction of farm ponds, afforestation, diversification of crops etc.

3. It is proposed to construct 59 Check dams and 60 ponds under supply side management.

4. In demand side management there is urgent need to promote piped and pressurized irrigation practices, n-farm Activities, Water Use Efficiency Measures (Drip & Sprinkler + HDPE Pipes). It is proposed to initiate these measures initially in 30% of area. The measures adopted for supply side and demand side management in district will substantially bring down stage of ground water development.

5. It is also proposed to adopt new water saving agricultural practices. Such practices have the potential of saving 35-40% irrigation water thereby drastically reducing the draft for irrigation leading the change of category of the district from semi-critical to safe.

6. Agriculture department should promote to conserve the soil moisture by reducing ET losses through cultivation of ‘Green Manure’.

7. Alternate cropping pattern system having lower requirement of water should be encouraged in accordance to the irrigation water availability.

8. Furrow irrigation with raised bed planting in wide row crops should be practiced.

9. Drip irrigation in wide row crops should be practiced with mulch in the area.

10. Multiuse of water through integrated farming system.

11. Conjunctive use of surface and groundwater should be encouraged in the block.

12. Besides the above, there is urgent need for participatory ground water management in the area which will further help in bringing more awareness among the common farmers which will reduce the ground water draft and bring down the stage of ground water development.

13. All efforts should be taken to ensure treatment of waste disposal both solid and liquid from industries and urban areas to prevent pollution of ground water and surface water.

REFERENCES

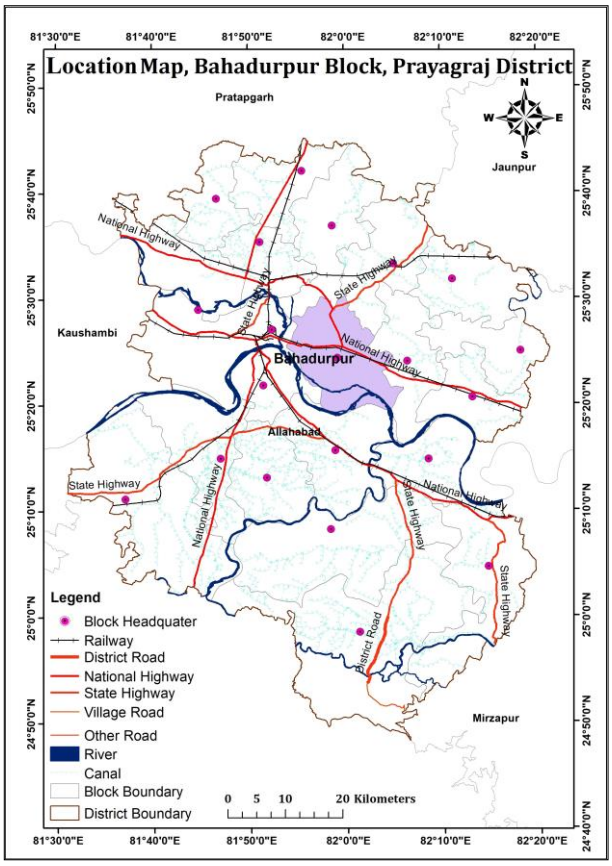
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- Report on the Selection of Tubewell sites at M.E.S new cantonment, Allahabad, Uttar Pradesh by Shri Arun Kumar, Asstt. Hydrogeologist, Jan, 1994, Lucknow

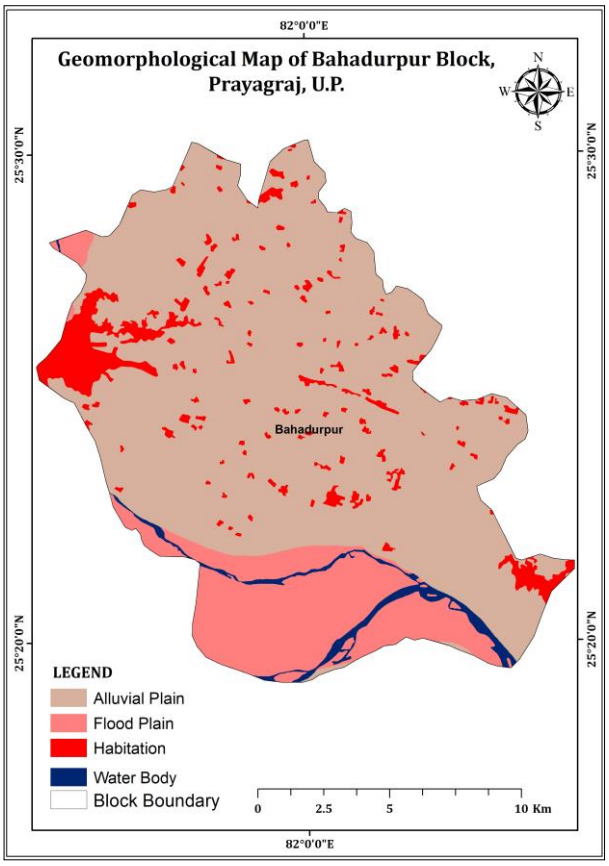
Aquifer Map and Management plan of Bahadurpur Block, Prayagraj District, U.P.

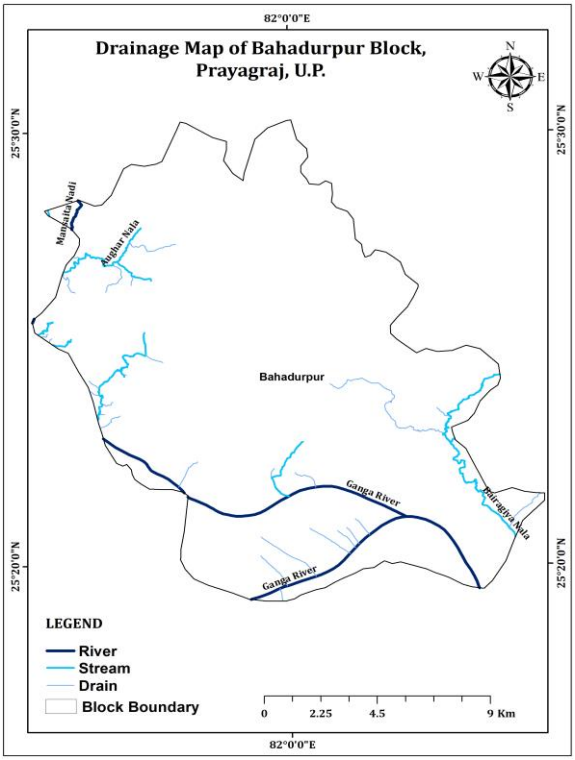
BAHADURPUR BLOCK AT A GLANCE

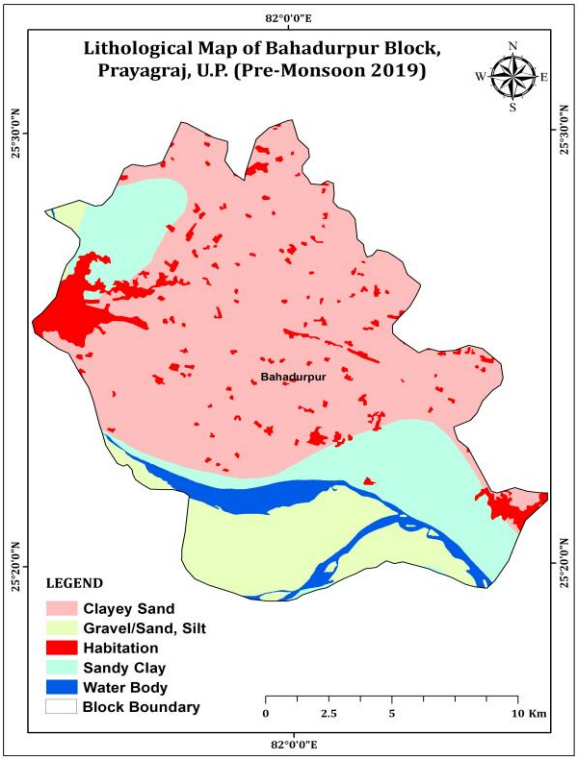
General Information

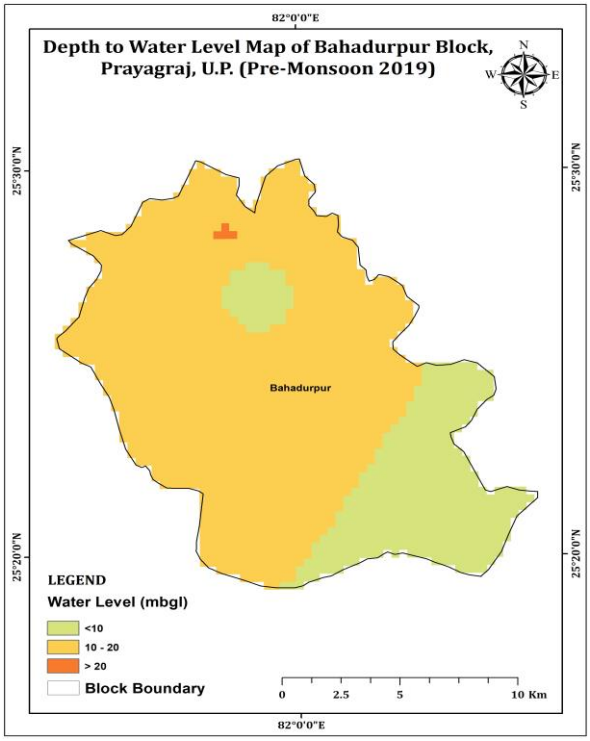
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Bahadurpur
Geographical area	260.59 Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm

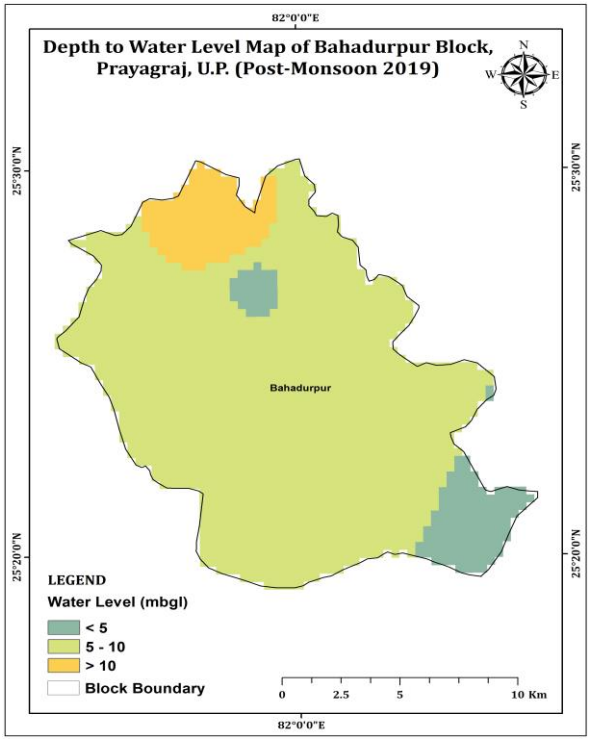





Drainage Map of Bahadurpur Block,
Prayagraj, U.P.
82°0'0"E
25°30'0"N
25°20'0"N
Bahadurpur
Ganga River
Ganga River
LEGEND
River
Stream
Drain
Block Boundary
0 2.25 4.5 9 Km

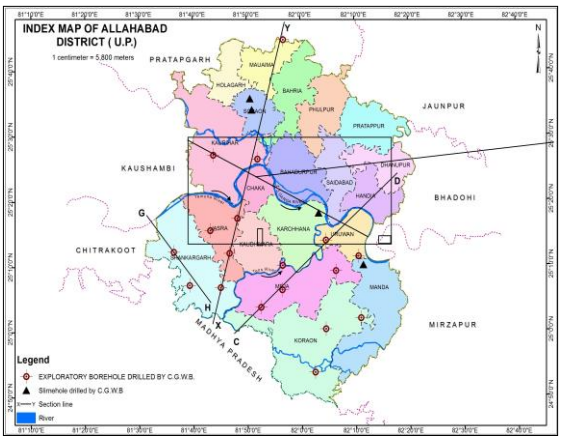

Lithological Map of Bahadurpur Block,
Prayagraj, U.P. (Pre-Monsoon 2019)
82°0'0"E
25°30'0"N
25°20'0"N
Bahadurpur
LEGEND
Clayey Sand
Gravel/Sand, Silt
Habitation
Sandy Clay
Water Body
Block Boundary
0 2.5 5 10 Km

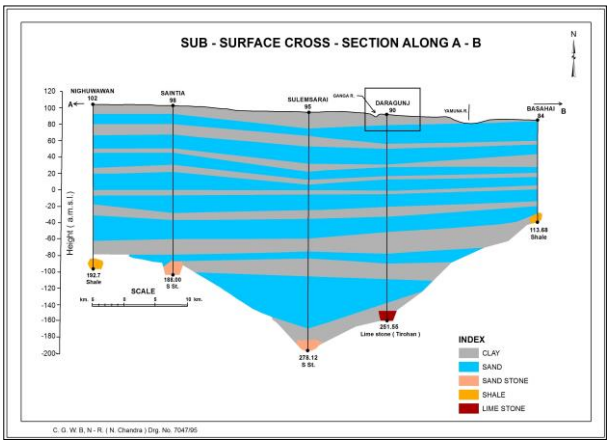

Depth to Water Level Map of Bahadurpur Block,
Prayagraj, U.P. (Pre-Monsoon 2019)
82°0'0"E
25°30'0"N
25°20'0"N
Bahadurpur
LEGEND
Water Level (mbgl)
<10
10 - 20
> 20
Block Boundary
0 2.5 5 10 Km


Depth to Water Level Map of Bahadurpur Block,
Prayagraj, U.P. (Post-Monsoon 2019)
82°0'0"E
25°30'0"N
25°20'0"N
Bahadurpur
LEGEND
Water Level (mbgl)
< 5
5 - 10
> 10
Block Boundary
0 2.5 5 10 Km

Aquifer Disposition

Aquifer Disposition	Three aquifer group upto 250 meter depth. • Aquifer Group I: upto 115 m. • Aquifer Group II: from 125 upto 185 m • Aquifer Group II: Beyond 202m Transmissivity 3330 m2/day and Storativity- 1.79*10-4 Discharge 3050 lpm
Groundwater Monitoring Status	2
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	3050 lpm
Groundwater Resource	Annual Extractable GW Resource : 80.50MCM GW Draft :67.70 MCM Stage of GW Development : 84.10 %; Semi- Critical
Existing and Future Water Demand	Present demand for All Usage 67.70MCM Future Demand for Domestic Use: 9.62 MCM





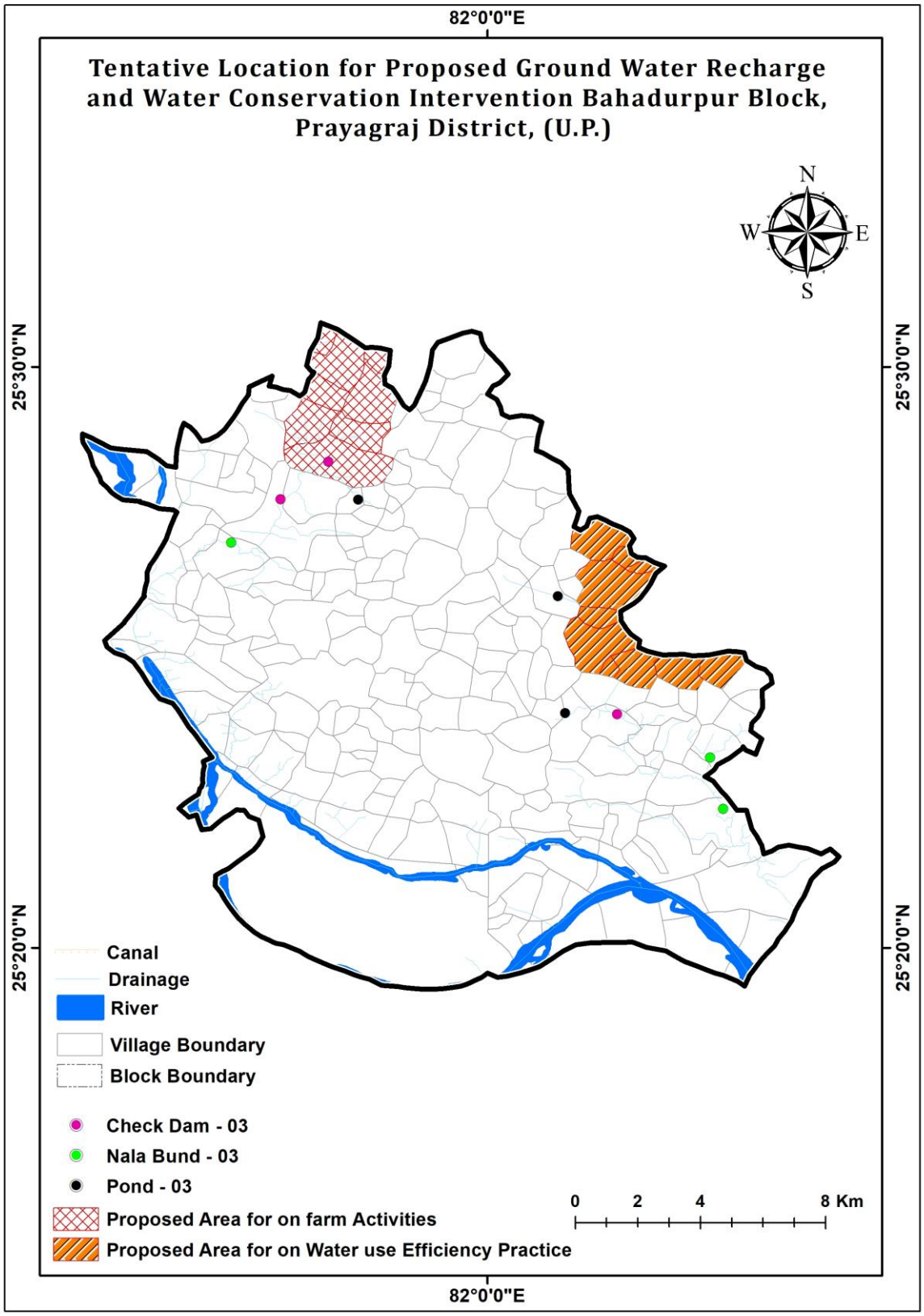
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	Exploratory Wells : 01 Observation Wells : 01
Aquifer Characteristics	Transmissivity 3330 m2/day and Storativity- 1.79*10-4 Discharge 3050 lpm

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
Bahadurpur	3	3	3	3	1123	1123	0.21	0.21	5.27	0.21	5.48	84.10	77.10

Blocks	Area (Sq. Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
Bahadurpur	260.59	80.51	67.70	84.10	0.21	5.48	80.71	62.23	**77.10**

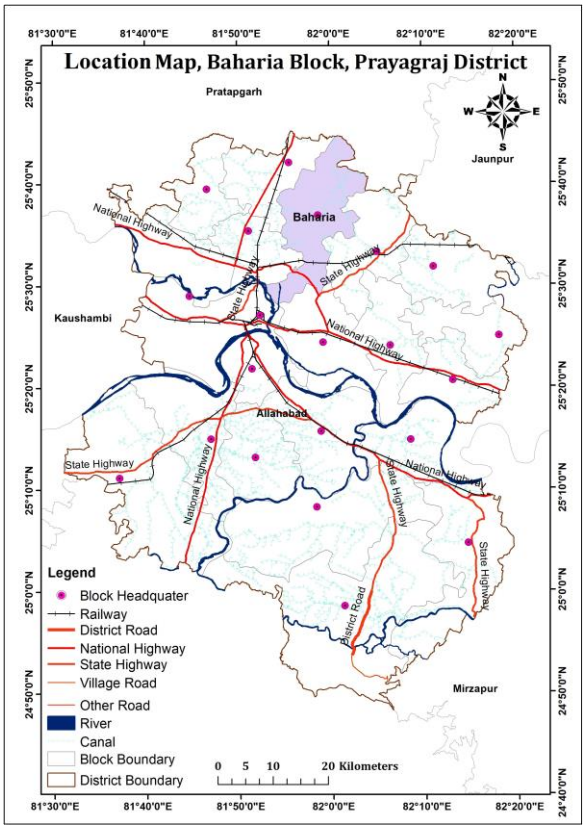

82°0'0"E
Tentative Location for Proposed Ground Water Recharge and Water Conservation Intervention Bahadurpur Block, Prayagraj District, (U.P.)
N
W
E
S
25°30'0"N
25°20'0"N
Canal
Drainage
River
Village Boundary
Block Boundary
Check Dam - 03
Nala Bund - 03
Pond - 03
Proposed Area for on farm Activities
Proposed Area for on Water use Efficiency Practice
0 2 4 8 Km
82°0'0"E

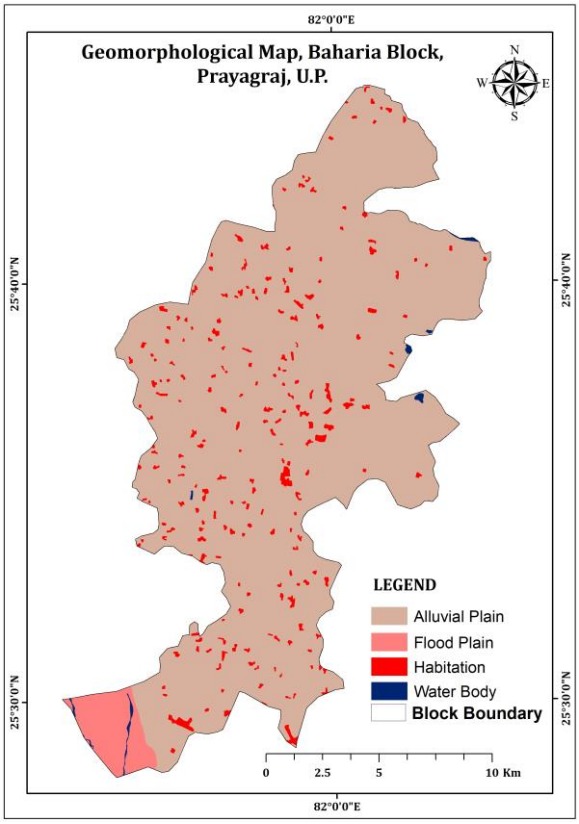
Aquifer Map and Management plan of Bahariya Block, Prayagraj District, U.P.

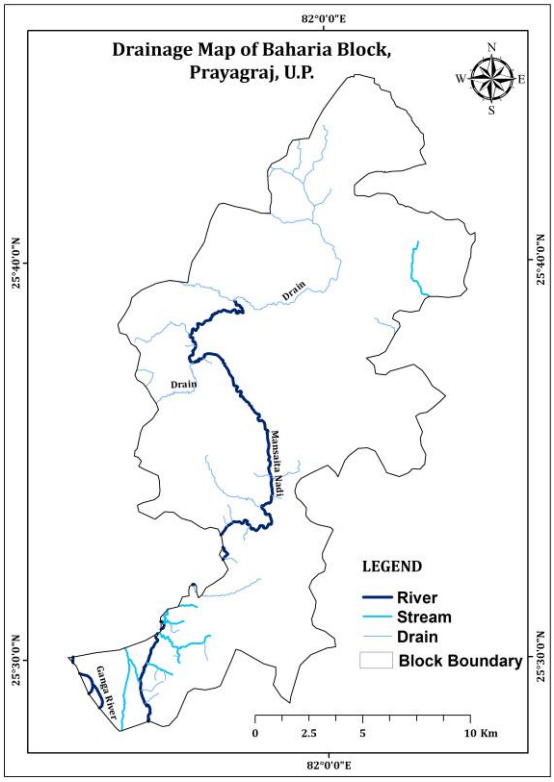
BAHARIYA BLOCK AT A GLANCE

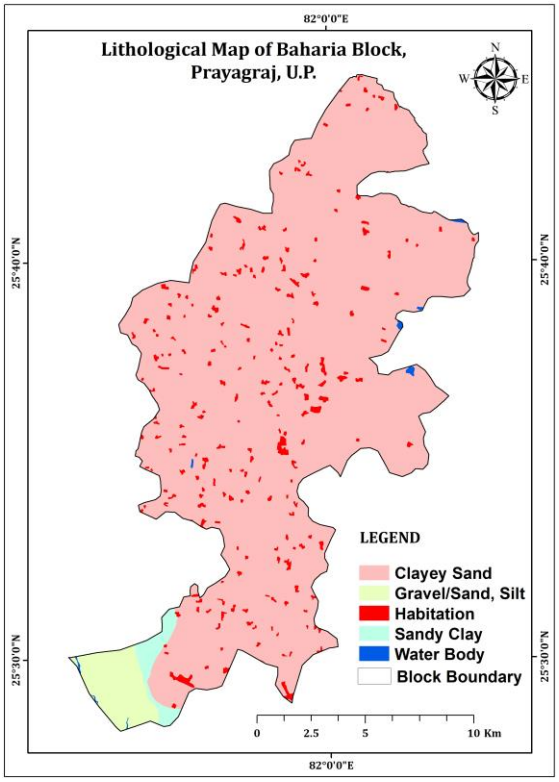
General Information

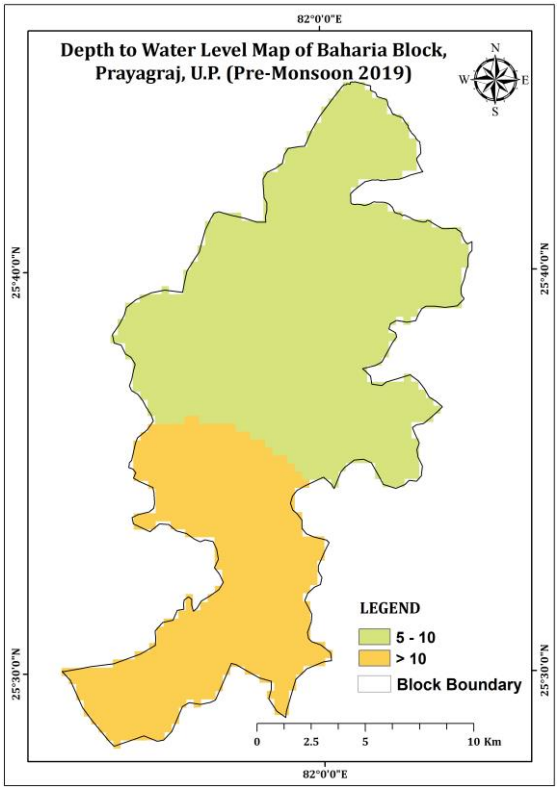
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Bahariya
Geographical area	248.75 Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm

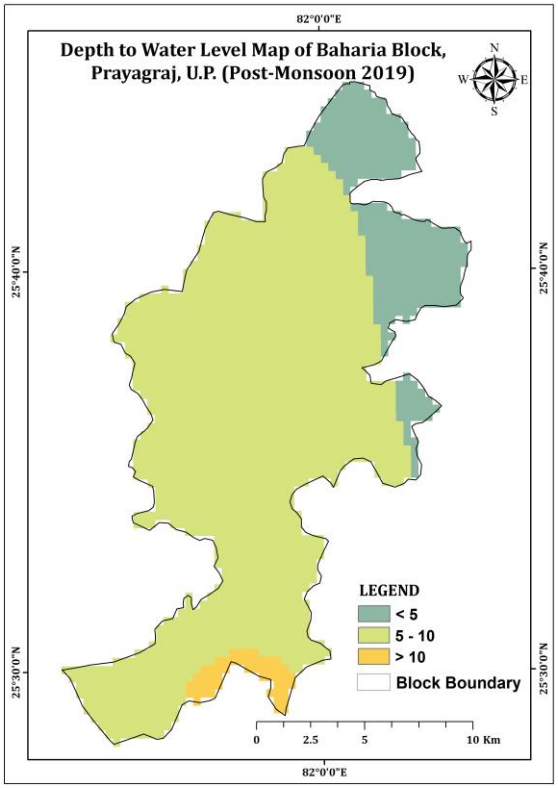





Drainage Map of Baharia Block,
Prayagraj, U.P.
82°0'0"E
25°40'0"N
25°30'0"N
Drain
Drain
Mansatta Nadi
Ganga River
LEGEND
River
Stream
Drain
Block Boundary
0
2.5
5
10 Km

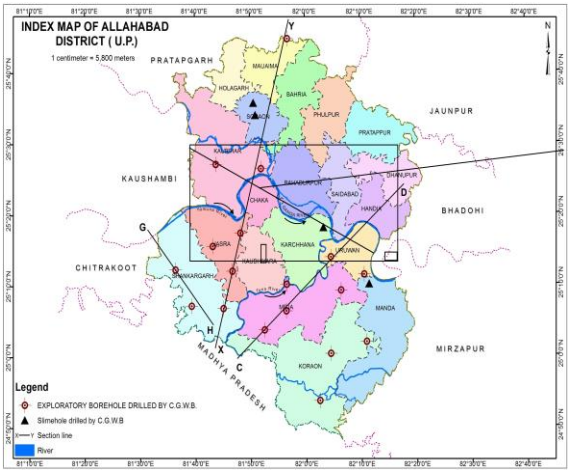

Lithological Map of Baharia Block,
Prayagraj, U.P.
82°0'0"E
25°40'0"N
25°30'0"N
LEGEND
Clayey Sand
Gravel/Sand, Silt
Habitation
Sandy Clay
Water Body
Block Boundary
0
2.5
5
10 Km

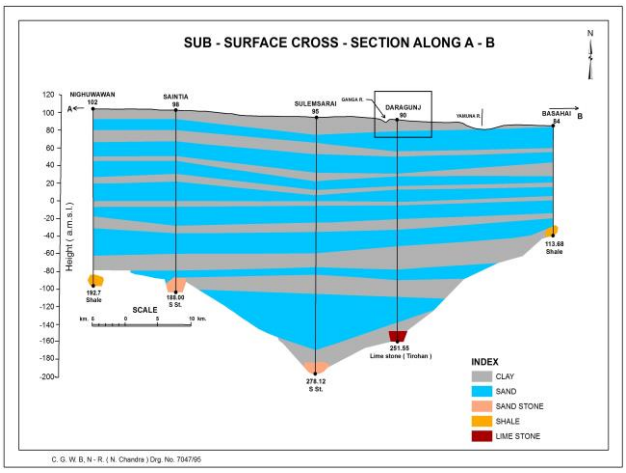

Depth to Water Level Map of Baharia Block,
Prayagraj, U.P. (Pre-Monsoon 2019)
82°0'0"E
25°40'0"N
25°30'0"N
LEGEND
5 - 10
> 10
Block Boundary
0
2.5
5
10 Km


Depth to Water Level Map of Baharia Block,
Prayagraj, U.P. (Post-Monsoon 2019)
82°0'0"E
25°40'0"N
25°30'0"N
LEGEND
< 5
5 - 10
> 10
Block Boundary
0
2.5
5
10 Km

Aquifer Disposition

Aquifer Disposition	Three aquifer group up to 250 meter depth. • Aquifer Group I: upto 115 m. • Aquifer Group II: from 125 upto 185 m • Aquifer Group II: Beyond 202m Transmissivity 3330 m2/day and Storativity- 1.79*10-4 Discharge 3050lpm
Groundwater Monitoring Status	1
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	Aquifer Group I : 3050lpm
Groundwater Resource	Annual Extractable GW Resource : 85.30 MCM GW Draft :83.38 MCM Stage of GW Development :97.75 % Critical
Existing and Future Water Demand	Present demand for All Usage 83.38MCM Future Demand for Domestic use: 7.909 MCM





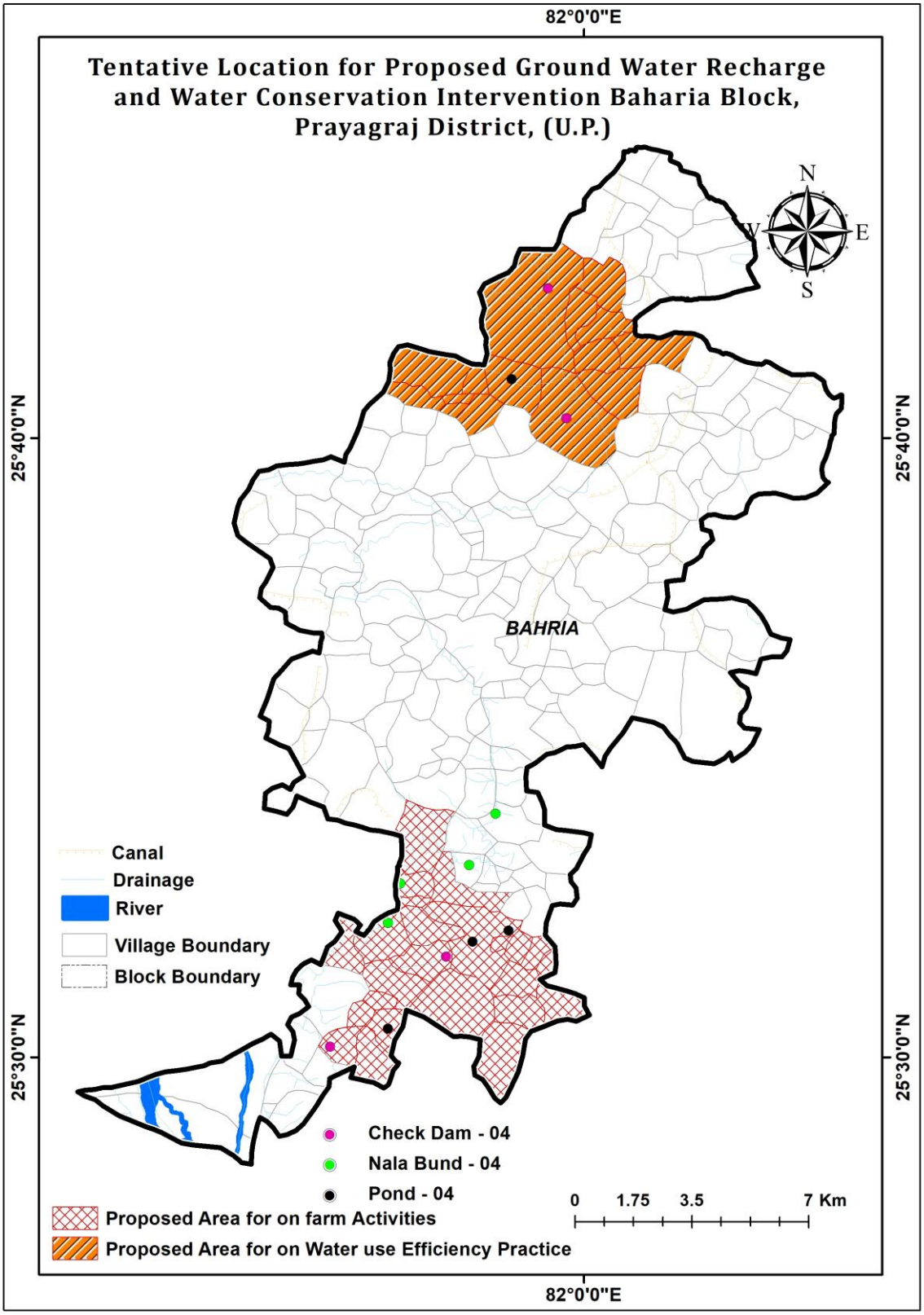
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	Exploratory Wells : 01 Observation Wells : 01
Aquifer Characteristics	Transmissivity 3330 m2/day and Storativity- 1.79*10-4 Discharge 3050lpm

Management Plan

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
BAHARIA	248.75	85.31	83.39	97.75	0.28	10.93	85.58	72.45	**84.66**

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
BAHARIA	4	4	4	5	2719	2719	0.28	0.28	10.66	0.28	10.93	97.75	84.66

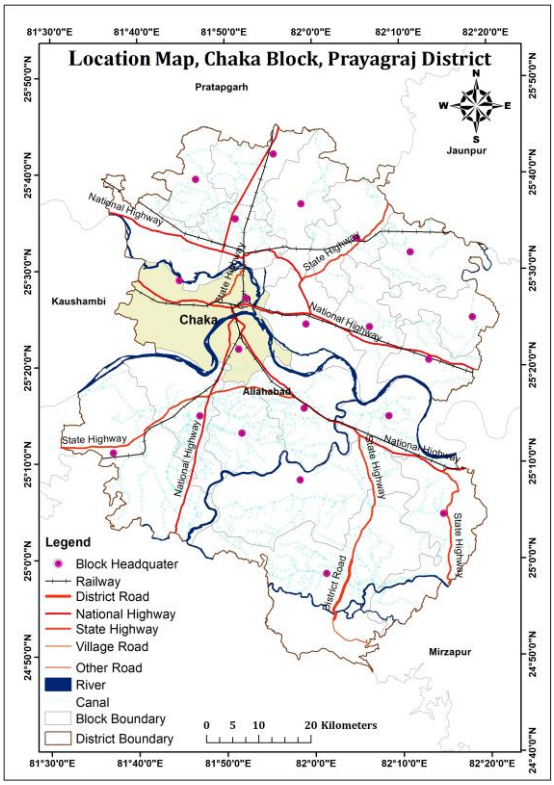

Tentative Location for Proposed Ground Water Recharge and Water Conservation Intervention Baharia Block, Prayagraj District, (U.P.)
82°0'0"E
25°40'0"N
25°30'0"N
N
W
E
S
BAHRIA
Canal
Drainage
River
Village Boundary
Block Boundary
Check Dam - 04
Nala Bund - 04
Pond - 04
Proposed Area for on farm Activities
Proposed Area for on Water use Efficiency Practice
0 1.75 3.5 7 Km

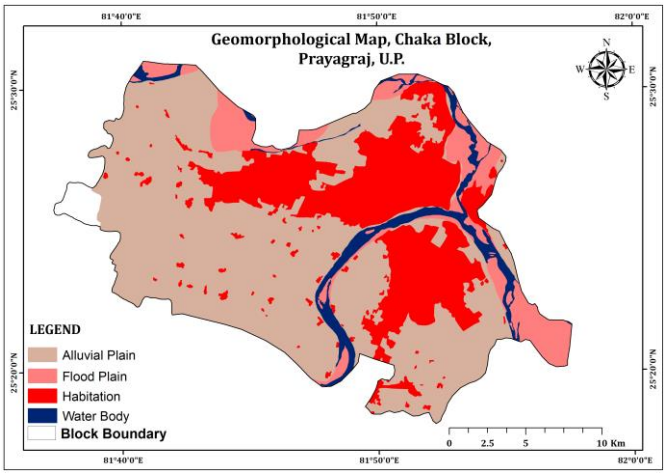
Aquifer Map and Management plan of Chaka Block, Prayagraj District, U.P.

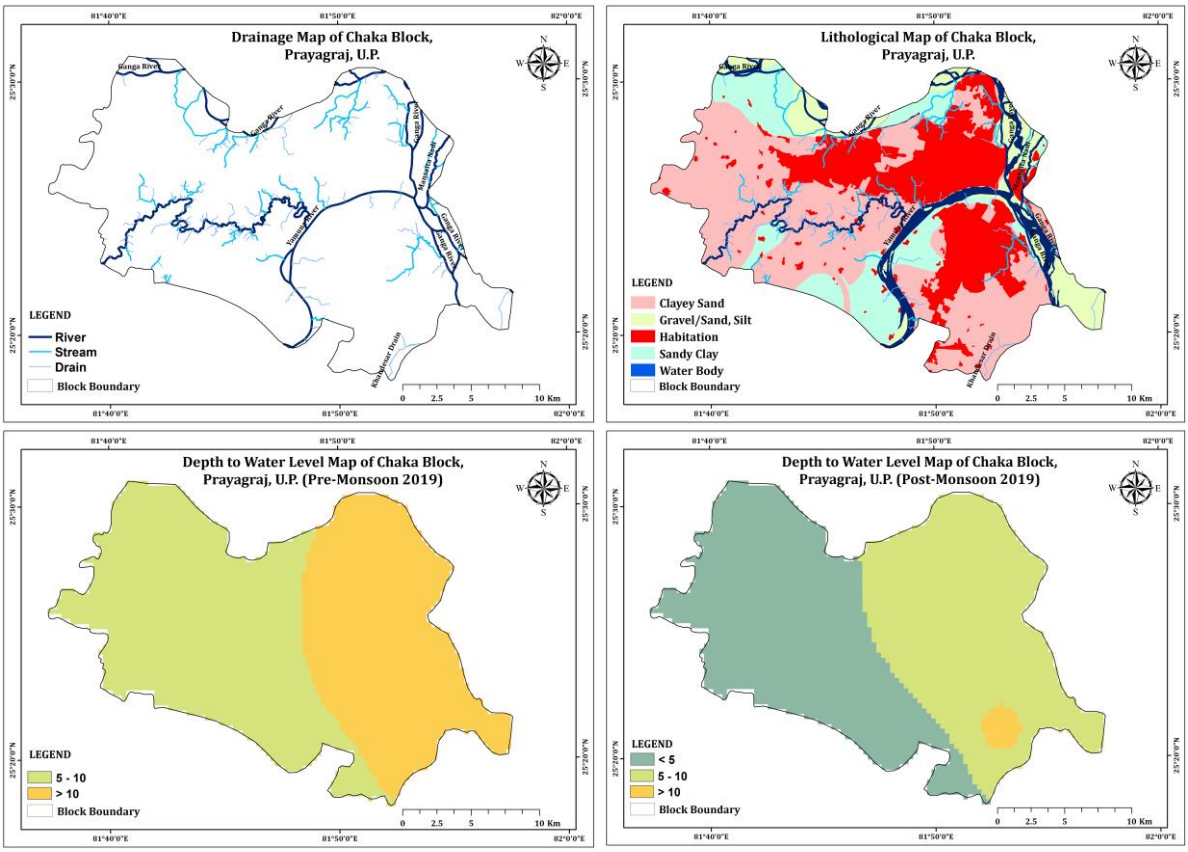
CHAKA BLOCK AT A GLANCE

General Information

State Name	Uttar Pradesh
District name	prayagraj
Block Name	Chaka
Geographical area	135.75Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042 mm

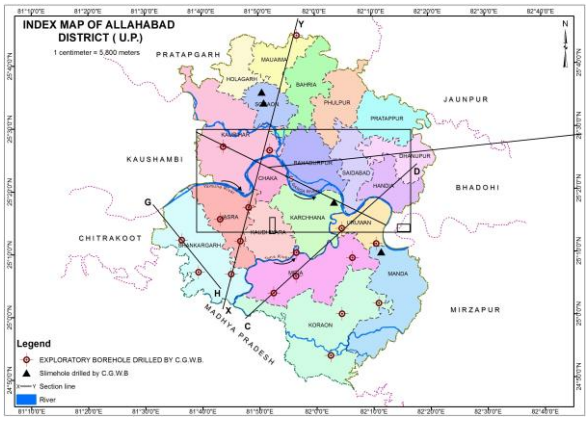


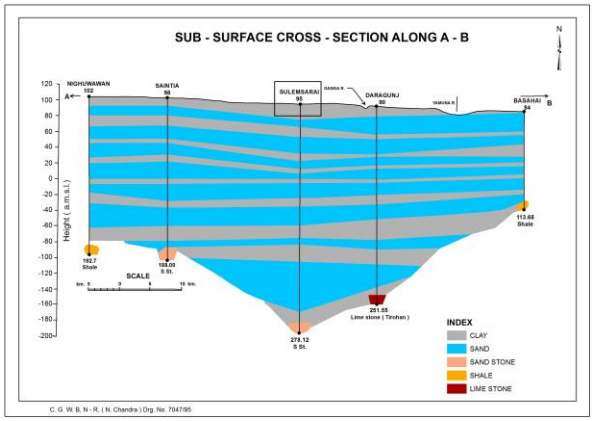




Aquifer Disposition

Aquifer Disposition	Three aquifer group upto 200 meter depth. • Aquifer Group I: upto 115 m. • Aquifer Group II: from 122-200 m Transmissivity 2790 m2/day and Storativity- 4.13*10-4 Discharge 3600 lpm
Groundwater Monitoring Status	2
Ground Water Quality	Suitable for domestic, drinking, irrigation and Industrial purpose
Aquifer Potential	Aquifer Group I : 2790 lpm
Groundwater Resource	Annual Extractable GW Resource : 44.04 MCM GW Draft :41.09 MCM Stage of GW Development :93.29 % Critical
Existing and Future Water Demand	Present demand for All Usage 41.09MCM Future Demand for Domestic use:5.9199 MCM





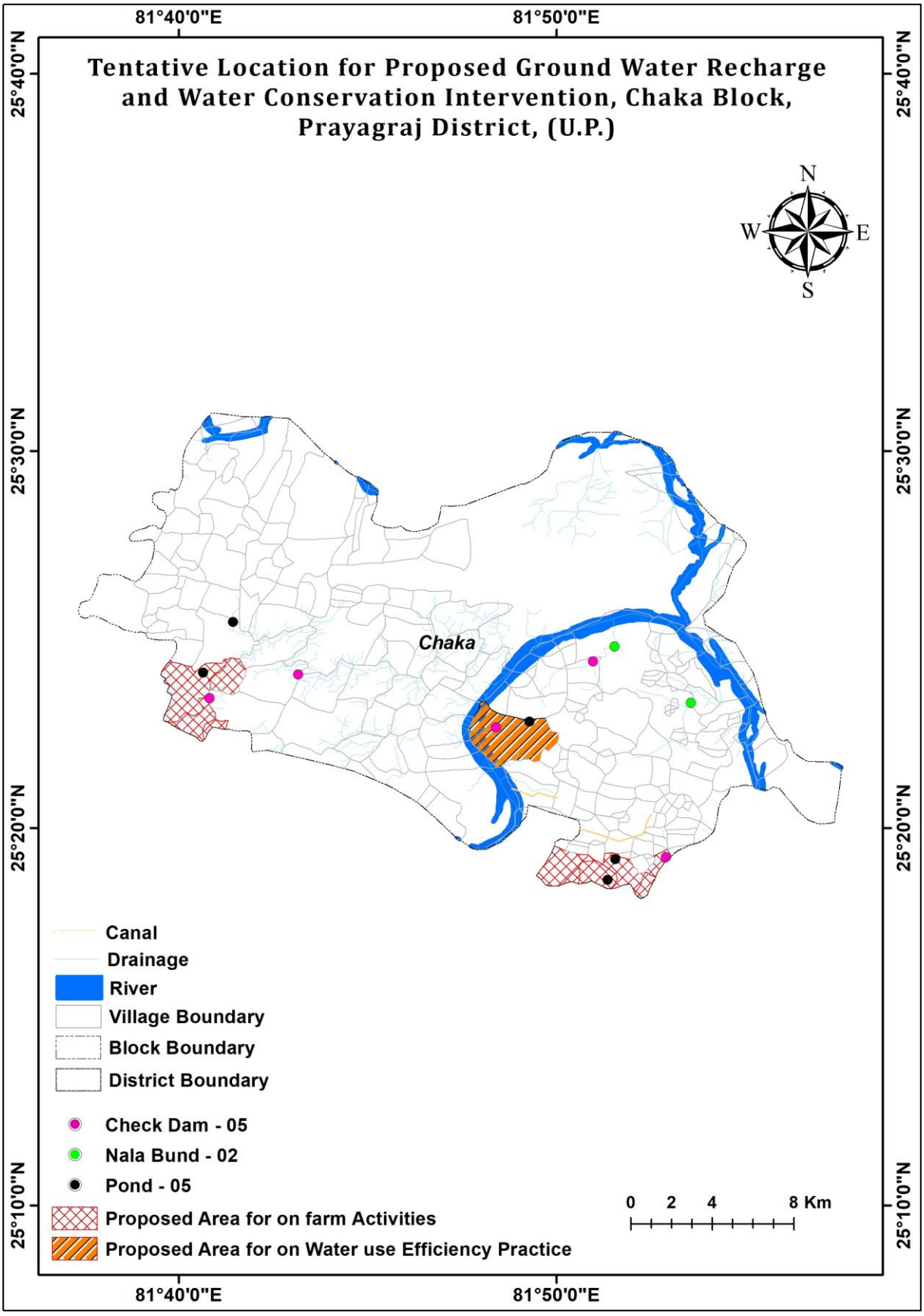
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water Use Efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	Exploratory Wells : 05 Observation Wells : 01
Aquifer Characteristics	Transmissivity 2790 m2/day and Storativity- 4.13*10-4 Discharge 3600 lpm

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
CHAKA	2	2	2	3	786	786	0.15	0.15	3.76	0.15	3.19	93.29	84.12

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
CHAKA	135.75	44.04	41.09	93.29	0.15	3.91	44.20	37.18	**84.12**

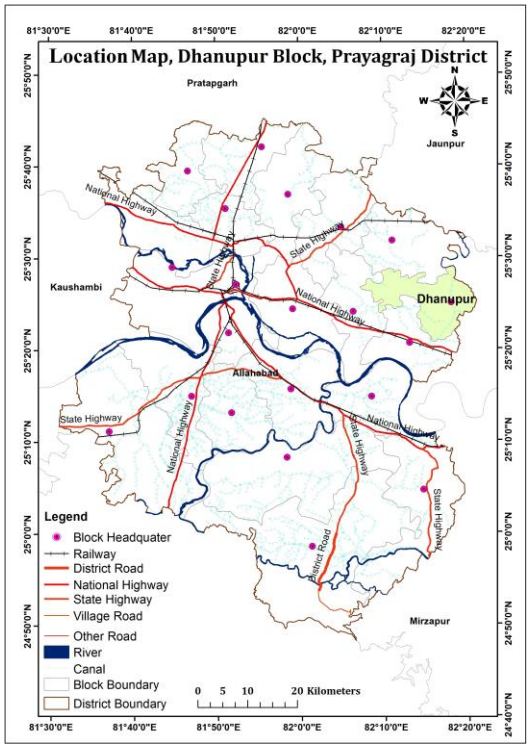

Tentative Location for Proposed Ground Water Recharge and Water Conservation Intervention, Chaka Block, Prayagraj District, (U.P.)
81°40'0"E
81°50'0"E
25°40'0"N
25°30'0"N
25°20'0"N
25°10'0"N
N
W
E
S
Chaka
Canal
Drainage
River
Village Boundary
Block Boundary
District Boundary
Check Dam - 05
Nala Bund - 02
Pond - 05
Proposed Area for on farm Activities
Proposed Area for on Water use Efficiency Practice
0 2 4 8 Km

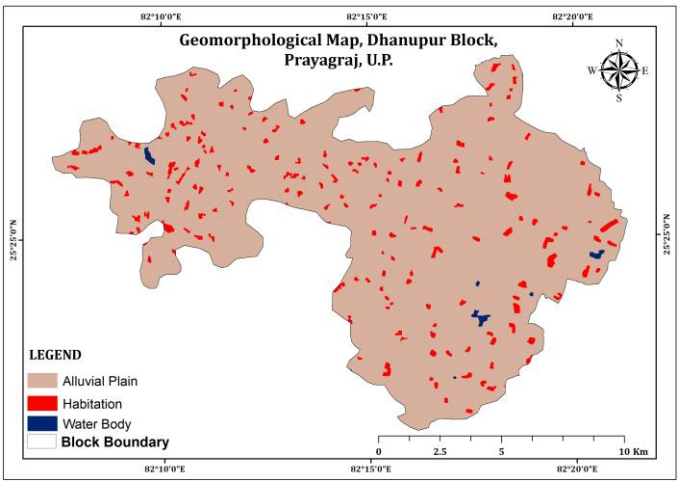
Aquifer Map and Management plan of Dhanupur Block, Prayagraj District, U.P.

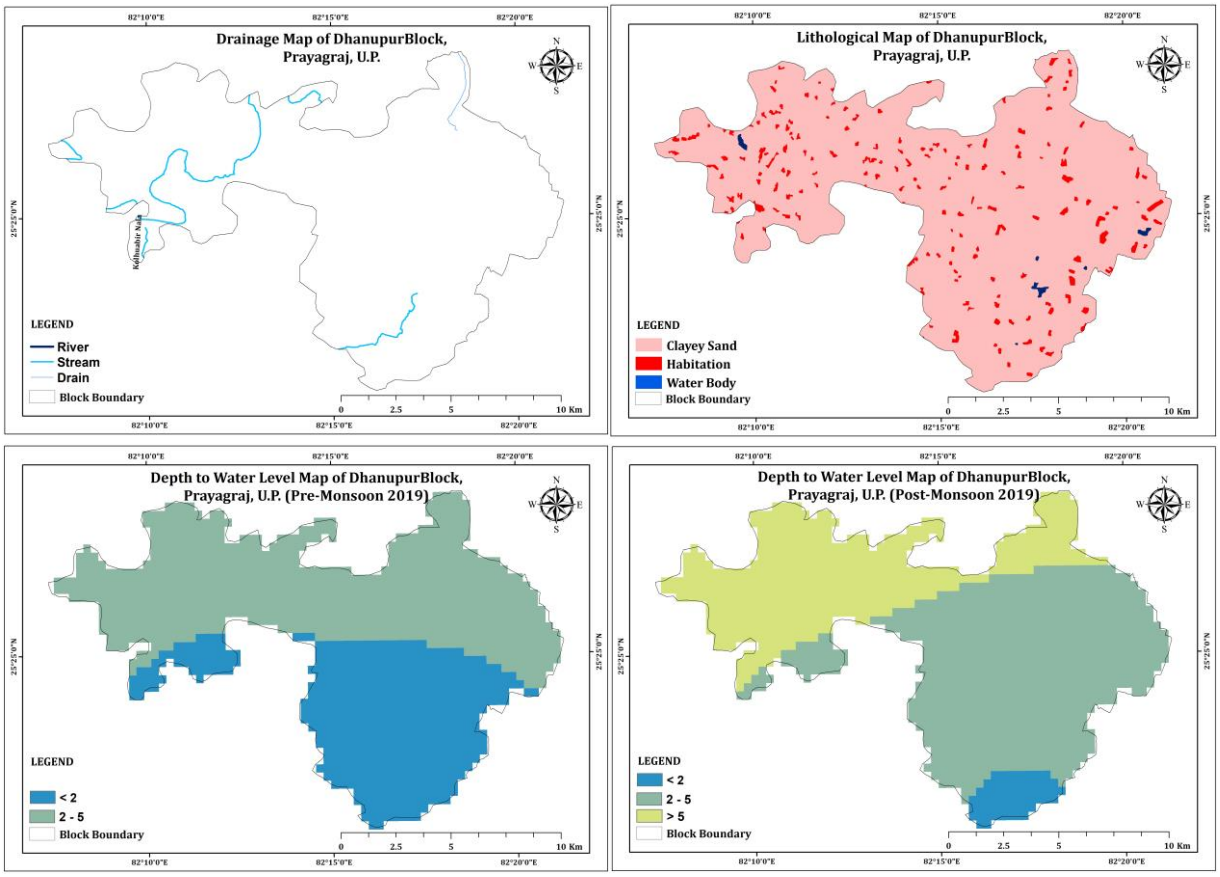
DHANUPUR BLOCK AT A GLANCE

General Information

State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Dhanupur
Geographical area	173.22 Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm







Aquifer Disposition

Aquifer Disposition	-
Groundwater Monitoring Status	1
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable GW Resource : 62.22MCM GW Draft :53.37MCM Stage of GW Development :85.78% Semi- Critical
Existing and Future Water Demand	Present demand for All Usage :53.37MCM Future Demand for Domestic Use: 6.91 MCM

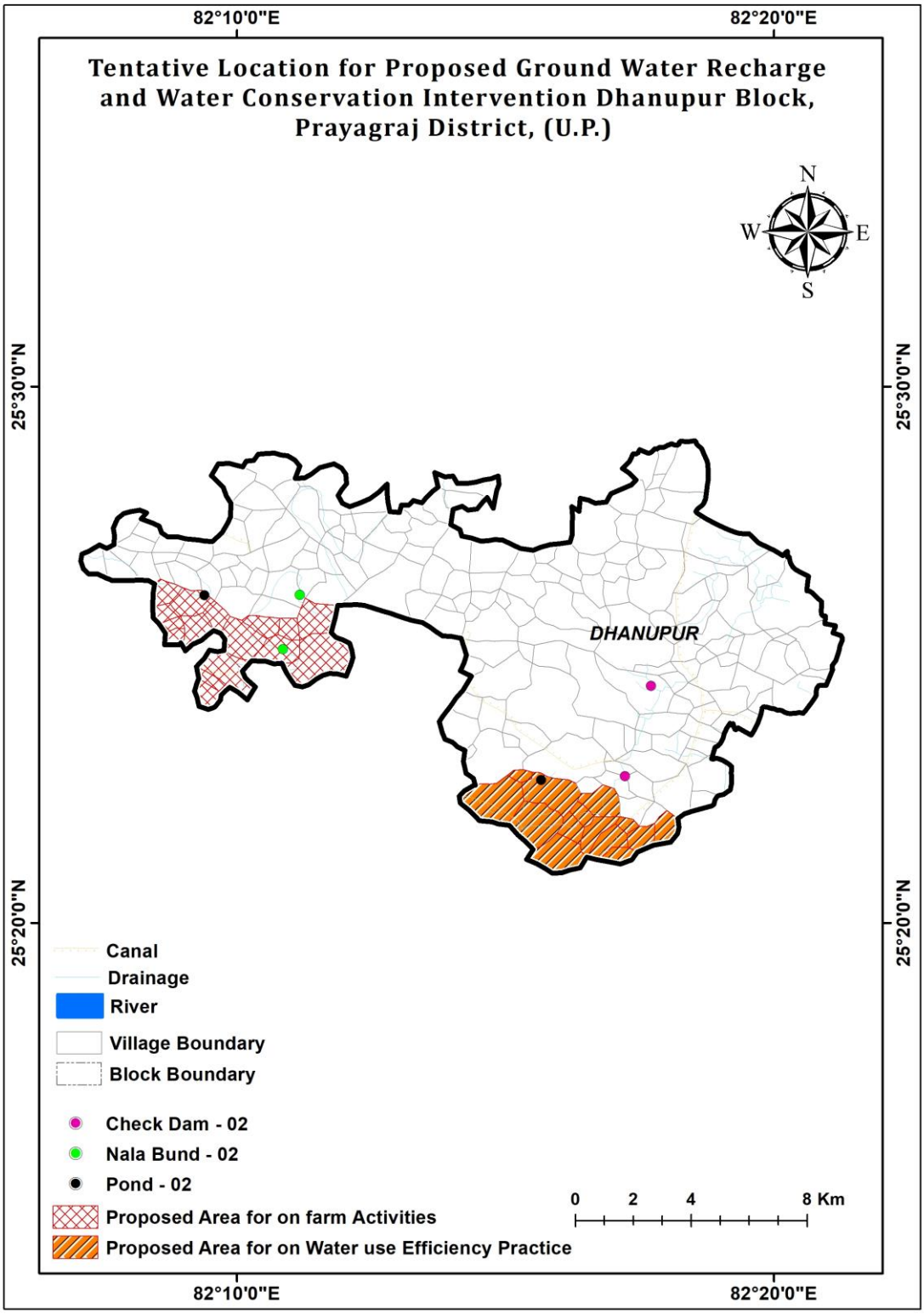
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water Use Efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
DHANUPUR	2	2	2	2	1248	1248	0.14	0.14	4.44	0.14	4.58	85.78	78.25

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
DHANUPUR	173.22	62.23	53.38	85.78	0.14	4.58	62.37	48.80	**78.25**

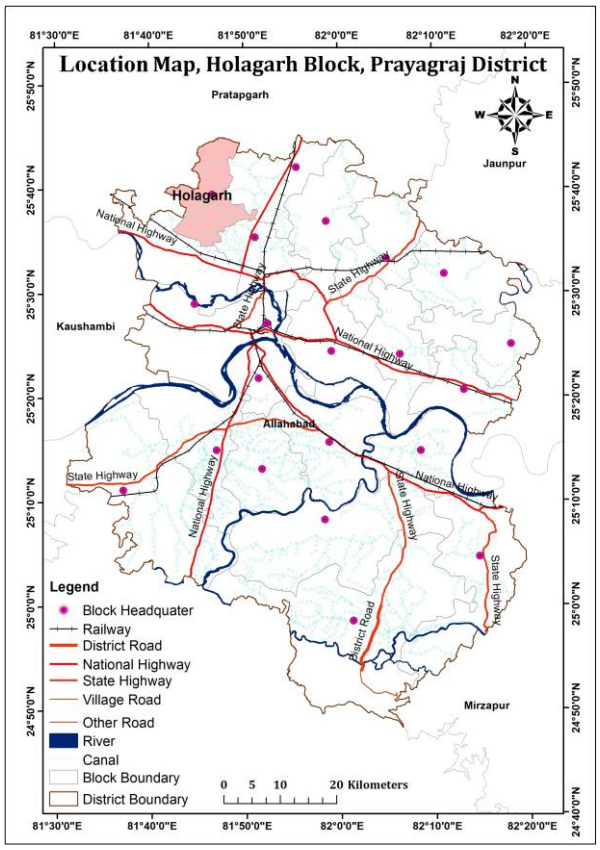

Tentative Location for Proposed Ground Water Recharge and Water Conservation Intervention Dhanupur Block, Prayagraj District, (U.P.)
82°10'0"E
82°20'0"E
25°30'0"N
25°20'0"N
N
W
E
S
DHANUPUR
Canal
Drainage
River
Village Boundary
Block Boundary
Check Dam - 02
Nala Bund - 02
Pond - 02
Proposed Area for on farm Activities
Proposed Area for on Water use Efficiency Practice
0 2 4 8 Km

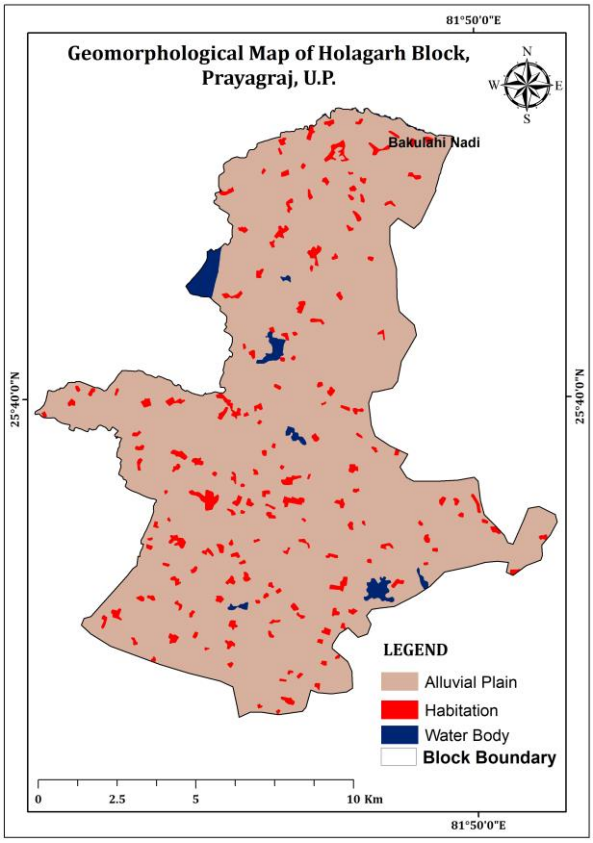
Aquifer Map and Management plan of Holagarh Block, Prayagraj District, U.P.

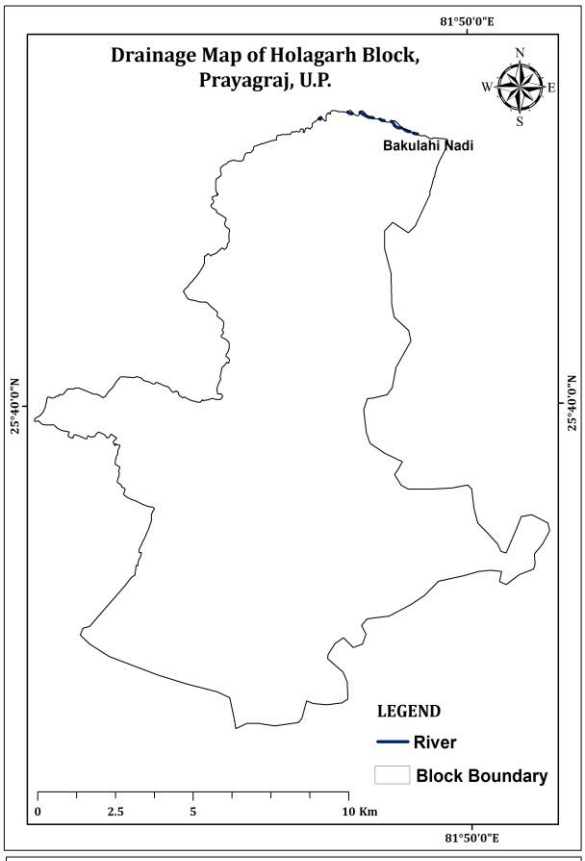
HOLAGARH BLOCK AT A GLANCE

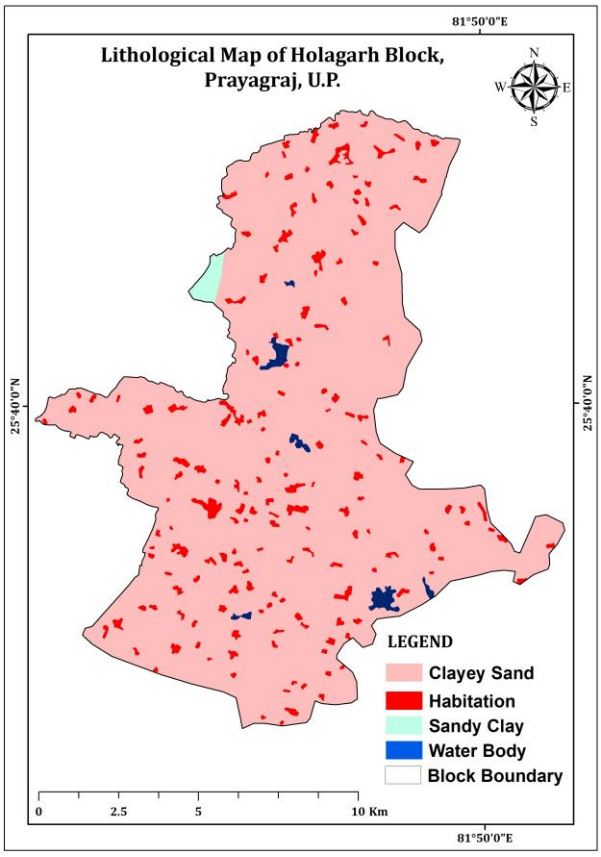
General Information

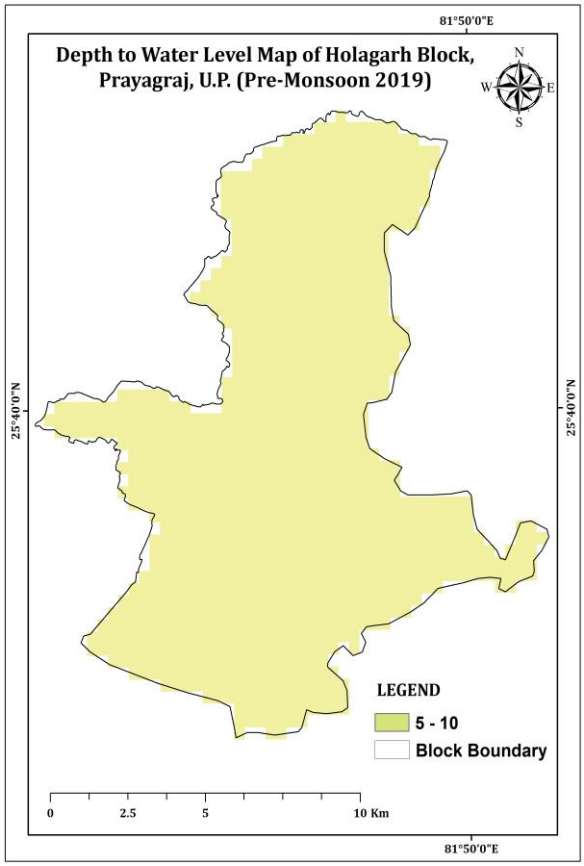
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Holagarh
Geographical area	148.46 Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm

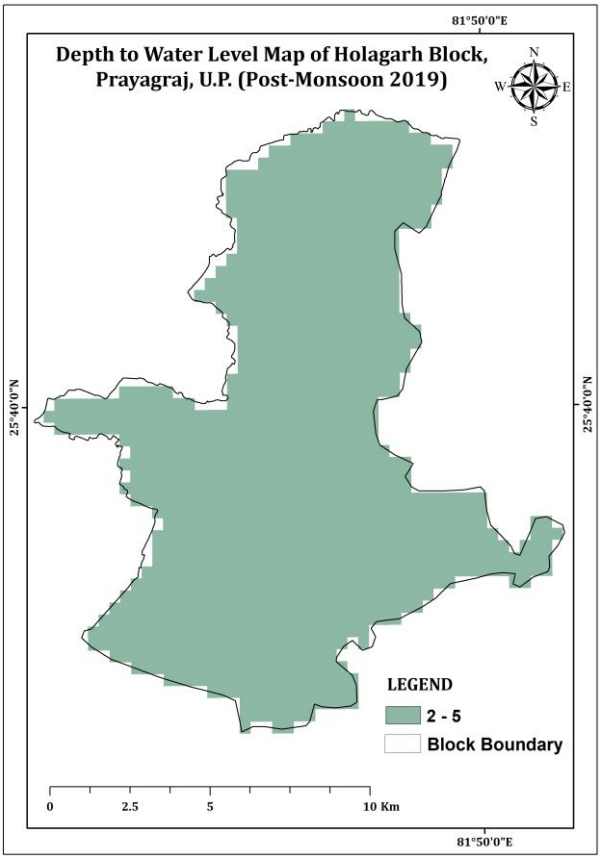





Drainage Map of Holagarh Block,
Prayagraj, U.P.
81°50'0"E
25°40'0"N
N
W
E
S
Bakulahi Nadi
LEGEND
River
Block Boundary
0
2.5
5
10 Km


Lithological Map of Holagarh Block,
Prayagraj, U.P.
81°50'0"E
25°40'0"N
N
W
E
S
LEGEND
Clayey Sand
Habitation
Sandy Clay
Water Body
Block Boundary
0
2.5
5
10 Km


Depth to Water Level Map of Holagarh Block,
Prayagraj, U.P. (Pre-Monsoon 2019)
81°50'0"E
25°40'0"N
N
W
E
S
LEGEND
5 - 10
Block Boundary
0
2.5
5
10 Km


Depth to Water Level Map of Holagarh Block,
Prayagraj, U.P. (Post-Monsoon 2019)
81°50'0"E
25°40'0"N
N
W
E
S
LEGEND
2 - 5
Block Boundary
0
2.5
5
10 Km

Aquifer Disposition

Aquifer Disposition	-
Groundwater Monitoring Status	1
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	
Groundwater Resource	Annual Extractable GW Resource : 60.70MCM GW Draft :47.26MCM Stage of GW Development :77.86% Semi- Critical
Existing and Future Water Demand	Present demand for All Usage :47.26MCM Future Demand for Domestic Use: 5.18 MCM

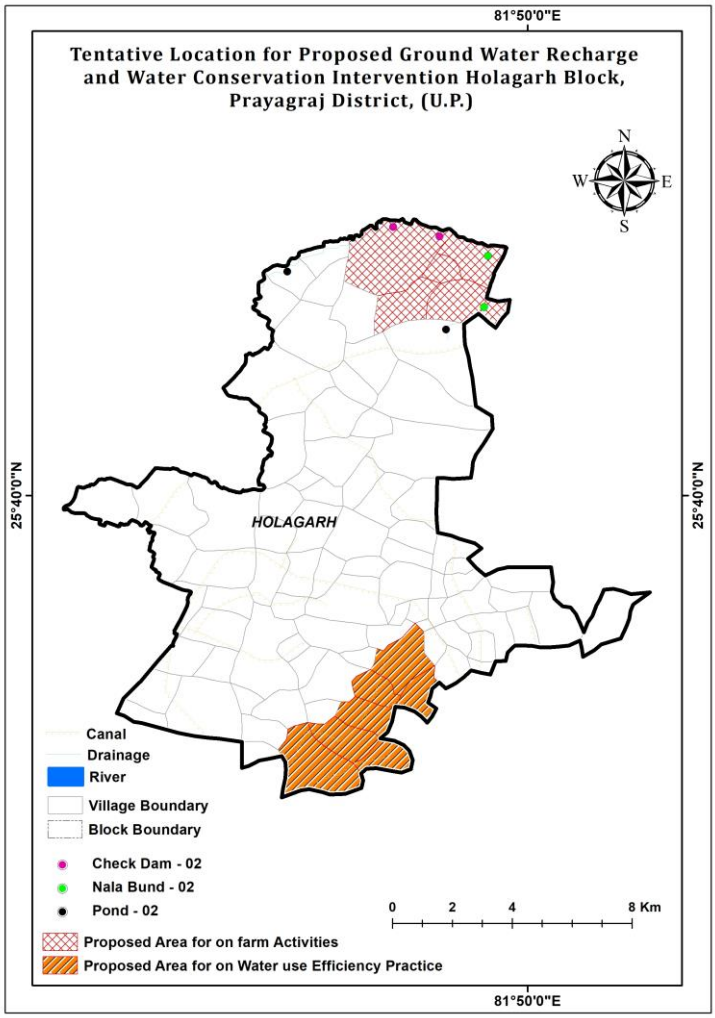
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water Use Efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
Holagarh	2	2	2	2	1208	1208	0.12	0.12	3.67	0.12	3.79	77.86	71.47

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
Holagarh	148.46	60.70	47.26	77.86	0.12	3.79	60.82	43.47	**71.47**

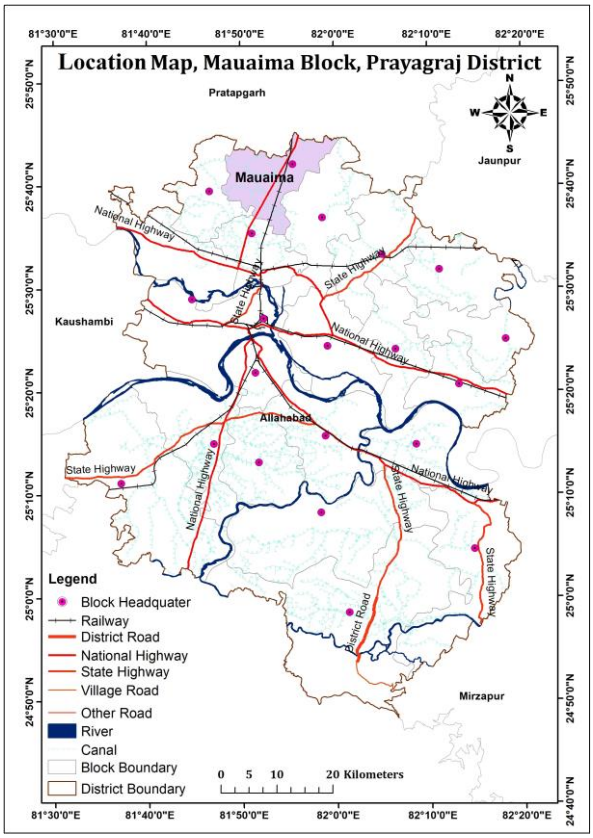


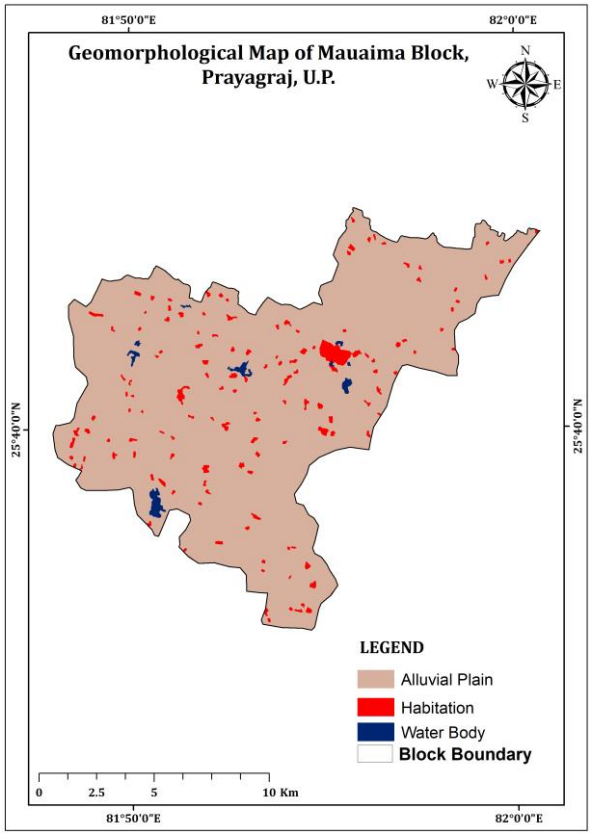
Aquifer Map and Management plan of Mauaima Block, Prayagraj District, U.P.

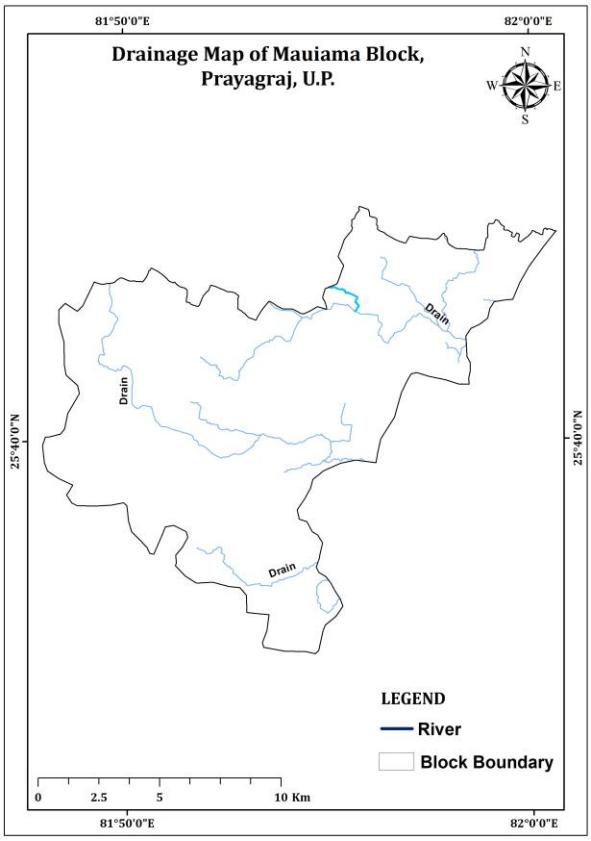
MAUAIMA BLOCK AT A GLANCE

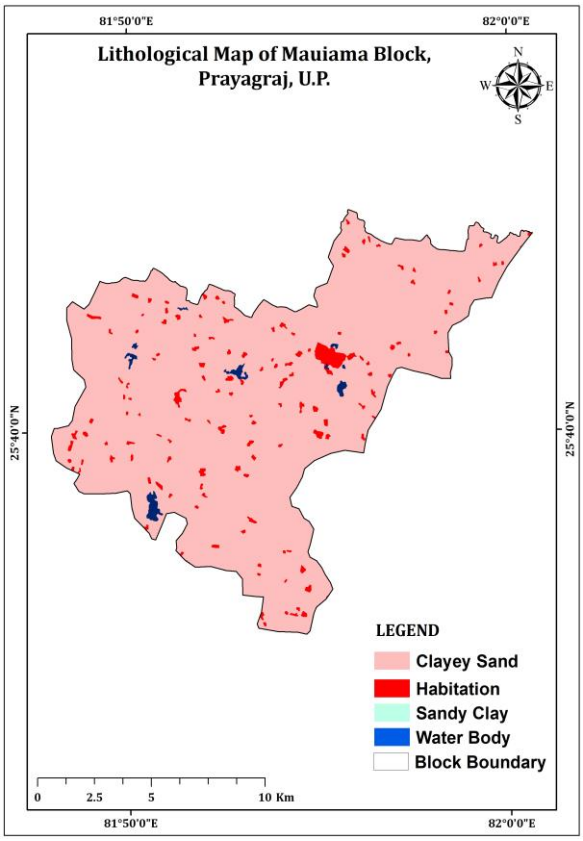
General Information

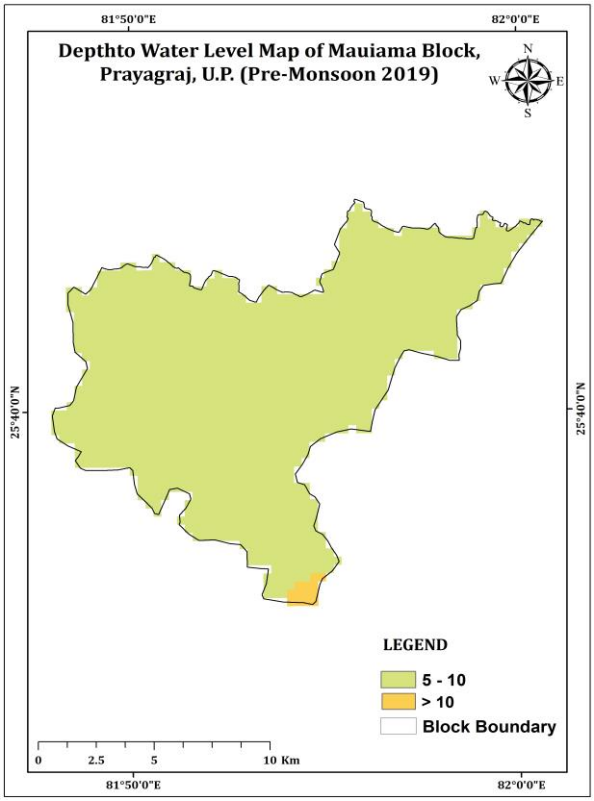
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Mauaima
Geographical area	155.60Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm

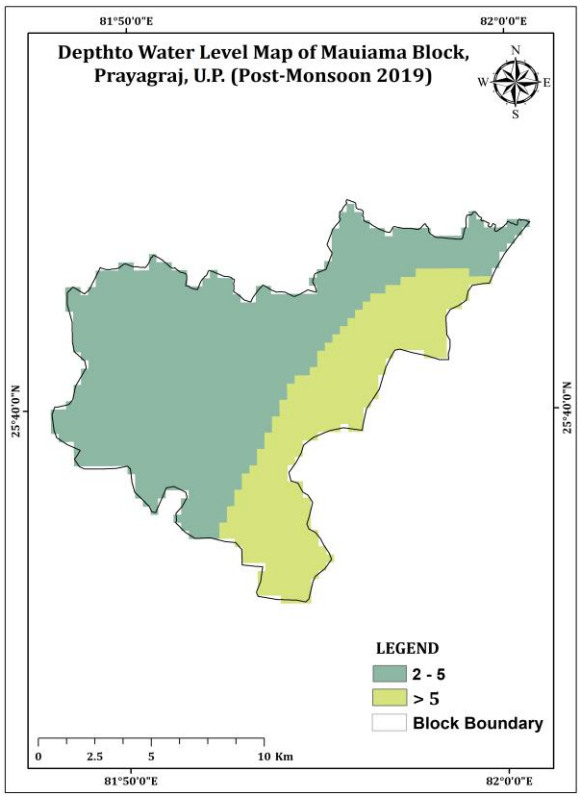





Drainage Map of Mauiama Block,
Prayagraj, U.P.
81°50'0"E
82°0'0"E
25°40'0"N
Drain
Drain
Drain
LEGEND
River
Block Boundary
0
2.5
5
10 Km


Lithological Map of Mauiama Block,
Prayagraj, U.P.
81°50'0"E
82°0'0"E
25°40'0"N
LEGEND
Clayey Sand
Habitation
Sandy Clay
Water Body
Block Boundary
0
2.5
5
10 Km


Depthto Water Level Map of Mauiama Block,
Prayagraj, U.P. (Pre-Monsoon 2019)
81°50'0"E
82°0'0"E
25°40'0"N
LEGEND
5 - 10
> 10
Block Boundary
0
2.5
5
10 Km


Depthto Water Level Map of Mauiama Block,
Prayagraj, U.P. (Post-Monsoon 2019)
81°50'0"E
82°0'0"E
25°40'0"N
LEGEND
2 - 5
> 5
Block Boundary
0
2.5
5
10 Km

Aquifer Disposition

Aquifer Disposition	-
Groundwater Monitoring Status	-
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable GW Resource : 55.75MCM GW Draft :45.62MCM Stage of GW Development :81.83% Semi- Critical
Existing and Future Water Demand	Present demand for All Usage :45.62MCM Future Demand for Domestic Use: 6.07 MCM

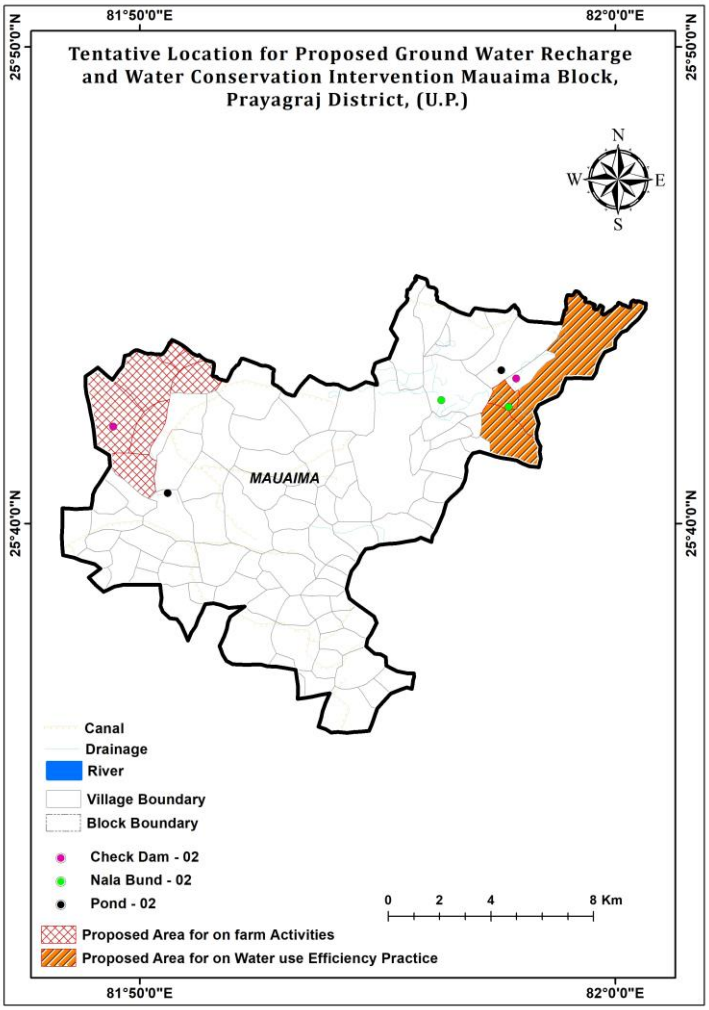
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water Use Efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
Mauiama	2	2	2	2	1271	1271	0.12	0.12	3.45	0.12	3.57	81.83	75.26

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
Mauiama	155.6	55.75	45.62	81.83	0.12	3.57	55.88	42.05	**75.26**

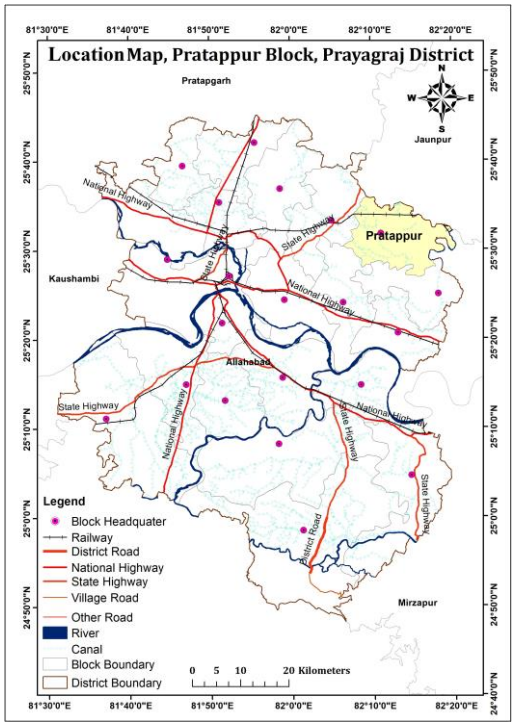


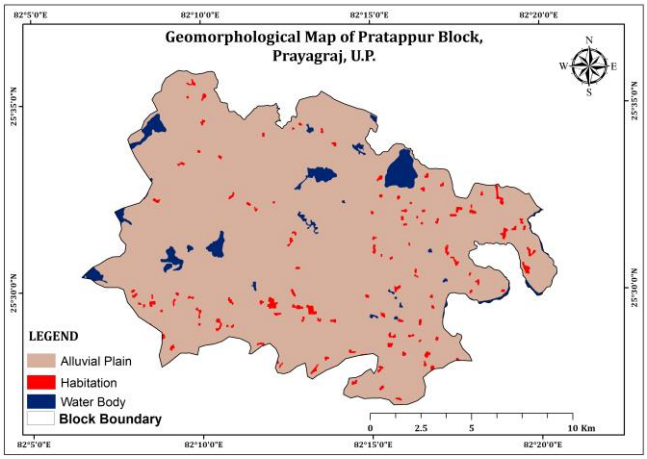
Aquifer Map and Management plan of Pratappur Block, Prayagraj District, U.P.

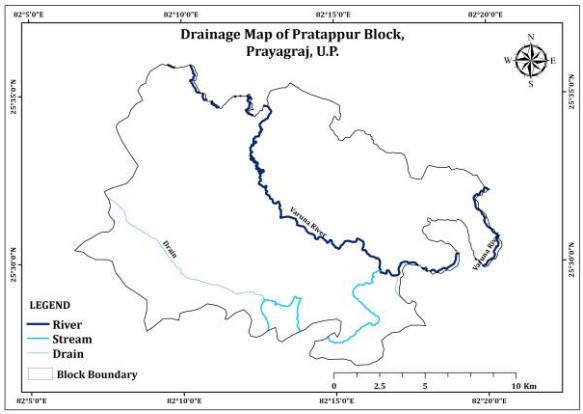
PRATAPPUR BLOCK AT A GLANCE

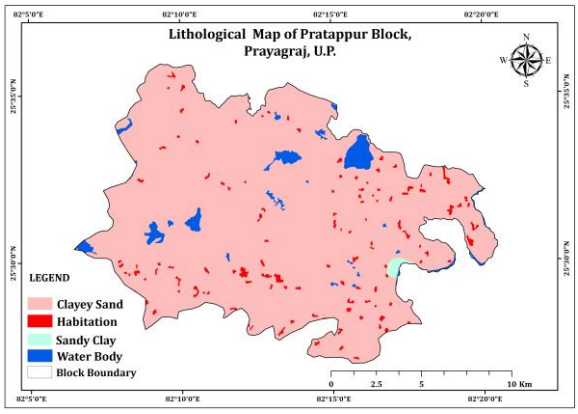
General Information

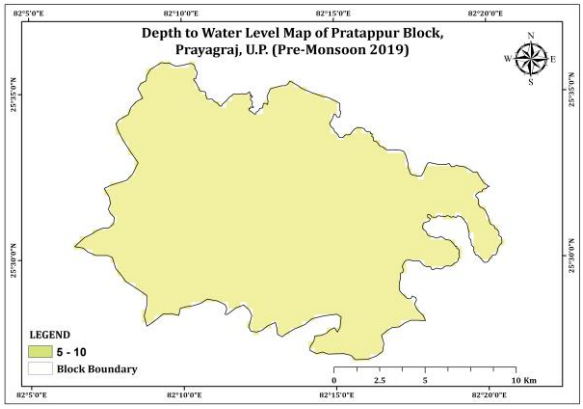
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Pratappur
Geographical area	211.07 Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm

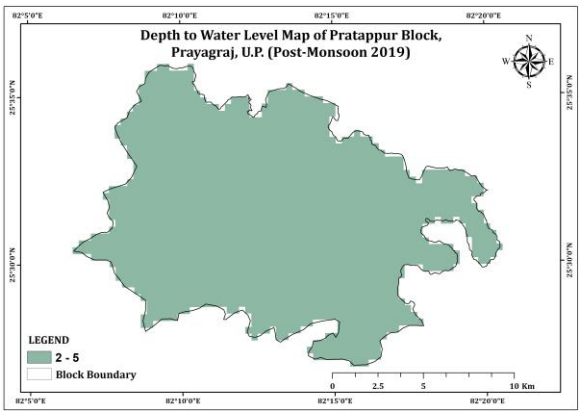












Aquifer Disposition

Aquifer Disposition	-
Groundwater Monitoring Status	-
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	
Groundwater Resource	Annual Extractable GW Resource : 66.75MCM GW Draft :58.07MCM Stage of GW Development :86.99% Semi- Critical
Existing and Future Water Demand	Present demand for All Usage :58.07MCM Future Demand for Domestic Use: 7.36 MCM

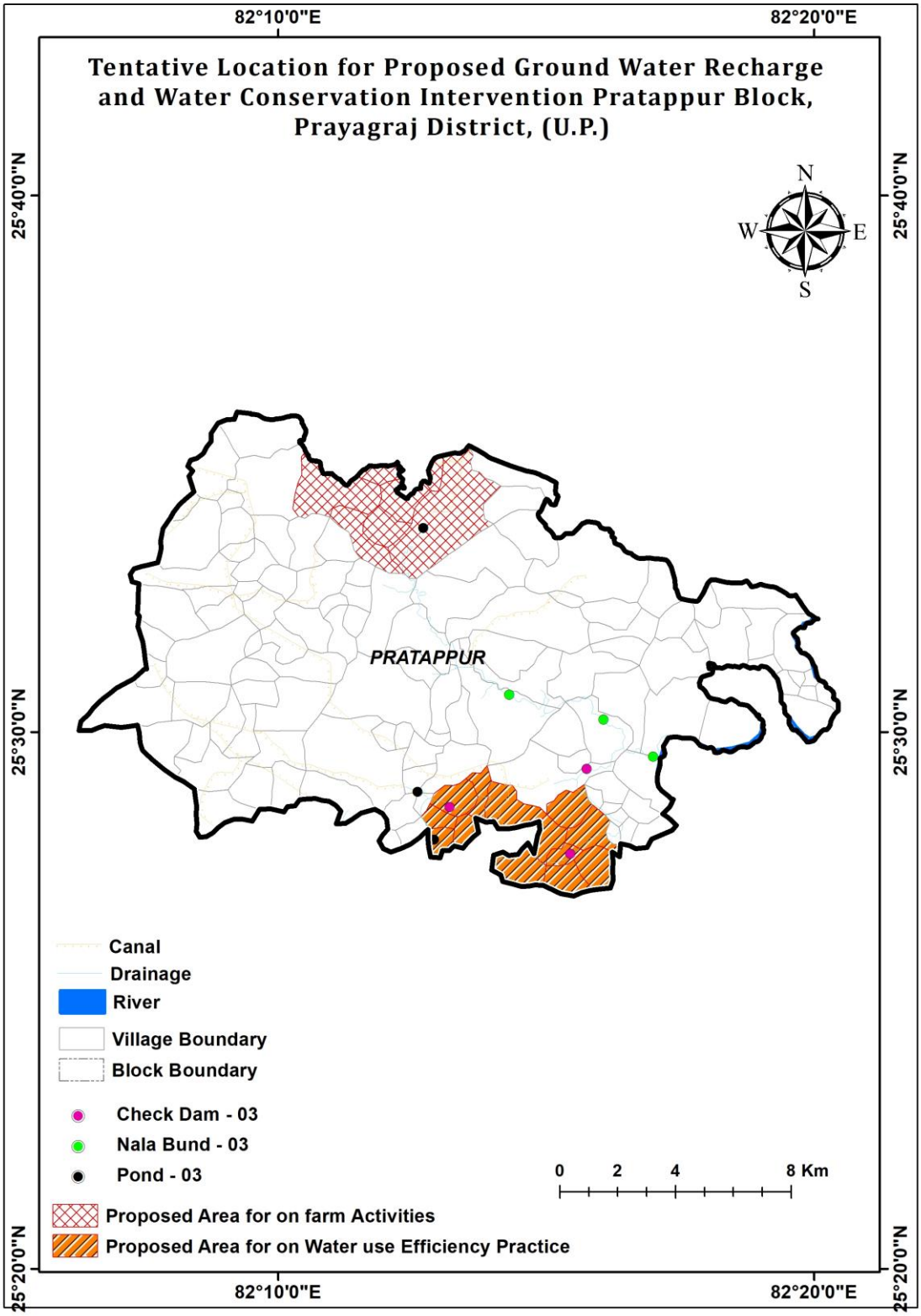
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water Use Efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
Pratappur	3	3	3	3	1326	1326	0.17	0.17	4.79	0.17	4.95	86.99	79.37

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
Pratappur	211.07	66.76	58.07	86.99	0.17	4.95	66.93	53.12	**79.37**

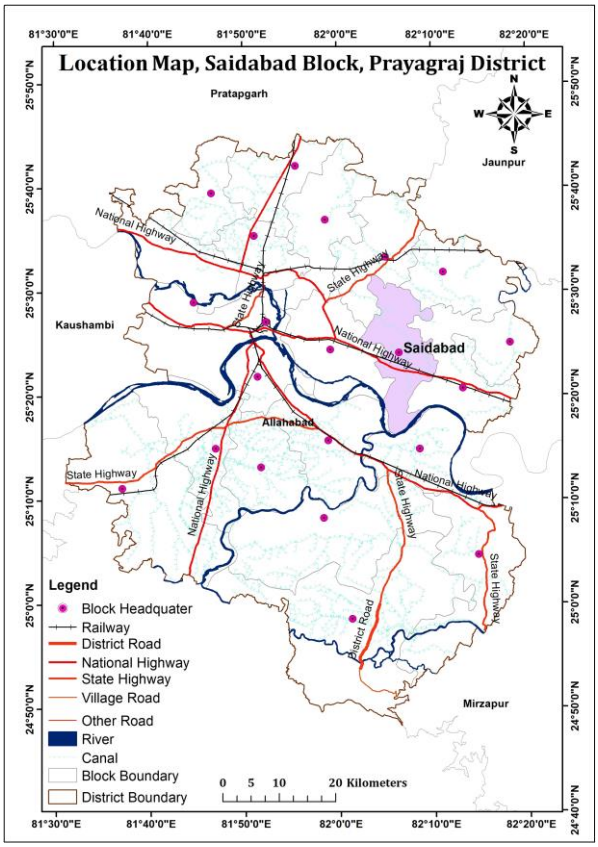

Tentative Location for Proposed Ground Water Recharge and Water Conservation Intervention Pratappur Block, Prayagraj District, (U.P.)
82°10'0"E
82°20'0"E
25°40'0"N
25°30'0"N
25°20'0"N
N
W
E
S
PRATAPPUR
Canal
Drainage
River
Village Boundary
Block Boundary
Check Dam - 03
Nala Bund - 03
Pond - 03
0 2 4 8 Km
Proposed Area for on farm Activities
Proposed Area for on Water use Efficiency Practice

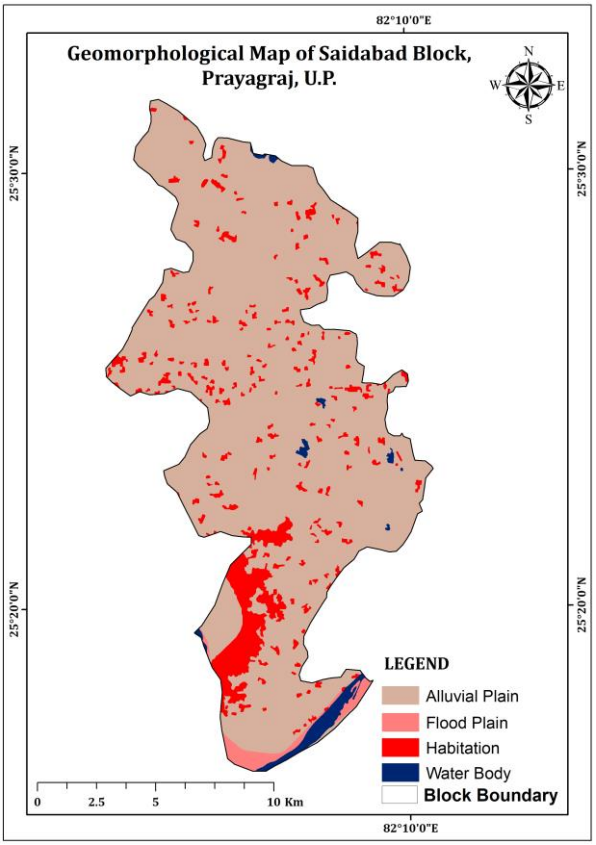
Aquifer Map and Management plan of Saidabad Block, Prayagraj District, U.P.

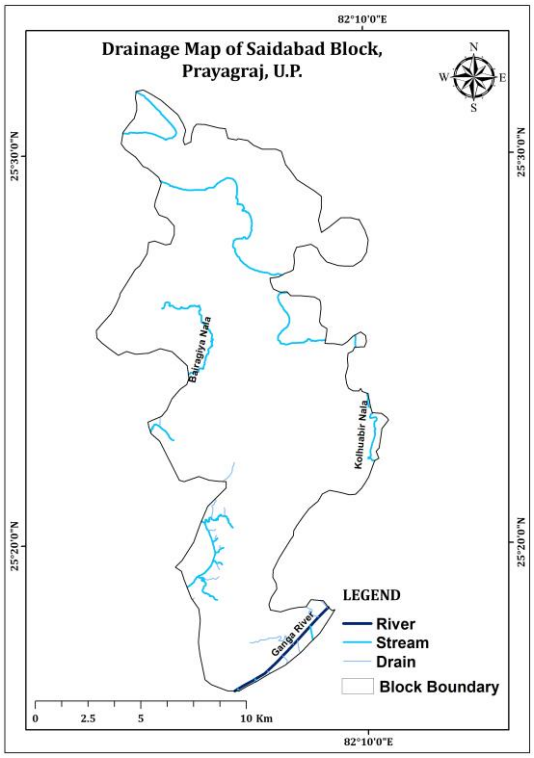
SAIDABAD BLOCK AT A GLANCE

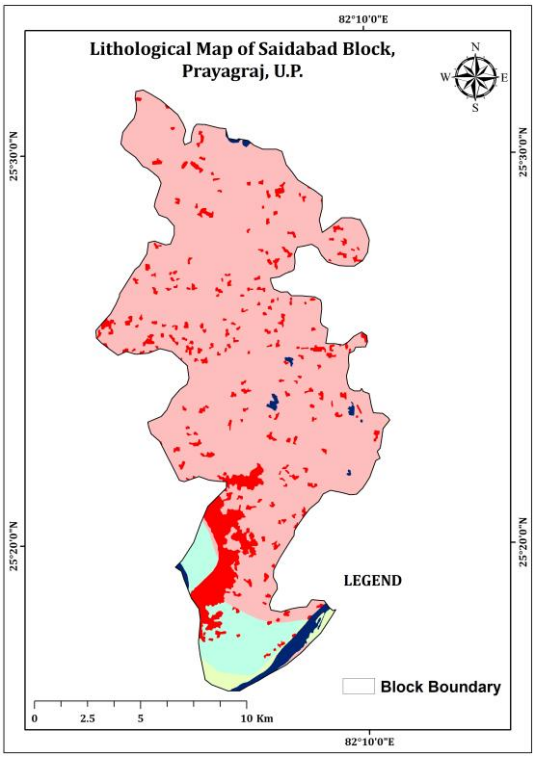
General Information

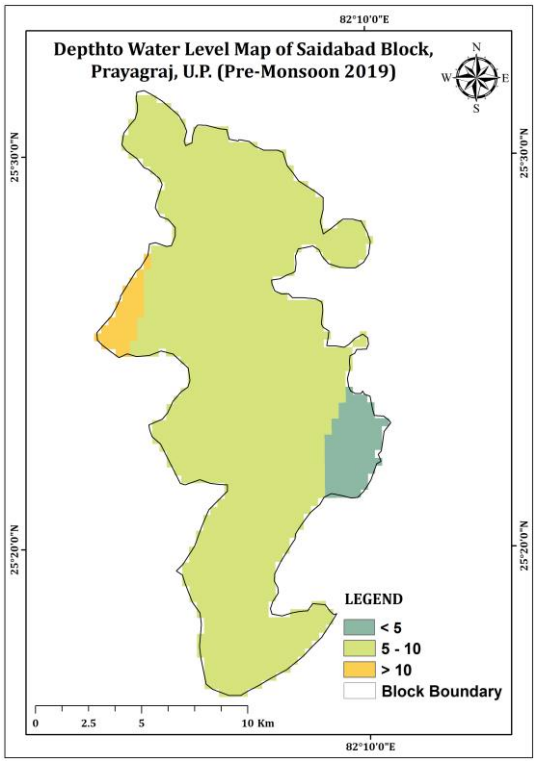
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Saidabad
Geographical area	191.42Sq.Km.
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00 mm

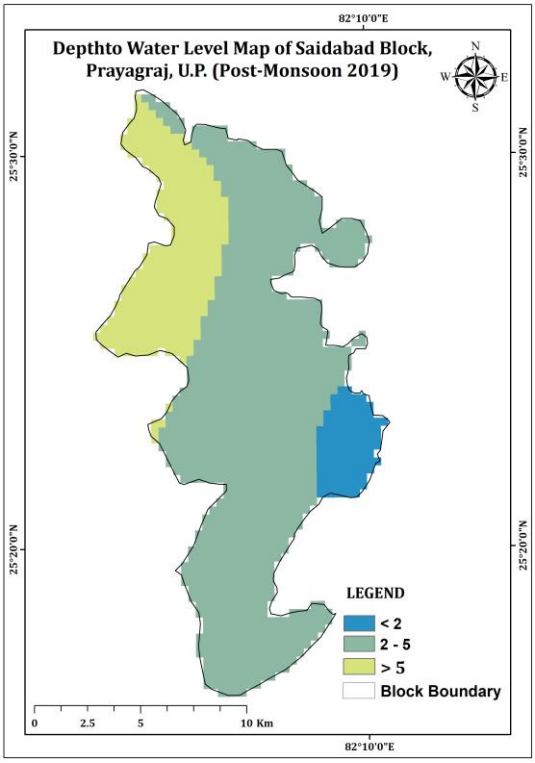





Drainage Map of Saidabad Block,
Prayagraj, U.P.
82°10'0"E
25°30'0"N
25°20'0"N
N
S
E
W
Bhargava Nala
Kolhuabir Nala
Ganga River
LEGEND
River
Stream
Drain
Block Boundary
0
2.5
5
10 Km


Lithological Map of Saidabad Block,
Prayagraj, U.P.
82°10'0"E
25°30'0"N
25°20'0"N
N
S
E
W
LEGEND
Block Boundary
0
2.5
5
10 Km


Depthto Water Level Map of Saidabad Block,
Prayagraj, U.P. (Pre-Monsoon 2019)
82°10'0"E
25°30'0"N
25°20'0"N
N
S
E
W
LEGEND
< 5
5 - 10
> 10
Block Boundary
0
2.5
5
10 Km


Depthto Water Level Map of Saidabad Block,
Prayagraj, U.P. (Post-Monsoon 2019)
82°10'0"E
25°30'0"N
25°20'0"N
N
S
E
W
LEGEND
< 2
2 - 5
> 5
Block Boundary
0
2.5
5
10 Km

Aquifer Disposition

Aquifer Disposition	-
Groundwater Monitoring Status	-
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable GW Resource : 57.96MCM GW Draft :47.65MCM Stage of GW Development :82.21% Semi- Critical
Existing and Future Water Demand	Present demand for All Usage :47.65MCM Future Demand for Domestic Use: 5.52 MCM

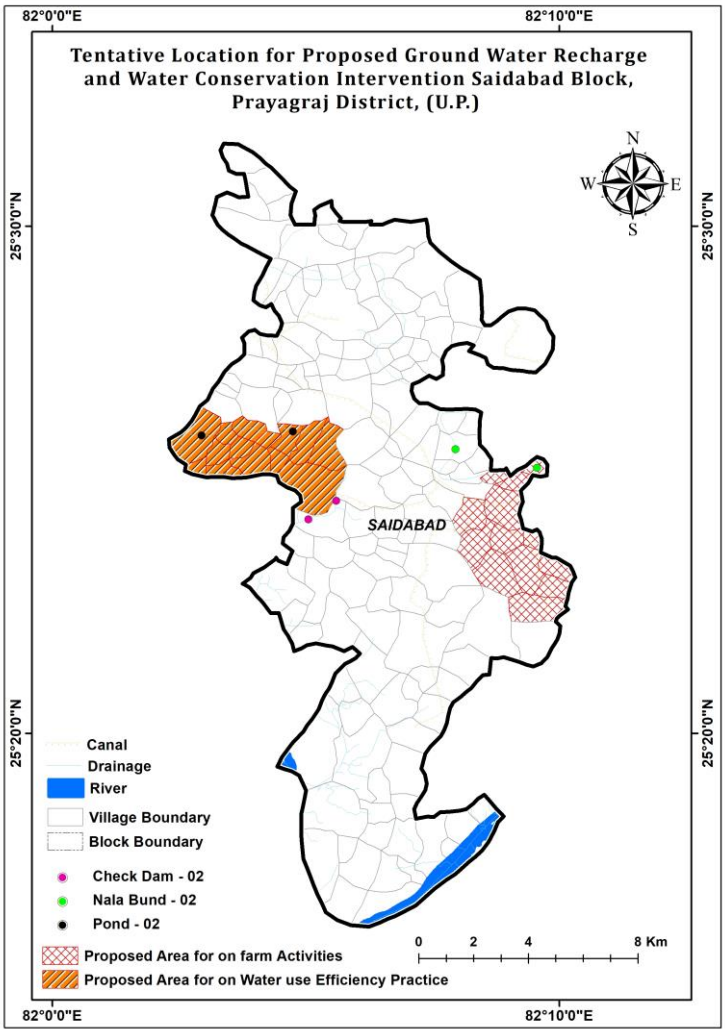
Aquifer Management Plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial recharge and water conservation structures, On-farm activities and adoption of water use efficiency measures.
Groundwater Management Plan	**Supply Side Management:** Water conservation and Artificial Recharge to ground water and on Farm Activities. **Demand Side Management:** Adoption of techniques to enhance water Use Efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern.
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
Saidabad	2	2	2	2	1172	1172	0.15	0.15	3.76	0.15	3.91	82.21	75.26

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
Saidabad	191.42	57.97	47.65	82.21	0.15	3.91	58.12	43.74	**75.26**

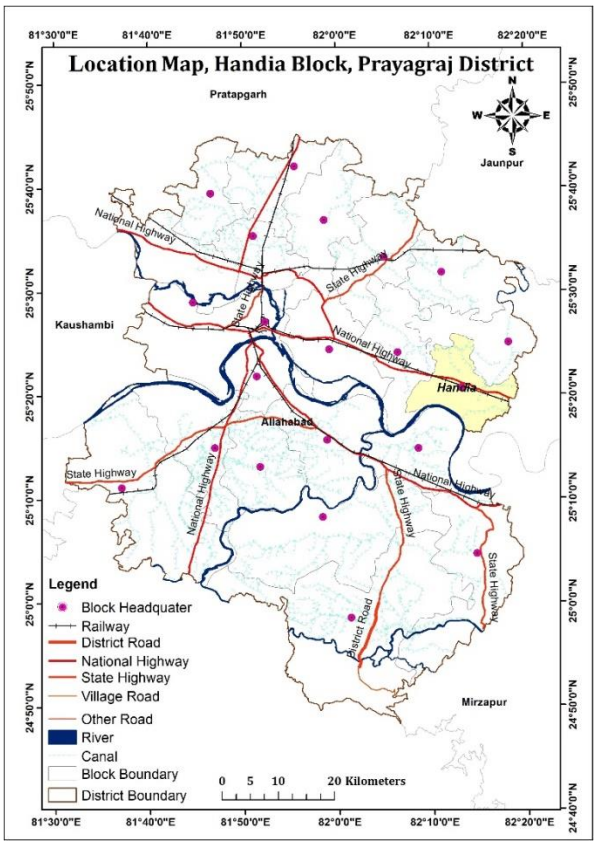


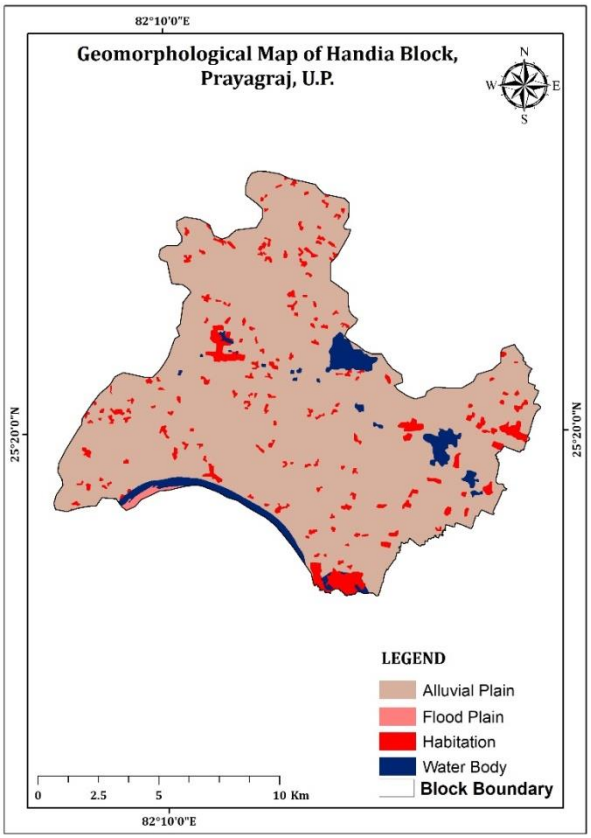
Aquifer Map and Management plan of Handia_Block, Prayagraj District, U.P.

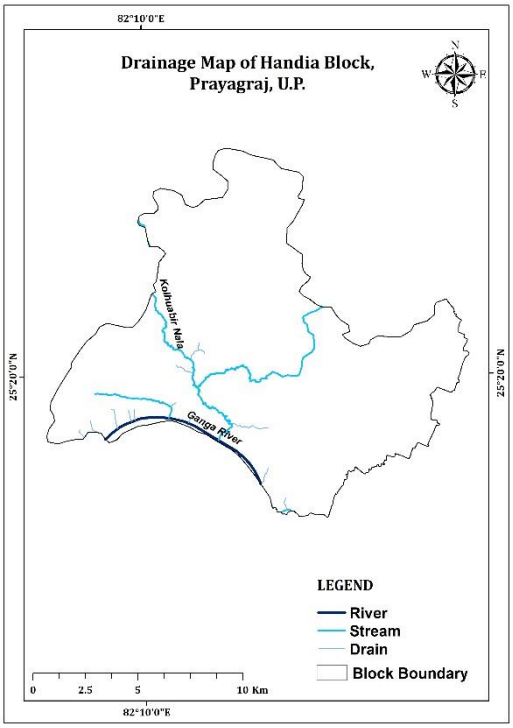
HANDIA_BLOCK AT A GLANCE

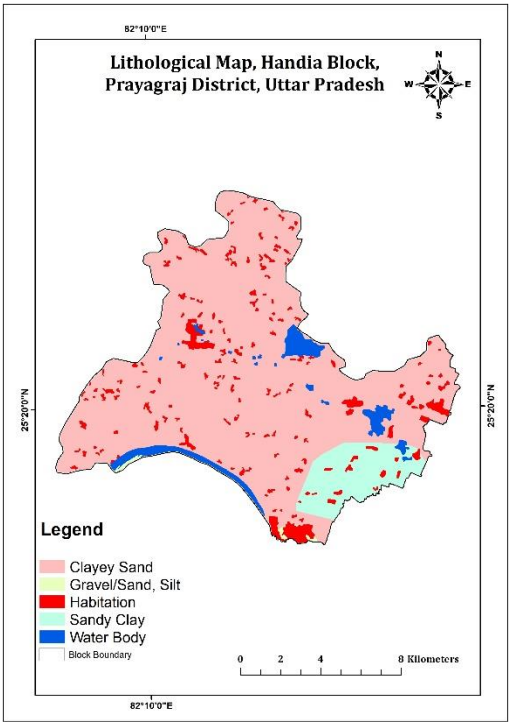
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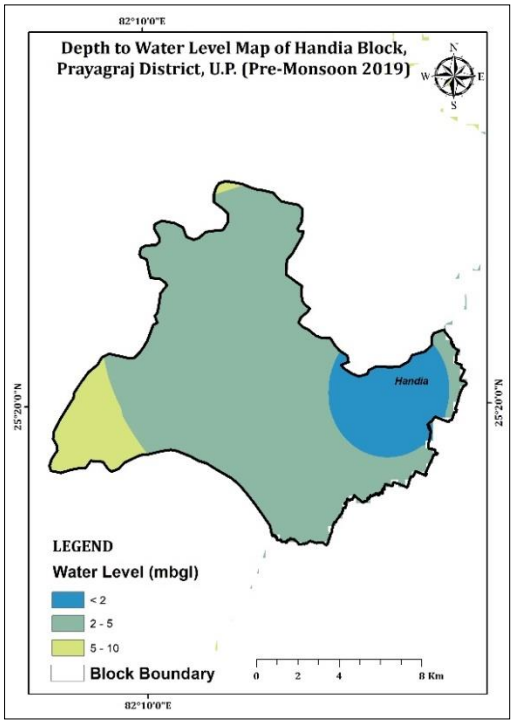
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Handia
Geographical area	170.52km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

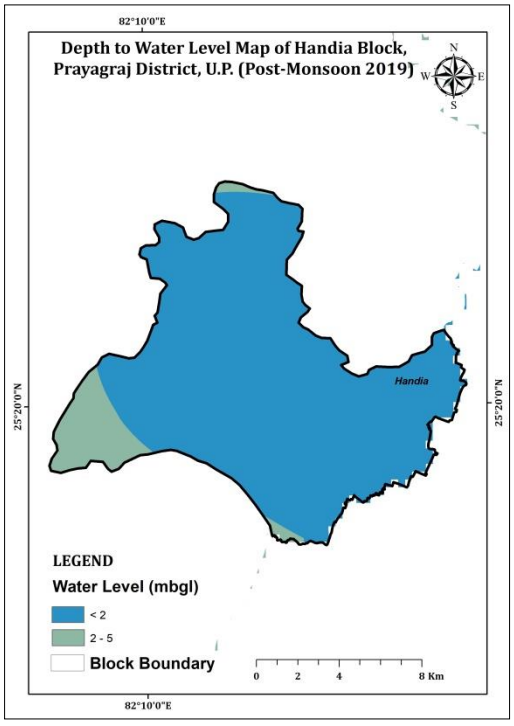





Drainage Map of Handia Block,
Prayagraj, U.P.
82°10'0"E
25°20'0"N
Kolhuabir Nala
Ganga River
LEGEND
River
Stream
Drain
Block Boundary
0 2.5 5 10 Km


Lithological Map, Handia Block,
Prayagraj District, Uttar Pradesh
82°10'0"E
25°20'0"N
Legend
Clayey Sand
Gravel/Sand, Silt
Habitation
Sandy Clay
Water Body
Block Boundary
0 2 4 8 Kilometers


Depth to Water Level Map of Handia Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
82°10'0"E
25°20'0"N
Handia
LEGEND
Water Level (mbgl)
< 2
2 - 5
5 - 10
Block Boundary
0 2 4 8 Km


Depth to Water Level Map of Handia Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
82°10'0"E
25°20'0"N
Handia
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 2 4 8 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	2
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource: 46.71 Ground water Draft: 31.51 Stage of Ground water development: 67.46% Category: safe
Existing and Future Water Demand	Present demand for all Usages:31.51mcm Future Demand for Domestic and Industrial Use:14.2895mcm

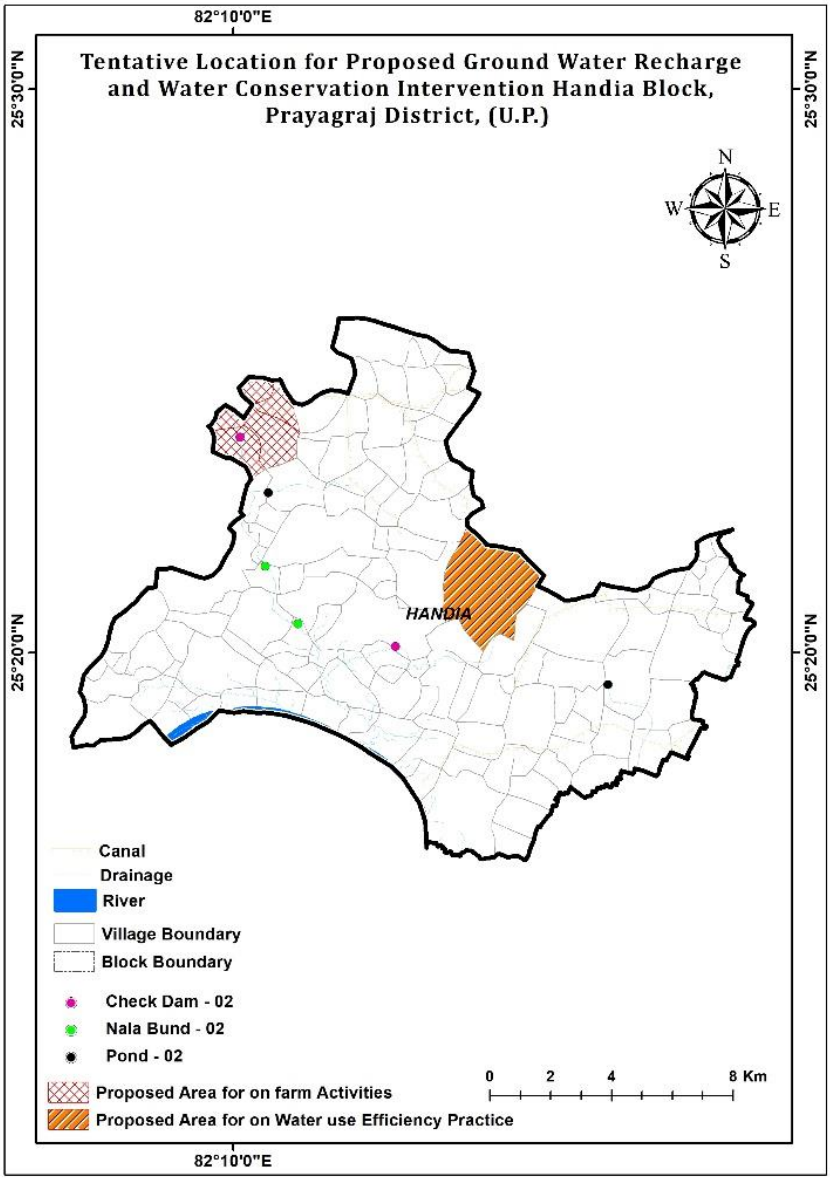
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
HANDIA	2	2	2	2	488	488	0.11	0.11	0.14	0.11	1.25	67.46	64.62

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
HANDIA	170.52	46.71	31.51	67.46	0.11	1.25	46.82	30.26	**64.62**

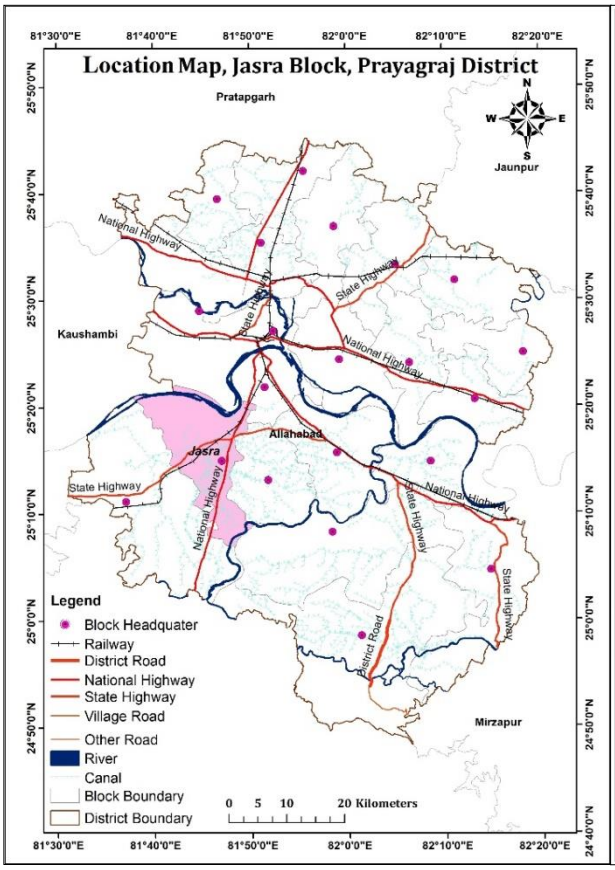


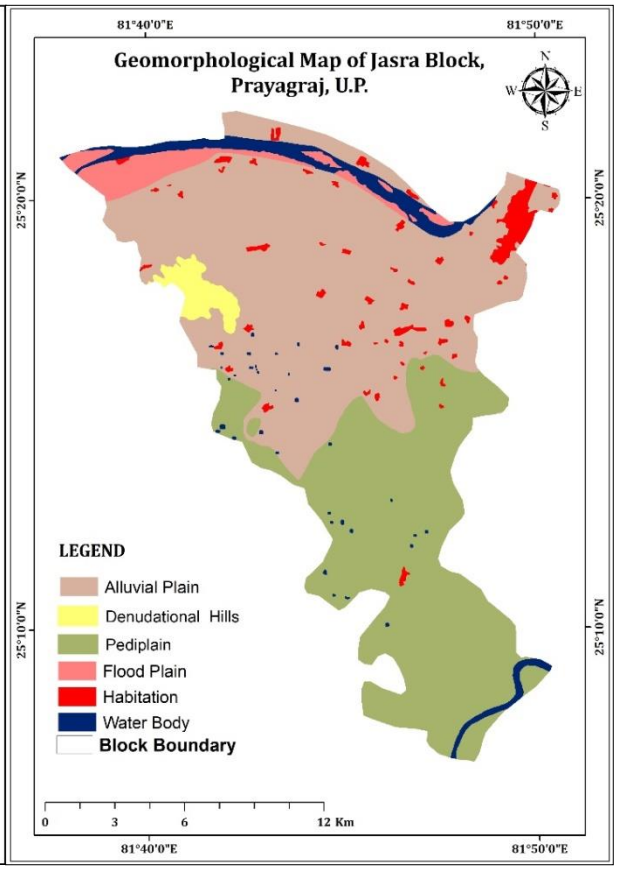
Aquifer Map and Management plan of Jasra Block, Prayagraj District, U.P.

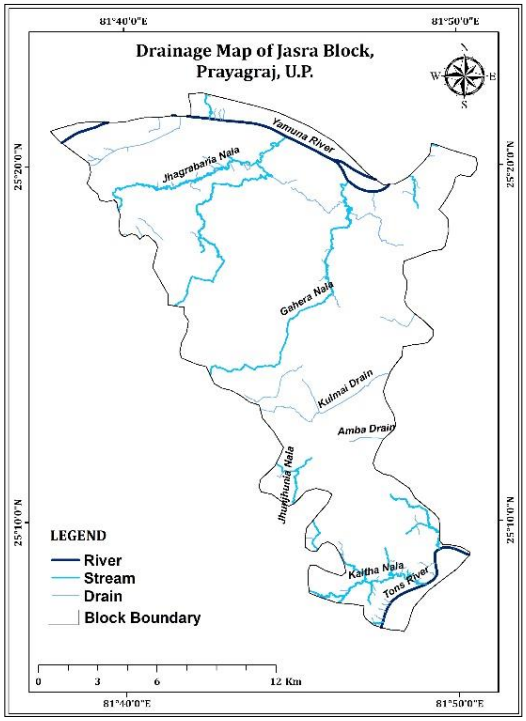
JASRA BLOCK AT A GLANCE

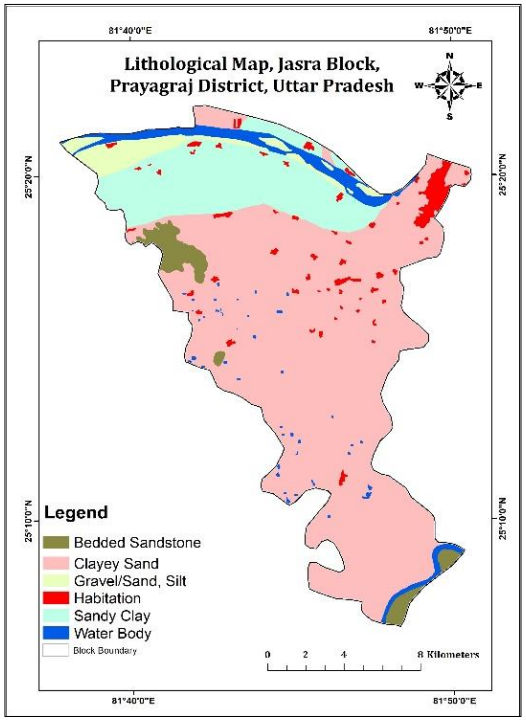
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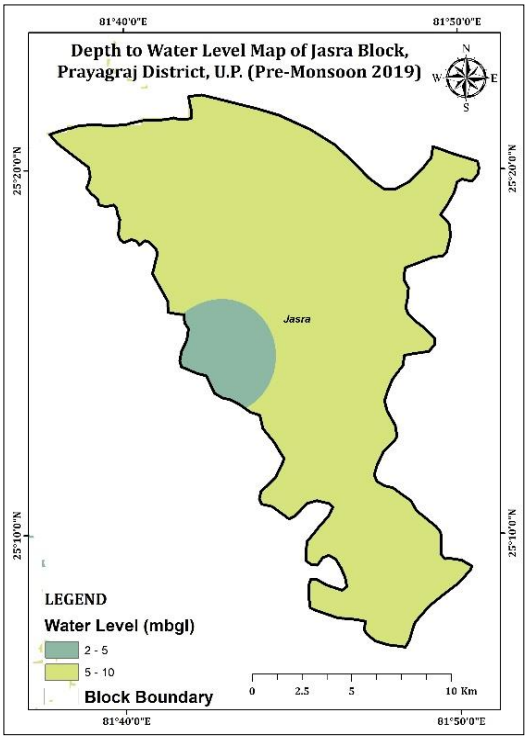
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Jasra
Geographical area	268.22km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Major Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Normal Annual Rainfall	1042.00mm

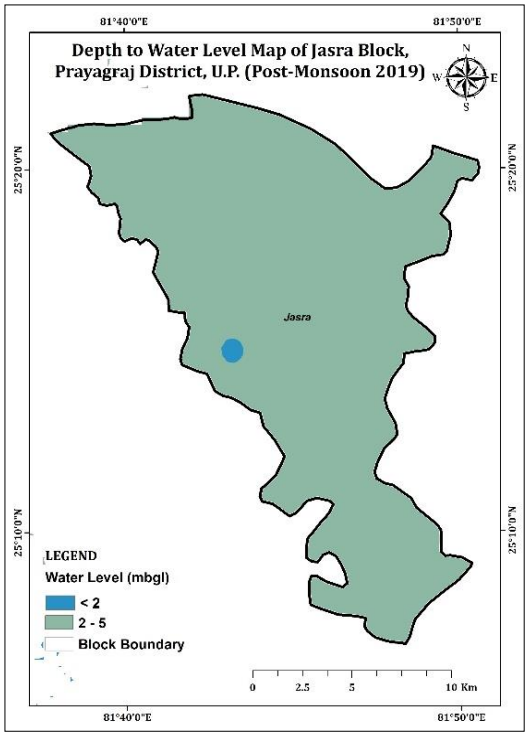





Drainage Map of Jasra Block,
Prayagraj, U.P.
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
Yamuna River
Jhagrabaria Nala
Gehera Nala
Kulmai Drain
Amba Drain
Jhudhuna Nala
Kaitha Nala
Tons River
LEGEND
River
Stream
Drain
Block Boundary
0 3 6 12 Km


Lithological Map, Jasra Block,
Prayagraj District, Uttar Pradesh
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
Legend
Bedded Sandstone
Clayey Sand
Gravel/Sand, Silt
Habitation
Sandy Clay
Water Body
Block Boundary
0 2 4 8 Kilometers


Depth to Water Level Map of Jasra Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
Jasra
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2.5 5 10 Km


Depth to Water Level Map of Jasra Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
Jasra
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 2.5 5 10 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	4
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource: 75.86mcm Ground water Draft: 52.87mcm Stage of Ground water development: 69.69**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages: 52.86mcm Future Demand for Domestic and Industrial Use: 22.43mcm

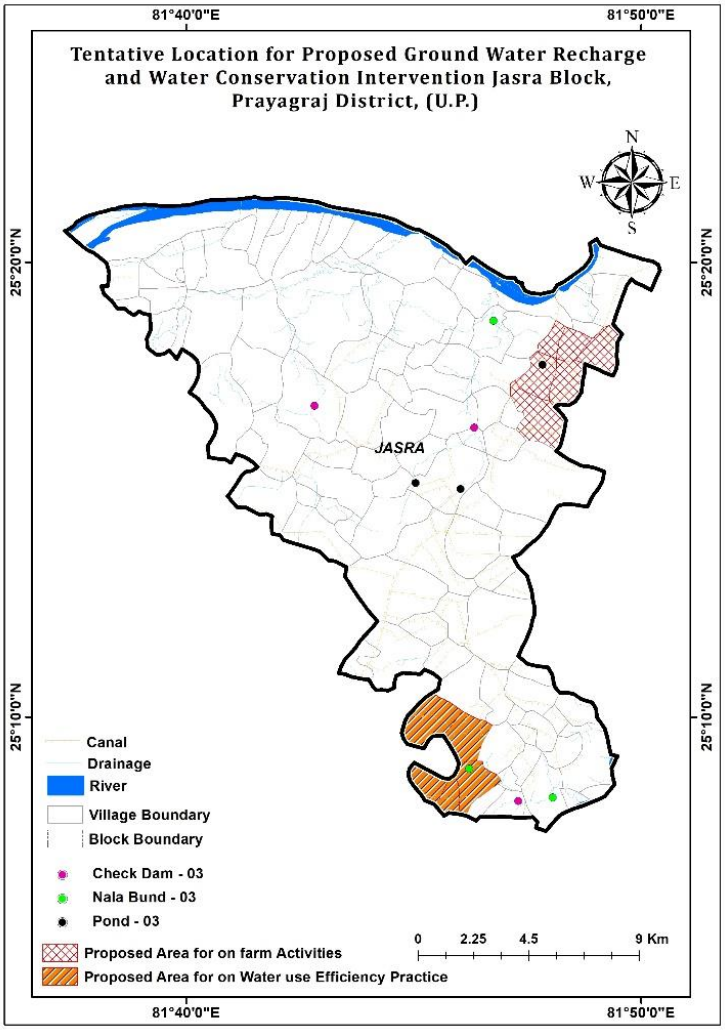
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
JASRA	3	3	3	3	910	910	0.17	0.17	2.01	0.17	2.18	69.69	66.66

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
JASRA	268.22	75.86	52.87	69.69	0.17	2.18	76.03	50.69	66.66

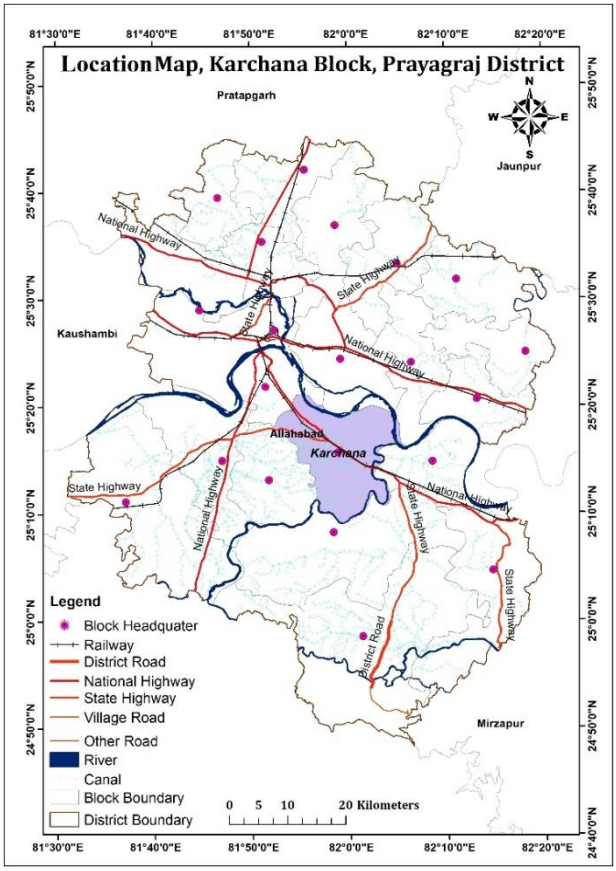


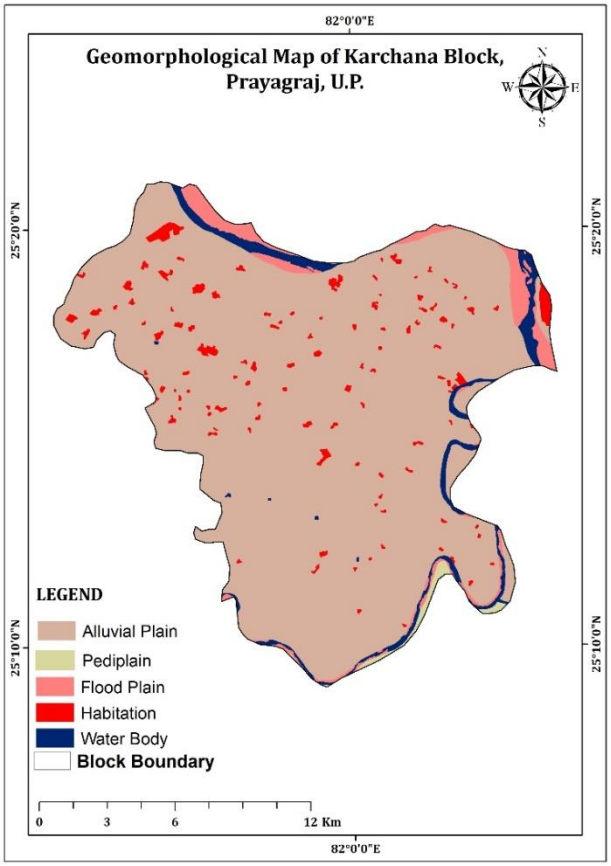
Aquifer Map and Management plan of Karchana Block, Prayagraj District, U.P.

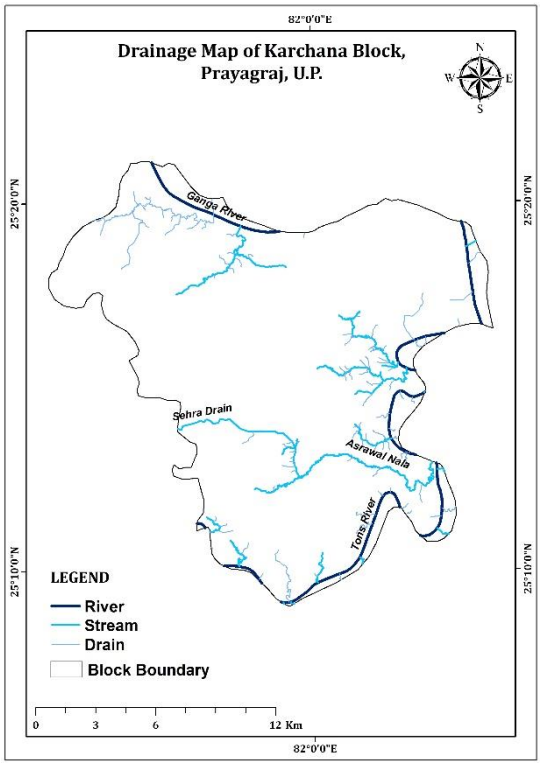
KARCHANA BLOCK AT A GLANCE

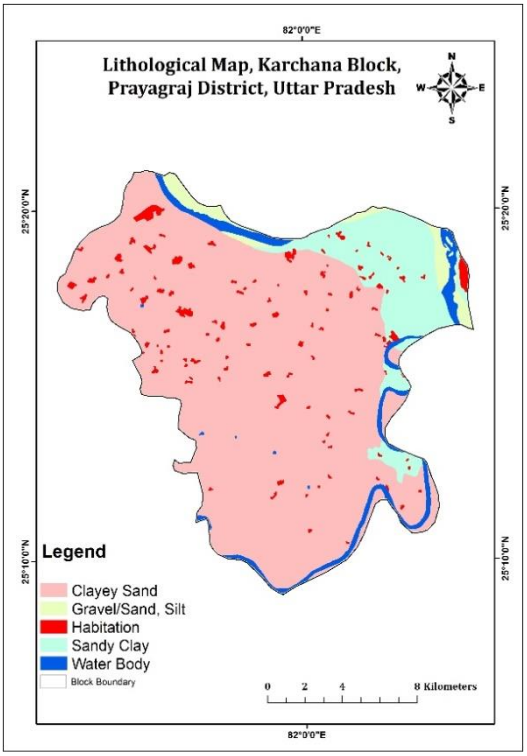
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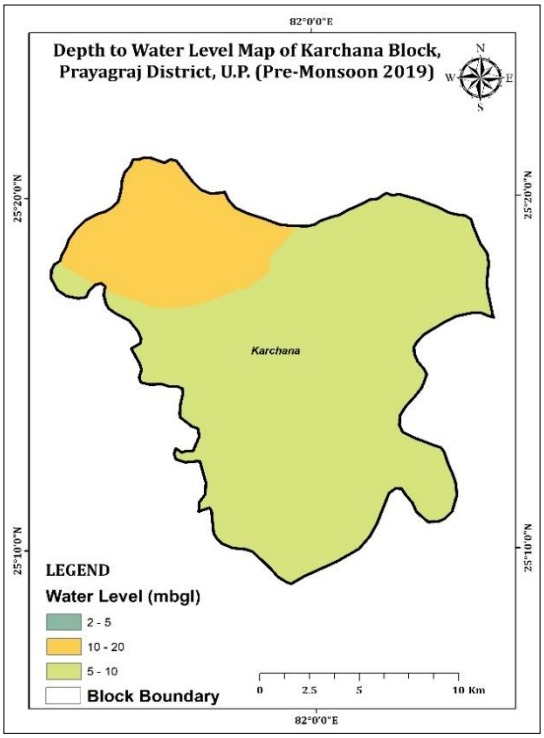
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Karchana
Geographical area	232.81km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

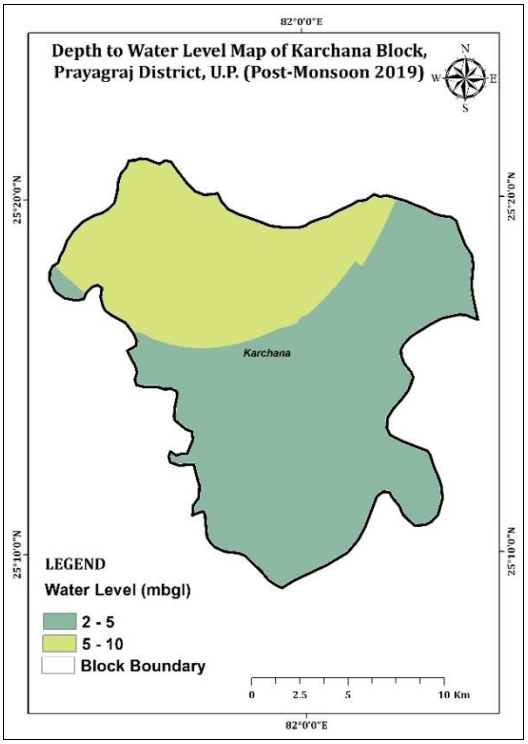





Drainage Map of Karchana Block,
Prayagraj, U.P.
82°0'0"E
25°20'0"N
25°10'0"N
Ganga River
Sehra Drain
Asrawal Nala
Tons River
LEGEND
River
Stream
Drain
Block Boundary
0 3 6 12 Km


Lithological Map, Karchana Block,
Prayagraj District, Uttar Pradesh
82°0'0"E
25°20'0"N
25°10'0"N
Legend
Clayey Sand
Gravel/Sand, Silt
Habitation
Sandy Clay
Water Body
Block Boundary
0 2 4 8 Kilometers


Depth to Water Level Map of Karchana Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
82°0'0"E
25°20'0"N
25°10'0"N
Karchana
LEGEND
Water Level (mbgl)
2 - 5
10 - 20
5 - 10
Block Boundary
0 2.5 5 10 Km


Depth to Water Level Map of Karchana Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
82°0'0"E
25°20'0"N
25°10'0"N
Karchana
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2.5 5 10 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	2
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource:79.47mcm Ground water Draft: 52.16mcm Stage of Ground water development: 65.63**%** Category: Safe
Existing and Future Water Demand	Present demand for all Usages: 52.15mcm Future Demand for Domestic and Industrial Use: 26.59mcm

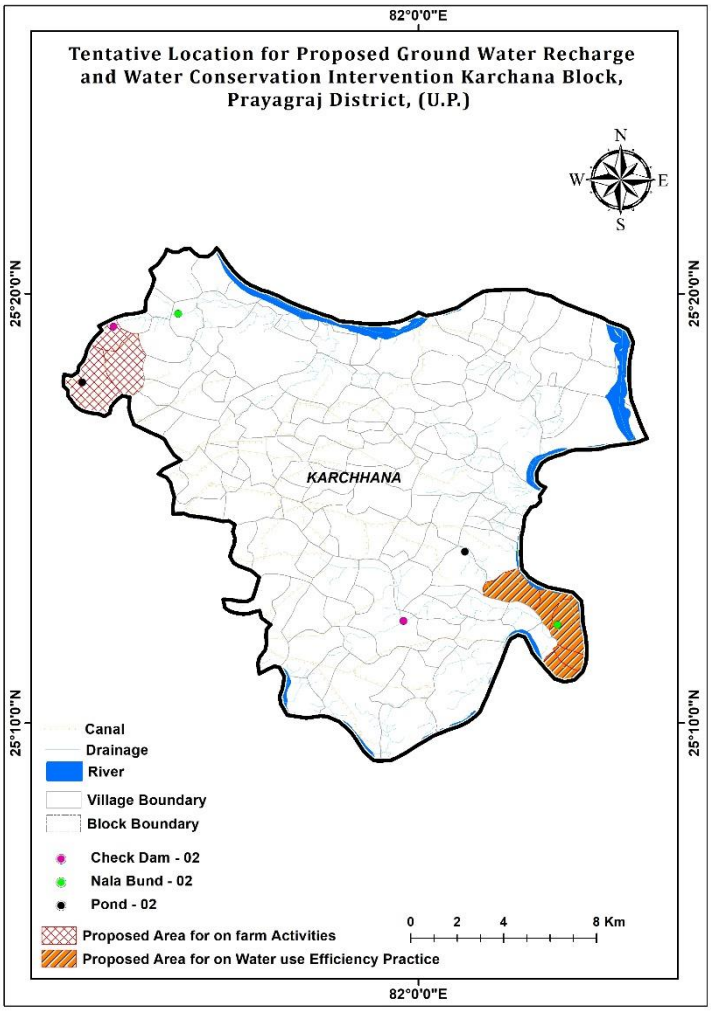
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
Karchana	2	2	2	2	791	791	0.15	0.15	1.71	0.15	1.86	65.63	63.17

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
Karchana	232.81	79.47	52.16	65.63	0.15	1.86	79.62	50.30	**63.17**

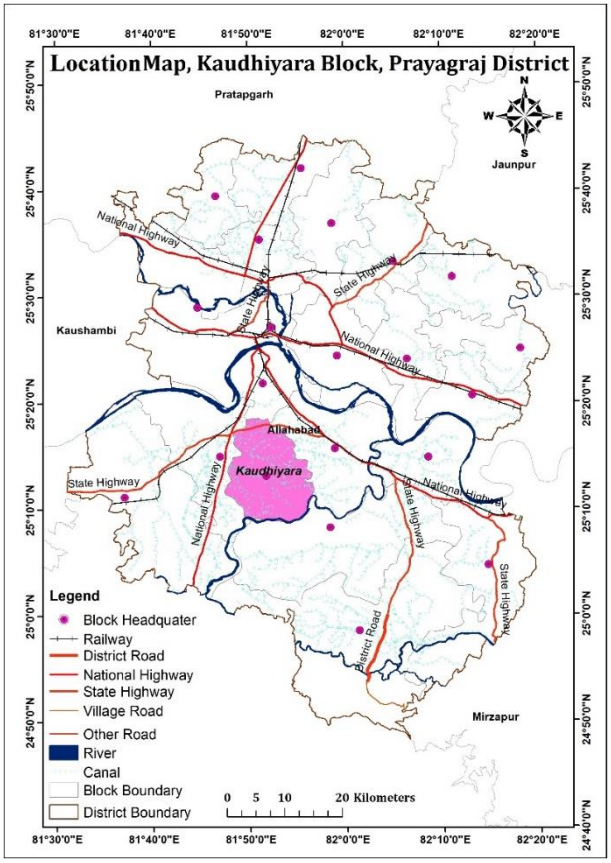


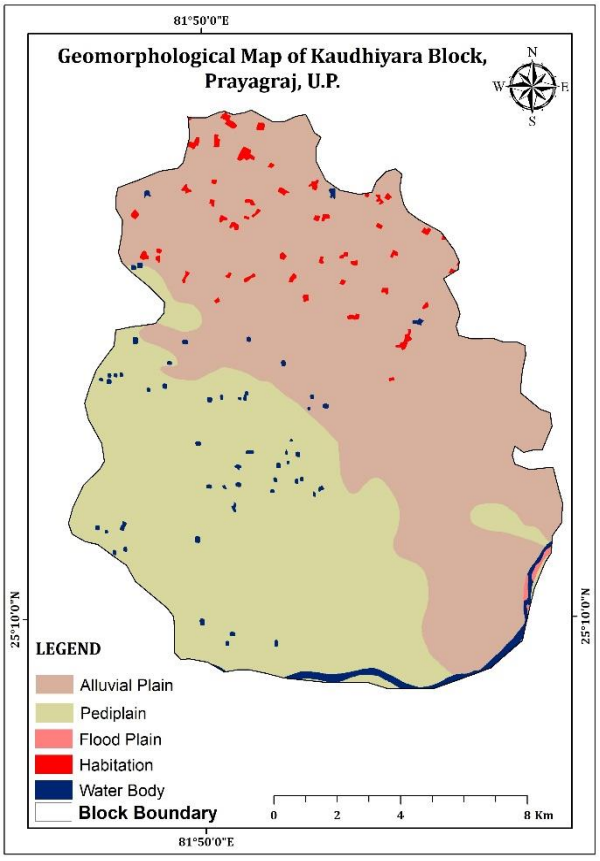
Aquifer Map and Management plan of Kaudhiyara Block, Prayagraj District, U.P.

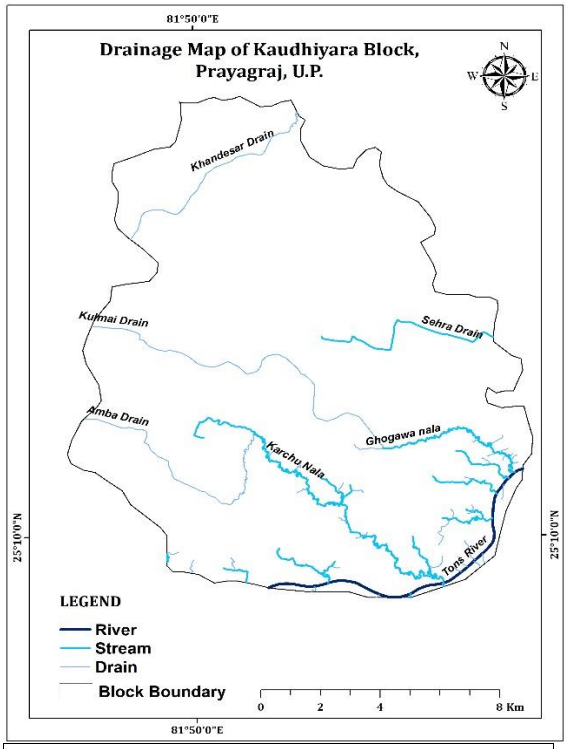
KAUDHIYARA BLOCK AT A GLANCE

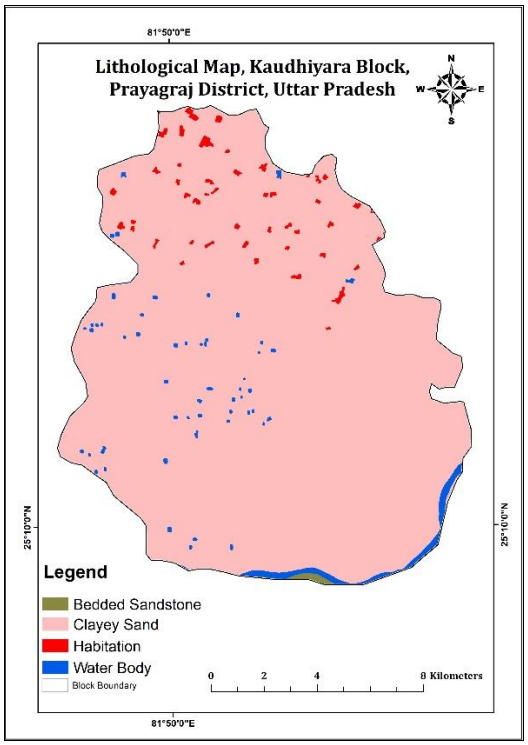
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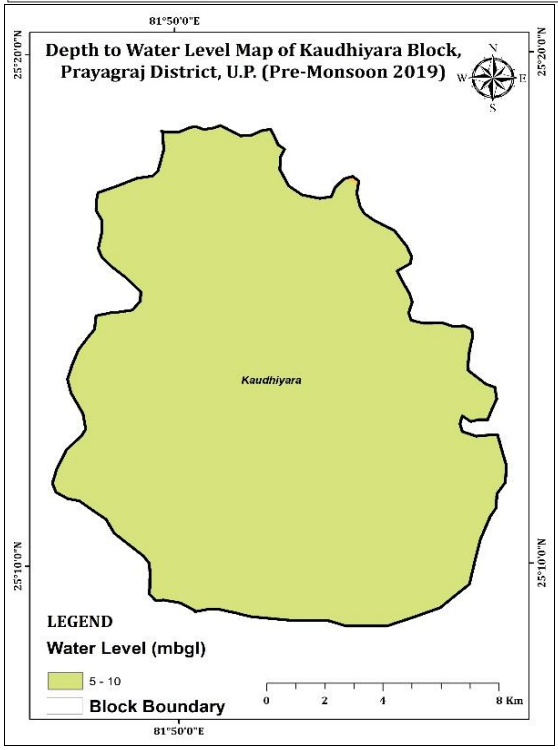
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Jasra
Geographical area	268.22km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

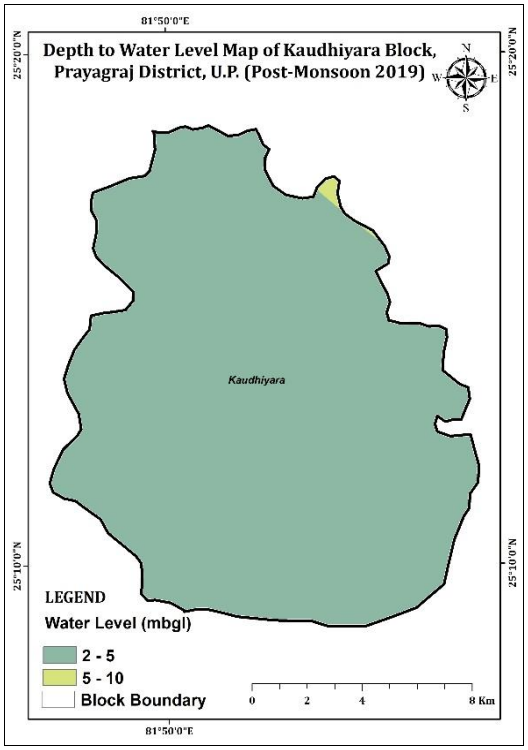





Drainage Map of Kaudhiyara Block,
Prayagraj, U.P.
81°50'0"E
25°10'0"N
Khandesar Drain
Kuhnai Drain
Sehra Drain
Amba Drain
Karchu Nala
Ghogawa nala
Tons River
LEGEND
River
Stream
Drain
Block Boundary
0 2 4 8 Km


Lithological Map, Kaudhiyara Block,
Prayagraj District, Uttar Pradesh
81°50'0"E
25°10'0"N
Legend
Bedded Sandstone
Clayey Sand
Habitation
Water Body
Block Boundary
0 2 4 8 Kilometers


Depth to Water Level Map of Kaudhiyara Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
81°50'0"E
25°20'0"N
25°10'0"N
Kaudhiyara
LEGEND
Water Level (mbgl)
5 - 10
Block Boundary
0 2 4 8 Km


Depth to Water Level Map of Kaudhiyara Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
81°50'0"E
25°10'0"N
25°0'0"N
Kaudhiyara
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2 4 8 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	2
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Recharge:68.37 Ground water Draft: 41.17 Stage of Ground water development: 60.22 Category: safe
Existing and Future Water Demand	Present demand for all Usages: 41.17 Future Demand for Domestic and Industrial Use: 26.42

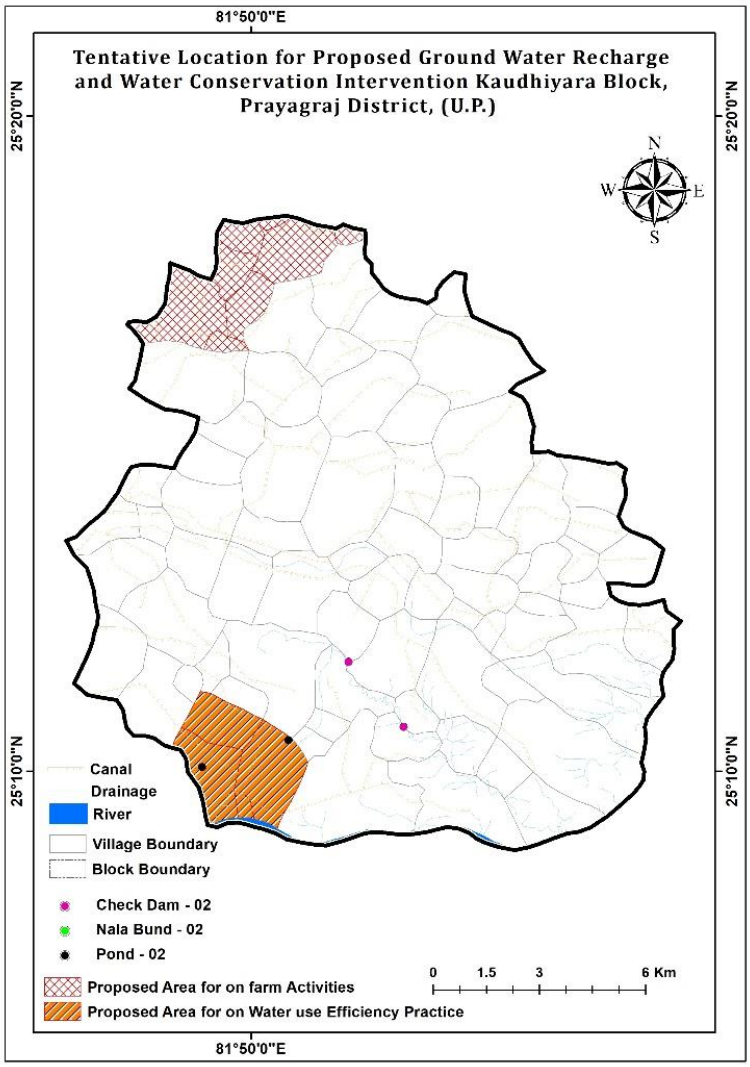
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
KAUDHIYARA	2	2	2	2	782	782	0.13	0.13	1.39	0.13	1.52	60.22	57.88

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
KAUDHIYARA	200.46	68.37	41.17	60.22	0.13	1.52	68.50	39.65	**57.88**

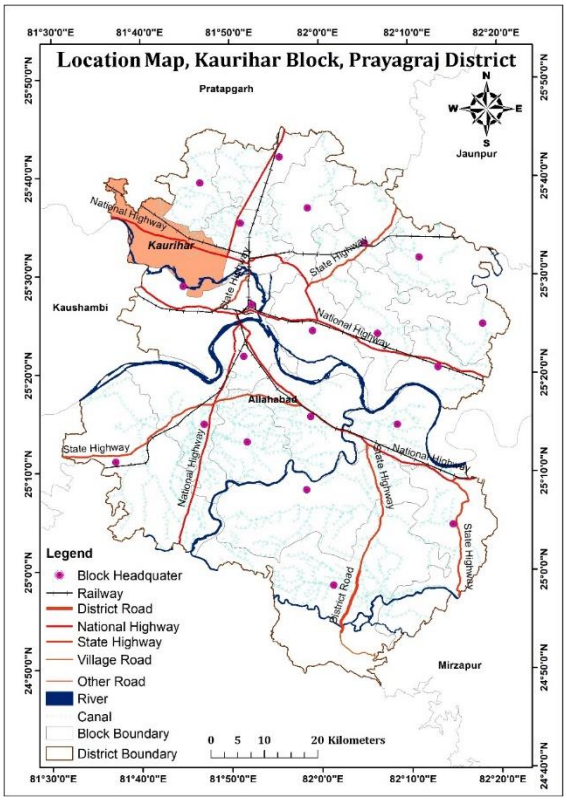


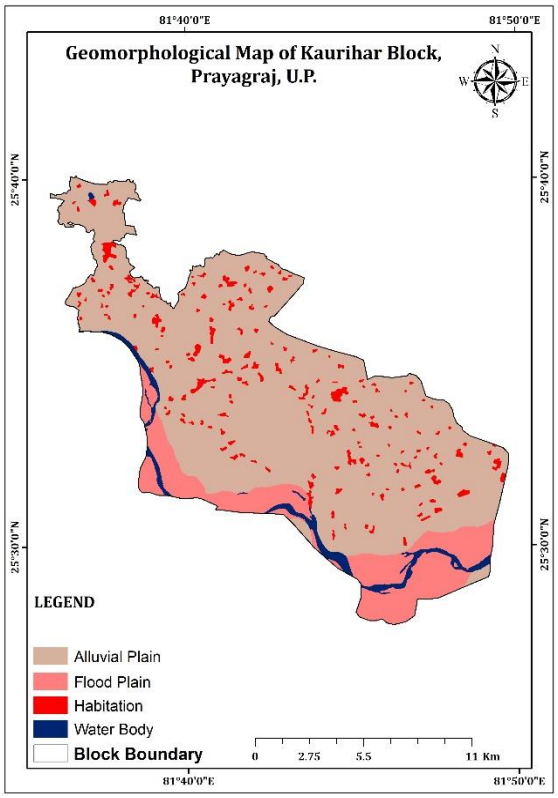
Aquifer Map and Management plan of Kaurihar Block, Prayagraj District, U.P.

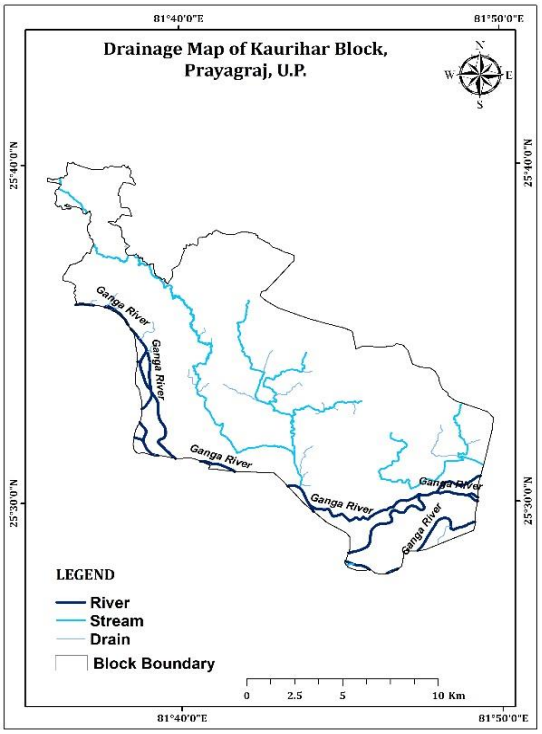
KAURIHAR BLOCK AT A GLANCE

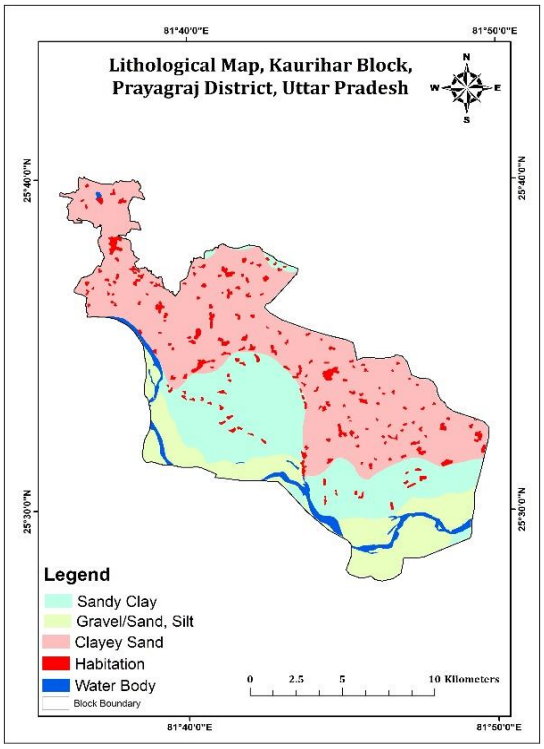
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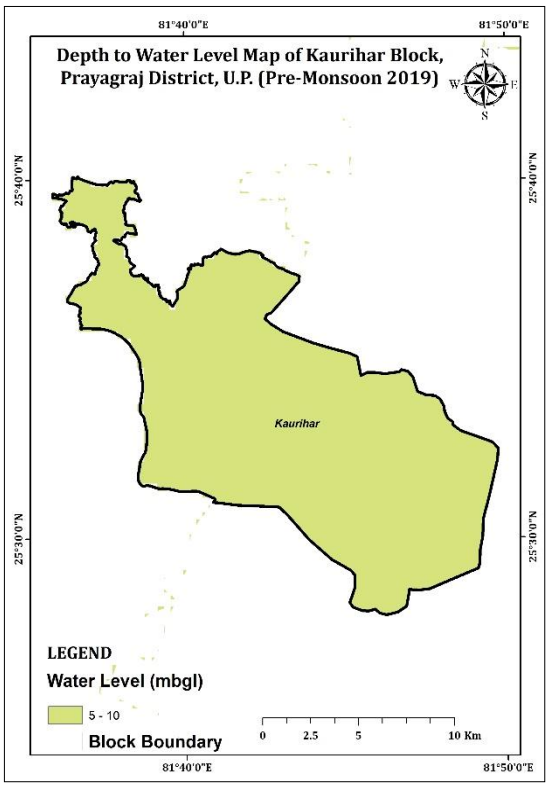
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Kaurihar
Geographical area	414.14km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

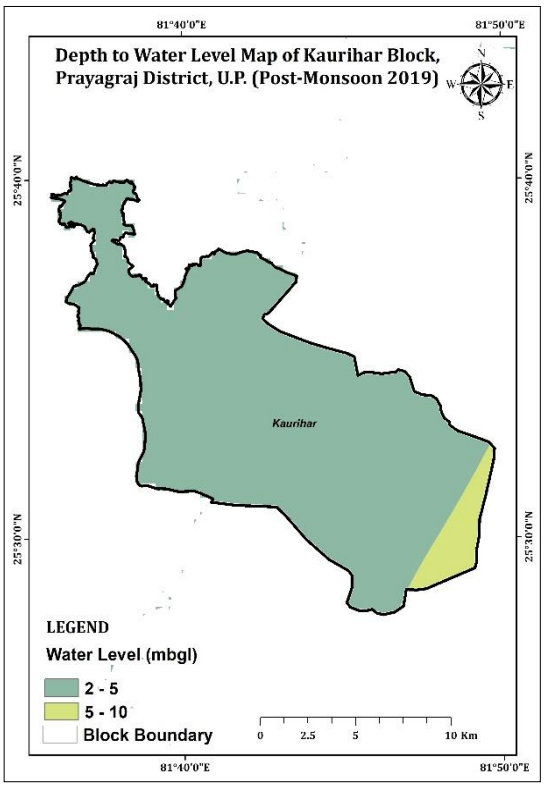





Drainage Map of Kaurihar Block,
Prayagraj, U.P.
81°40'0"E
81°50'0"E
25°40'0"N
25°30'0"N
Ganga River
LEGEND
River
Stream
Drain
Block Boundary
0 2.5 5 10 Km


Lithological Map, Kaurihar Block,
Prayagraj District, Uttar Pradesh
81°40'0"E
81°50'0"E
25°40'0"N
25°30'0"N
Legend
Sandy Clay
Gravel/Sand, Silt
Clayey Sand
Habitation
Water Body
Block Boundary
0 2.5 5 10 Kilometers


Depth to Water Level Map of Kaurihar Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
81°40'0"E
81°50'0"E
25°40'0"N
25°30'0"N
Kaurihar
LEGEND
Water Level (mbgl)
5 - 10
Block Boundary
0 2.5 5 10 Km


Depth to Water Level Map of Kaurihar Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
81°40'0"E
81°50'0"E
25°40'0"N
25°30'0"N
Kaurihar
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2.5 5 10 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	2
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Recharge:85.40mcm Ground water Draft:56.61mcm Stage of Ground water development:66.28**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages:56.61mcm Future Demand for Domestic and Industrial Use:27.80mcm

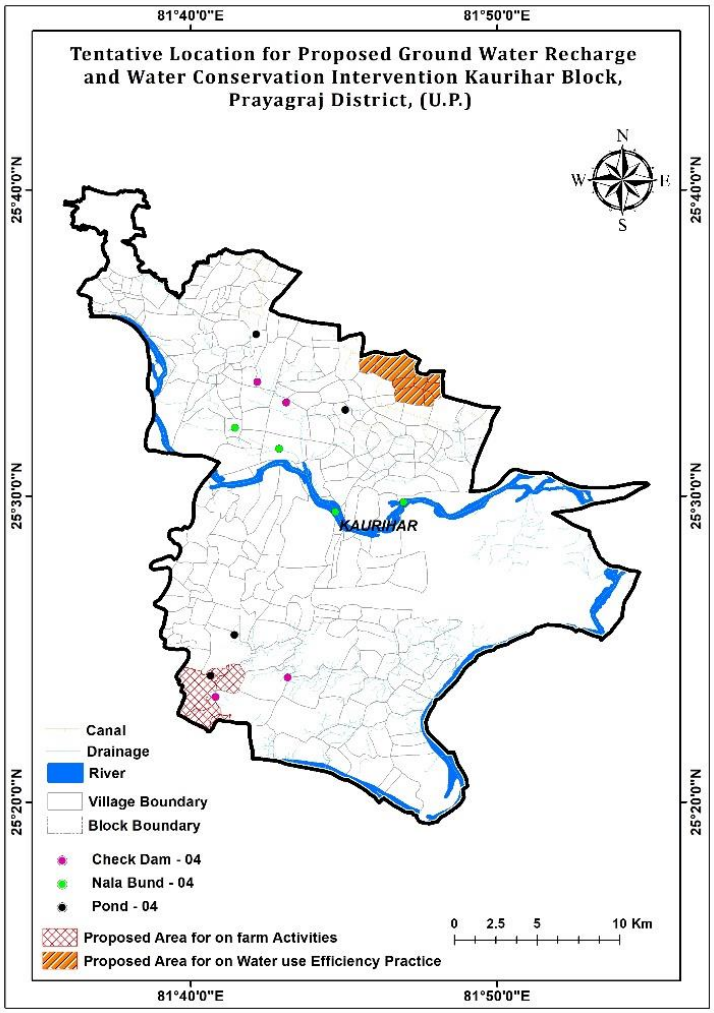
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
KAURIHAR	3	3	3	3	910	910	0.17	0.17	2.01	0.17	2.18	69.28	66.66

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
KAURIHAR	414.14	85.40	56.61	66.28	0.26	2.12	85.67	54.49	**63.60**

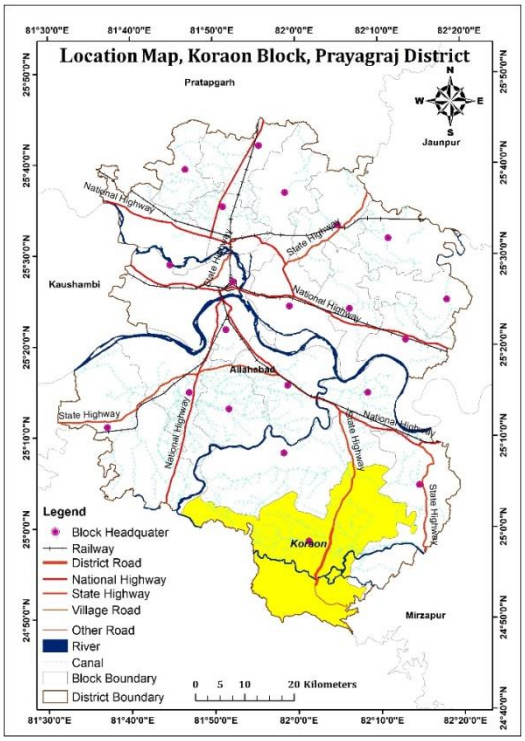


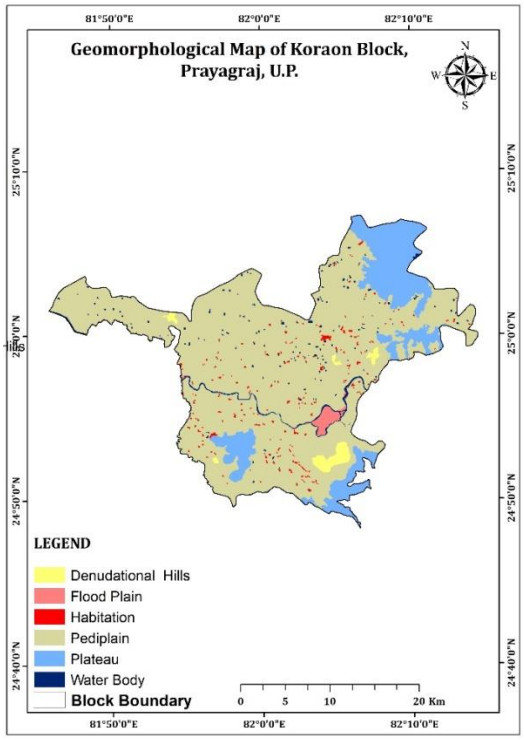
Aquifer Map and Management plan of Koraon Block, Prayagraj District, U.P.

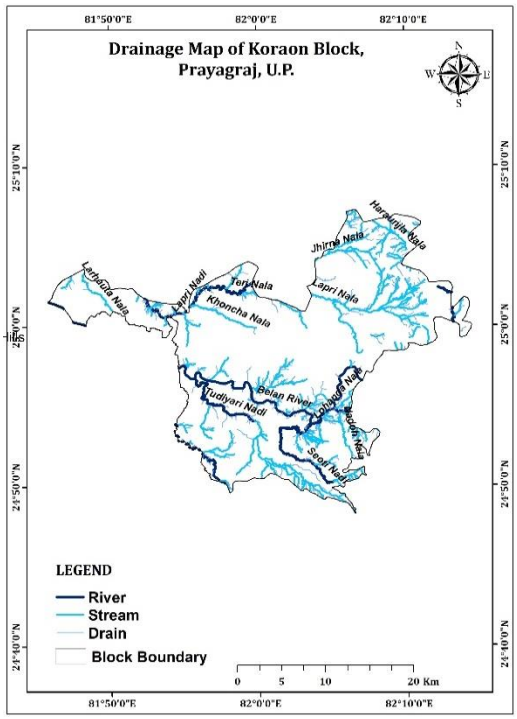
KORAON BLOCK AT A GLANCE

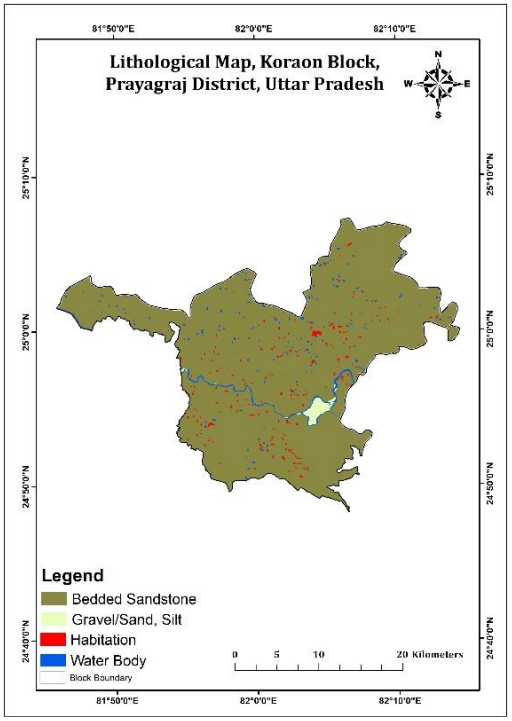
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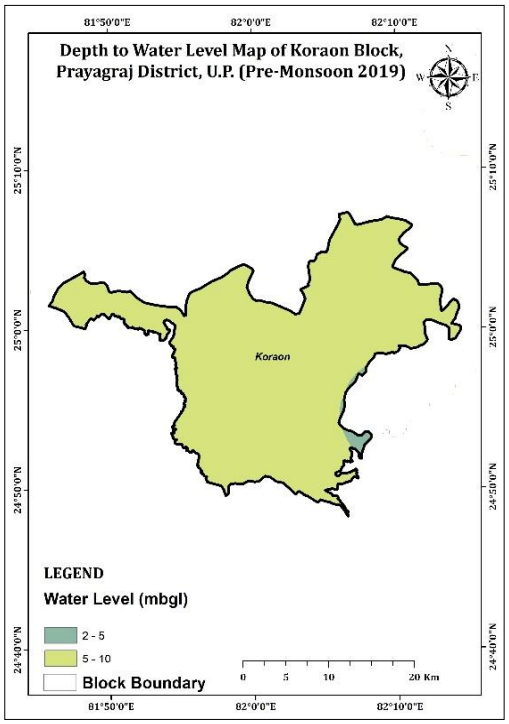
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Koraon
Geographical area	726.42km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Major Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Normal Annual Rainfall	1042.00mm

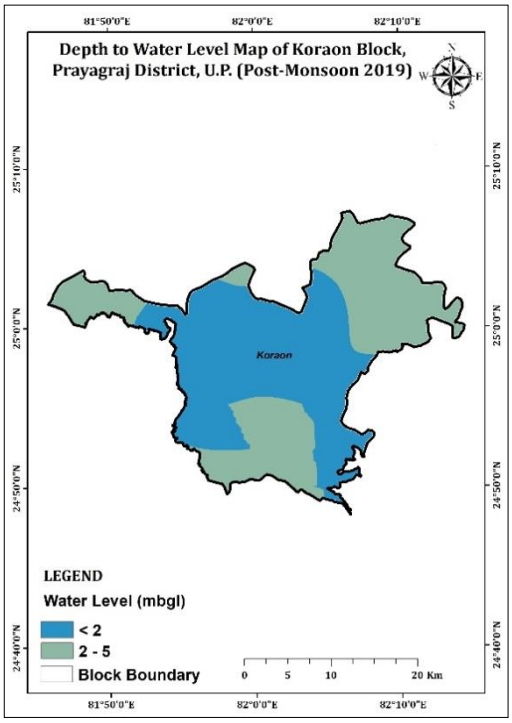





Drainage Map of Koraon Block,
Prayagraj, U.P.
Harpuriya Nala
Jhirna Nala
Lathiata Nala
Lapri Nadi
Teri Nala
Khoncha Nala
Lapri Nala
Kudiyari Nadi
Belan River
Sendi Nala
LEGEND
River
Stream
Drain
Block Boundary
0 5 10 20 Km


Lithological Map, Koraon Block,
Prayagraj District, Uttar Pradesh
Legend
Bedded Sandstone
Gravel/Sand, Silt
Habitation
Water Body
Block Boundary
0 5 10 20 Kilometers


Depth to Water Level Map of Koraon Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
Koraon
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 5 10 20 Km


Depth to Water Level Map of Koraon Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
Koraon
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 5 10 20 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	5
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource :96.56 Ground water Draft: 55.80 Stage of Ground water development:57.79**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages:55.80mcm Future Demand for Domestic and Industrial Use:39.73mcm

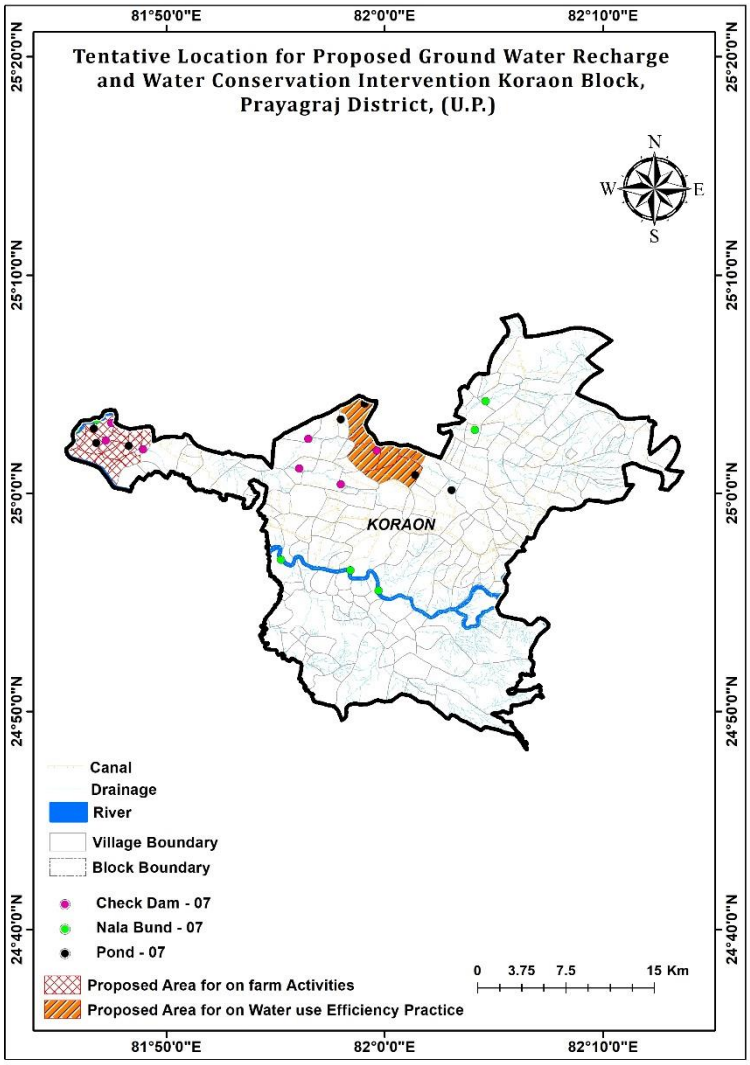
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
KORAON	7	7	7	7	2254	2254	0.46	0.46	1.83	0.46	2.29	57.79	55.15

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
KORAON	726.42	96.56	55.80	57.79	0.46	2.29	97.02	53.51	**55.15**

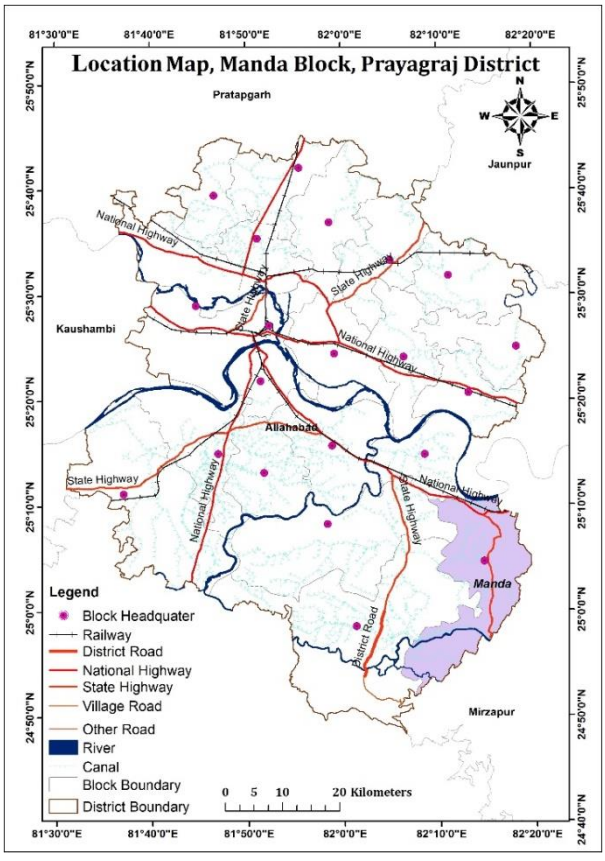


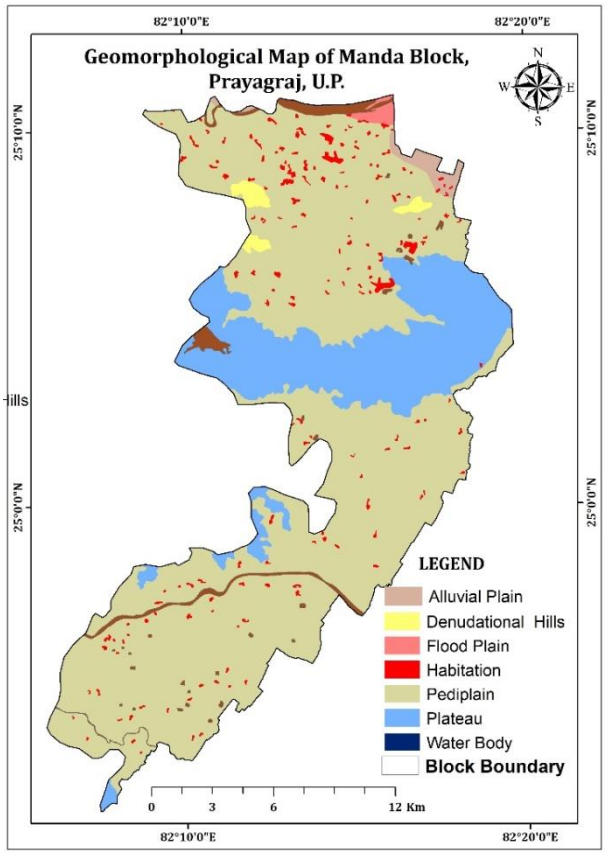
Aquifer Map and Management plan of Manda Block, Prayagraj District, U.P.

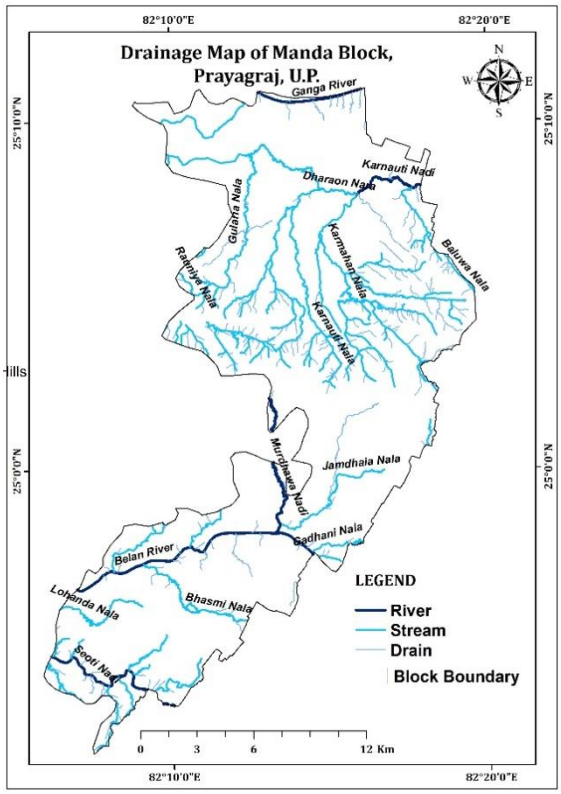
MANDA BLOCK AT A GLANCE

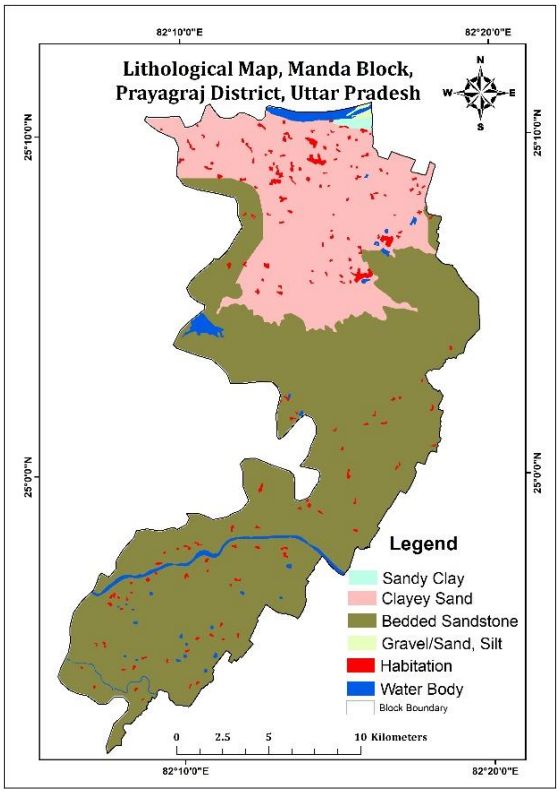
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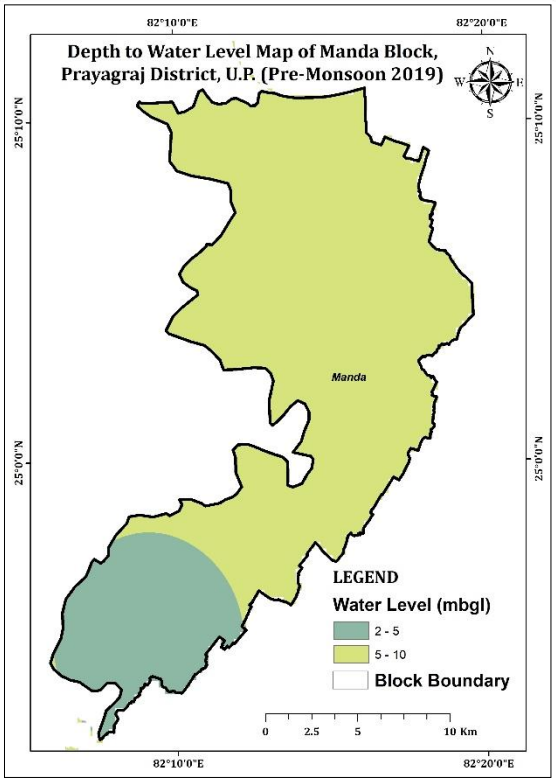
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Manda
Geographical area	349.43km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Major Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Normal Annual Rainfall	1042.00mm

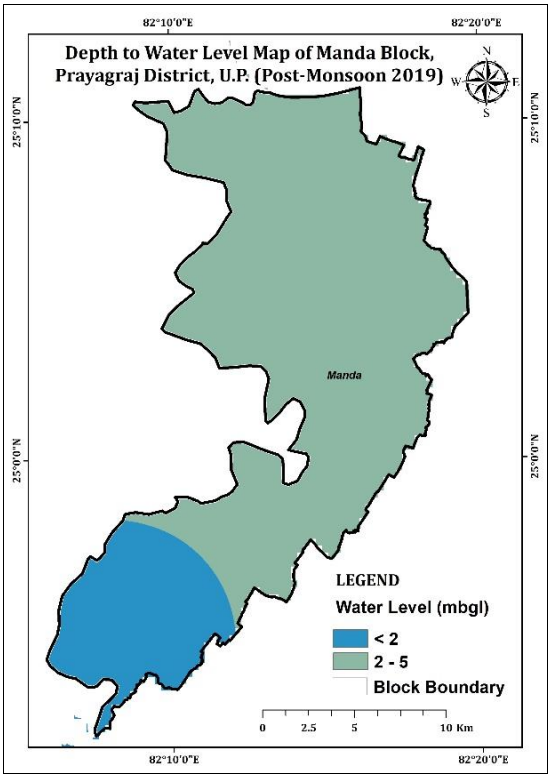





Drainage Map of Manda Block,
Prayagraj, U.P.
Ganga River
Karnauti Nadi
Dharaon Nala
Gulaha Nala
Karnahari Nala
Baluwa Nala
Ramiya Nala
Karnauti Nala
Murdawa Nadi
Jamdhala Nala
Sadhani Nala
Belan River
Lohanda Nala
Bhasmi Nala
Soti Nala
LEGEND
River
Stream
Drain
Block Boundary
0 3 6 12 Km
82°10'0"E
82°20'0"E
25°10'0"N
25°0'0"N


Lithological Map, Manda Block,
Prayagraj District, Uttar Pradesh
Legend
Sandy Clay
Clayey Sand
Bedded Sandstone
Gravel/Sand, Silt
Habitation
Water Body
Block Boundary
0 2.5 5 10 Kilometers
82°10'0"E
82°20'0"E
25°10'0"N
25°0'0"N


Depth to Water Level Map of Manda Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
Manda
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2.5 5 10 Km
82°10'0"E
82°20'0"E
25°10'0"N
25°0'0"N


Depth to Water Level Map of Manda Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
Manda
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 2.5 5 10 Km
82°10'0"E
82°20'0"E
25°10'0"N
25°0'0"N

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	2
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource: 51.25mcm Ground water Draft:32.95mcm Stage of Ground water development:64.3**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages:32.95mcm Future Demand for Domestic and Industrial Use:17.43mcm

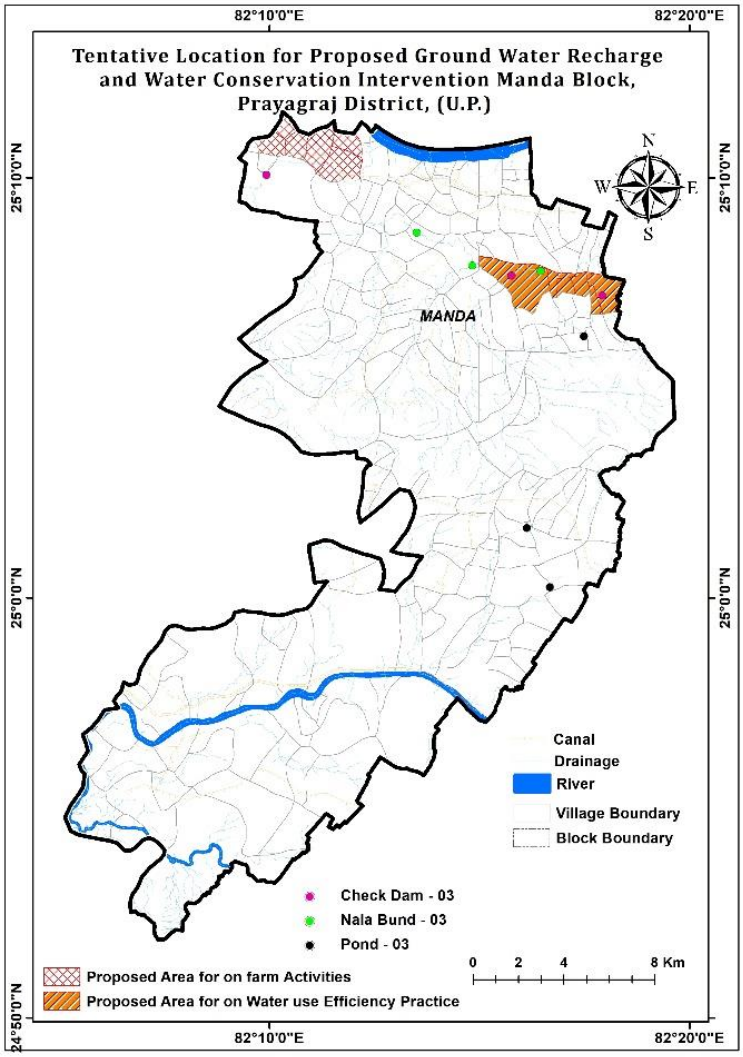
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
MANDA	3	3	3	3	689	689	0.22	0.22	1.06	0.22	1.29	64.31	61.53

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
MANDA	349.43	51.25	32.95	64.31	0.22	1.29	51.47	31.67	**61.53**

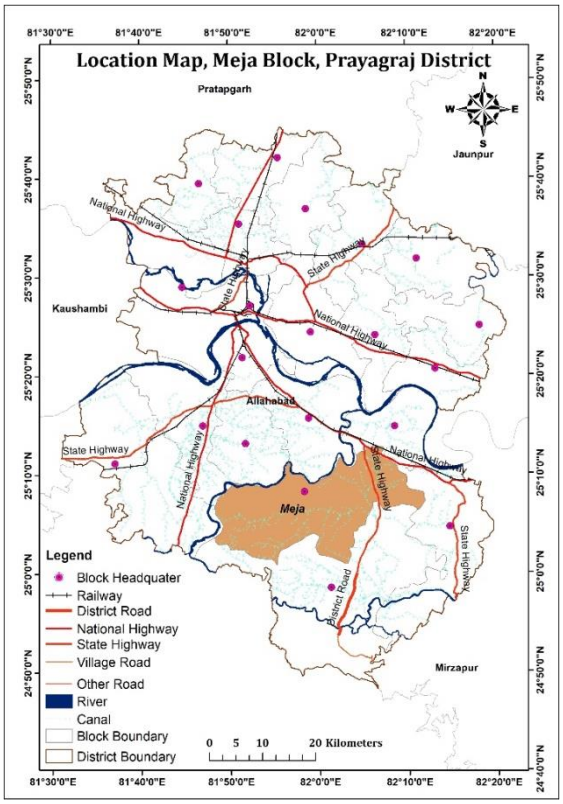


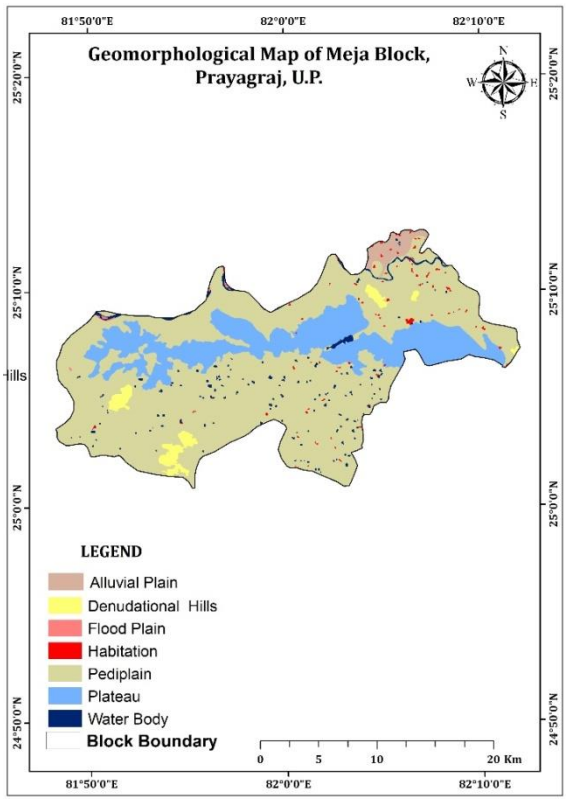
Aquifer Map and Management plan of Meja Block, Prayagraj District, U.P.

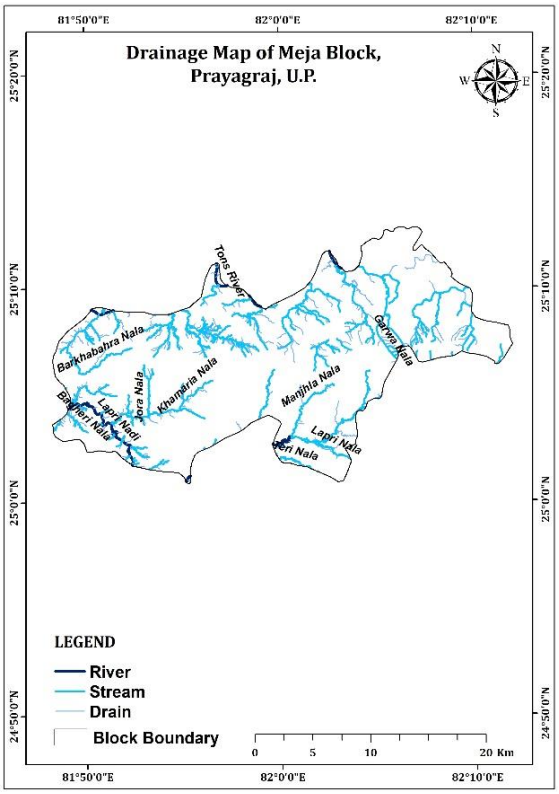
MEJA BLOCK AT A GLANCE

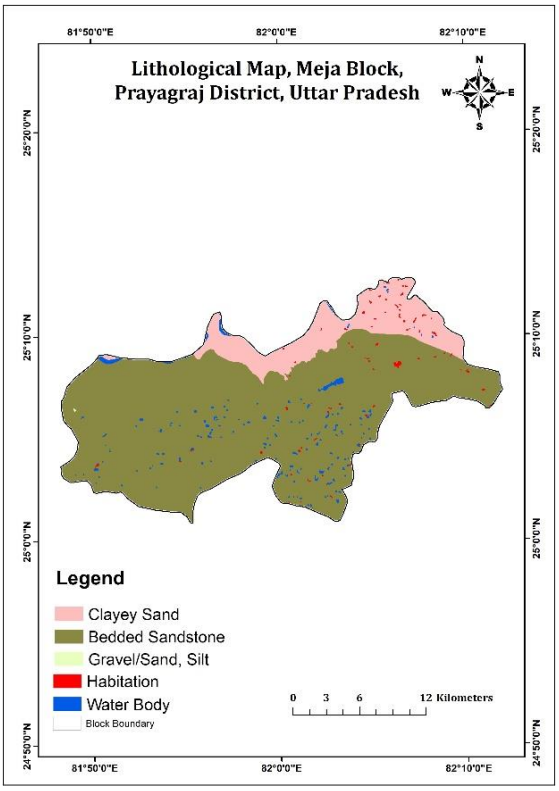
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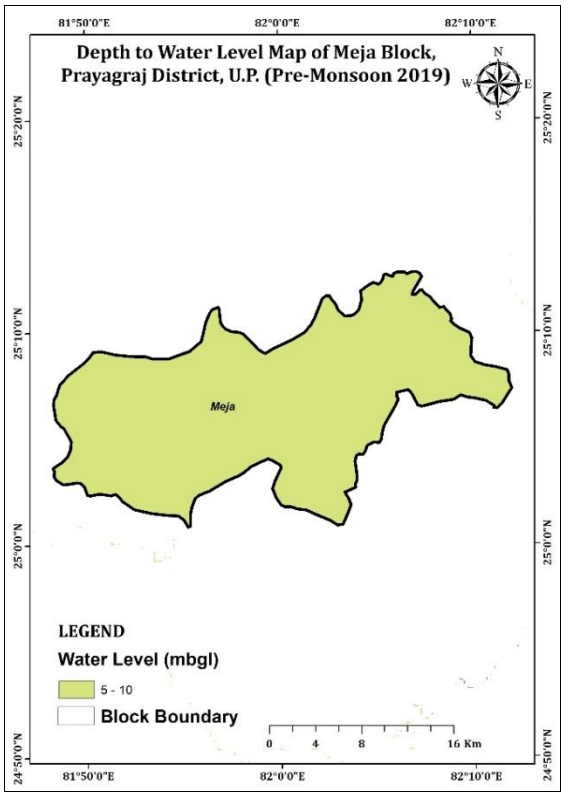
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Meja
Geographical area	444.21km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Major Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Normal Annual Rainfall	1042.00mm

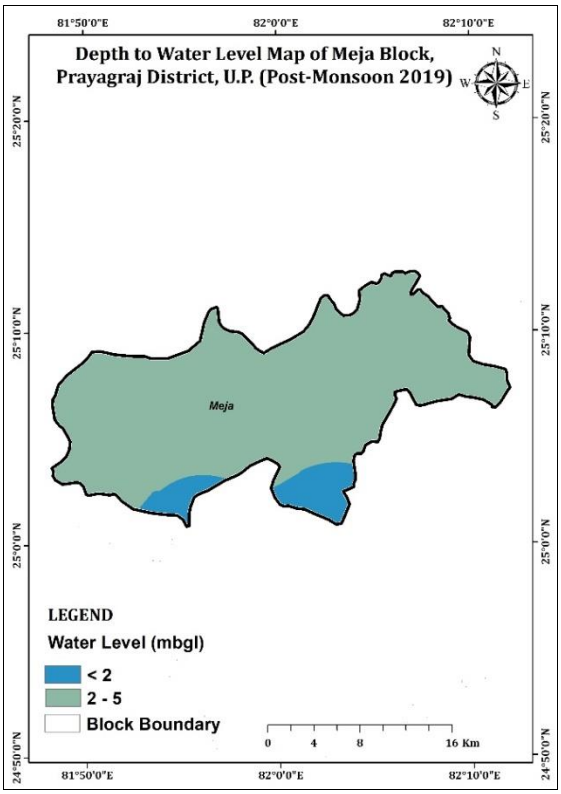





Drainage Map of Meja Block,
Prayagraj, U.P.
81°50'0"E
82°0'0"E
82°10'0"E
25°20'0"N
25°10'0"N
25°0'0"N
24°50'0"N
Tons River
Bankhabahra Nala
Bandheri Nala
Lapri Nadi
Tora Nala
Khamaria Nala
Manjhla Nala
Lapri Nala
Jeri Nala
Garwa Nala
LEGEND
River
Stream
Drain
Block Boundary
0 5 10 20 Km


Lithological Map, Meja Block,
Prayagraj District, Uttar Pradesh
81°50'0"E
82°0'0"E
82°10'0"E
25°20'0"N
25°10'0"N
25°0'0"N
24°50'0"N
Legend
Clayey Sand
Bedded Sandstone
Gravel/Sand, Silt
Habitation
Water Body
Block Boundary
0 3 6 12 Kilometers


Depth to Water Level Map of Meja Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
81°50'0"E
82°0'0"E
82°10'0"E
25°20'0"N
25°10'0"N
25°0'0"N
24°50'0"N
Meja
LEGEND
Water Level (mbgl)
5 - 10
Block Boundary
0 4 8 16 Km


Depth to Water Level Map of Meja Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
81°50'0"E
82°0'0"E
82°10'0"E
25°20'0"N
25°10'0"N
25°0'0"N
24°50'0"N
Meja
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 4 8 16 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	1
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource:75.53mcm Ground water Draft: 38.71mcm Stage of Ground water development:51.25**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages:38.71mcm Future Demand for Domestic and Industrial Use:35.80mcm

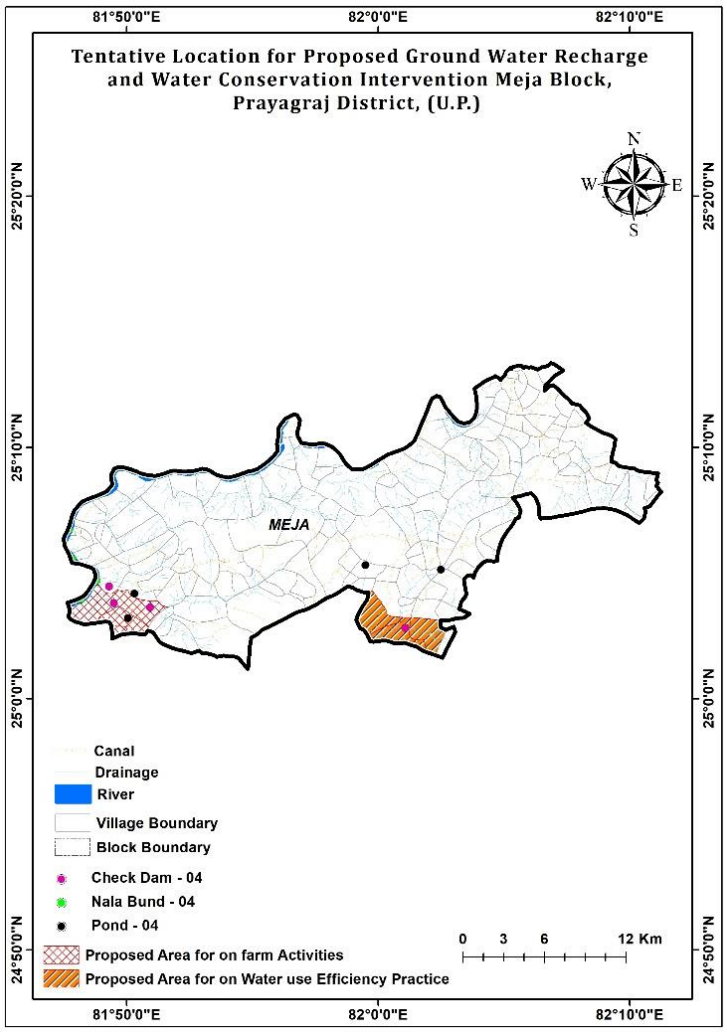
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
MEJA	4	4	4	4	1204	1204	0.28	0.28	1.19	0.28	1.47	51.25	49.12

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
MEJA	444.21	75.53	38.71	51.25	0.28	1.47	75.82	37.24	**49.12**

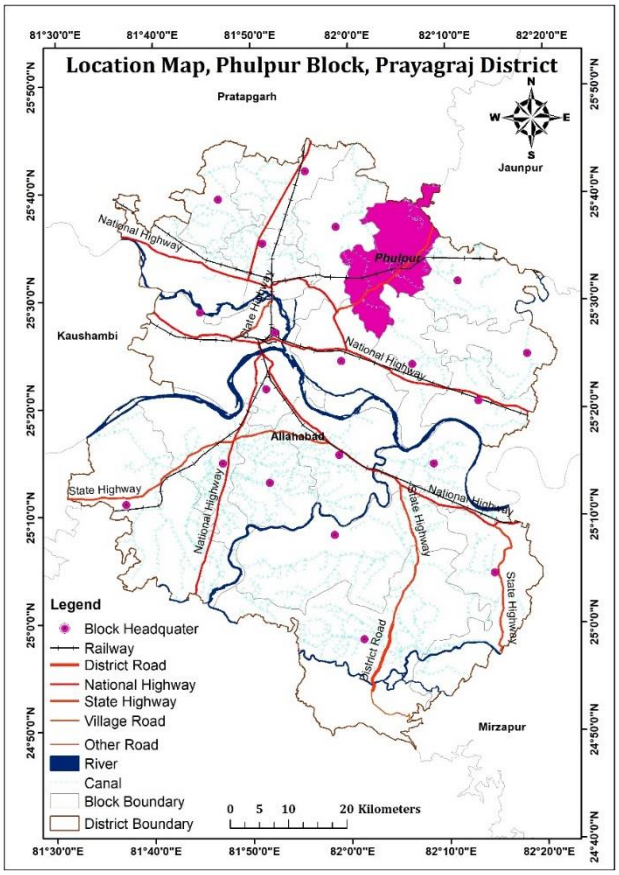


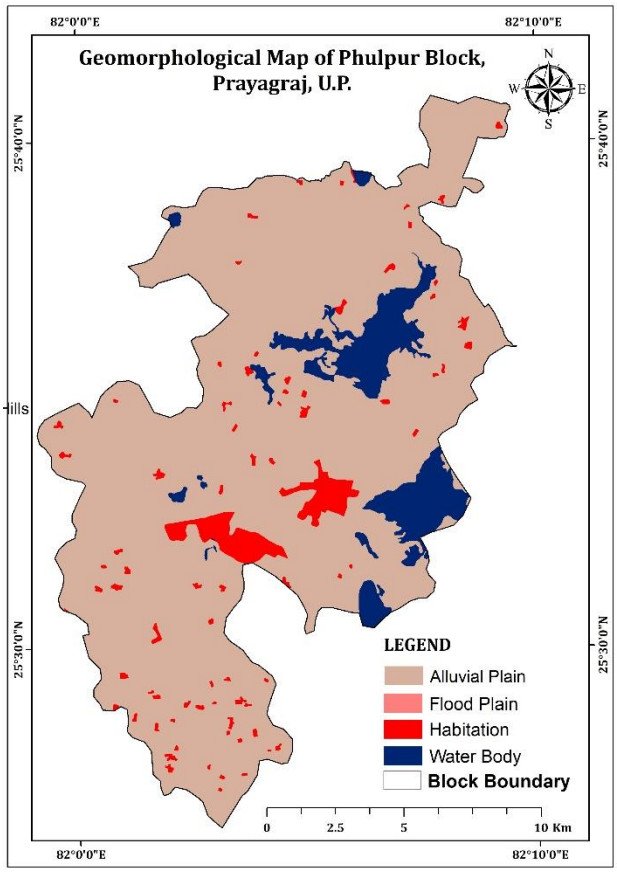
Aquifer Map and Management plan of Phulpur Block, Prayagraj District, U.P.

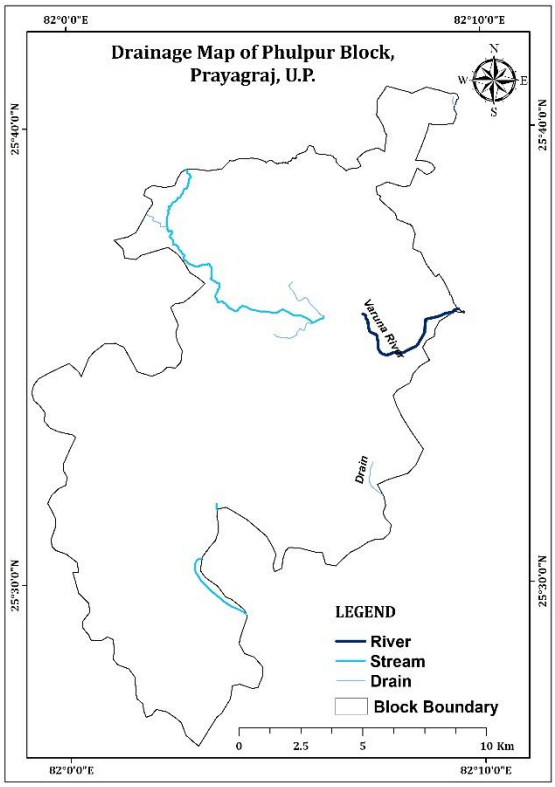
PHULPUR BLOCK AT A GLANCE

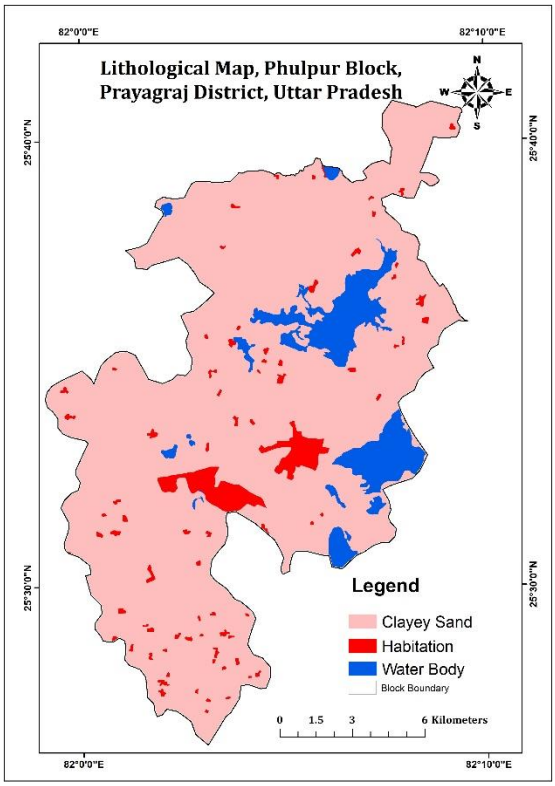
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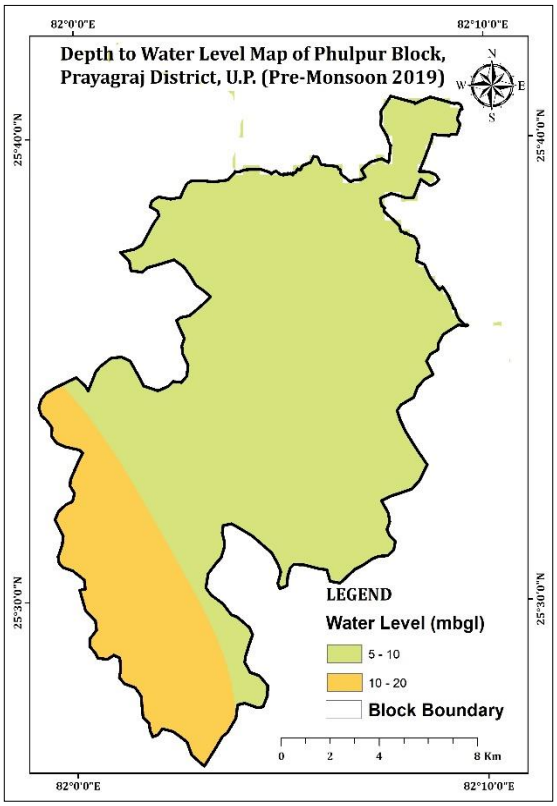
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Phulpur
Geographical area	231.54km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

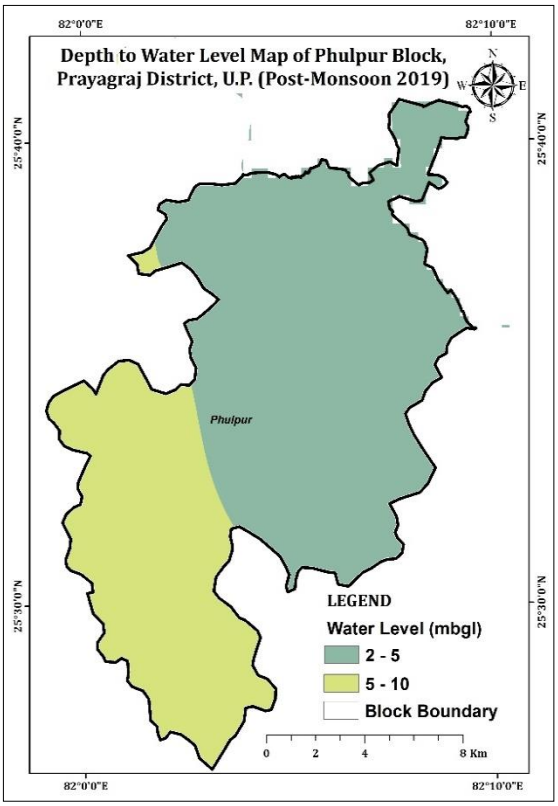





Drainage Map of Phulpur Block,
Prayagraj, U.P.
82°0'0"E
82°10'0"E
25°40'0"N
25°30'0"N
Varuna River
Drain
LEGEND
River
Stream
Drain
Block Boundary
0 2.5 5 10 Km


Lithological Map, Phulpur Block,
Prayagraj District, Uttar Pradesh
82°0'0"E
82°10'0"E
25°40'0"N
25°30'0"N
Legend
Clayey Sand
Habitation
Water Body
Block Boundary
0 1.5 3 6 Kilometers


Depth to Water Level Map of Phulpur Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
82°0'0"E
82°10'0"E
25°40'0"N
25°30'0"N
LEGEND
Water Level (mbgl)
5 - 10
10 - 20
Block Boundary
0 2 4 8 Km


Depth to Water Level Map of Phulpur Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
82°0'0"E
82°10'0"E
25°40'0"N
25°30'0"N
Phulpur
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2 4 8 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	3
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource:64.33mcm Ground water Draft:42.94mcm Stage of Ground water development:40.98**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages:42.93mcm Future Demand for Domestic and Industrial Use:20.77mcm

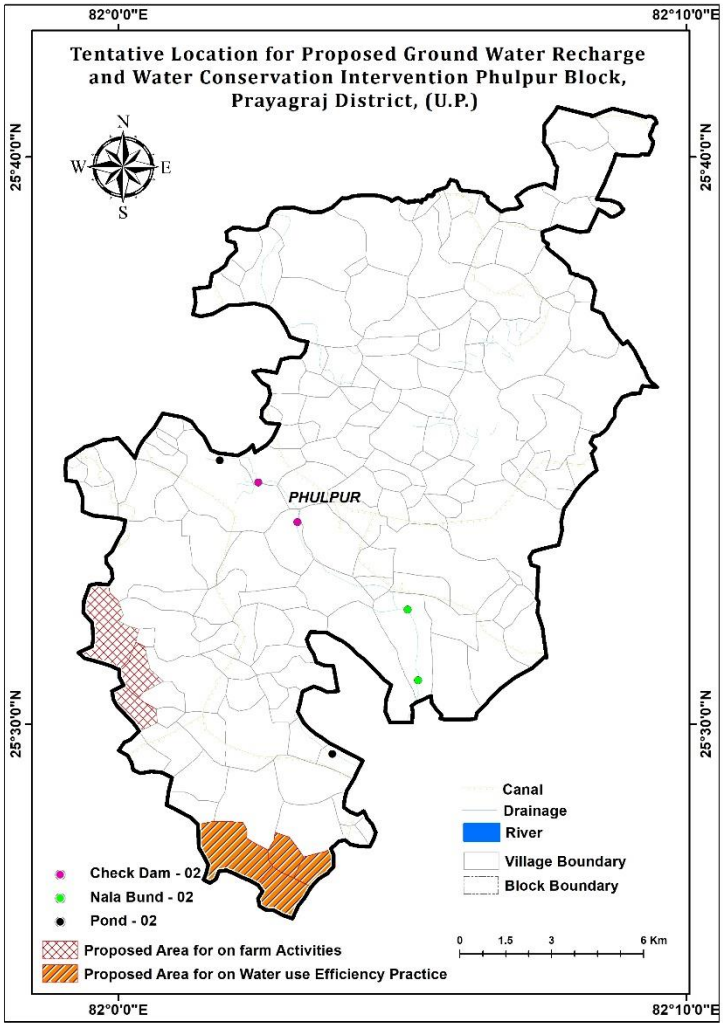
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
PHULPUR	2	2	2	2	551	551	0.15	0.15	1.81	0.15	1.96	66.75	63.55

Blocks	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
PHULPUR	231.54	64.33	42.94	66.75	0.15	1.96	64.48	40.98	**63.55**

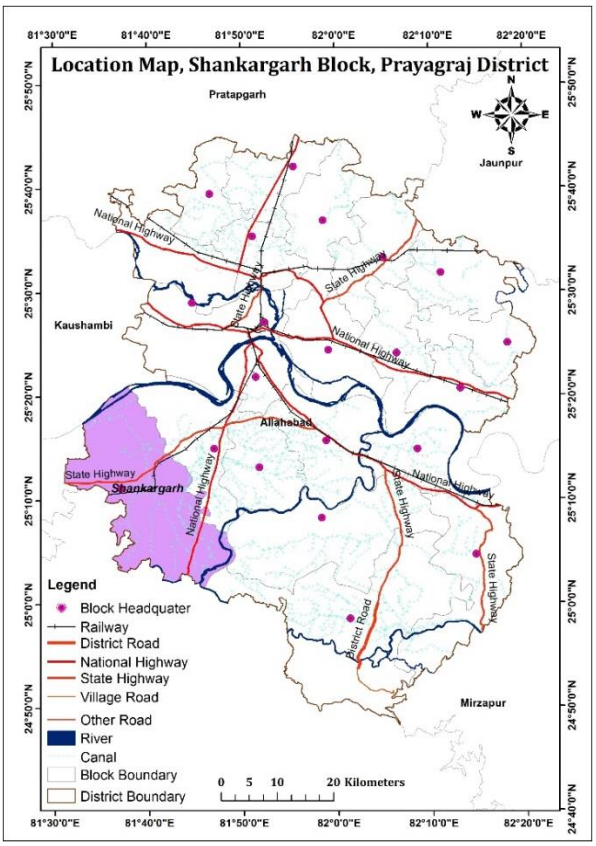


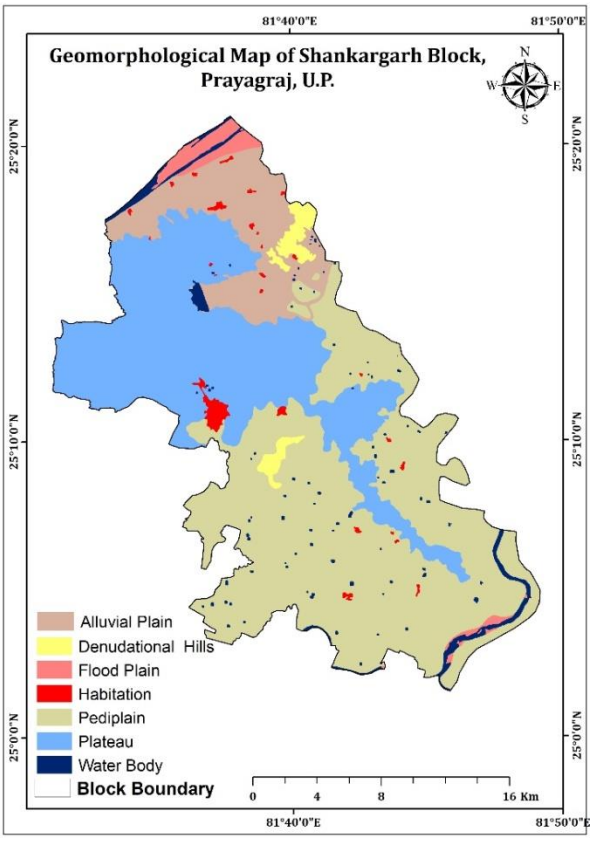
Aquifer Map and Management plan of Shankargarh Block, Prayagraj District, U.P.

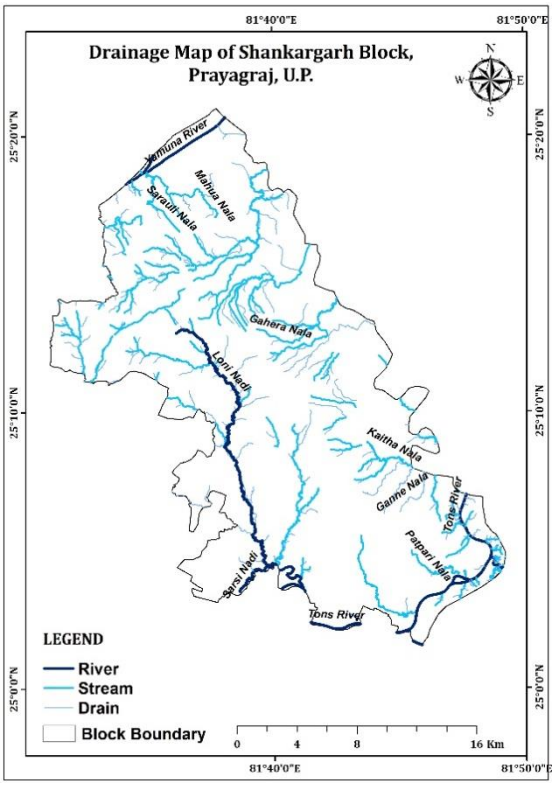
SHANKARGARH BLOCK AT A GLANCE

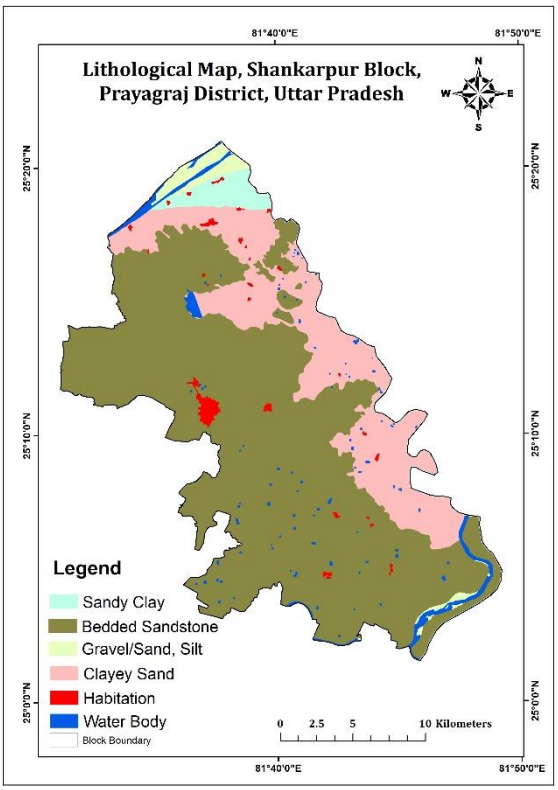
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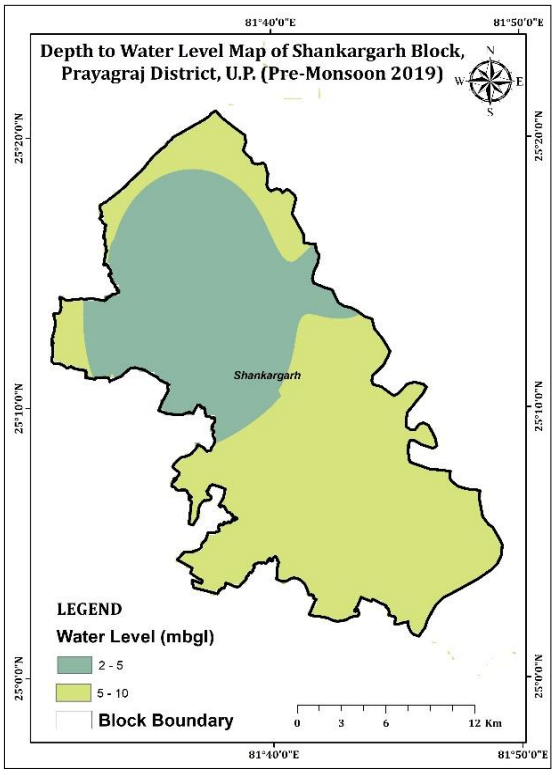
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Shankargarh
Geographical area	472.08km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Major Aquifer System	Fractured and jointed Sandstone formations of Vindhyan Super Group
Normal Annual Rainfall	1042.00mm

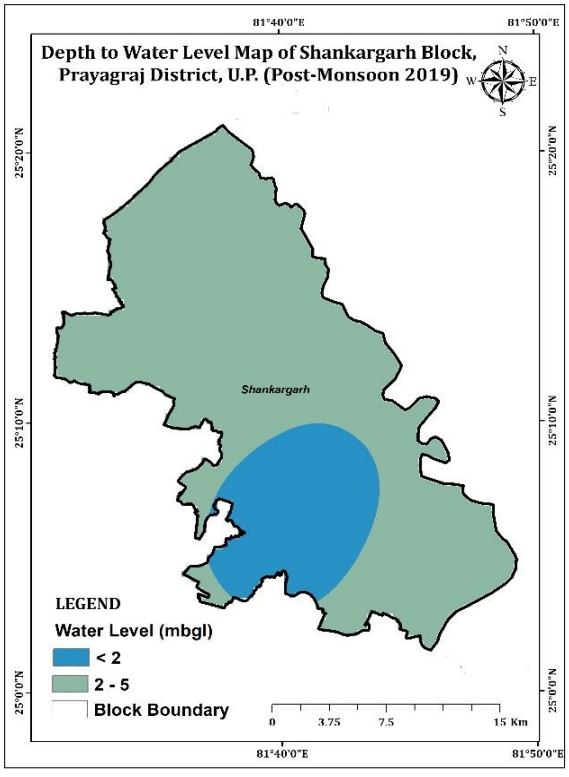





Drainage Map of Shankargarh Block,
Prayagraj, U.P.
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
25°0'0"N
Yamuna River
Mahua Nala
Sarauli Nala
Gahera Nala
Lani Nadi
Kaitha Nala
Ganne Nala
Tons River
Patpari Nala
Sirsi Nadi
LEGEND
River
Stream
Drain
Block Boundary
0 4 8 16 Km


Lithological Map, Shankarpur Block,
Prayagraj District, Uttar Pradesh
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
25°0'0"N
Legend
Sandy Clay
Bedded Sandstone
Gravel/Sand, Silt
Clayey Sand
Habitation
Water Body
Block Boundary
0 2.5 5 10 Kilometers


Depth to Water Level Map of Shankargarh Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
25°0'0"N
Shankargarh
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 3 6 12 Km


Depth to Water Level Map of Shankargarh Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
81°40'0"E
81°50'0"E
25°20'0"N
25°10'0"N
25°0'0"N
Shankargarh
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 3.75 7.5 15 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	5
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Recharge:68.15mcm Ground water Draft:40.60mcm Stage of Ground water development:59.58**%** Category: safe
Existing and Future Water Demand	Present demand for all Usages:40.60mcm Future Demand for Domestic and Industrial Use:26.86mcm

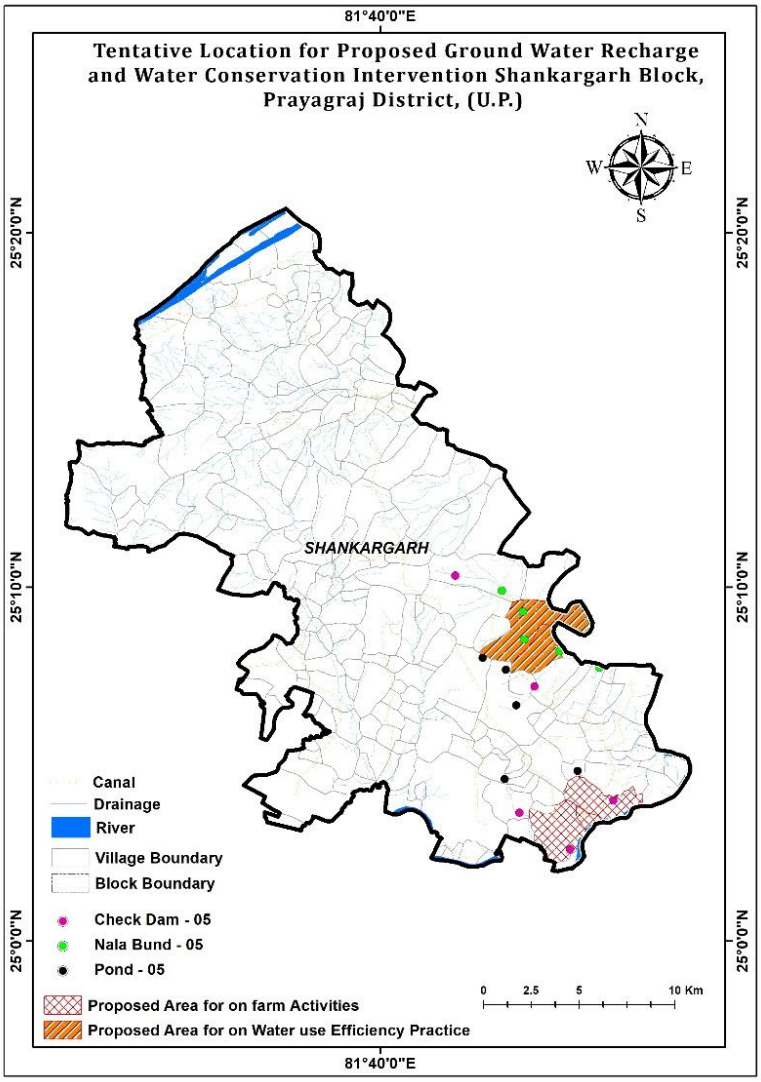
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
SHANKARGARH	5	5	5	5	1114	1114	0.30	0.30	1.50	0.30	1.80	59.58	56.68

Block	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
SHANKARGARH	472.08	68.15	40.60	59.58	0.30	1.80	68.45	38.80	**56.68**

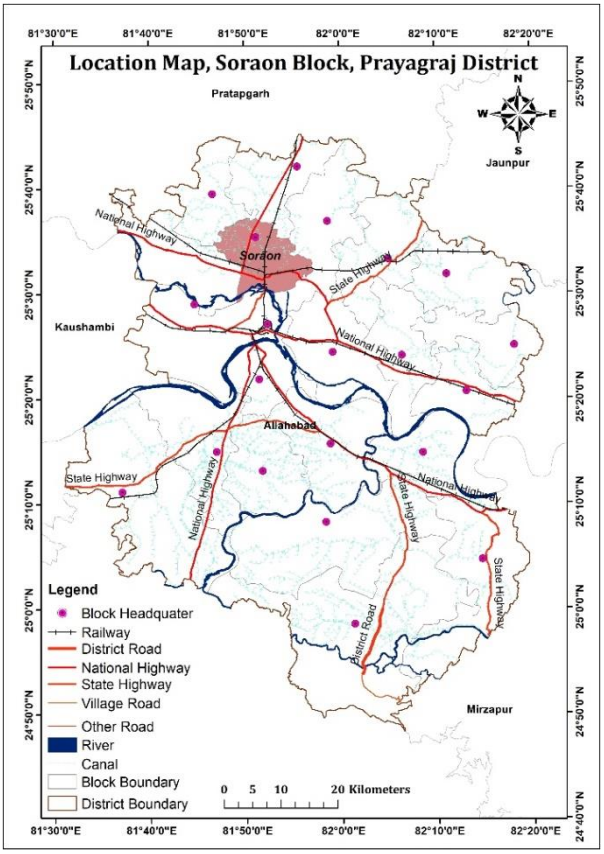


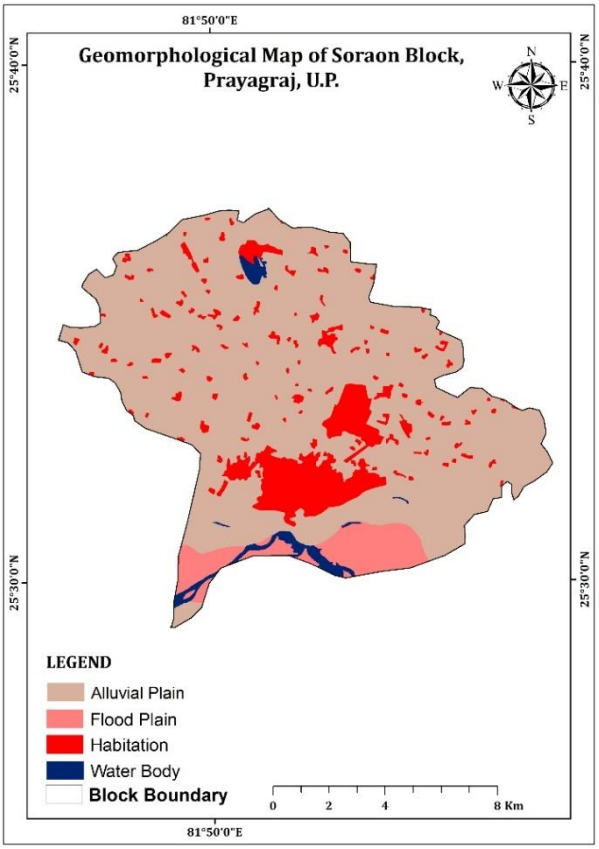
Aquifer Map and Management plan of Meja Block, Prayagraj District, U.P.

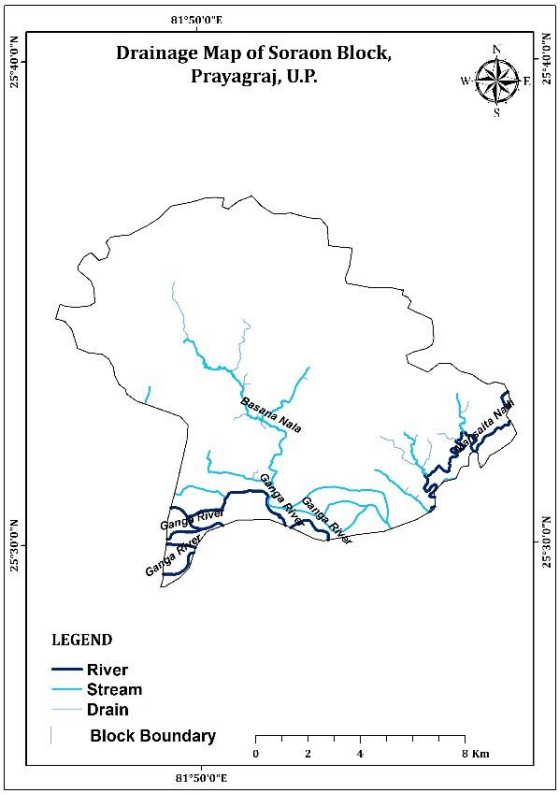
SORAON BLOCK AT A GLANCE

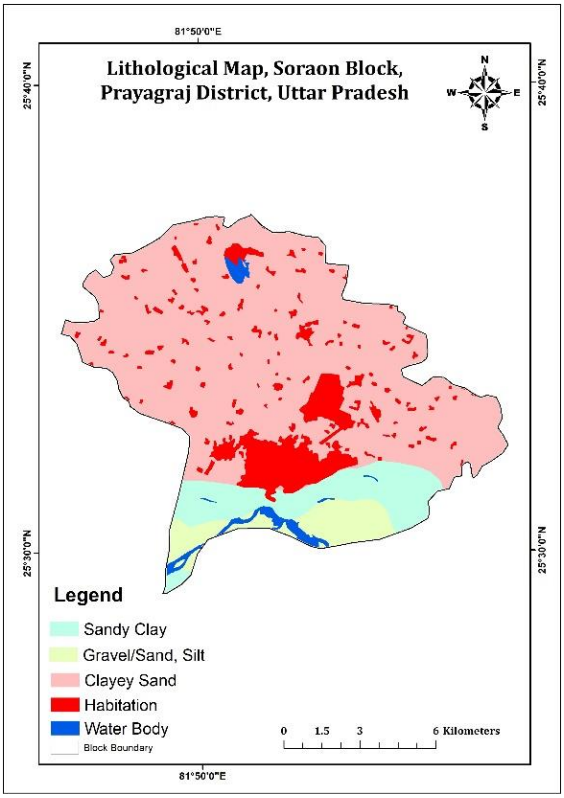
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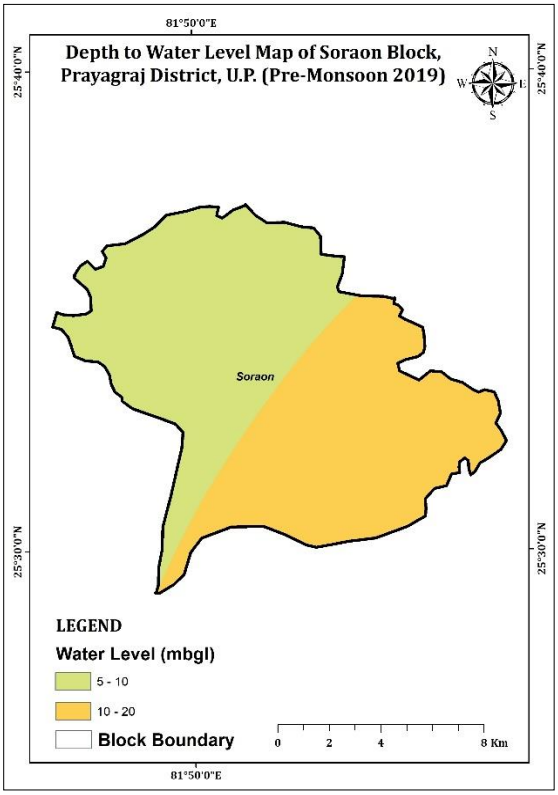
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Soraon
Geographical area	147.75km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

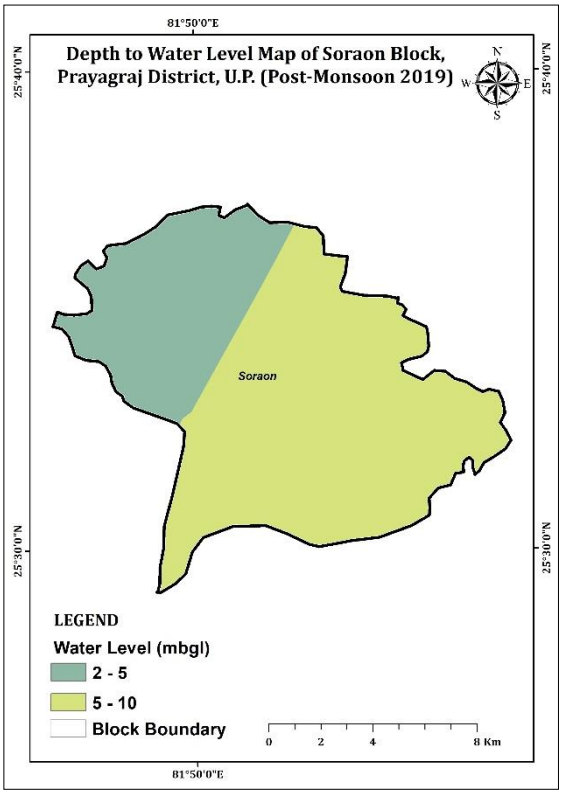





Drainage Map of Soraon Block,
Prayagraj, U.P.
81°50'0"E
25°40'0"N
25°30'0"N
Basaha Nala
Ganga River
Ganga River
Ganga River
Danda River
LEGEND
River
Stream
Drain
Block Boundary
0 2 4 8 Km


Lithological Map, Soraon Block,
Prayagraj District, Uttar Pradesh
81°50'0"E
25°40'0"N
25°30'0"N
Legend
Sandy Clay
Gravel/Sand, Silt
Clayey Sand
Habitation
Water Body
Block Boundary
0 1.5 3 6 Kilometers


Depth to Water Level Map of Soraon Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
81°50'0"E
25°40'0"N
25°30'0"N
Soraon
LEGEND
Water Level (mbgl)
5 - 10
10 - 20
Block Boundary
0 2 4 8 Km


Depth to Water Level Map of Soraon Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
81°50'0"E
25°40'0"N
25°30'0"N
Soraon
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2 4 8 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	-
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Recharge: 43.08mcm Ground water Draft:29.45mcm Stage of Ground water development:68.36**%** Category: Safe
Existing and Future Water Demand	Present demand for all Usages:29.45mcm Future Demand for Domestic and Industrial Use:13.16mcm

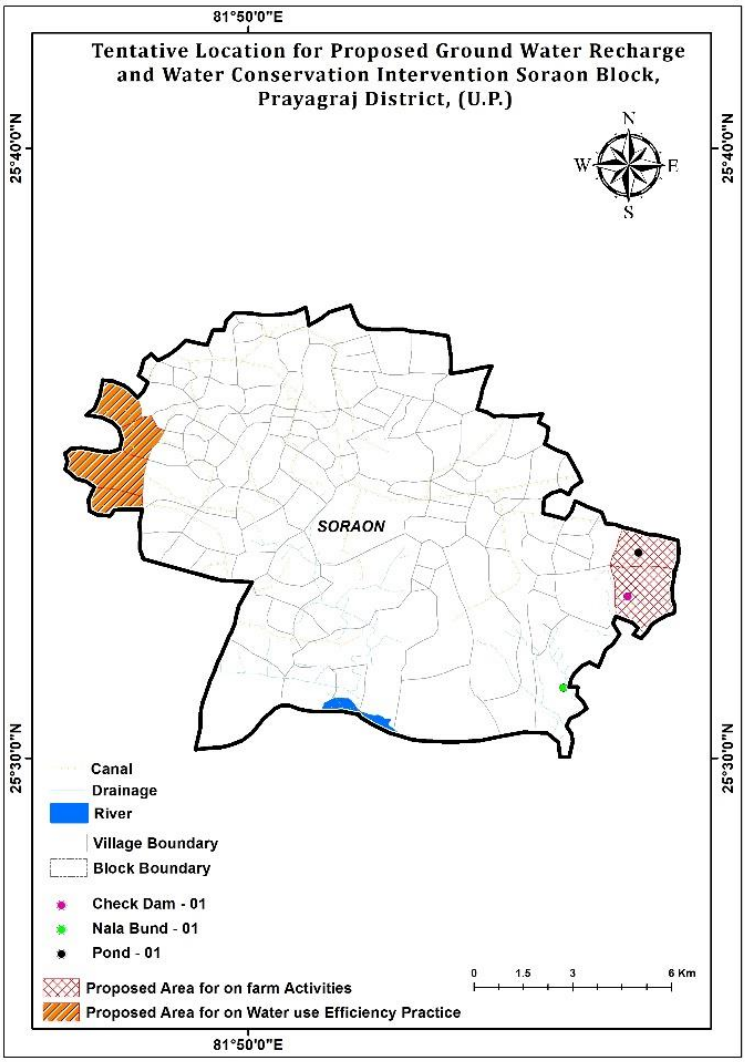
Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
SORAON	1	1	1	1	479	479	0.09	0.09	1.03	0.09	1.13	68.36	65.59

Block	Area (Sq.Km)	Net Annual Ground Water Avail-ability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
SORAON	147.75	43.08	29.45	68.36	0.09	1.13	43.18	28.32	**65.59**

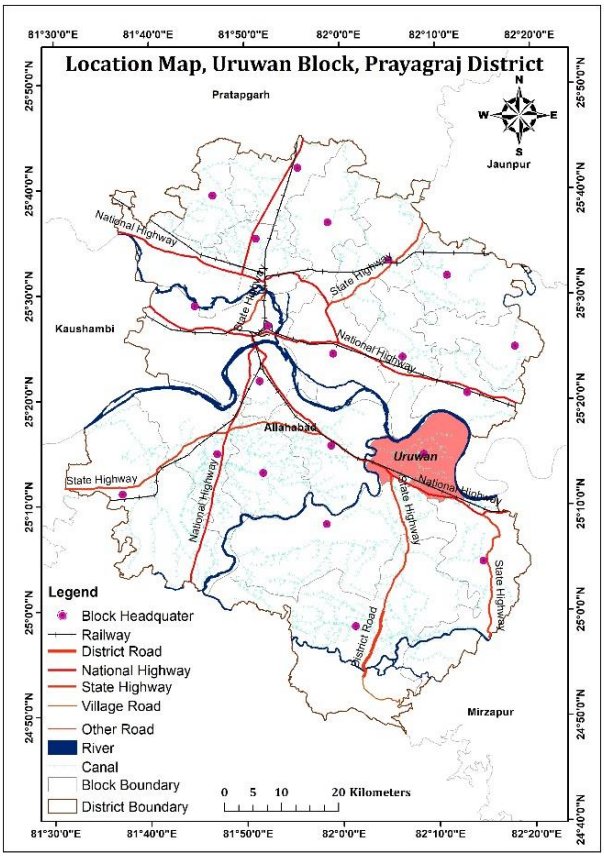


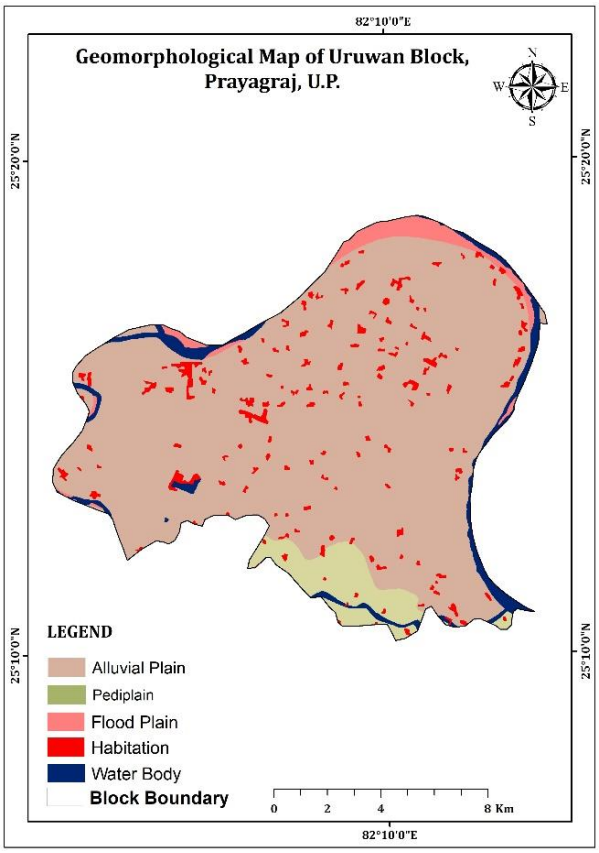
Aquifer Map and Management plan of Uruwan Block, Prayagraj District, U.P.

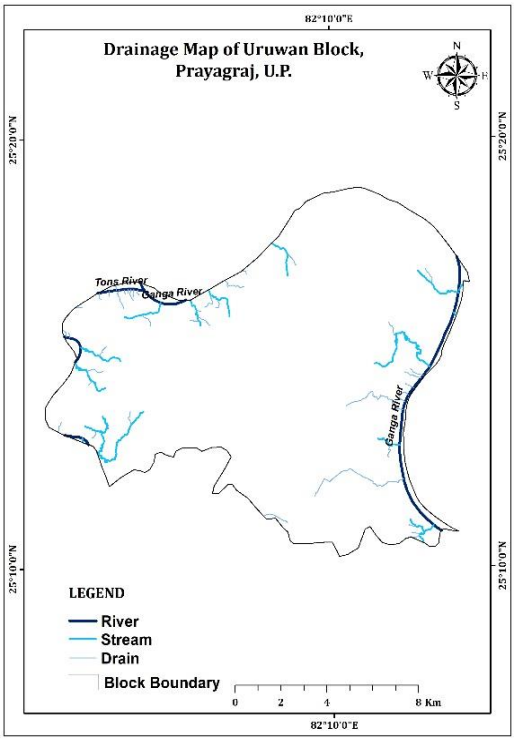
URUWAN BLOCK AT A GLANCE

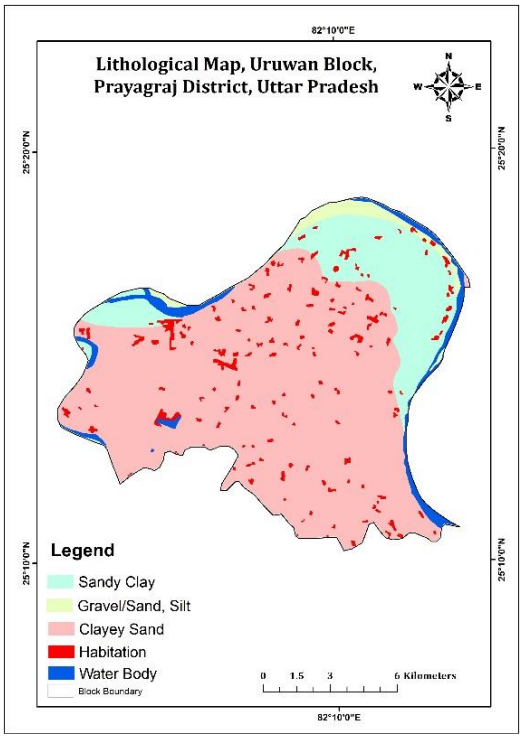
General Information:

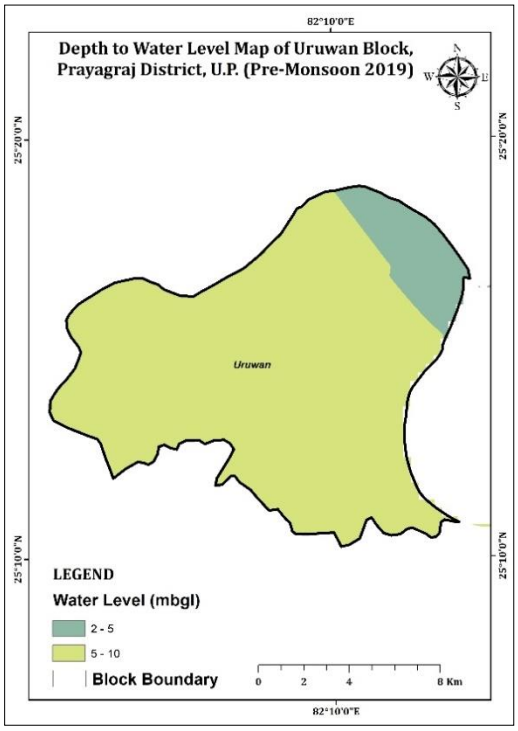
State Name	Uttar Pradesh
District name	Prayagraj
Block Name	Uruwan
Geographical area	168.9km^2
Basin/Sub-basin	Ganga, Yamuna and Tons
Principal Aquifer System	Alluvium
Major Aquifer System	Alluvium
Normal Annual Rainfall	1042.00mm

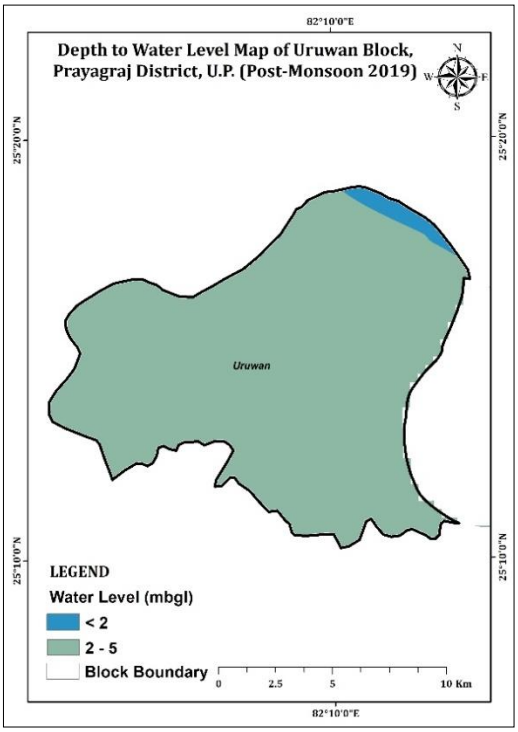





Drainage Map of Uruwan Block,
Prayagraj, U.P.
Tons River
Ganga River
Ganga River
LEGEND
River
Stream
Drain
Block Boundary
0 2 4 8 Km


Lithological Map, Uruwan Block,
Prayagraj District, Uttar Pradesh
Legend
Sandy Clay
Gravel/Sand, Silt
Clayey Sand
Habitation
Water Body
Block Boundary
0 1.5 3 6 Kilometers


Depth to Water Level Map of Uruwan Block,
Prayagraj District, U.P. (Pre-Monsoon 2019)
Uruwan
LEGEND
Water Level (mbgl)
2 - 5
5 - 10
Block Boundary
0 2 4 8 Km


Depth to Water Level Map of Uruwan Block,
Prayagraj District, U.P. (Post-Monsoon 2019)
Uruwan
LEGEND
Water Level (mbgl)
< 2
2 - 5
Block Boundary
0 2.5 5 10 Km

Aquifer Disposition:

Aquifer Disposition	-
Groundwater Monitoring Status	-
Ground Water Quality	It is observed that the ground water is suitable for drinking and domestic uses.
Aquifer Potential	-
Groundwater Resource	Annual Extractable Ground Water Resource:57.19mcm Ground water Draft:38.43mcm Stage of Ground water development:67.20% Category: Safe
Existing and Future Water Demand	Present demand for all Usages:29.45mcm Future Demand for Domestic and Industrial Use:13.16mcm

Aquifer management plan

Groundwater Management issues	Cropping pattern is the main reason as the Gross Ground Water Draft for Irrigation is in excess of Net Annual GW Availability Non-availability of Surface Water for irrigation. High decline observed in northern part (alluvium) because ground water draft exceeds GW recharge for agriculture purpose.
AR & Conservation Possibilities	Construction of Artificial Recharge structure and water conservation structures, On- Farm activities and adoption of water use efficiency practices
Groundwater Management Plan	**Supply side management:** Water conservation and Artificial recharge to ground water and on farm activities. **Demand Side Management:** Adoption of techniques to enhance water use efficiency viz. sprinkler, drip irrigation and diversification of cropping pattern
Status of GW Exploration	-
Aquifer Characteristics	-

Management Plan

Block	CDs (Nos)	NBs (Nos)	Str Dev (Km)	Ponds (Nos)	On-farm (ha)	WUE (ha)	Rech from Str MCM	Saving from Str MCM	Saving from On-farm & WUE MCM	Total Rech MCM	Total Saving MCM	Present Stage of Ground Water Development (%)	Projected Stage of Development (%) After Interventions
URUWAN	2	2	2	2	482	482	0.11	0.11	1.36	0.11	1.47	67.20	64.51

Block	Area (Sq.Km)	Net Annual Ground Water Availability (MCM)	Existing Gross Ground Water Draft for All Uses (MCM)	Present Stage of Ground Water Extraction (%)	TOT RECH through interventions (MCM)	Tot GW Saving through interventions (MCM)	Projected Net GW Availability (MCM)	Projected Gross GW Draft (MCM)	Projected Stage of Extraction (%) After Interventions
URUWAN	168.9	57.19	38.43	67.20	0.11	1.47	57.30	36.96	**64.51**

